

8		7		6		5		4		3		2		1		
1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%. 2. ALL CAPACITANCE VALUES ARE IN MICROFARADS. 3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.												REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE	
												9	0008409760	ENGINEERING RELEASED	2017-04-05	
X891/X893 MLB Top: EVT																
LAST_MODIFICATION=Mon Apr 3 13:03:06 2017																
D	PAGE	CSA	CONTENTS	SYNC	DATE	PAGE	CSA	CONTENTS	SYNC	DATE						
	1	1	TABLE OF CONTENTS			46	61	I/O: Accessory Buck 附件供电电路	test_mlb	10/17/2016						
	2	2	SYSTEM:BOM Tables	test_mlb	10/13/2016	47	62	I/O: USB PD 快充芯片	test_mlb	10/13/2016						
	3	4	SYSTEM: Mechanical Components			48	63	I/O: Hydra USB芯片	test_mlb	10/13/2016						
	4	5	SYSTEM: Testpoints (Top) 测试点	test_mlb	10/13/2016	49	64	I/O: B2B Dock 尾插接口	test_mlb	10/13/2016						
	5	6	BOOTSTRAPPING 主板版本配置	test_mlb	10/13/2016	50	65	I/O: Interposer (Bottom) 尾插接口	test_mlb	10/13/2016						
	6	10	SOC: JTAG,USB,XTAL CPU的JTAG模块、USB模块、时钟电路	test_mlb	10/17/2016	51	80	RADIOS		06/04/2015						
	7	11	SOC: PCIE CPU的PCIE模块	test_mlb	10/17/2016											
	8	12	SOC: MIPI & ISP CPU的MIPI总线(显示和前相头)和ISP_I2C总线	test_mlb	10/13/2016											
	9	13	SOC: LPDP CPU的LPDP总线(后相头)	test_mlb	10/13/2016											
	10	14	SOC: Serial CPU的串行总线(I2C总线)	test_mlb	10/17/2016											
	11	15	SOC: GPIO & UART CPU可编程引脚+UART总线	test_mlb	10/13/2016											
	12	16	SOC: AOP CPU内部集成的传感器模块(协处理器)	test_mlb	10/17/2016											
	13	17	SOC: Power (1/3) CPU的供电	test_mlb	10/17/2016											
	C	14	18	SOC: Power (2/3) CPU的供电	test_mlb	10/17/2016										
		15	19	SOC: Power (3/3) CPU的供电	test_mlb	10/17/2016										
16		26	NAND 硬盘	test_mlb	10/13/2016											
17		27	SYSTEM POWER: PMU Bucks (1/4) 电源芯片的PWM供电输出	test_mlb	10/13/2016											
18		28	SYSTEM POWER: PMU Bucks (2/4) 电源芯片的PWM供电输出	test_mlb	10/13/2016											
19		29	SYSTEM POWER: PMU LD0s (3/4) 电源芯片的线性供电输出	test_mlb	10/13/2016											
20		30	SYSTEM POWER: PMU (4/4) 电源芯片的控制信号部分	test_mlb	11/01/2016											
21		31	SYSTEM POWER: Boost PP_VDD_BOOST 产生电路	test_mlb	10/13/2016											
22		32	SYSTEM POWER: B2B Battery 电池接口	test_mlb	10/13/2016											
23		33	SYSTEM POWER: Charger 充电芯片	test_mlb	10/13/2016											
24		34	SYSTEM POWER: Iktara 无线充电芯片													
25		35	SYSTEM POWER: B2B Cyclone + Button 无线充电接收电路	test_mlb	10/13/2016											
26		36	SENSORS 传感器	test_mlb	10/13/2016											
27		37	CAMERA: PMU (1/2) 摄像头的供电芯片	test_mlb	10/13/2016											
28		38	CAMERA: PMU (2/2) 摄像头的供电芯片	test_mlb	10/13/2016											
B		29	39	CAMERA: B2B Wide (WY) 后置摄像头接口(广角相头)	test_mlb	10/13/2016										
	30	40	CAMERA: B2B Tele (MT) 后置摄像头接口(长焦相头)	test_mlb	10/13/2016											
	31	41	CAMERA: Strobe Drivers 闪光灯驱动芯片	test_mlb	10/13/2016											
	32	42	CAMERA: B2B FCAM 前置摄像头, 听筒, 感光感距等	test_mlb	10/13/2016											
	33	43	CAMERA: B2B Strobe + Hold Button 闪光灯接口、按键、麦克风	test_mlb	10/13/2016											
	34	44	PEARL: Power	test_mlb	10/13/2016											
	35	45	PEARL: B2B Romeo + Juliet	test_mlb	10/13/2016											
	36	46	PEARL: B2B Rosaline + Misc	test_mlb	10/13/2016											
	37	47	AUDIO: CODEC (1/2) 大音频	test_mlb	10/13/2016											
	38	48	AUDIO: CODEC (2/2) 大音频	test_mlb	10/13/2016											
	39	49	AUDIO: Speaker Amp Bottom 小音频(铃声功放)		08/25/2015											
	40	50	AUDIO: Speaker Amp Top 小音频(听筒功放)		08/25/2015											
	41	51	ARC: Driver 振动芯片	test_mlb	10/13/2016											
	42	56	CG: Power Supplies - Touch & Display 显示触摸供电芯片	test_mlb	10/13/2016											
	43	57	CG: B2B Display 显示座子	test_mlb	10/13/2016											
	44	58	CG: B2B Orb & Touch 触摸座子		08/25/2015											
A	45	59	I/O: Overvoltage Cut-Off Circuit 充电电压过压保护电路	sync	01/10/2017											
	BOM:639-04583 (Ultimate)															
	BOM:639-03409 (Extreme)															
	MCO:056-04077															
8		7		6		5		4		3		2		1		
PART#	QTY	DESCRIPTION		REFERENCE DESIGNATOR(S)		CRITICAL	BOM OPTION									
051-02221	1	SCH,MLB_TOP,X891		SCH		NO	COMMON									
820-00863	1	PCB,MLB_TOP,X891		PCB		NO	COMMON									

EEEE Codes

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE FOR (MLB_TOP,639-84583,ULTIMATE)	EEEE_32WJ	NO	ULTIMATE
825-7691	1	EEEE FOR (MLB_TOP,639-83499,EXTREME)	EEEE_HP26	NO	EXTREME

SOC

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
339S00358	1	SKYE+3GB, B0, M, DEV	U1000	CRITICAL	COMMON

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339S00359	339S00358	BOM_TABLE_ALTS	U1000	DDR-H, 3G, B0
339S00360	339S00358	BOM_TABLE_ALTS	U1000	DDR-S-20, 3G, B0
339S00361	339S00358	BOM_TABLE_ALTS	U1000	DDR-S-18, 3G, B0

NAND Ultimate

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00287	1	HYNIX, 3DV3, ULTIMATE	U2600	CRITICAL	ULTIMATE

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00284	335S00287	BOM_TABLE_ALTS	U2600	TOSHIBA, 1Z, ULTIMATE
335S00285	335S00287	BOM_TABLE_ALTS	U2600	TOSHIBA, 8IC53, ULTIMATE
335S00286	335S00287	BOM_TABLE_ALTS	U2600	SANDISK, 8IC53, ULTIMATE
335S00288	335S00287	BOM_TABLE_ALTS	U2600	SAMSUNG, 3DV4, ULTIMATE

Extreme

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00240	1	HYNIX, 3DV3, EXTREME	U2600	CRITICAL	EXTREME

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00228	335S00240	BOM_TABLE_ALTS	U2600	TOSHIBA, 8IC53, EXTREME
335S00247	335S00240	BOM_TABLE_ALTS	U2600	SANDISK, 8IC53, EXTREME
335S00276	335S00240	BOM_TABLE_ALTS	U2600	SAMSUNG, 3DV4, EXTREME

Global Capacitors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00148	138S00149	BOM_TABLE_ALTS	ALL	0402-3T,10.5uF@1V, Kyocera
138S00150	138S00149	BOM_TABLE_ALTS	ALL	0402-3T,10.5uF@1V, SEMCO
138S00151	138S00149	BOM_TABLE_ALTS	ALL	0402-3T,10.5uF@1V, TY

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00143	138S00144	BOM_TABLE_ALTS	ALL	0402,16uF@1V, Kyocera
138S00163	138S00144	BOM_TABLE_ALTS	ALL	0402,16uF@1V, Taiyo

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00138	138S00139	BOM_TABLE_ALTS	ALL	0201,3uF@1V, Kyocera
138S00164	138S00139	BOM_TABLE_ALTS	ALL	0201,3uF@1V, Taiyo

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00145	138S00146	BOM_TABLE_ALTS	ALL	0402,5.1uF@3V, Kyocera
138S00165	138S00146	BOM_TABLE_ALTS	ALL	0402,5.1uF@3V, Taiyo

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00140	138S00141	BOM_TABLE_ALTS	ALL	0201,1.1uF@3V, Kyocera
138S00142	138S00141	BOM_TABLE_ALTS	ALL	0201,1.1uF@3V, SEMCO
138S00166	138S00141	BOM_TABLE_ALTS	ALL	0201,1.1uF@3V, Taiyo

Global Ferrites

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
155S00194	155S00610	BOM_TABLE_ALTS	ALL	FERR BD, 1500HM, TDK
155S00200	155S00610	BOM_TABLE_ALTS	ALL	FERR BD, 1500HM, TY

CRITICAL PART#	COMMENT
155S00610	FERR BD, 1500HM, 01005

Global R/C Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
118S0764	118S0717	BOM_TABLE_ALTS	ALL	RES, 3.92K, 0.1%, 0201
138S0648	138S0652	BOM_TABLE_ALTS	ALL	CAP,X5R,4.7UF,6.3V,0.65MM,0402,TAIYO
138S0739	138S0706	BOM_TABLE_ALTS	ALL	CAP,CER,X5R,0.22UF,20%,6.3V,20%
132S0436	132S0400	BOM_TABLE_ALTS	ALL	CAP,CER,X5R,0.22UF,20%,6.3V,C2005
138S00049	138S0831	BOM_TABLE_ALTS	ALL	CAP,CER,X5R,2.2UF,20%,6.3V,0201

CRITICAL PART#	COMMENT
118S0717	RES, 3.92K, 0.1%, 0201
138S0652	CAP,X5R,4.7UF,6.3V,0.65MM,0402
138S0706	CAP,CER,X5R,0.22UF,20%,6.3V,20%
132S0400	CAP,CER,X5R,0.22UF,20%,6.3V,01005
138S0831	CAP,CER,X5R,2.2UF,20%,6.3V,0201

Global Inductors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00710	152S00617	BOM_TABLE_ALTS	ALL	IND,MLD,0.1UH,20%,6.1A,2900HM,H=-.65,1608
152S00712	152S00620	BOM_TABLE_ALTS	ALL	IND,MLD,0.1UH,20%,7.2A,1700HM,H=0.8,2012
152S00713	152S00621	BOM_TABLE_ALTS	ALL	IND,MLD,0.47UH,20%,3.5A,5300,H=-.65,2012
152S00714	152S00622	BOM_TABLE_ALTS	ALL	IND,MLD,1.0UH,20%,2.1A,10000,H=-.65,2012
152S00716	152S00626	BOM_TABLE_ALTS	ALL	IND,MLD,1.5UH,20%,3.1A,10000,H=-.65,2012
152S00717	152S00631	BOM_TABLE_ALTS	ALL	IND,MLD,1.0UH,20%,3.1A,7000,H=0.8,2012
152S00718	152S00632	BOM_TABLE_ALTS	ALL	IND,MLD,1.0UH,20%,3.1A,6000,H=0.8,2016
152S00720	152S00640	BOM_TABLE_ALTS	ALL	IND,MLD,0.47UH,3.8A,5500,H=0.65MM,2012
152S00721	152S00641	BOM_TABLE_ALTS	ALL	IND,MLD,0.47UH,4A,4800,H=0.8MM,2012
152S00715	152S00623	BOM_TABLE_ALTS	ALL	IND,MLD,1UH,3.6A,6000,H=0.8MM,2016
152S00653	152S00651	BOM_TABLE_ALTS	ALL	IND,1.2UH,3A,2016,0.65Z
152S00649	152S00650	BOM_TABLE_ALTS	L3340, L3341	IND,0.47UH,6.6A,3225,0.8Z

CRITICAL PART#	COMMENT
152S00617	IND,MLD,0.1UH,20%,6.1A,2900HM,H=-.65,1608
152S00620	IND,MLD,0.1UH,20%,7.2A,1700HM,H=0.8,2012
152S00621	IND,MLD,0.47UH,20%,3.5A,5300,H=-.65,2012
152S00622	IND,MLD,1.0UH,20%,2.1A,10000,H=-.65,2012
152S00626	IND,MLD,1.5UH,20%,3.1A,10000,H=-.65,2012
152S00631	IND,MLD,1.0UH,20%,3.1A,7000,H=0.8,2012
152S00632	IND,MLD,1.0UH,20%,3.1A,6000,H=0.8,2016
152S00640	IND,MLD,0.47UH,3.8A,5500,H=0.65MM,2012
152S00641	IND,MLD,0.47UH,4A,4800,H=0.8MM,2012
152S00623	IND,MLD,1UH,3.6A,6000,H=0.8MM,2016
152S00651	IND,1.2UH,3A,2016,0.65Z
152S00650	IND,0.47UH,6.6A,3225,0.8Z

XTAL Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
197S0612	197S0446	BOM_TABLE_ALTS	Y1000	XTAL, 24M, 1612

CRITICAL PART#	COMMENT
197S0446	XTAL, 24M, 1612

Multi-Vendor Criticals

CRITICAL PART#	COMMENT
377S0106	SUPPR, TRANS, VARISTOR, 12V, 33PF, 01005
197S0446	XTAL, 24MHZ, 30PPM, 9.5PF, 50 OHM MAX, 1612
155S0576	FERR BD,10 OHM,50%,750MA,0.07 DCR,01005
155S00160	FLTR,NOISE,65 OHMZ,3.40MM,0.7-20HZ,0005
138S0979	CAP,CER,X5R,10UF,20%,10V,0402,H=0.65MM
138S0602	CAP,CER,X5R,1UF,20%,6.3V,0201
138S0603	CAP,CER,X5R,1UF,10%,25V,0402
138S0652	CAP,CER,X5R,4.7UF,20%,6.3V,H=0.65MM,0402
138S00070	CAP,X5R,4.7UF,20%,25V,0402
138S00014	CAP,CER,1UF,20%,10V,X5R,0201,H=0.30MM
132S0664	CAP,CER,0.047UF,10%,25V,X5R,0201
132S0663	CAP,CER,X5R,1UF,10%,25V,0402
132S06534	CAP,CER,X5R,0.1UF,10%,25V,0201
132S0436	CAP,CER,X5R,0.22UF,20%,6.3V,01005
132S0396	CAP,CER,X5R,1000PF,10%,10V,01005
132S0316	CAP,CER,X5R,0.1UF,20%,6.3V,01005
132S0304	CAP,CER,X5R,0.22UF,20%,6.3V,0201
132S0296	CAP,CER,X5R,1000PF,10%,6.3V,01005
132S0318	CAP,CER,X5R,820PF,10%,10V,01005

CRITICAL PART#	COMMENT
132S0288	CAP,CER,X5R,0.1UF,10%,10V,0201
132S0275	CAP,CER,X5R,470PF,10%,10V,01005
132S0249	CAP,CER,X7R,220PF,10%,10V,01005
132S0245	CAP,CER,X5R,0.01UF,10%,6.3V,01005
132S00093	CAP,X5R,0.022UF,20%,6.3V,01005
132S00025	CAP,CER,X5R,0.047UF,20%,6.3V,01005
132S00008	CAP,CER,0.1UF,10%,50V,X7R,0402
131S00083	CAP,CER,NP0/C0G,220PF,2%,50V,0201
131S00004	CAP,CER,27PF,5%,C0G,25V,0201
131S00307	CAP,CER,NP0/C0G,100PF,5%,10V,01005
131S0225	CAP,CER,NP0/C0G,15PF,5%,15V,01005
131S0223	CAP,CER,NP0/C0G,27PF,5%,10V,01005
131S0220	CAP,CER,NP0/C0G,12PF,5%,10V,01005
131S0216	CAP,CER,NP0/C0G,47PF,5%,10V,01005
131S00053	CAP,CER,C0G,220PF,5%,10V,01005
118S00068	RES,MF,1.3 MOHM,1%,2000PPM,1/20W,0201
117S0055	RES,MF,1/20W,2M OHM,5,0201,SMD
107S0257	THERMISTOR,NTC,10K OHM,1%,B=3435,01005

Soft-Term Cap Sub BOMs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00155	1	SUBBOM,MLB,TOP,CAP,TYPICAL,XB01	SUBBOM_CAP	CRITICAL	COMMON

Agnes Input

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	4	CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA	C2970,C2971,C2980,C2981	CRITICAL	SOFT_CAP
138S0831	4	CAP,TYPICAL,2.2UF,6.3V,0201,MURATA	C2970,C2971,C2980,C2981	CRITICAL	TYPICAL_CAP

Agnes Output

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	9	CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA	C3000,C3001,C3002,C3003,C3004,C3005,C3006,C3007,C3008,C3009	CRITICAL	SOFT_CAP
138S0831	9	CAP,TYPICAL,2.2UF,6.3V,0201,MURATA	C3000,C3001,C3002,C3003,C3004,C3005,C3006,C3007,C3008,C3009	CRITICAL	TYPICAL_CAP

Sensors

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	2	CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA	C3602,C3622	CRITICAL	SOFT_CAP
138S0831	2	CAP,TYPICAL,2.2UF,6.3V,0201,MURATA	C3602,C3622	CRITICAL	TYPICAL_CAP

RCAM B2Bs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	3	CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA	C3909,C3925,C4025	CRITICAL	SOFT_CAP
138S0831	3	CAP,TYPICAL,2.2UF,6.3V,0201,MURATA	C3909,C3925,C4025	CRITICAL	TYPICAL_CAP

Strobe B2B

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	1	CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA	C4303	CRITICAL	SOFT_CAP
138S0831	1	CAP,TYPICAL,2.2UF,6.3V,0201,MURATA	C4303	CRITICAL	TYPICAL_CAP

Audio

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	2	CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA	C4809,C4805	CRITICAL	SOFT_CAP
138S0831	2	CAP,TYPICAL,2.2UF,6.3V,0201,MURATA	C4809,C4805	CRITICAL	TYPICAL_CAP

Pearl B2B

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	1	CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA	C4613	CRITICAL	SOFT_CAP
138S0831	1	CAP,TYPICAL,2.2UF,6.3V,0201,MURATA	C4613	CRITICAL	TYPICAL_CAP

Acorn

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00160	2	CAP,SOFT-TERM,10UF,10V,0402,MURATA	C5641,C5653	CRITICAL	SOFT_CAP
138S0979	2	CAP,TYPICAL,10UF,10V,0402,MUR/KY0	C5641,C5653	CRITICAL	TYPICAL_CAP

CODEC

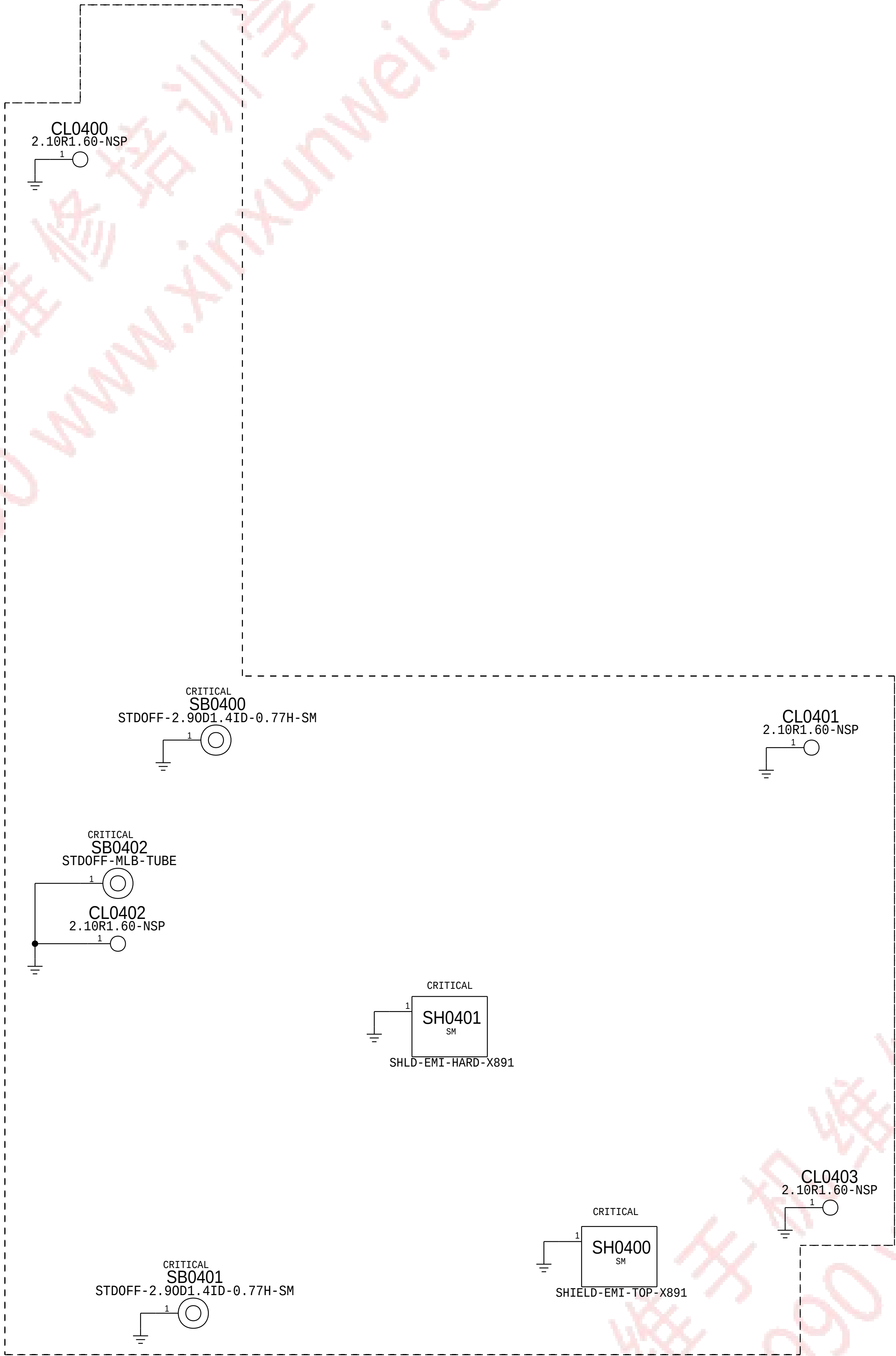
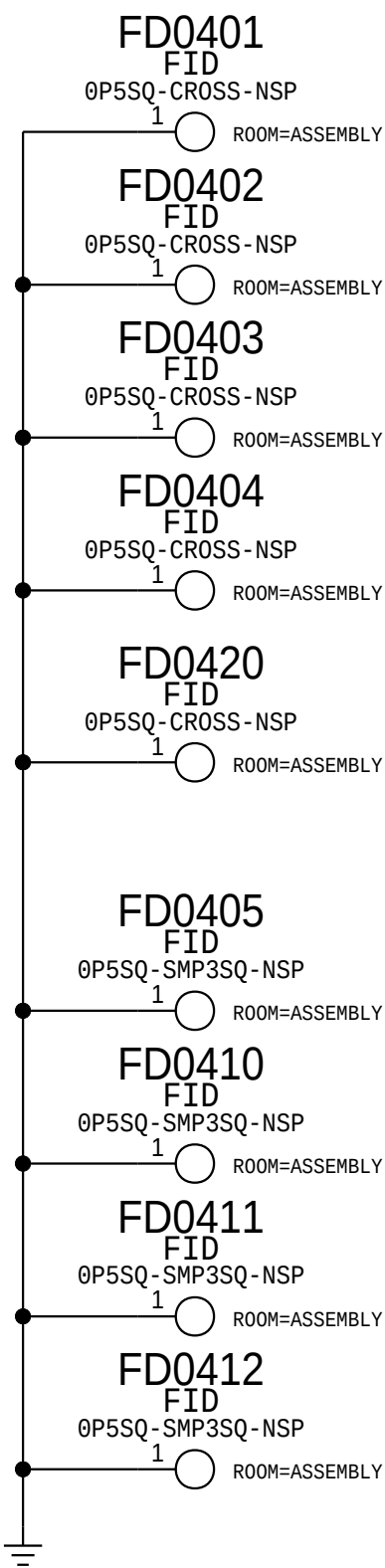
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00160	2	CAP,SOFT-TERM,10UF,10V,0402,MURATA	C4811,C4808	CRITICAL	SOFT_CAP
138S0979	2	CAP,TYPICAL,10UF,10V,0402,MUR/KY0	C4811,C4808	CRITICAL	TYPICAL_CAP

Ansel

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00160	1	CAP,SOFT-TERM,10UF,10V,0402,MURATA	C3710	CRITICAL	SOFT_CAP
138S0979	1	CAP,TYPICAL,10UF,10V,0402,MUR/KY0	C3710	CRITICAL	TYPICAL_CAP

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00156	685-00155	BOM_TABLE_ALTS	SUBBOM_CAP	SUBBOM,MLB,TOP,CAP,SOFT,XB01

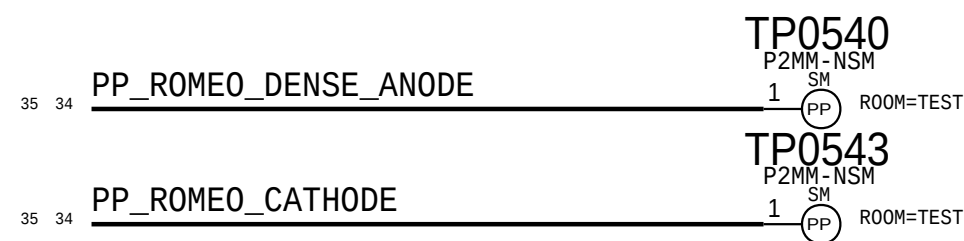
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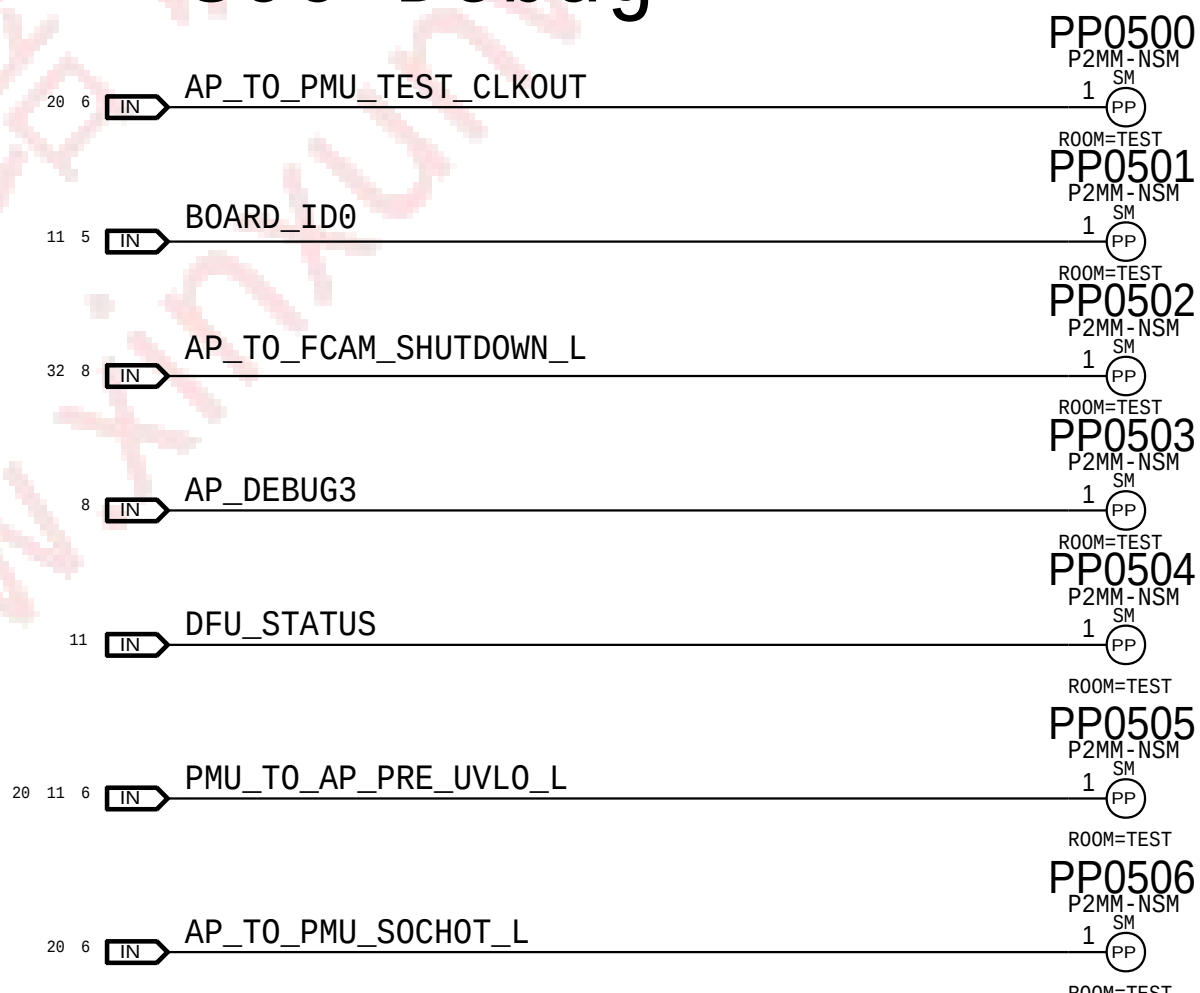


Test Points

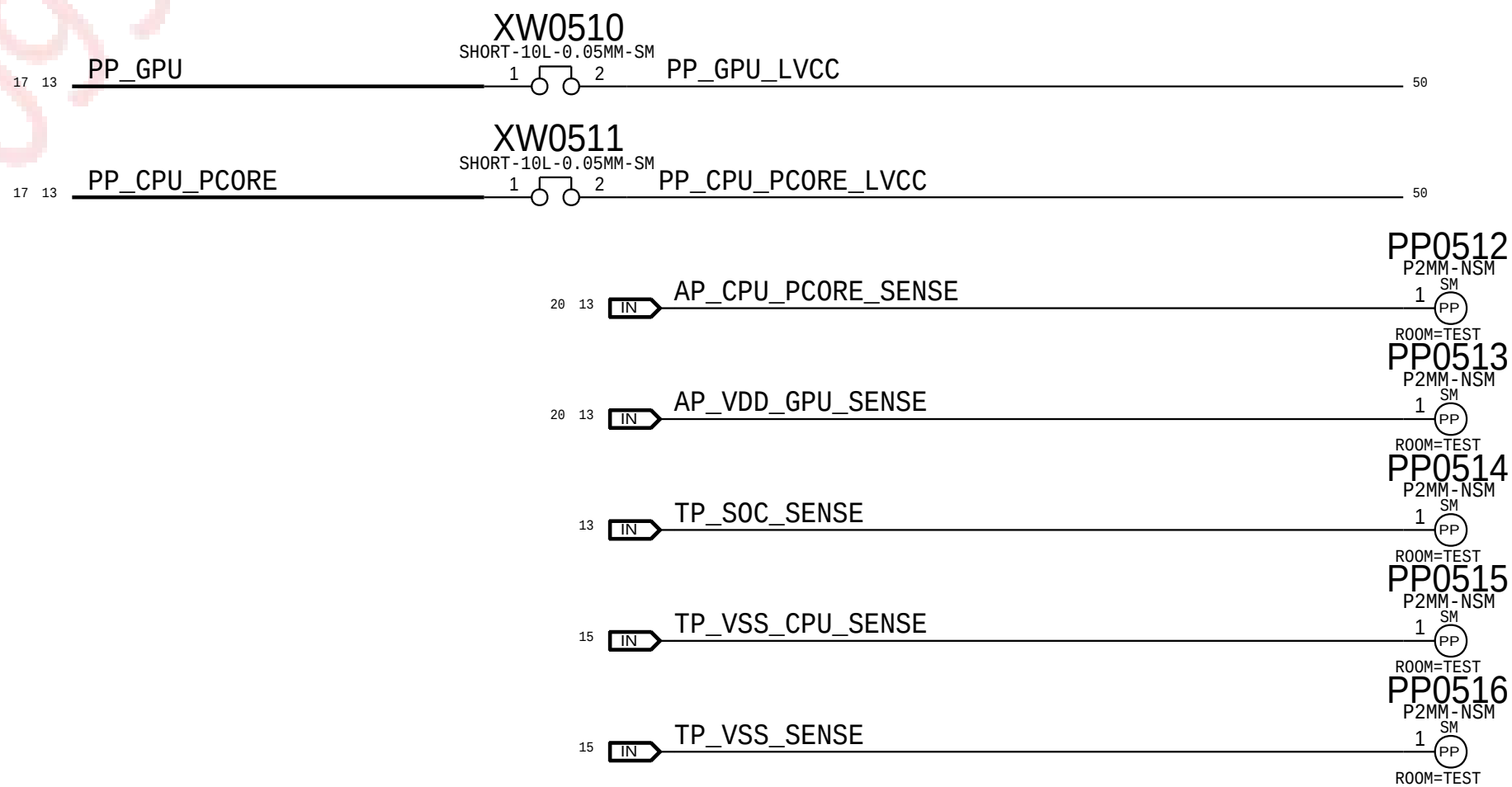
POWER



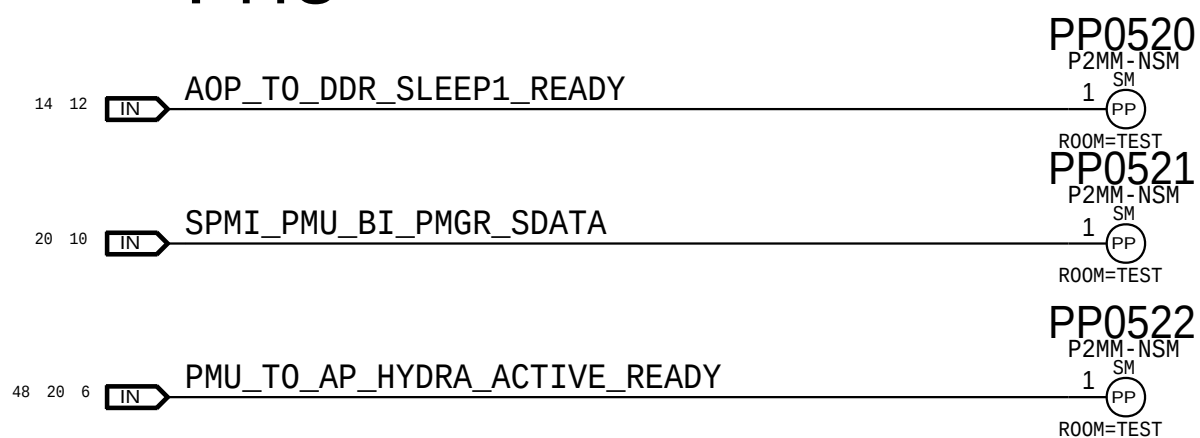
SOC Debug



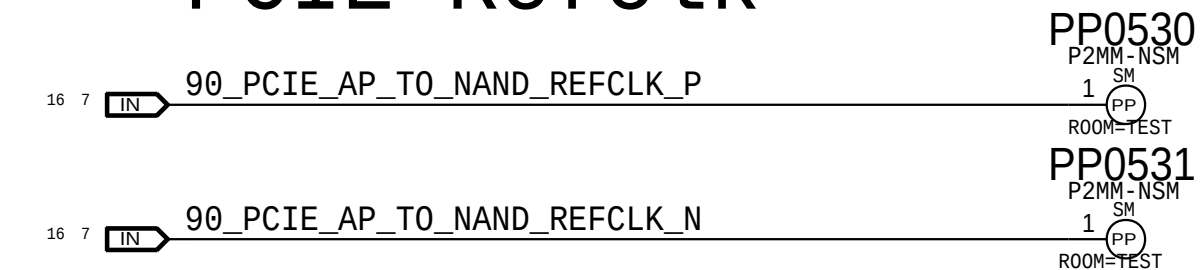
SOC CPU/GPU



PMU

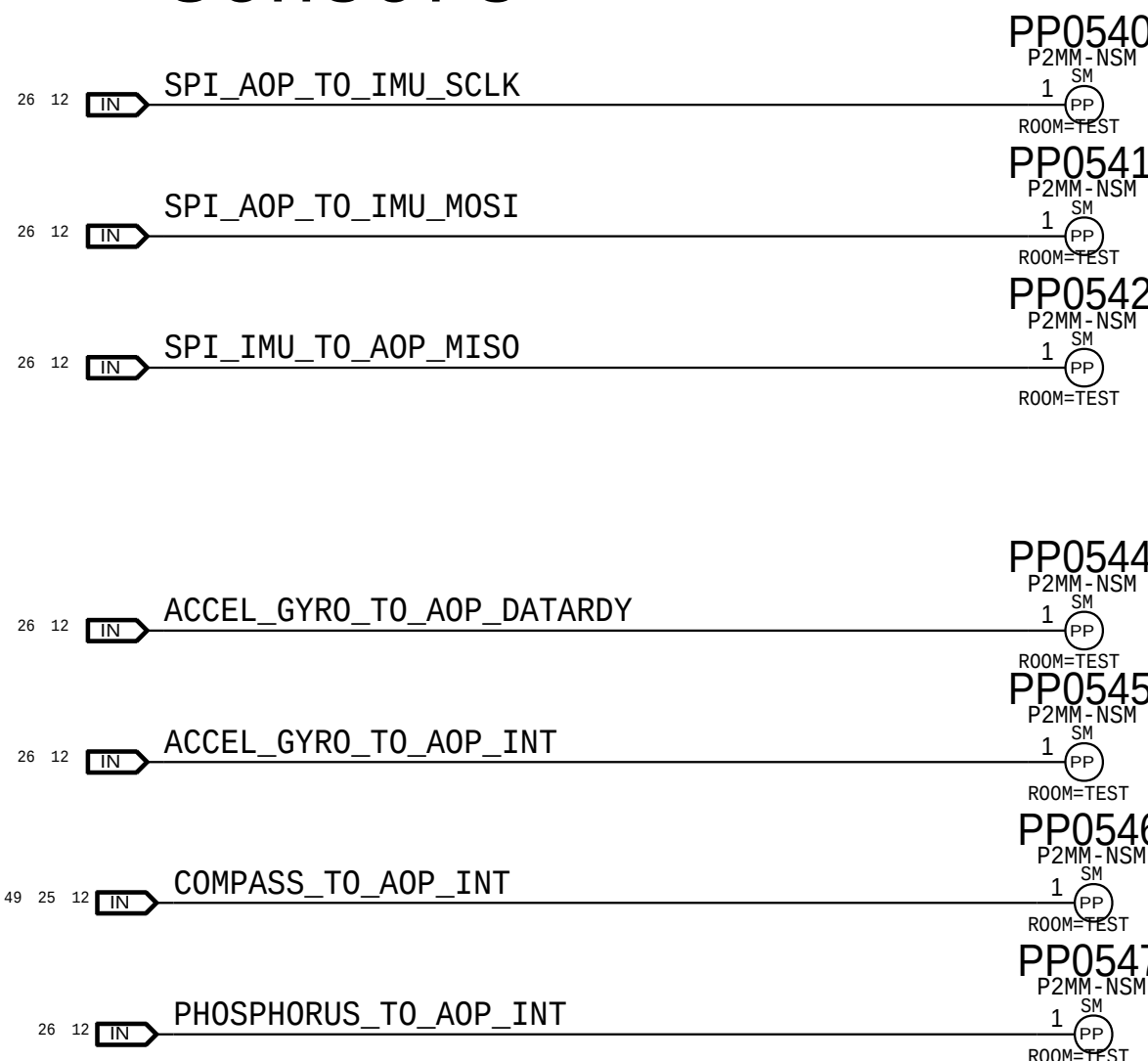


PCIE Refclk

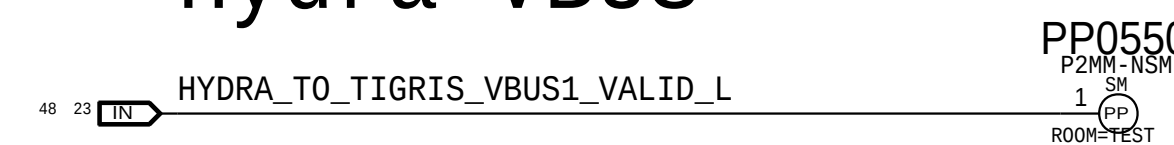


Probe Points

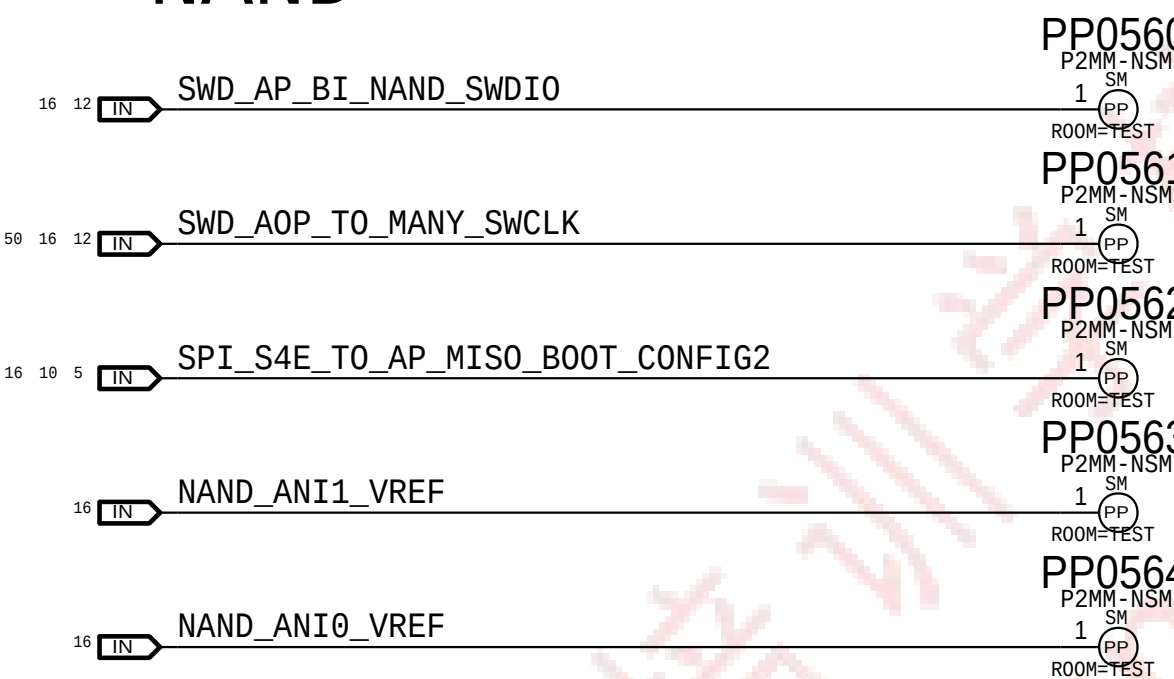
Sensors



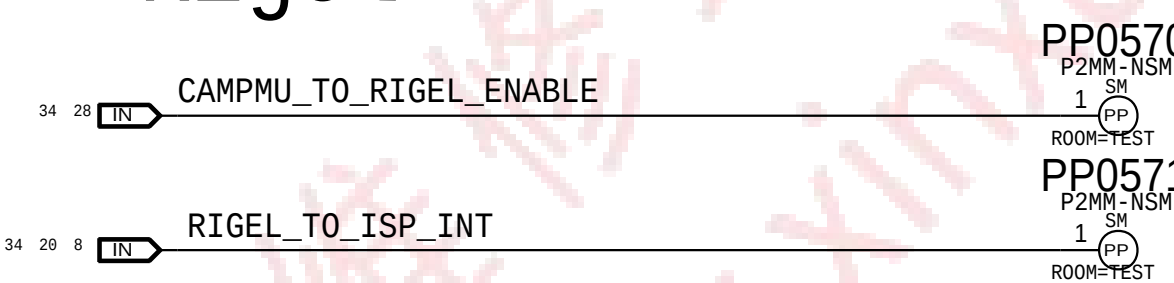
Hydra VBUS



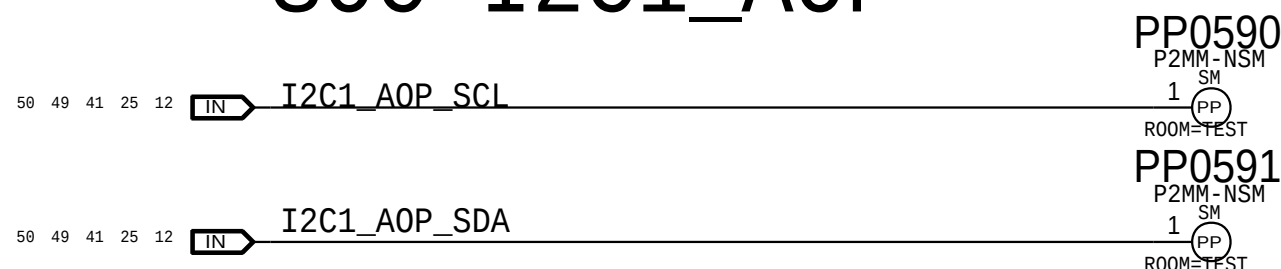
NAND



Rigel



SOC I2C1_AOP



CCG2



BOOTSTRAPPING:BOARD REV
BOARD ID
BOOT CONFIG



SELECTED -->

Board Rev[3:0]				
Float=Low I PU=High				
	3	2	1	0
Proto 1	1	1	1	1
Proto 1 (DOE6)	1	1	1	0
Proto 2	1	1	0	1
Proto 3	1	1	0	0
EVT	1	0	1	1
(Spare)	1	0	1	0
Carrier	0	1	1	1
(Spare)	0	1	1	0
DVT	0	0	1	1
(Spare)	0	0	1	0
(Spare)	0	0	0	1
PVT	0	0	0	0

SELECTED -->

Board ID[4:0]					
Float=Low I PU=High					
	4	3	2	1	0
	X	Baseband	Form Factor	MLB/Dev	
D20 MLB	0	0	0	1	0
D20 Dev	0	0	0	1	1
D21 MLB	0	0	1	0	0
D21 Dev	0	0	1	0	1
D22 MLB	0	0	1	1	0
D22 Dev	0	0	1	1	1
D201 MLB	0	1	0	1	0
D201 Dev	0	1	0	1	1
D211 MLB	0	1	1	0	0
D211 Dev	0	1	1	0	1
D221 MLB	0	1	1	1	0
D221 Dev	0	1	1	1	1

D221 Baseband Selected on RF Board

SELECTED -->

Boot Config[2:0]			
Float=Low I PU=High			
	2	1	0
SPI0	0	0	0
SPI0 Test Mode	0	0	1
SPI0 NAND	0	1	0
SPI0 NAND Test	0	1	1
Slow SPI0 Test	1	1	0

USB_DP AY6 90_USB_AP_DATA_P 48
USB_DM BA6 90_USB_AP_DATA_N 48 USB到CPU数据线

USB_VBUS AV7 USB插入检测 USB_VBUS_DETECT 23

USB_ID AW6 X NC

USB_REXT AU8 AP_USB_REXT 6 USB基准电源

CPU_TRIGGER0 AT22 PMU_TO_AP_THROTTLE_PCORE_L 20
CPU_TRIGGER1 AW21 PMU_TO_AP_THROTTLE_ECORE_L 20 PMU到CPU节能调节控制信号

GPU_TRIGGER0 AD2 PMU_TO_AP_THROTTLE_GPU0_L 20
GPU_TRIGGER1 AD3 PMU_TO_AP_THROTTLE_GPU1_L 20

SOCHOT1 A30 AP_TO_PMU_SOCHOT_L 4 CPU到电源复位信号+温控

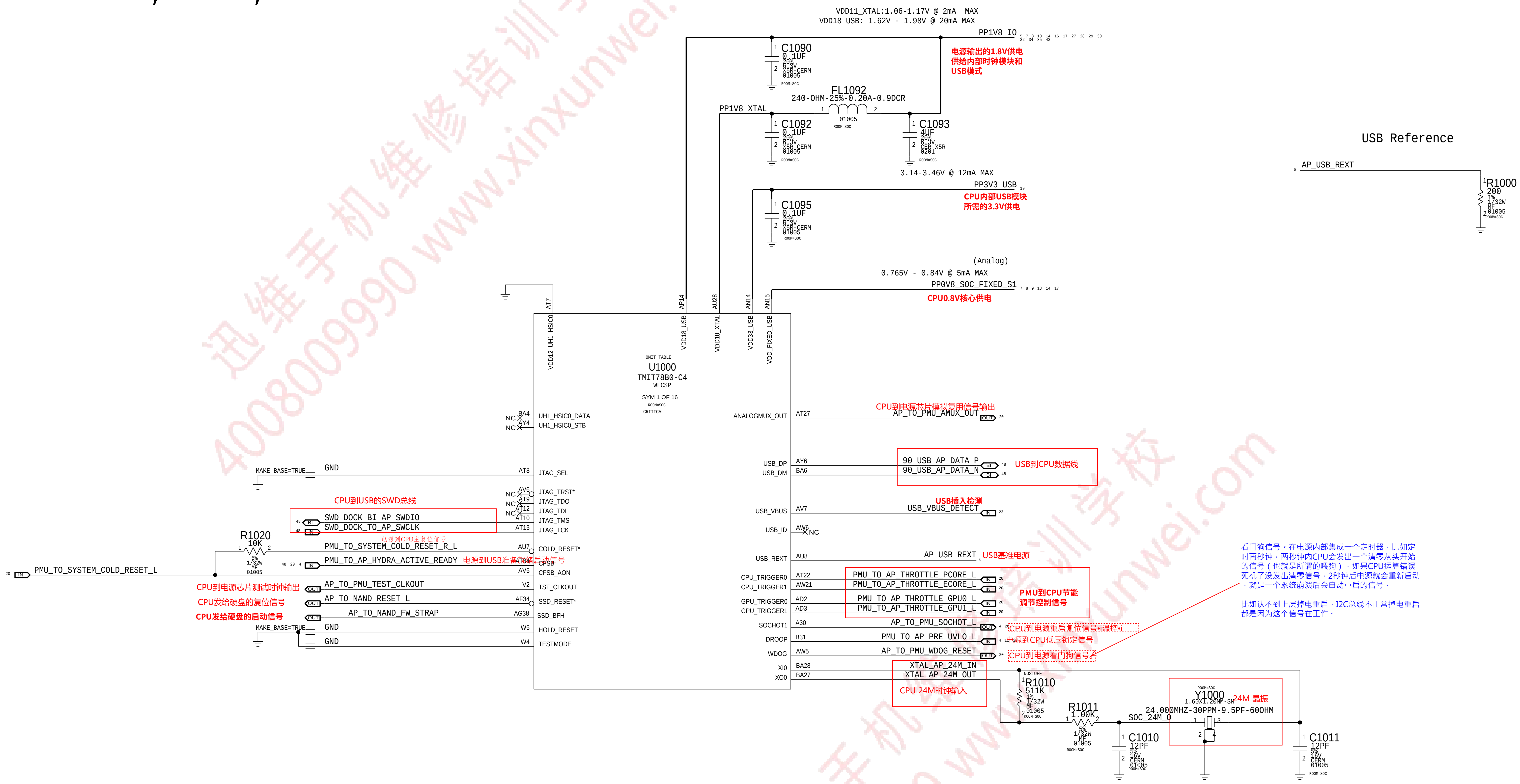
DROOP B31 PMU_TO_AP_PRE_UVLO_L 4 电源到CPU低压锁定信号

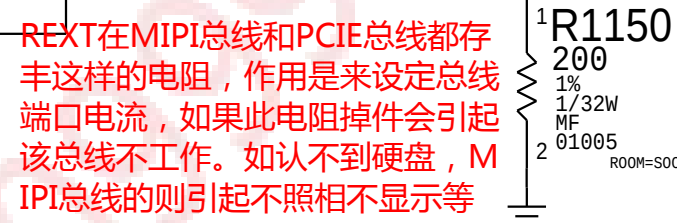
WDOG AW5 AP_TO_PMU_WDOG_RESET 20 CPU到电源看门狗信号

X10 BA28 XTAL_AP_24M_IN
X00 BA27 XTAL_AP_24M_OUT CPU 24M时钟输入

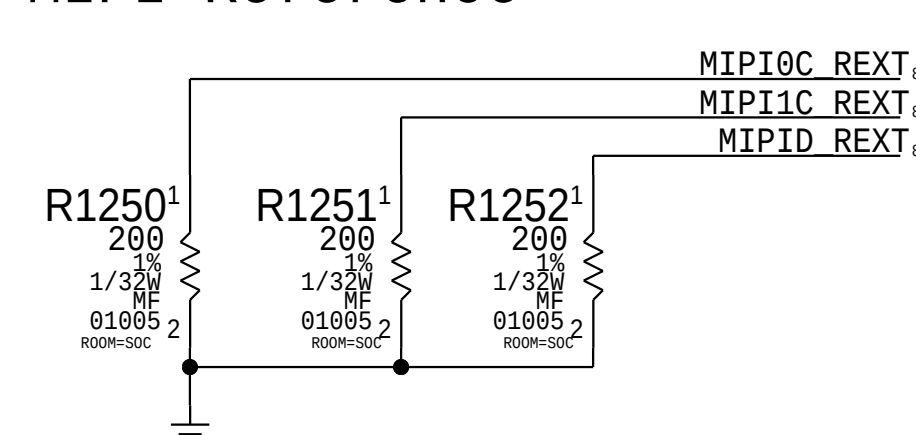
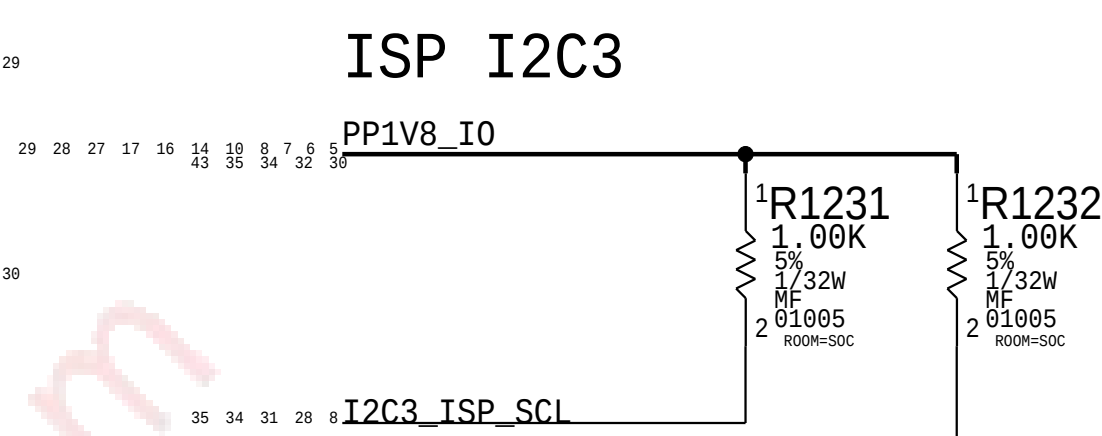
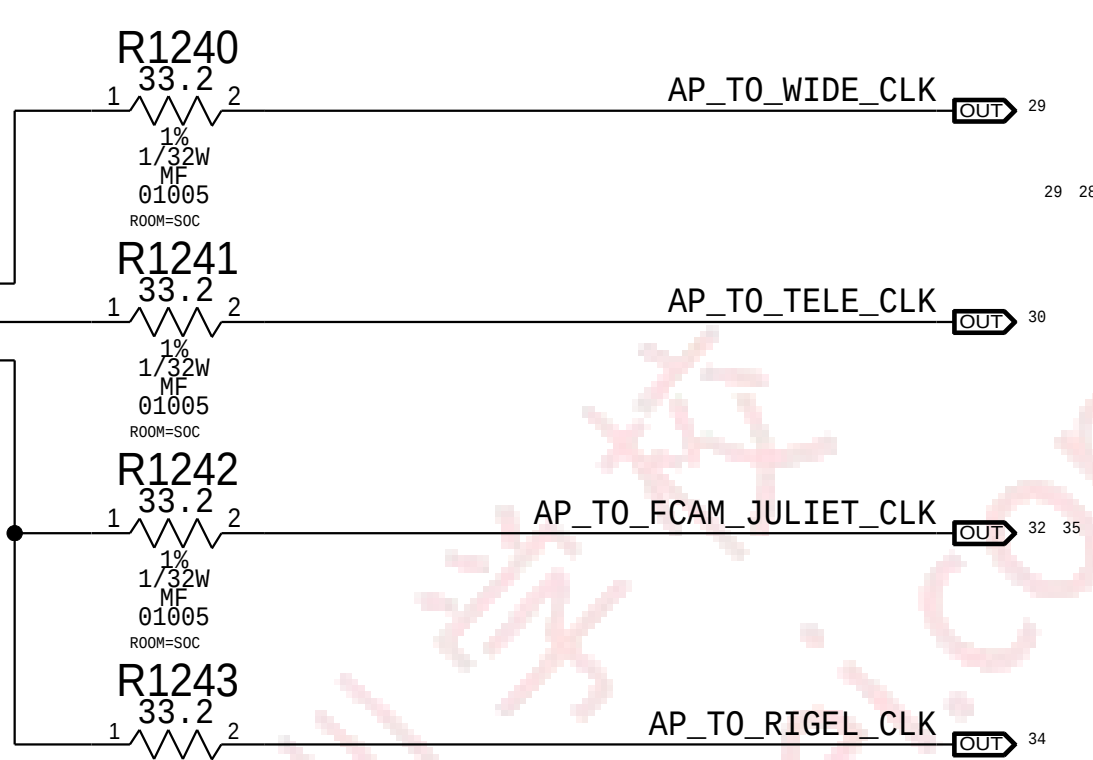
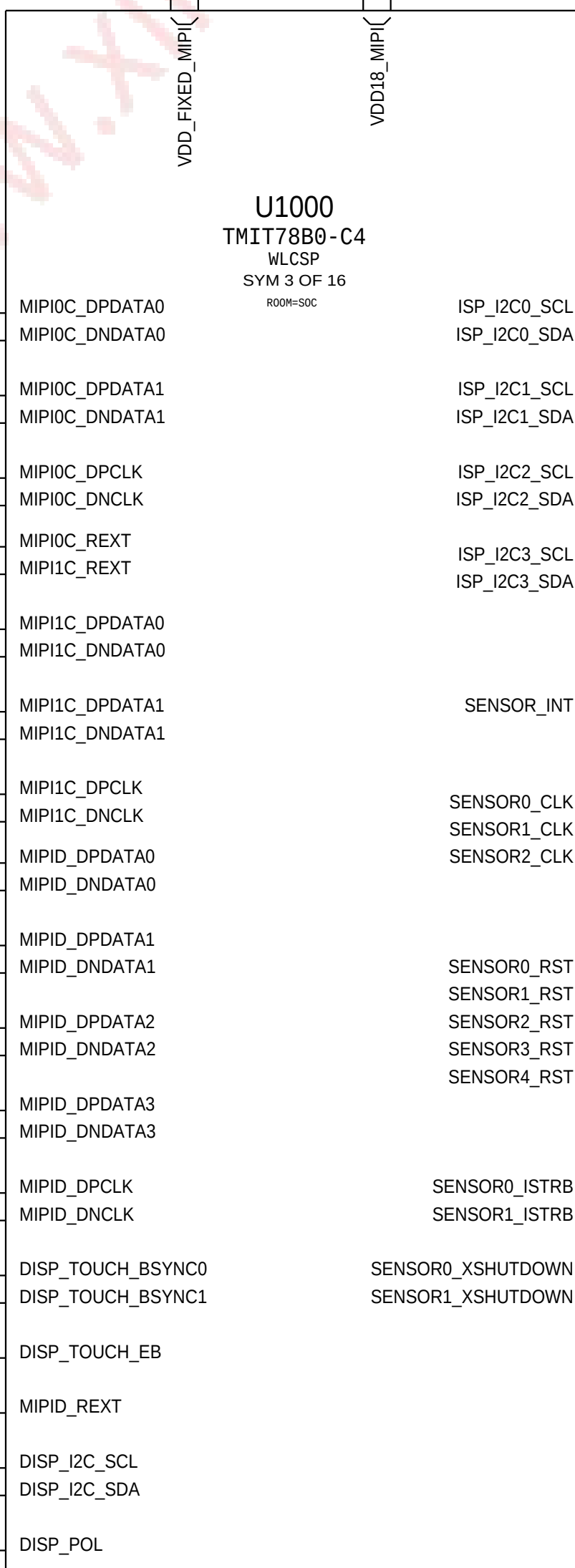
Y1000 1.60XT.25MM-SM 24M 晶振
S0C_24M_0 24.000MHZ-30PPM-9.5PF-600HM

R1010 511K
R1011 1.00K
C1010 12PF
C1011 100nF

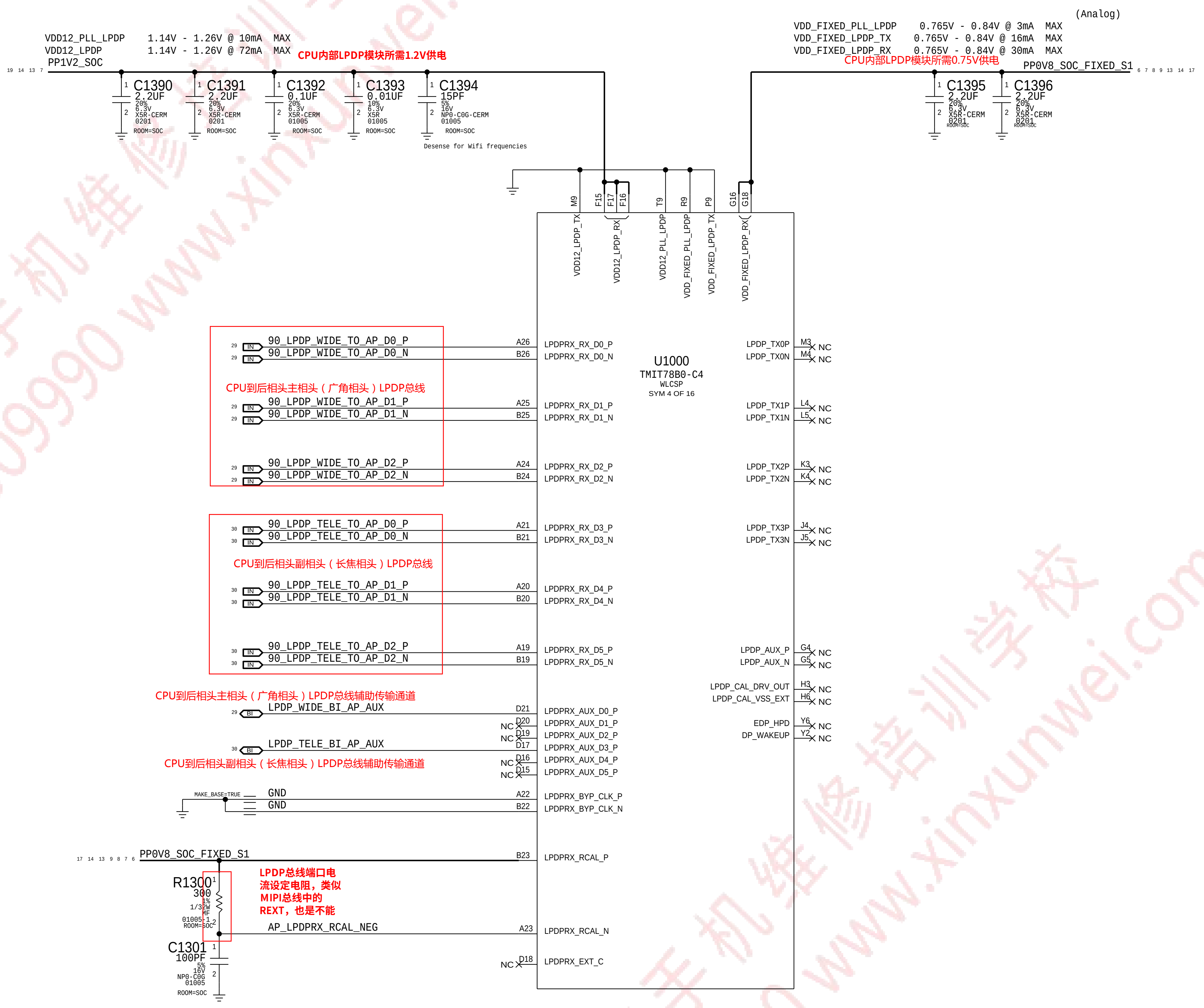




Program	Function	SOC_LANE	DISTAL_LANE	Lane Swap	Polarity Swap
XB81	Display	Clock_P	Clock_N	☐	☒
		Clock_N	Clock_P	☐	☒
		Lane_0_P	Lane_0_N	☐	☒
		Lane_0_N	Lane_0_P	☐	☒
		Lane_1_P	Lane_1_P	☐	☒
		Lane_1_N	Lane_1_N	☐	☒
		Lane_2_P	Lane_2_P	☒	☒
		Lane_2_N	Lane_2_N	☒	☒
		Lane_3_P	Lane_3_P	☒	☒
		Lane_3_N	Lane_3_N	☒	☒
		Clock_P	Clock_P	☐	☒
		Clock_N	Clock_N	☐	☒
	FCAM	Lane_0_P	Lane_0_P	☐	☒
		Lane_0_N	Lane_0_N	☐	☒
		Lane_1_P	Lane_1_N	☐	☒
		Lane_1_N	Lane_1_P	☐	☒
		Clock_P	Clock_N	☐	☒
		Clock_N	Clock_P	☐	☒
JULIET	Lane_0_P	Lane_0_P	☐	☒	
	Lane_0_N	Lane_0_N	☐	☒	
	Lane_1_P	Lane_1_P	☐	☒	
	Lane_1_N	Lane_1_N	☐	☒	
	Clock_P	Clock_N	☐	☒	
	Clock_N	Clock_P	☐	☒	



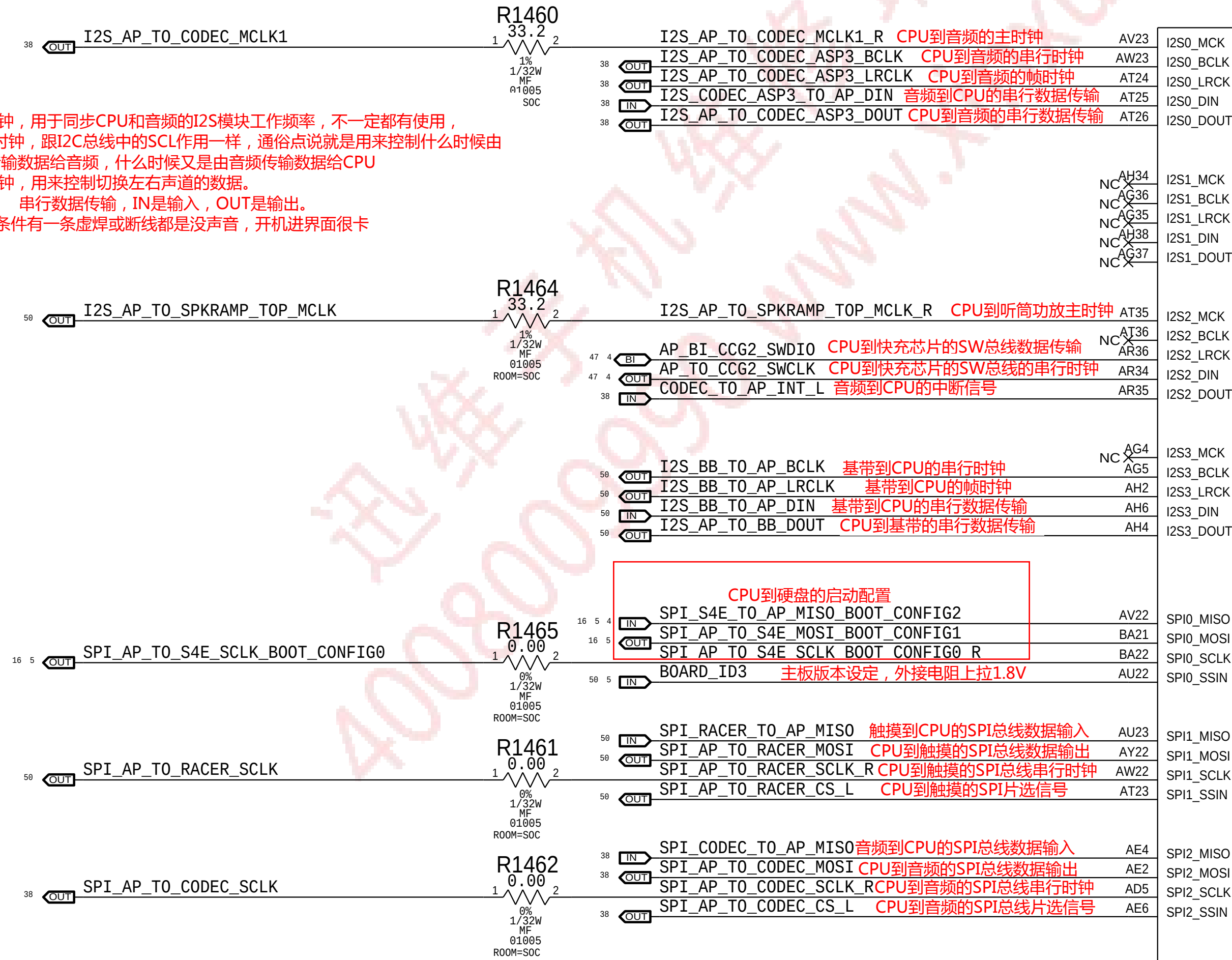
SOC - LPDP



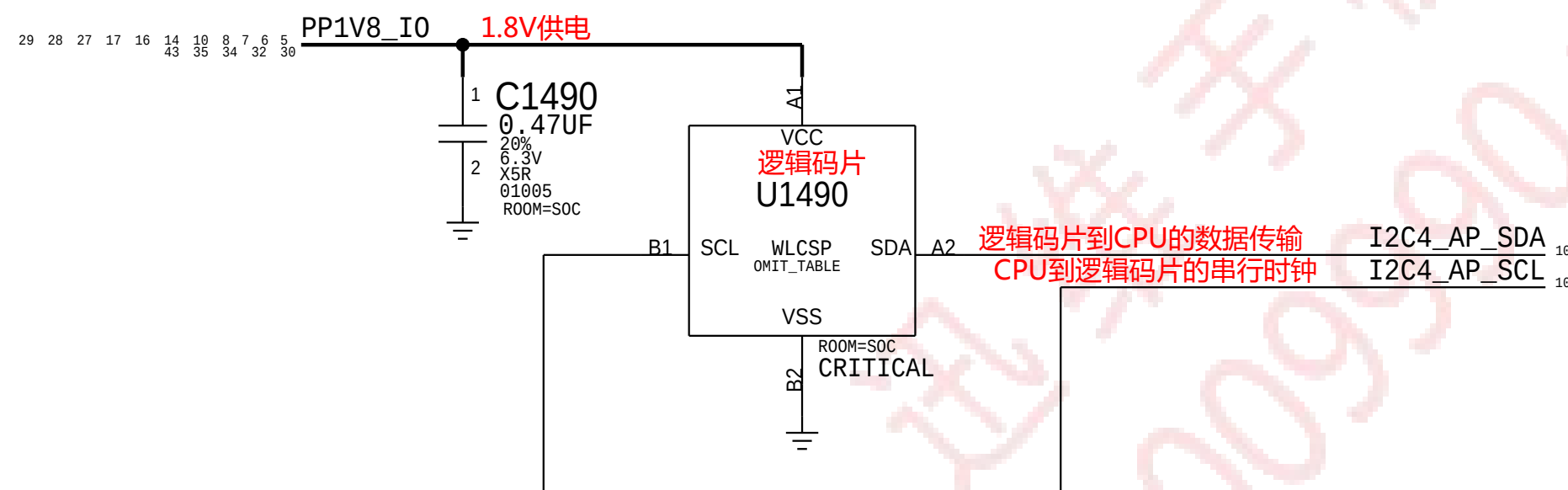
SOC - SERIAL INTERFACES

I2S总线：音频传输总线，用于芯片之音数字音频的传输，比如通话过程中，我方说的声音由麦克风采集后送给音频，音频通过I2S总线把声音发给CPU，CPU又通过I2S总线把声音发给基带，然后基带处理后发射出去。CPU到蓝牙之间也是用的I2S总线传输声间的。手机芯片之间的声音传输都是I2S

I2S总线有
MCLK：主时钟，用于同步CPU和音频的I2S模块工作频率，不一定都有使用，
BCLK：串行时钟，跟I2C总线中的SCL作用一样，通俗点说就是用来控制什么时候由CPU传输数据给音频，什么时候又是由音频传输数据给CPU
LRCLK：帧时钟，用来控制切换左右声道的数据。
DIN DOUT： 串行数据传输，IN是输入，OUT是输出。
到音频的这5条件有一条虚焊或断线都是没声音，开机进界面很卡

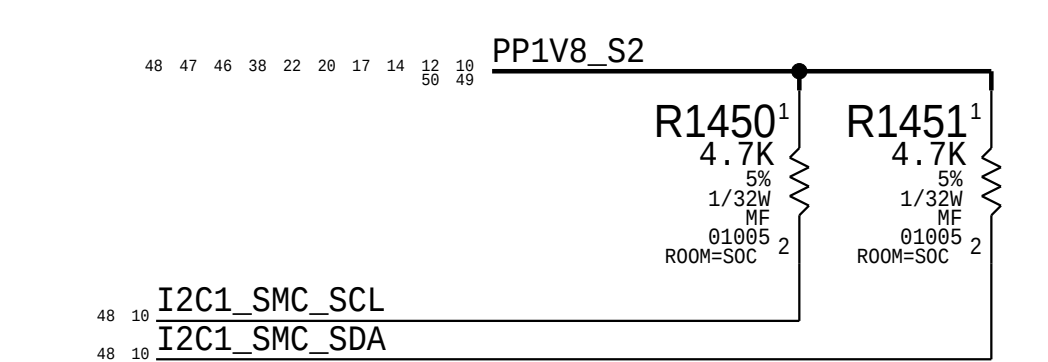
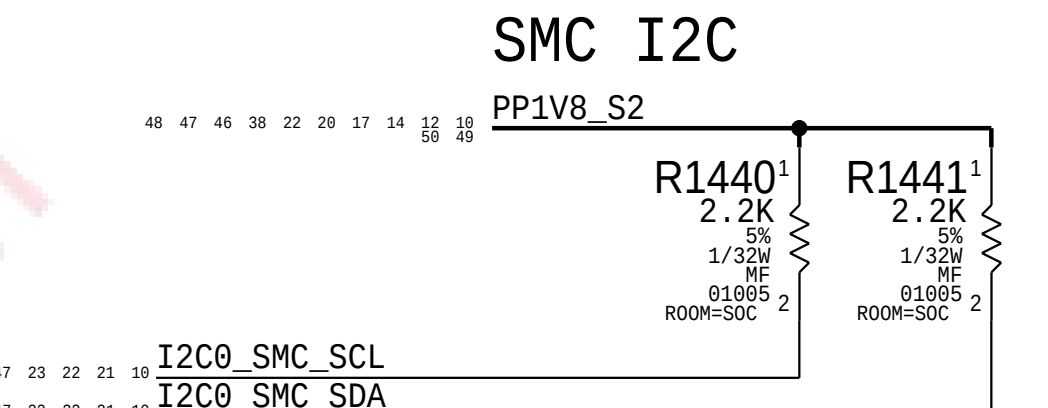
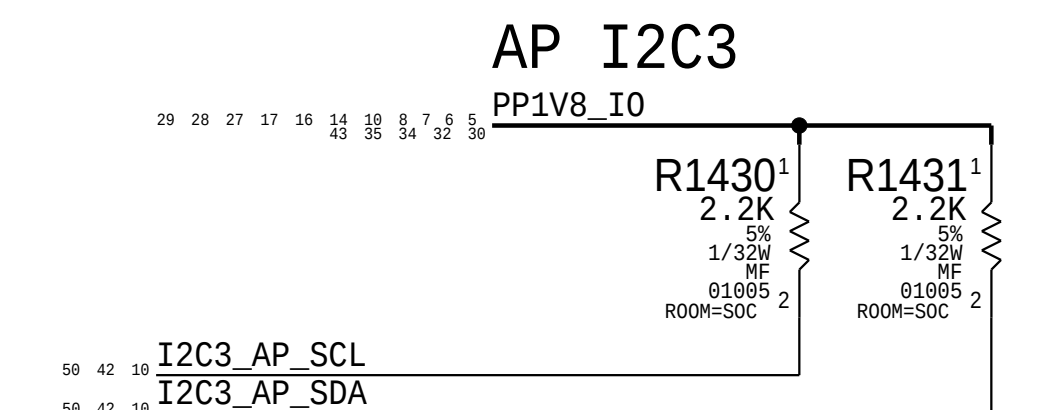
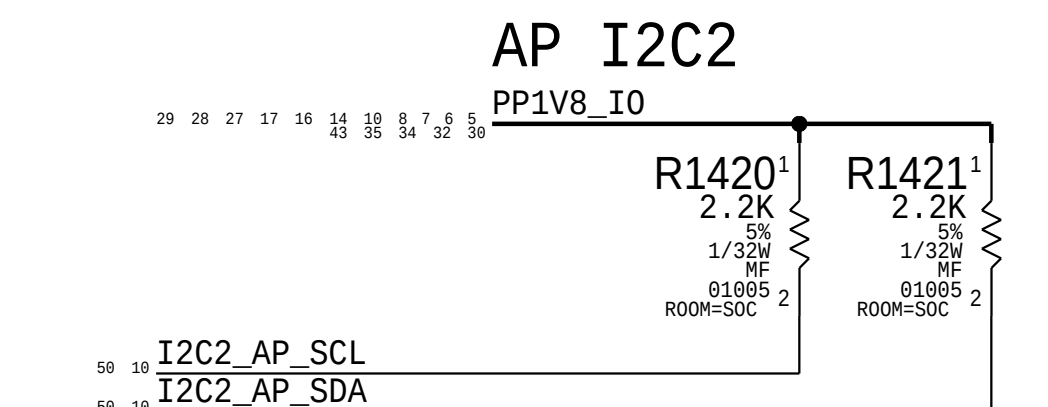
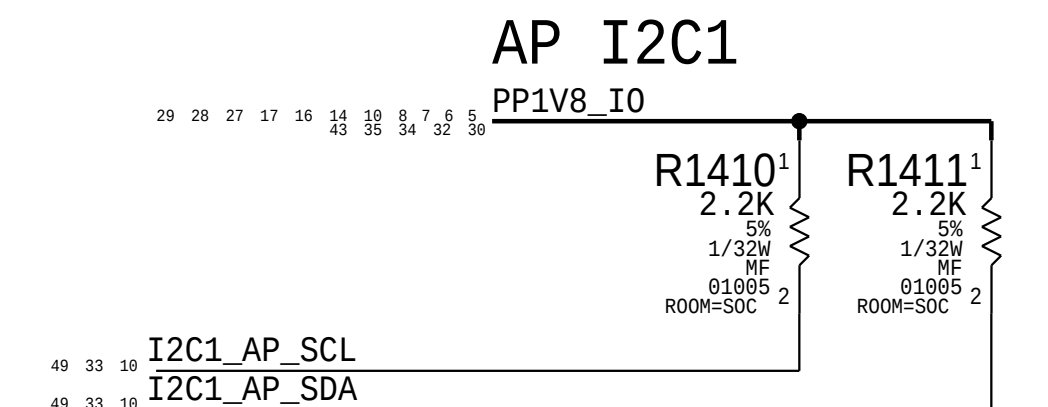
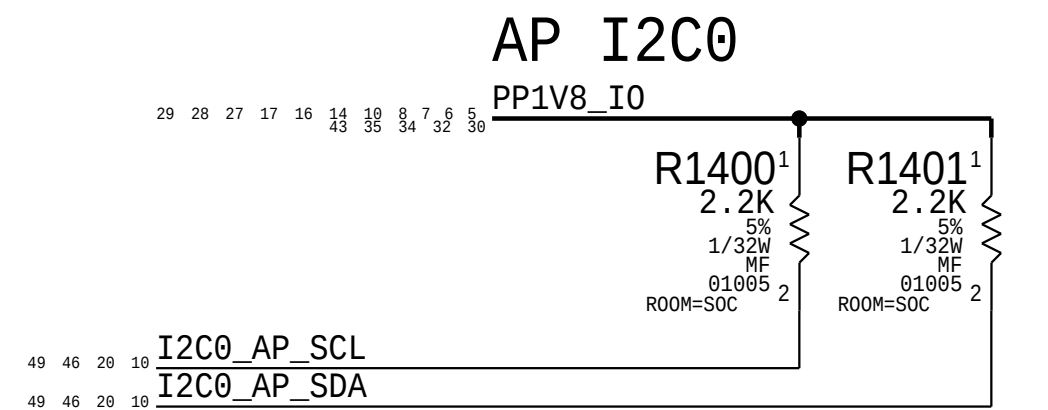


SPI: Route as Daisy-Chain. No T's Allowed
Place series terminations close to SoC Pins



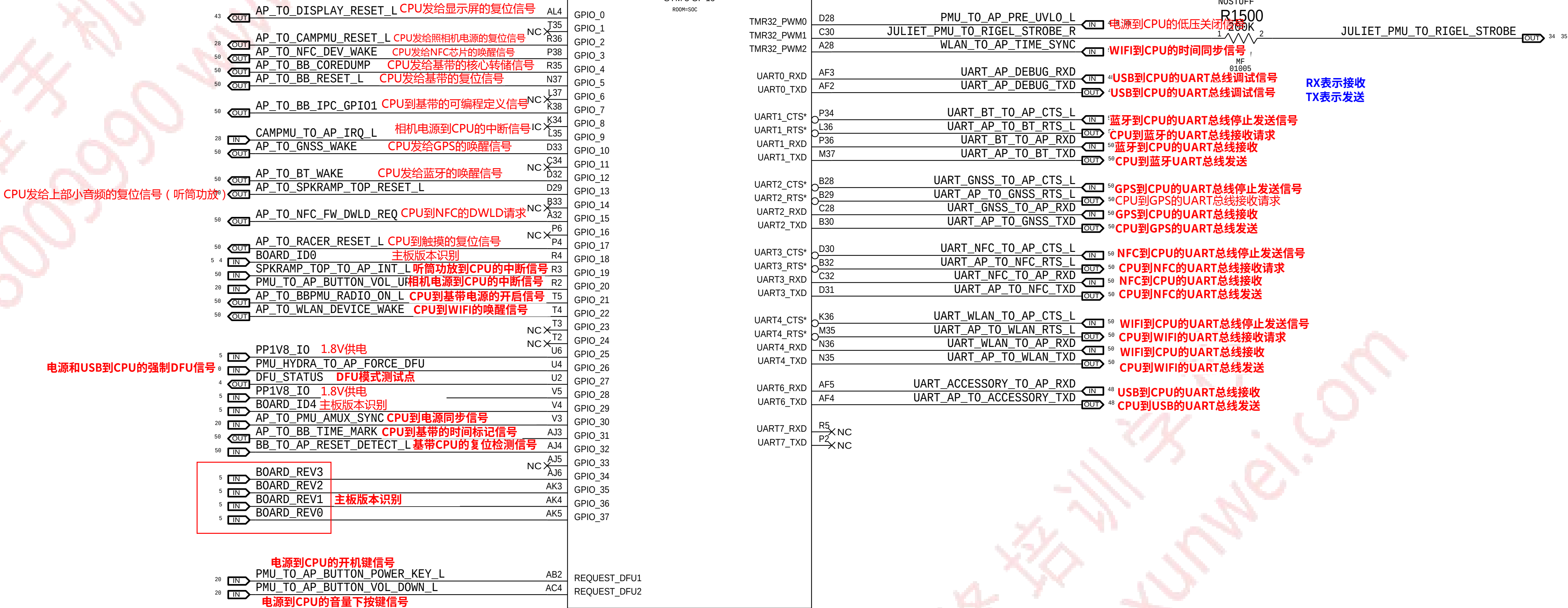
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00234	1	WLCSP	U1490	CRITICAL	COMMON

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00233	335S00234	BOM_TABLE_ALTS	U1490	U1490

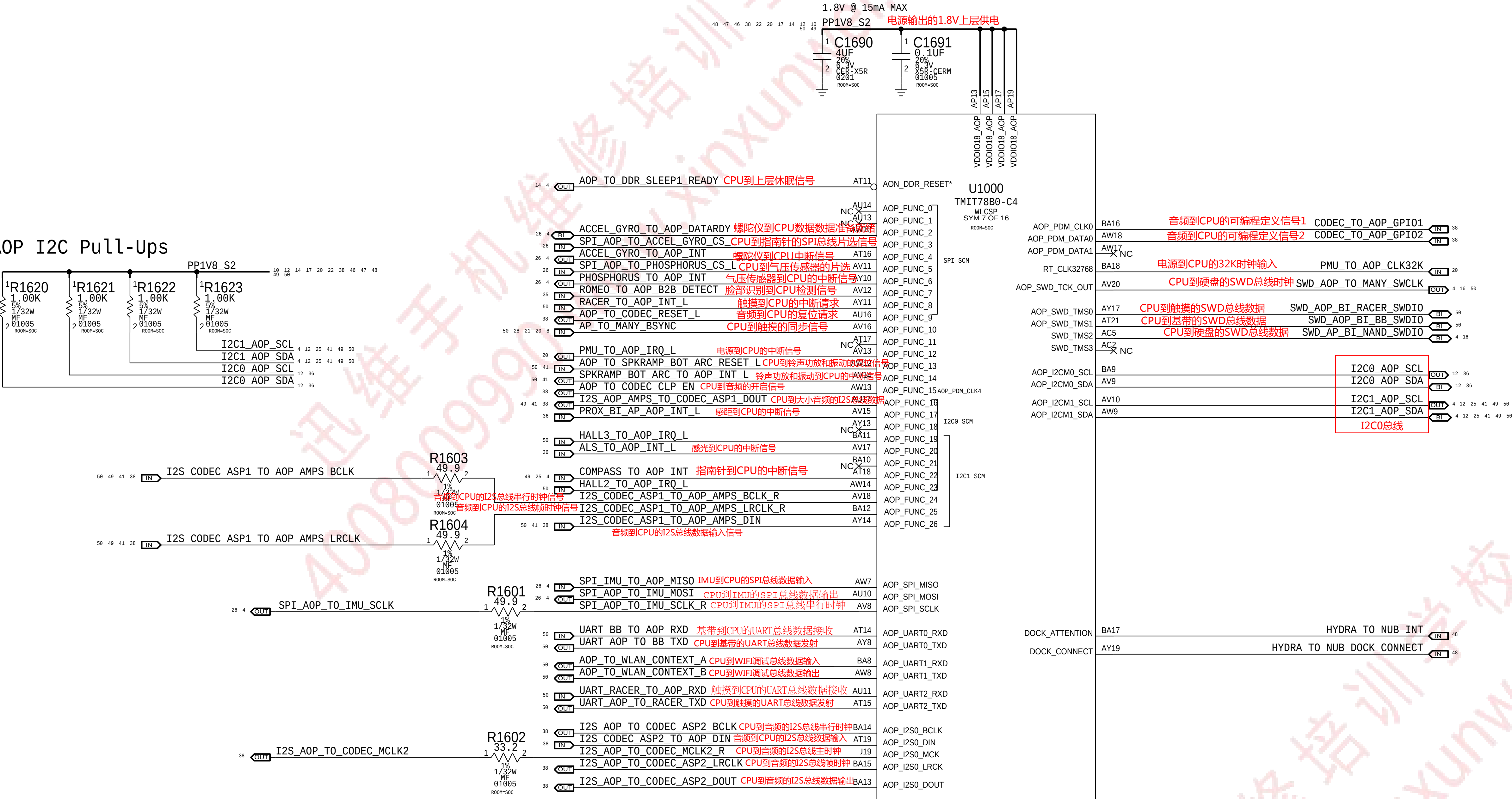


SOC - GPIO INTERFACES

DISPLAY：显示屏 RESET:复位信号，后面带L 低电平时复位
也就是说这个复位信号低电平时复位显示屏，正常工作的时候为高电平，手机中所有的复位信号，正常工作时都是高电平。
(低电平一般指0V，高电平一般是1.8V)



- I2C Pull-Ups

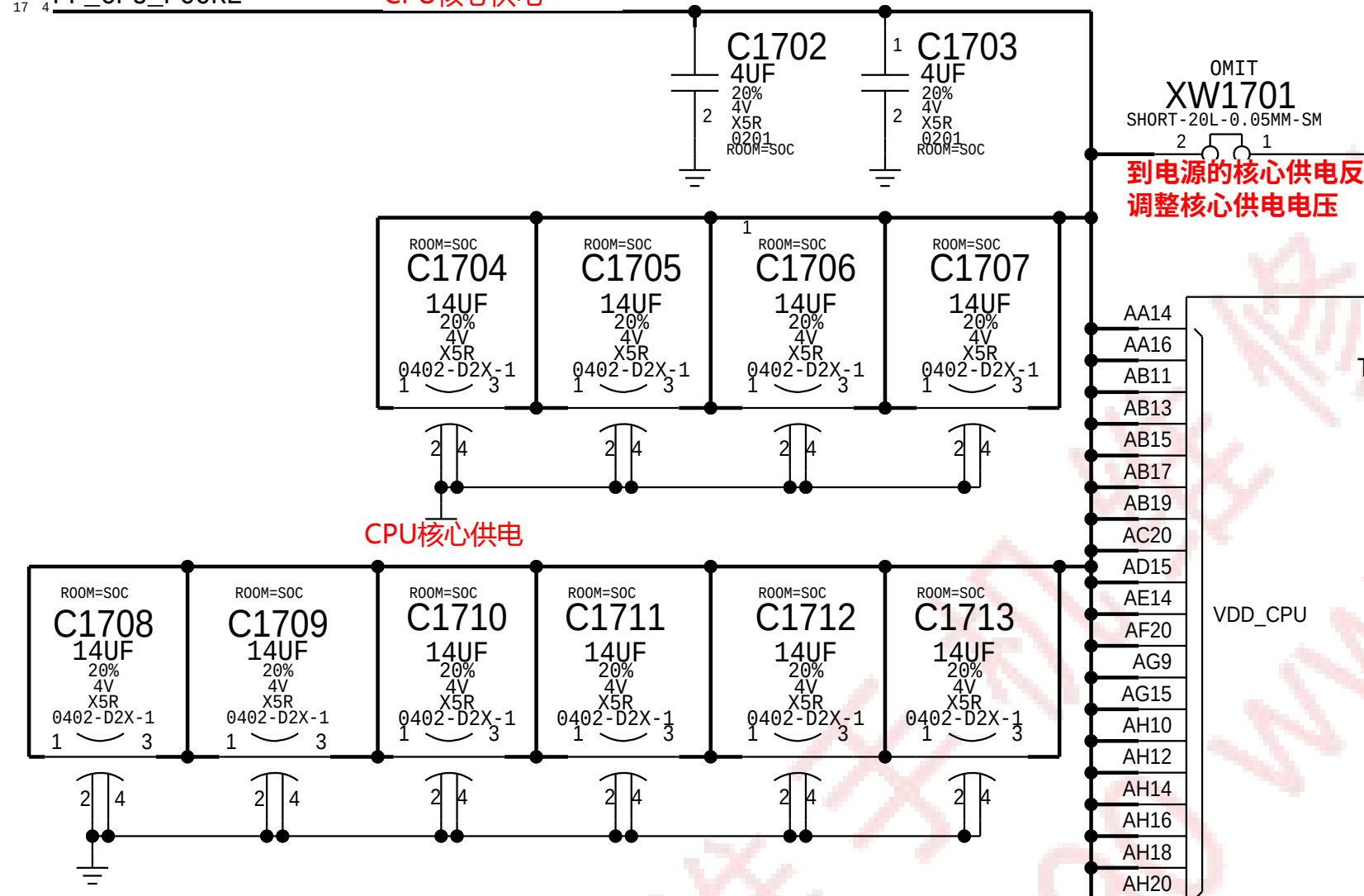


SOC - CPU, GPU & SOC RAILS

1.06V @ 11.0A MAX
0.8V @ 6A MAX
0.575V @ 2.7A MAX

PP_CPU_PCORE

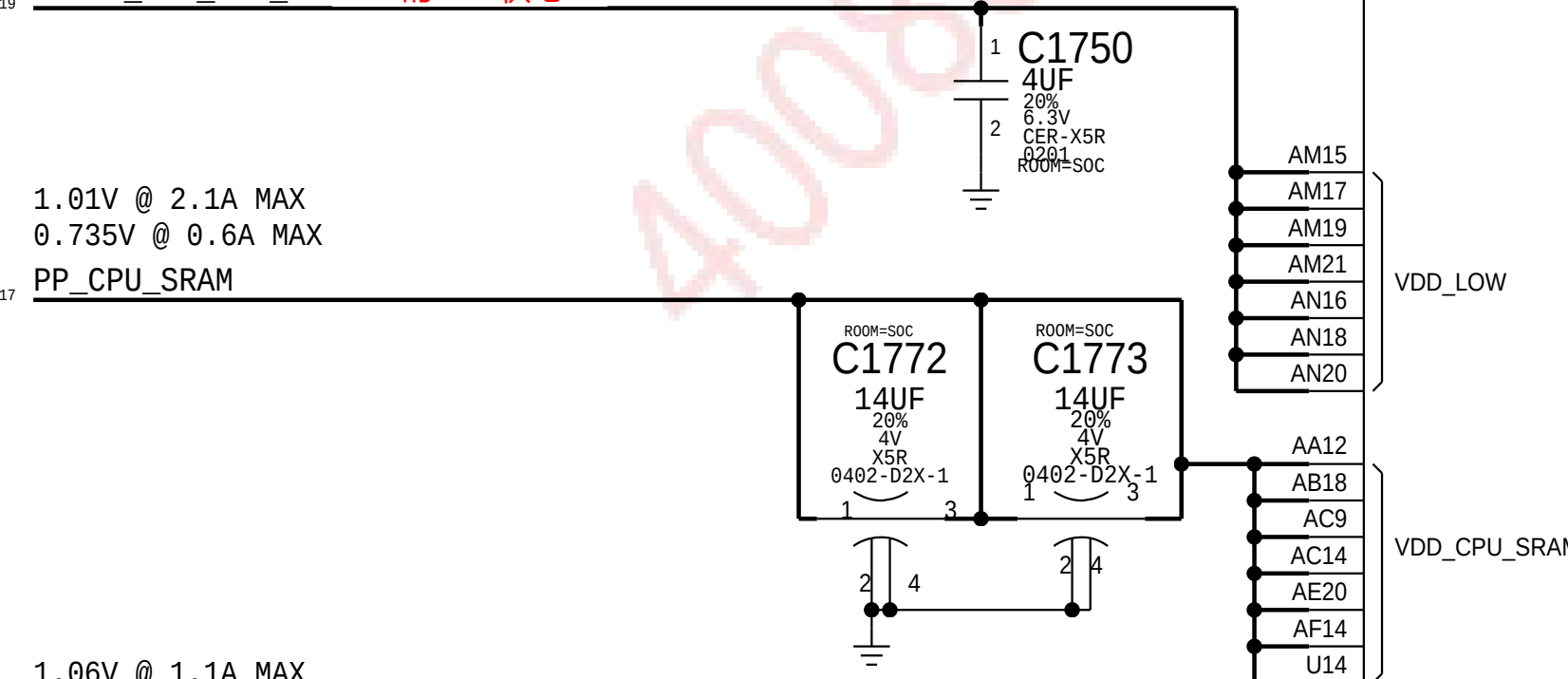
CPU核心供电



CPU核心供电

0.7V @ 75mA MAX

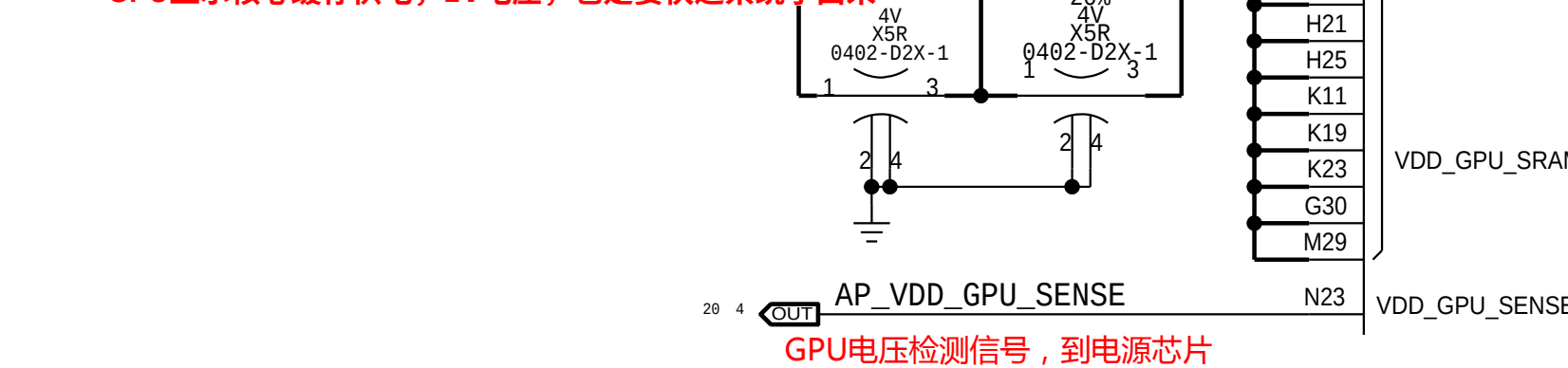
PP0V7_VDD_LOW_S2 CPU的0.7v供电



1.06V @ 1.1A MAX
0.80V @ 0.63A MAX
0.675V @ 0.19A MAX

PP_GPU_SRAM

GPU显示核心缓存供电, 1V电压, 也是要快进系统才出来

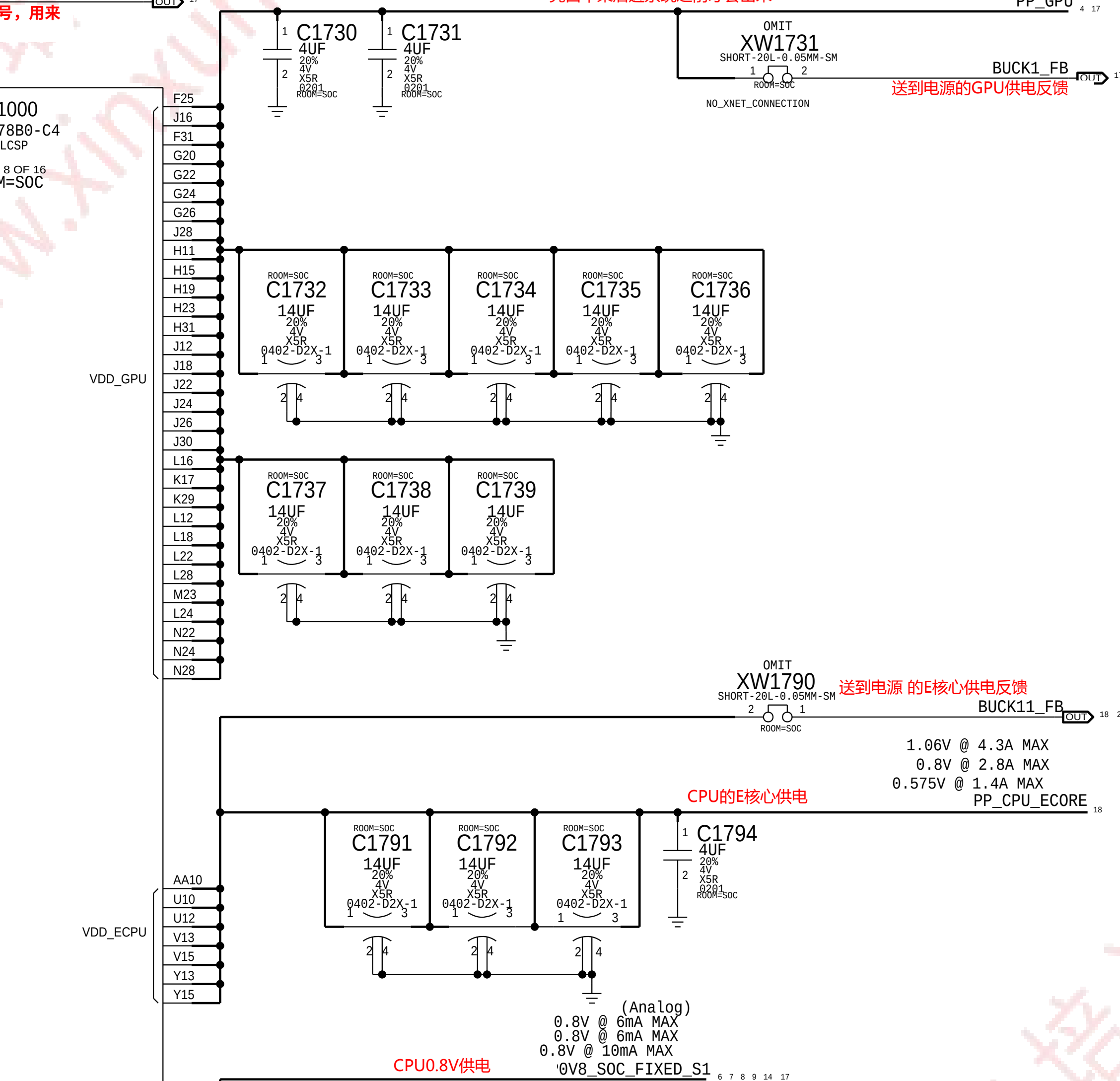


GPU电压检测信号, 到电源芯片

在苹果电路图中, 名称前面P表示供电, PP是正压, PN是负压
PP后面跟的数字表示电压值。
如PP0V7: 表示正的0.7V。
PN5V7: 表示负的5.7V。

1.06V @ 18.3A MAX
0.8V @ 10.6A MAX
0.575V @ 3.4A MAX

CPU内部显示核心供电, 电压0.8V左右此供电
亮白苹果后进系统之前才会出来



送到电源的E核心供电反馈

送到电源的E核心供电

1.06V @ 4.3A MAX
0.8V @ 2.8A MAX
0.575V @ 1.4A MAX
PP_CPU_ECORE

0.8V @ 6mA MAX (Analog)
0.8V @ 6mA MAX (CPU)
0.8V @ 10mA MAX (GPU)
0.8V @ 10mA MAX (SOC)
0V8_SOC_FIXED_S1

CPU0.8V供电

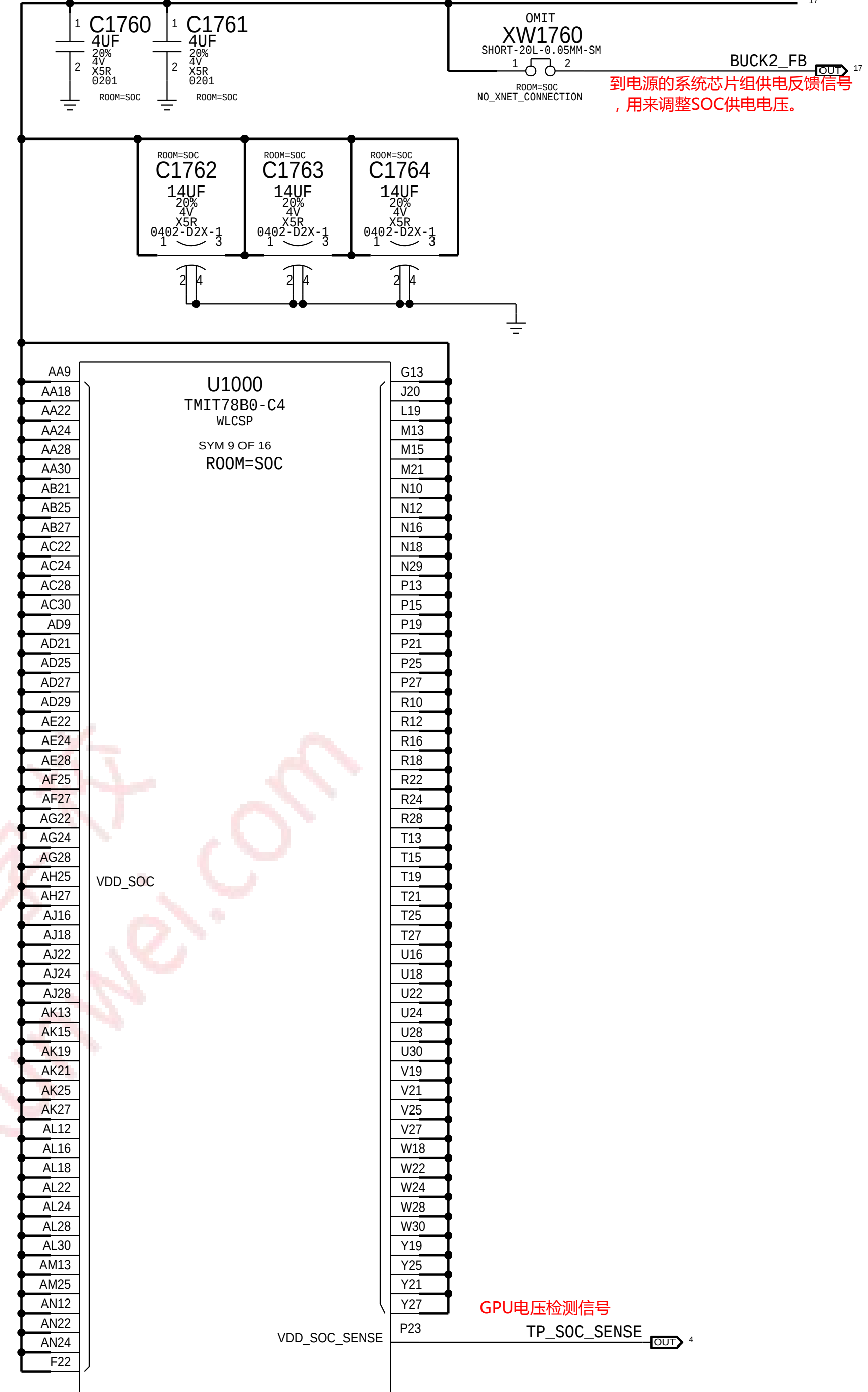
系统芯片组1.2V供电

1.2V @ 7mA MAX (Analog)
1.2V @ 7mA MAX (CPU)
1.2V @ 20mA MAX (GPU)
1.2V @ 20mA MAX (SOC)
PP1V2_SOC

系统芯片组供电, 0.8V左右

0.765V @ 4.9A MAX
0.635V @ 2.6A MAX

PP_SOC_S1

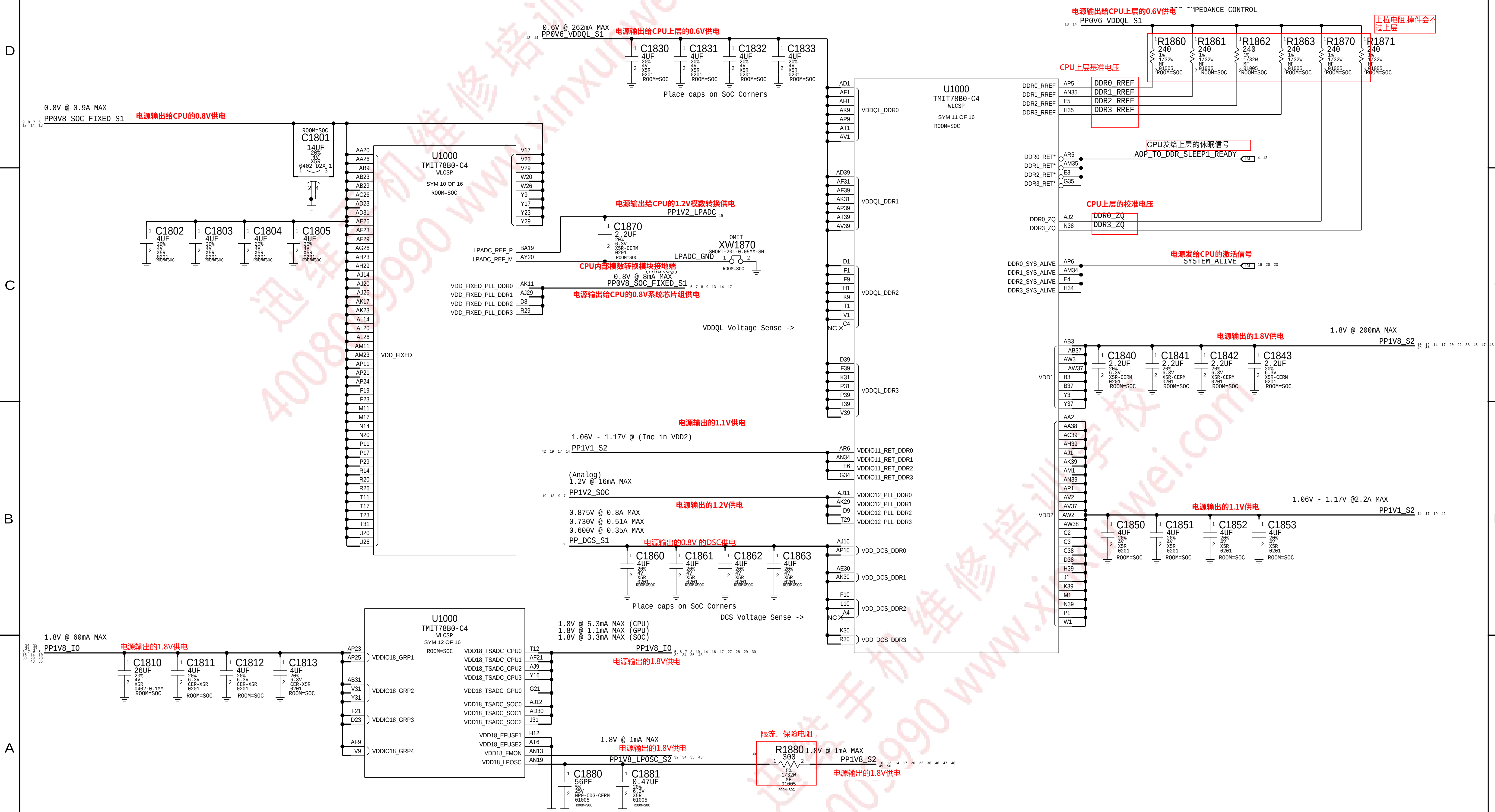


到电源的系统芯片组供电反馈信号
用来调整SOC供电电压。

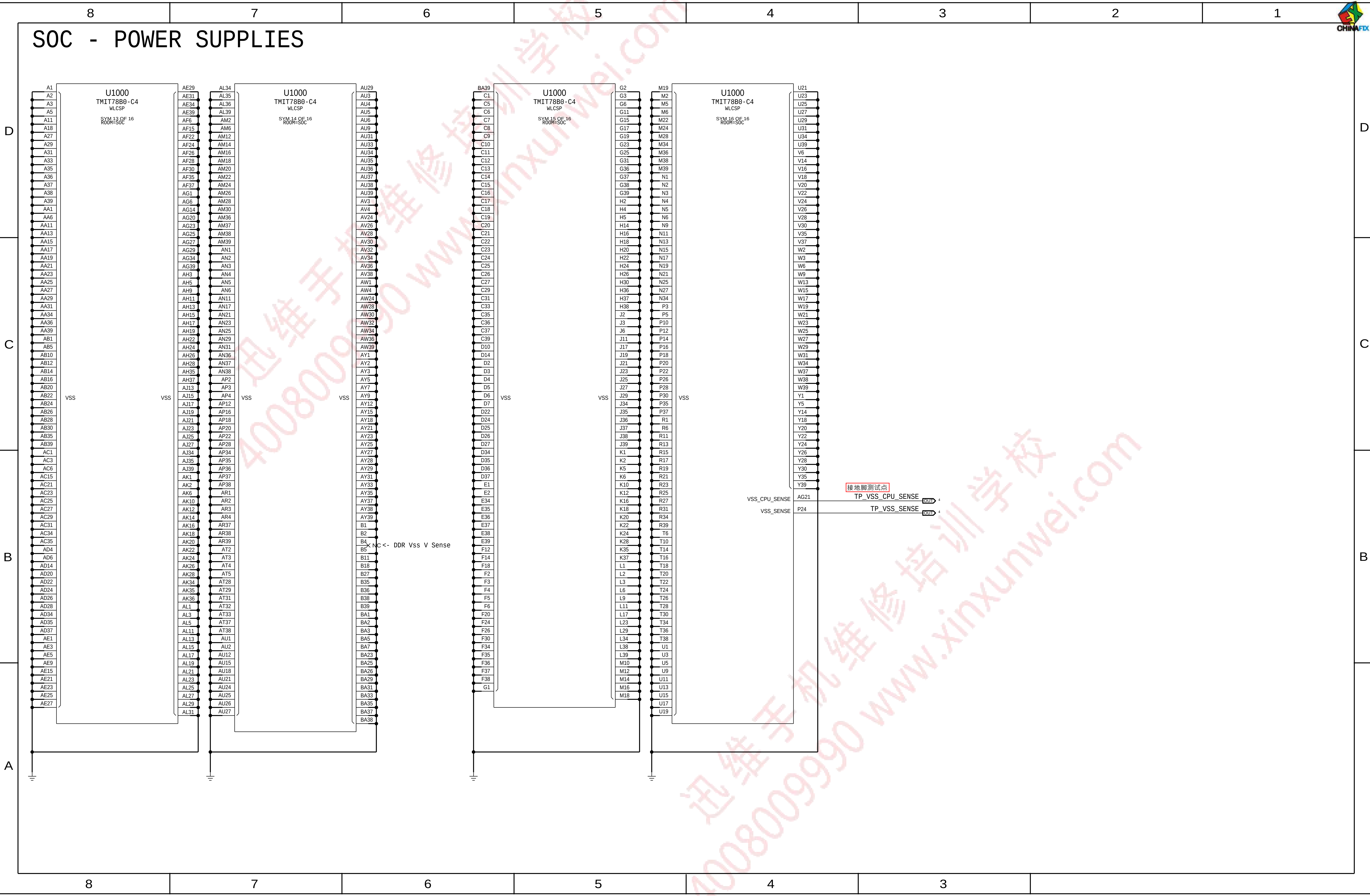
GPU电压检测信号

TP_SOC_SENSE

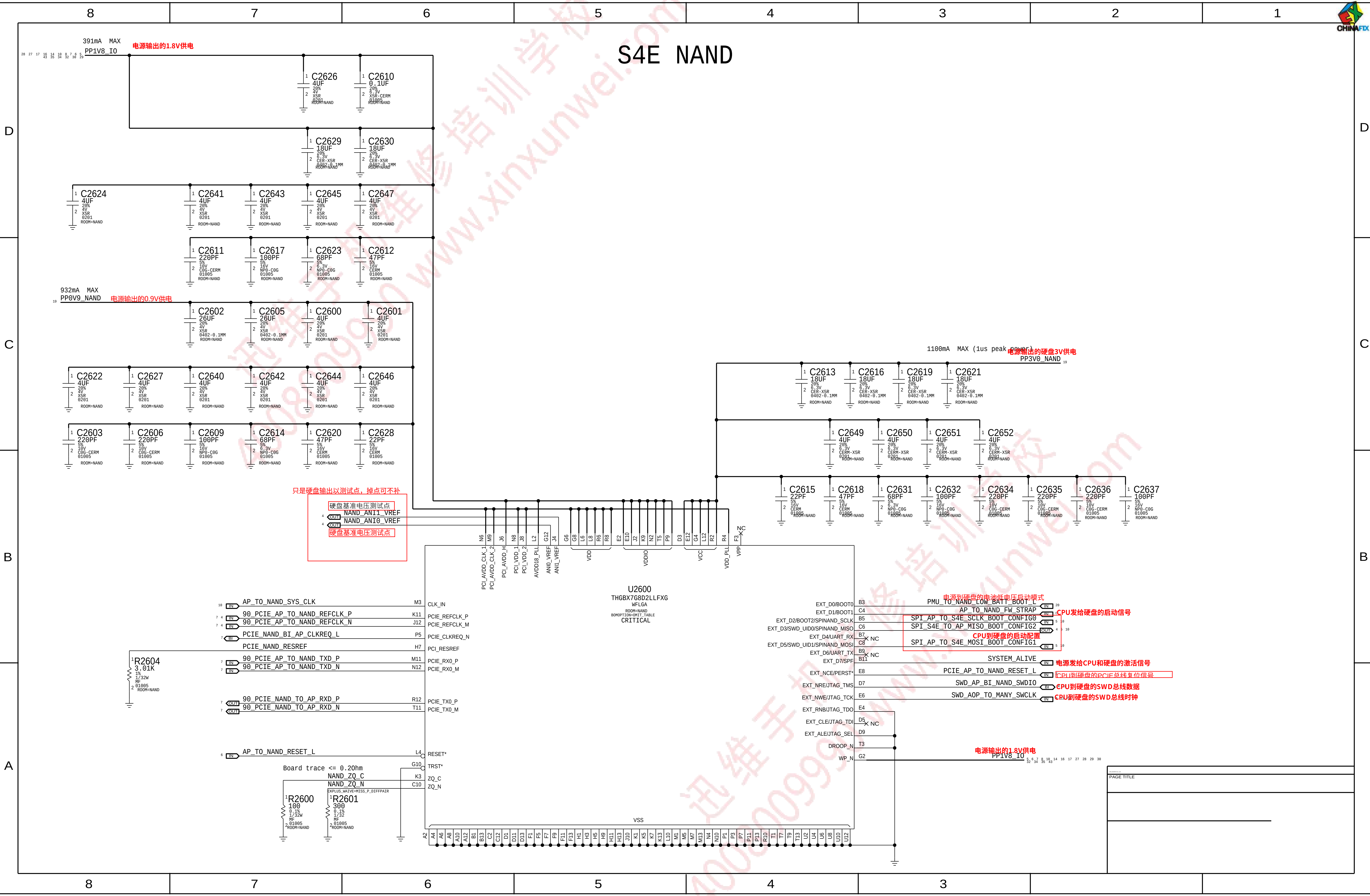
SOC - POWER SUPPLIES

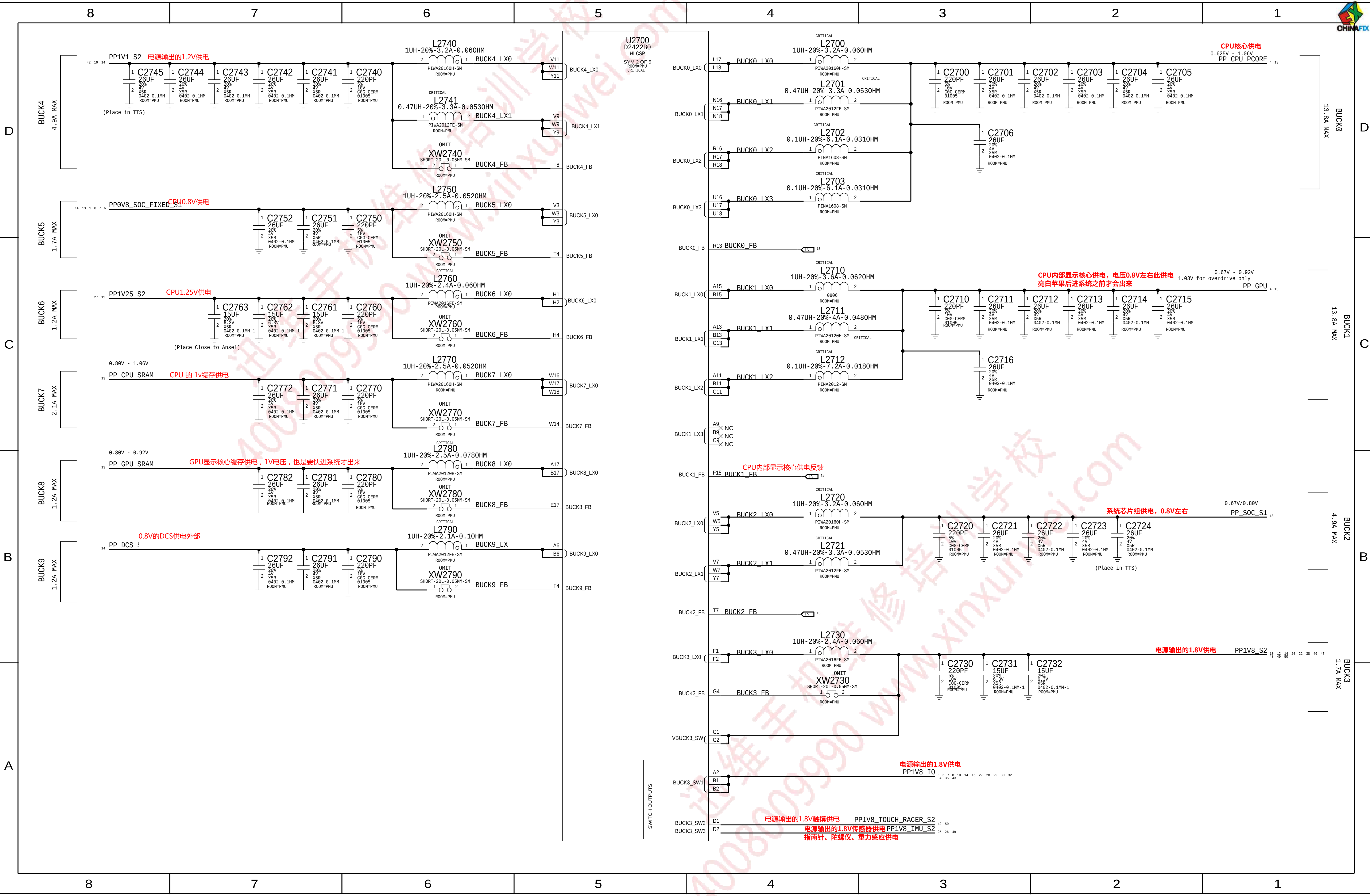


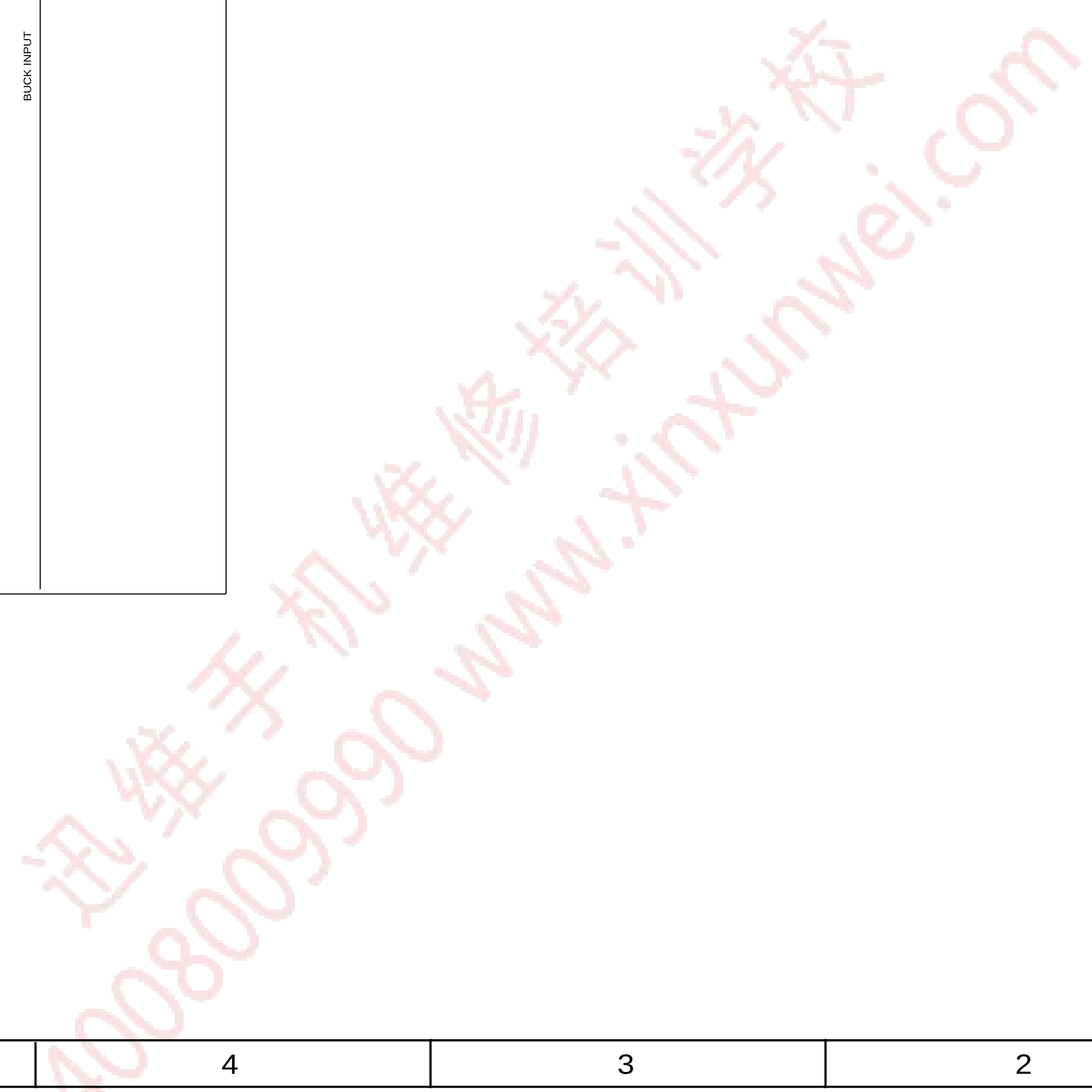
SOC - POWER SUPPLIES



S4E NAND

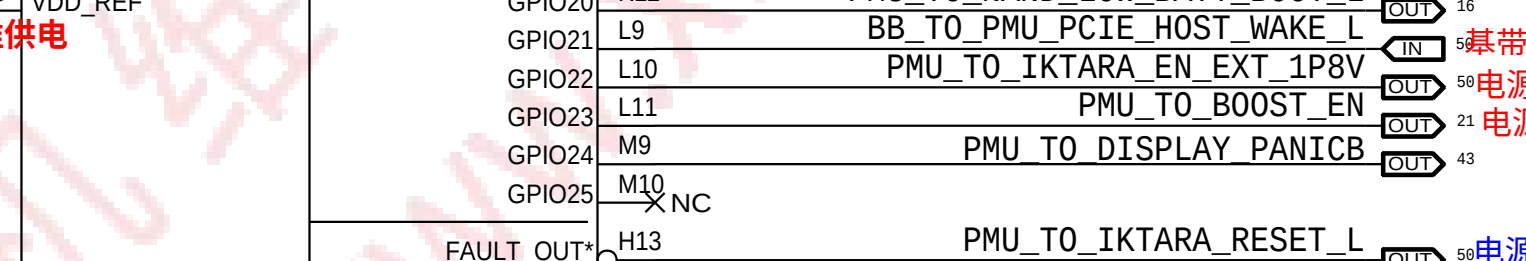
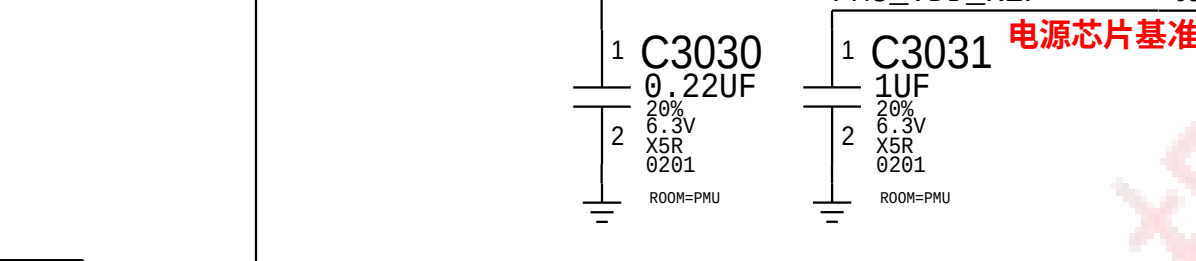
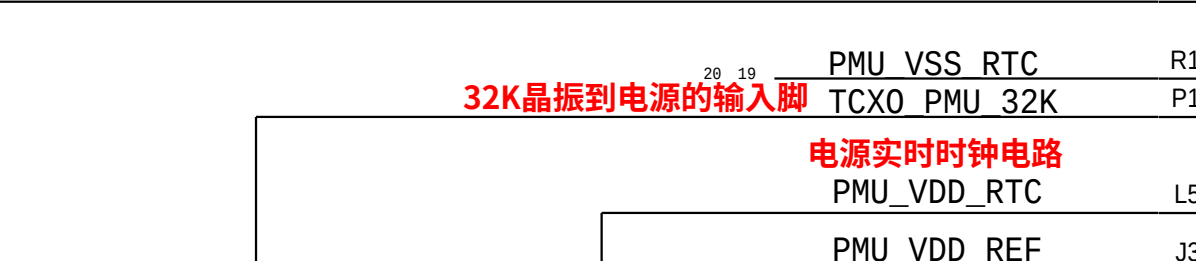
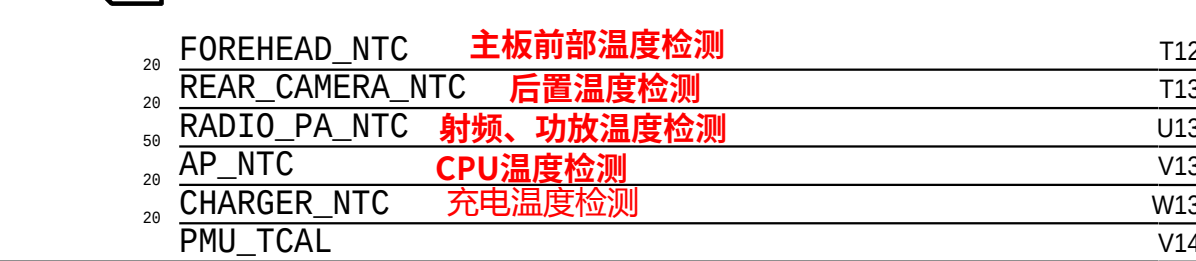
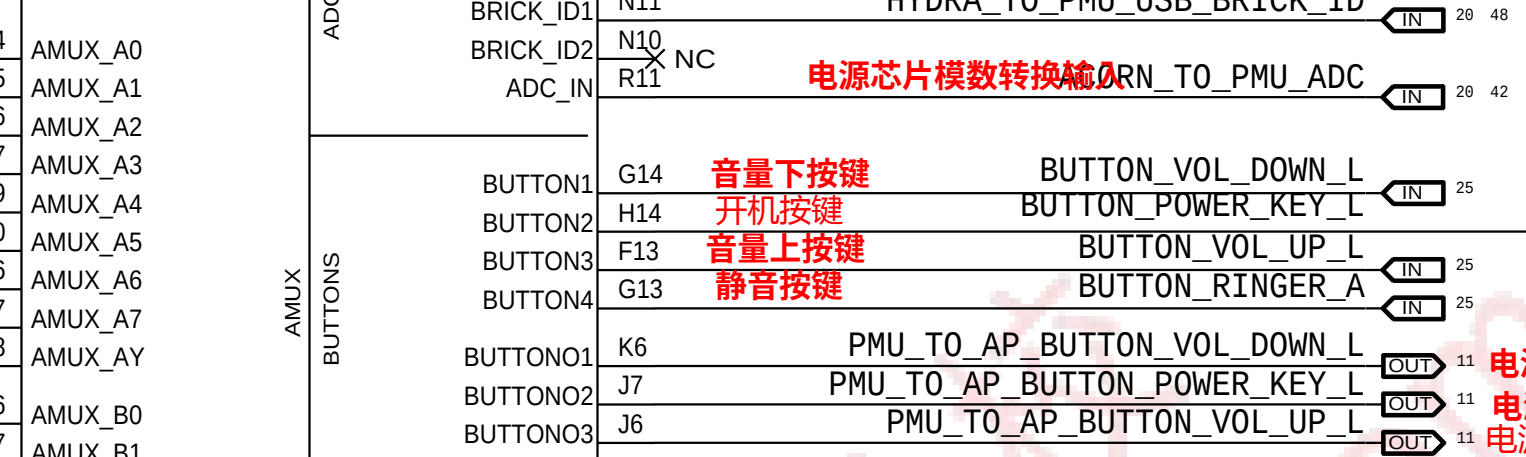
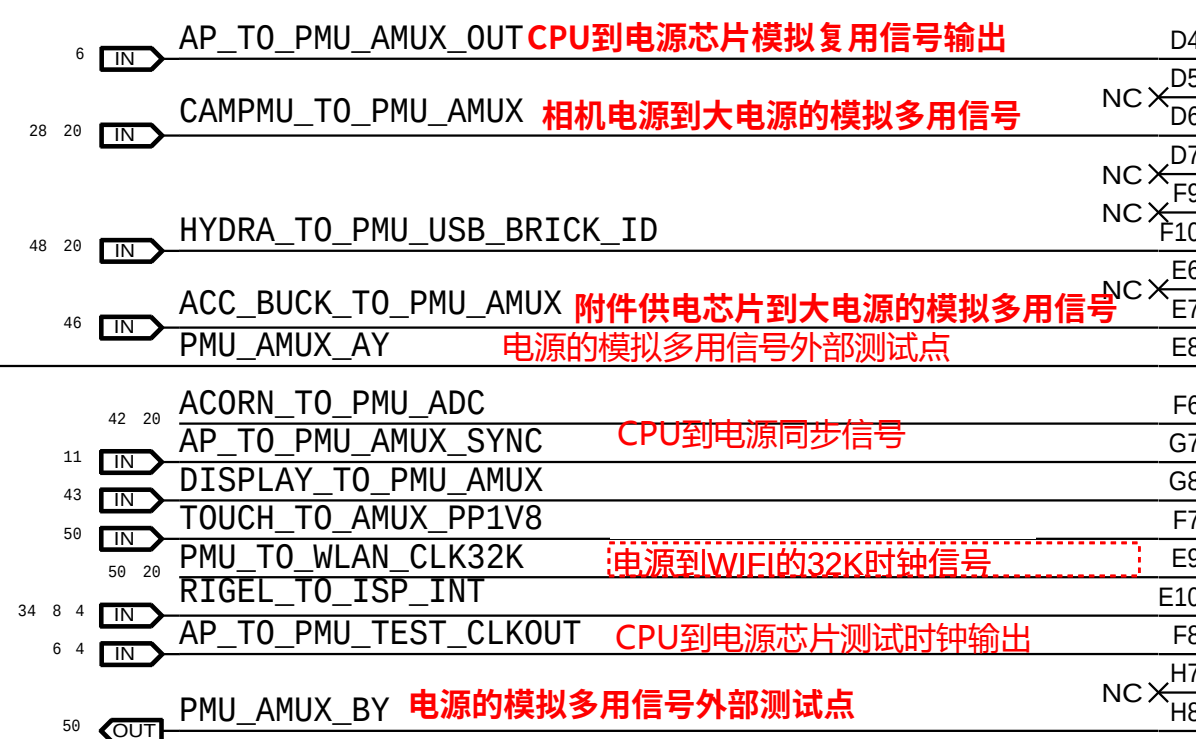
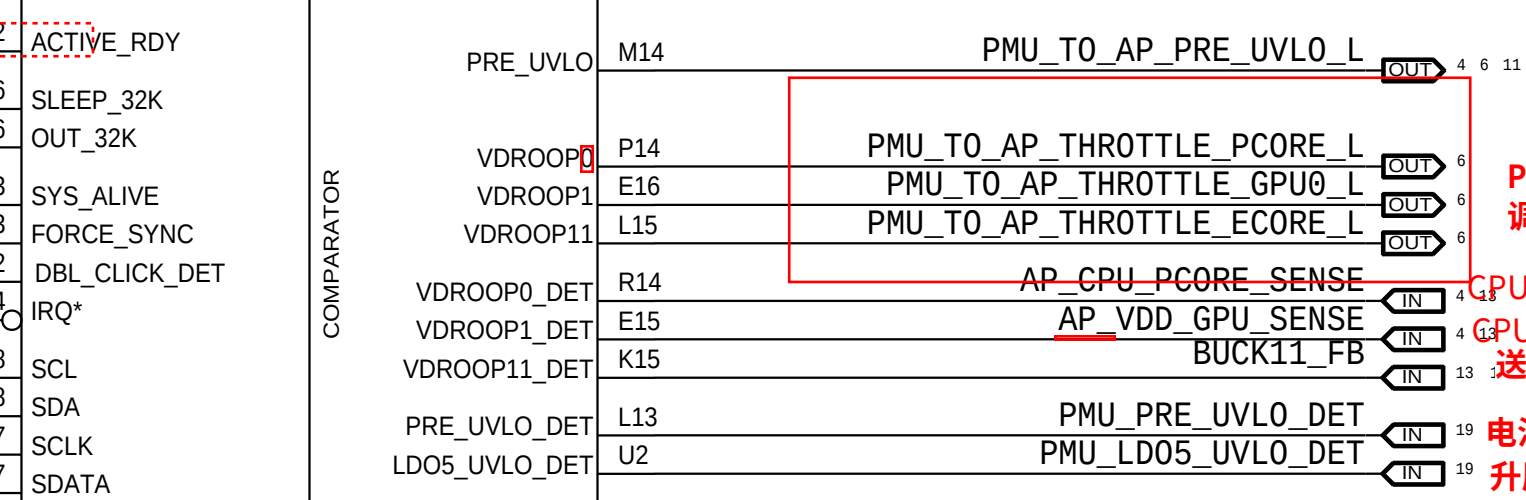
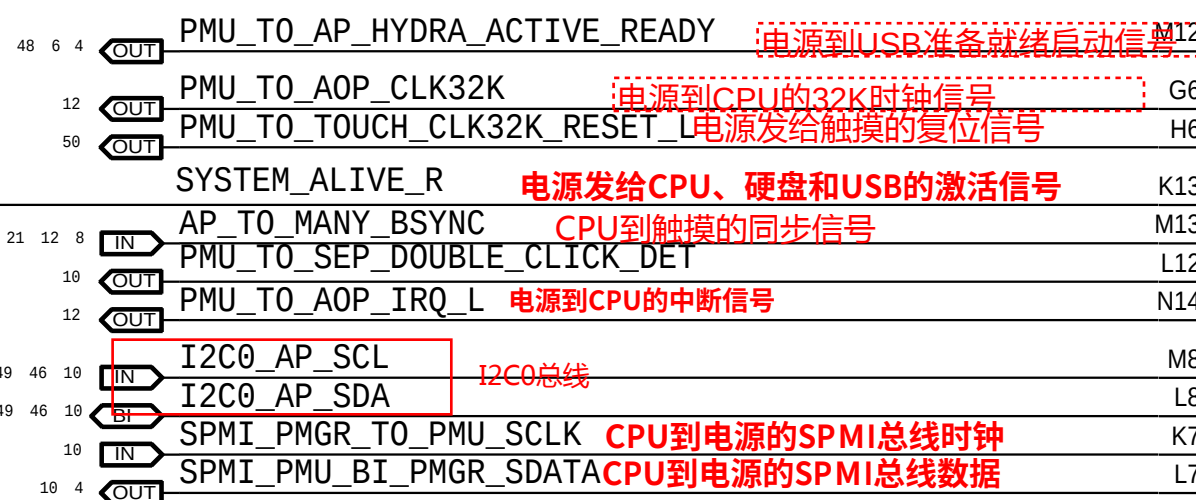
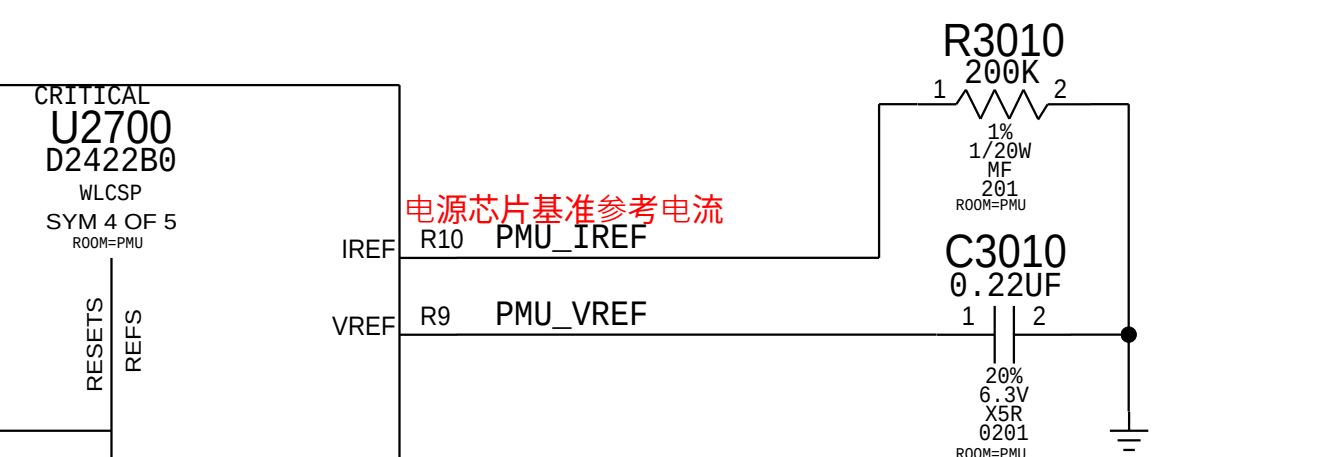
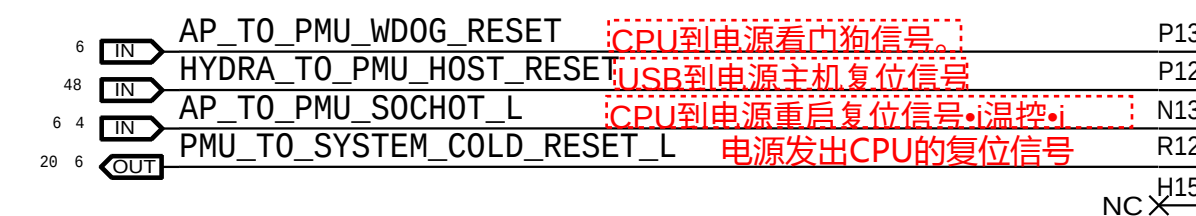
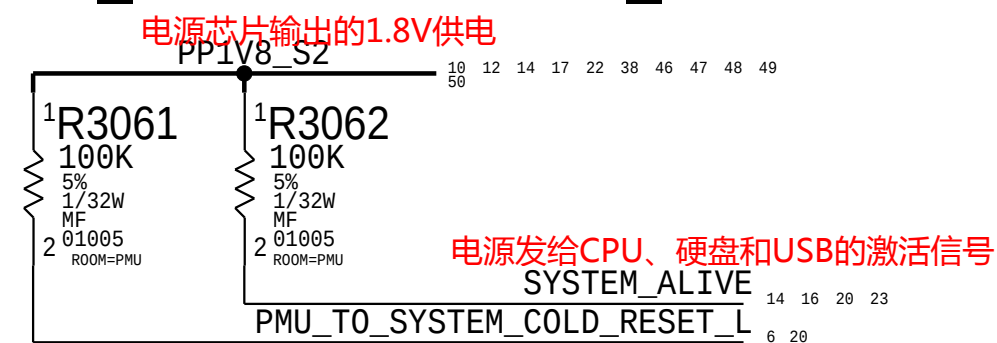
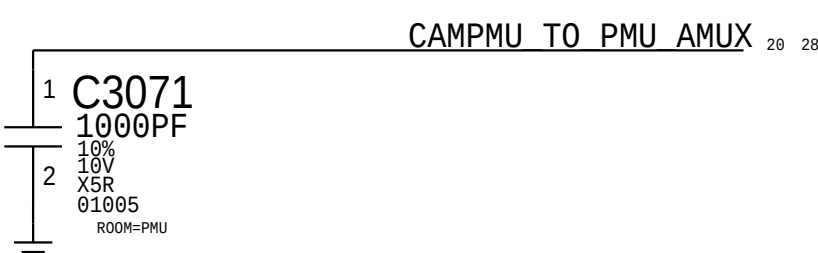






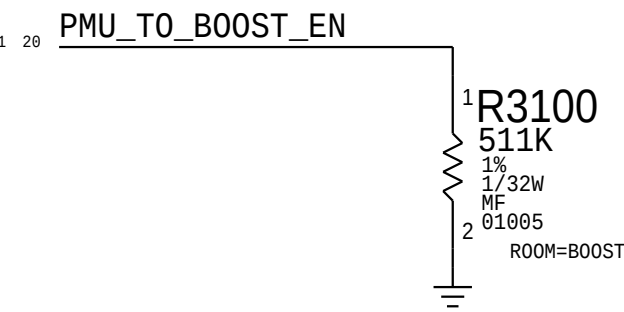


COLD_RESET & SYSTEM_ALIVE

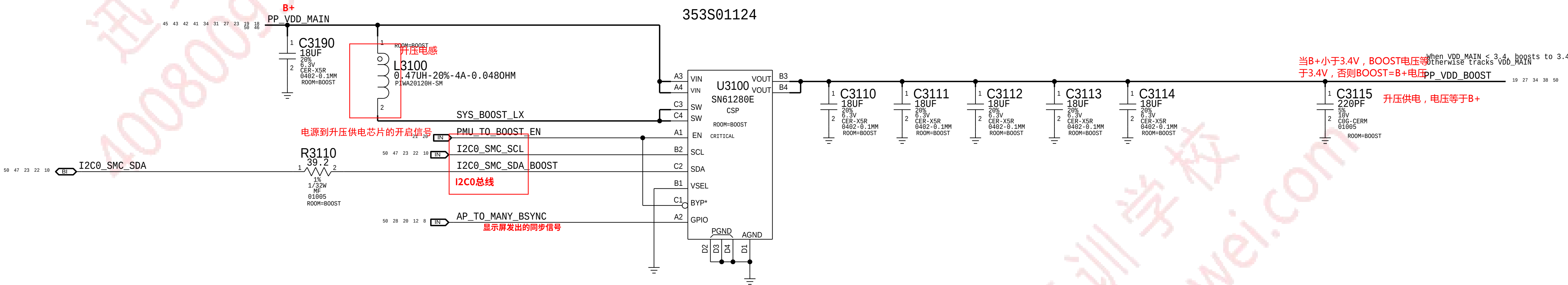


NOTE:100PF CAPS ARE THE SAMPLING CAPS FOR PMU ADC

Boost Enable Pull

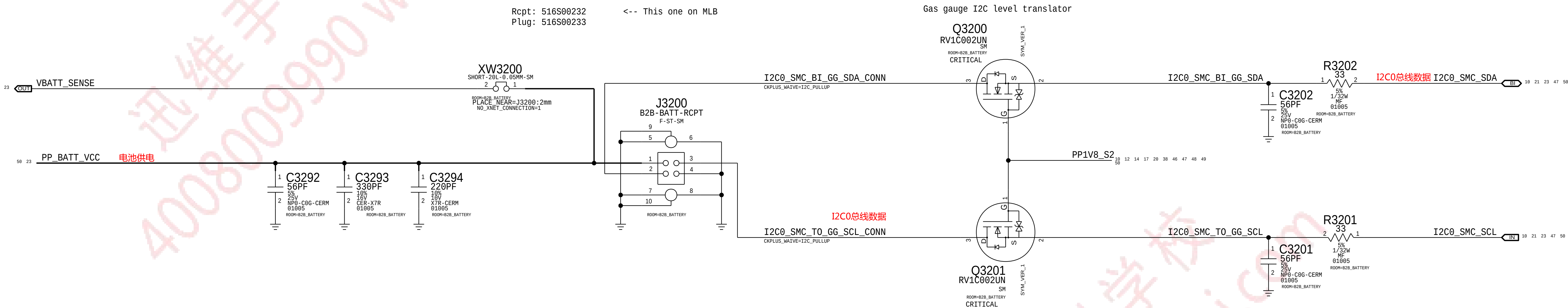


升压供电芯片
BOOST
353S01124

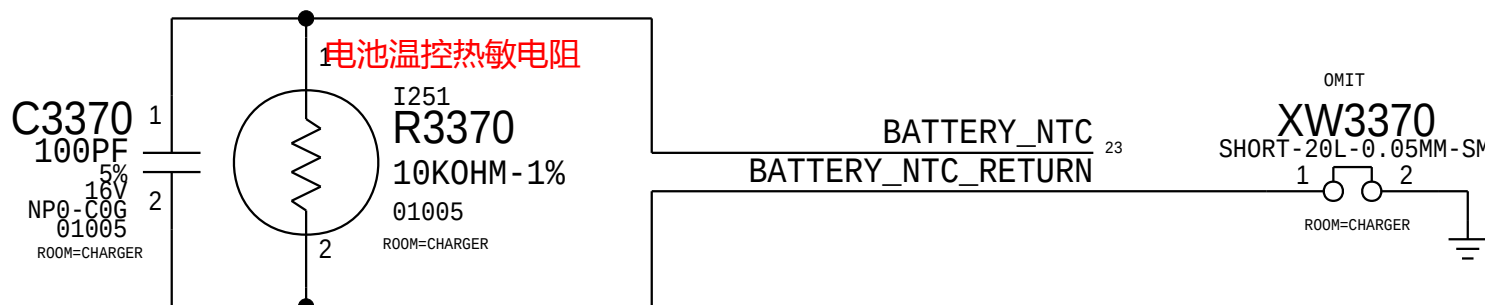


BATTERY CONNECTOR

Rcpt: 516S00232 <-- This one on MLB
Plug: 516S00233



BATTERY NTC





D

C

B

A

1

2

3

4

5

6

7

8

Iktara Components on MLB Bottom

1

2

3

4

5

6

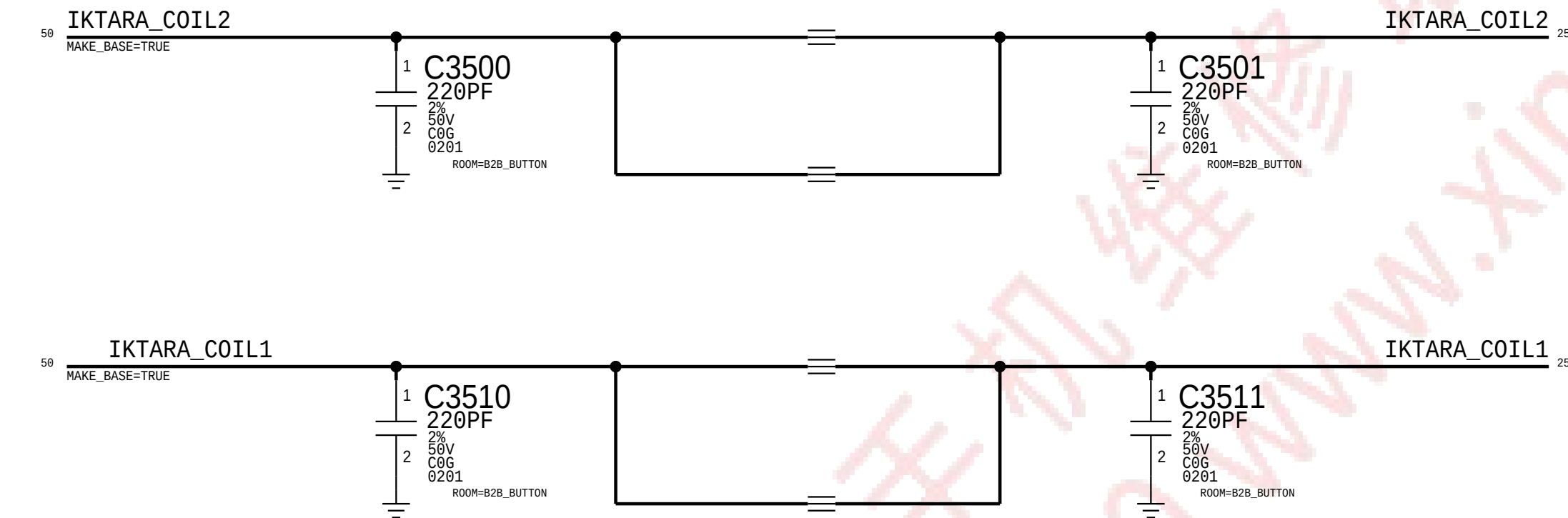
7

8

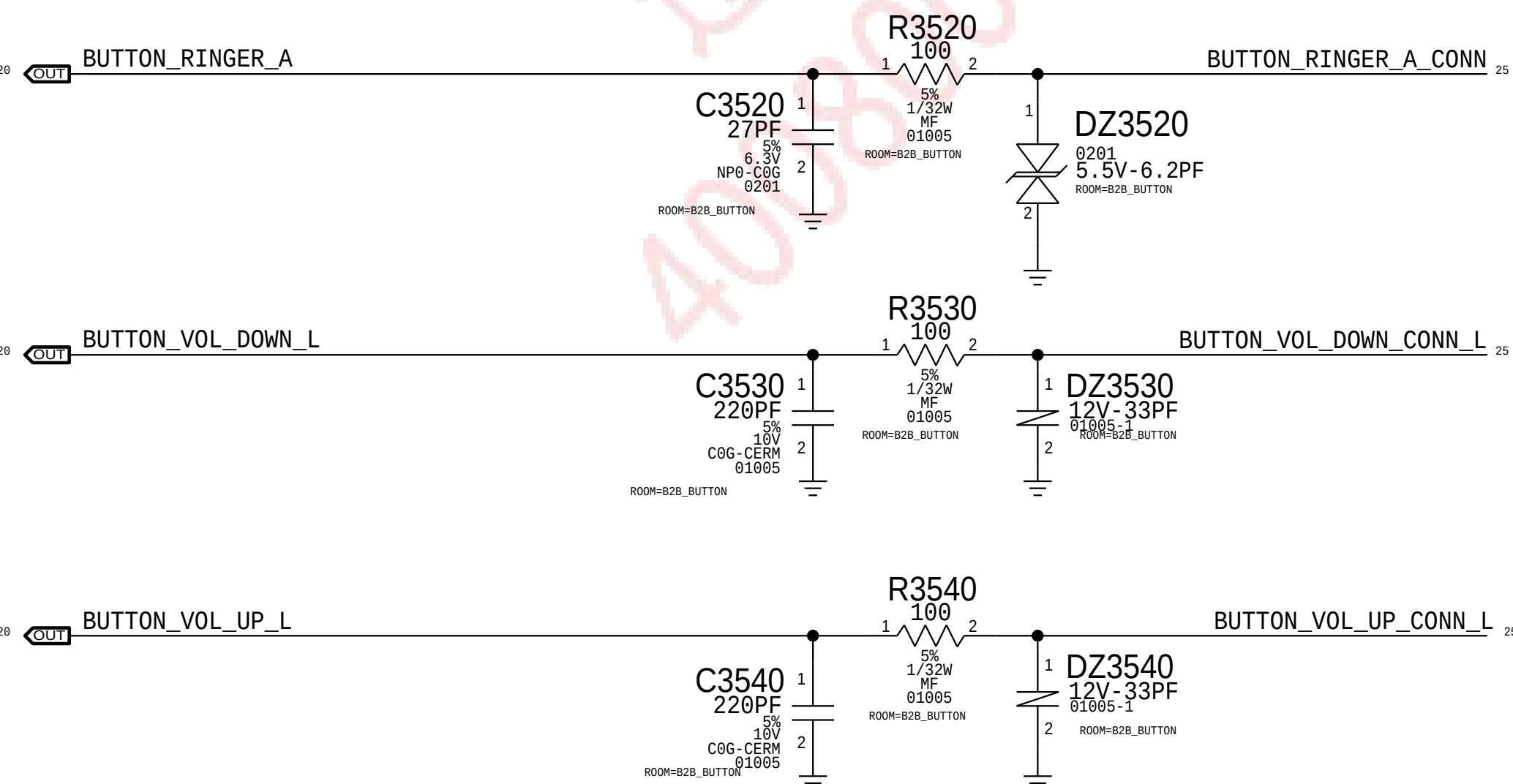
Cyclone + Button Connector

Rcpt: 516S00289 <-- This one on MLB
Plug: 516S00290

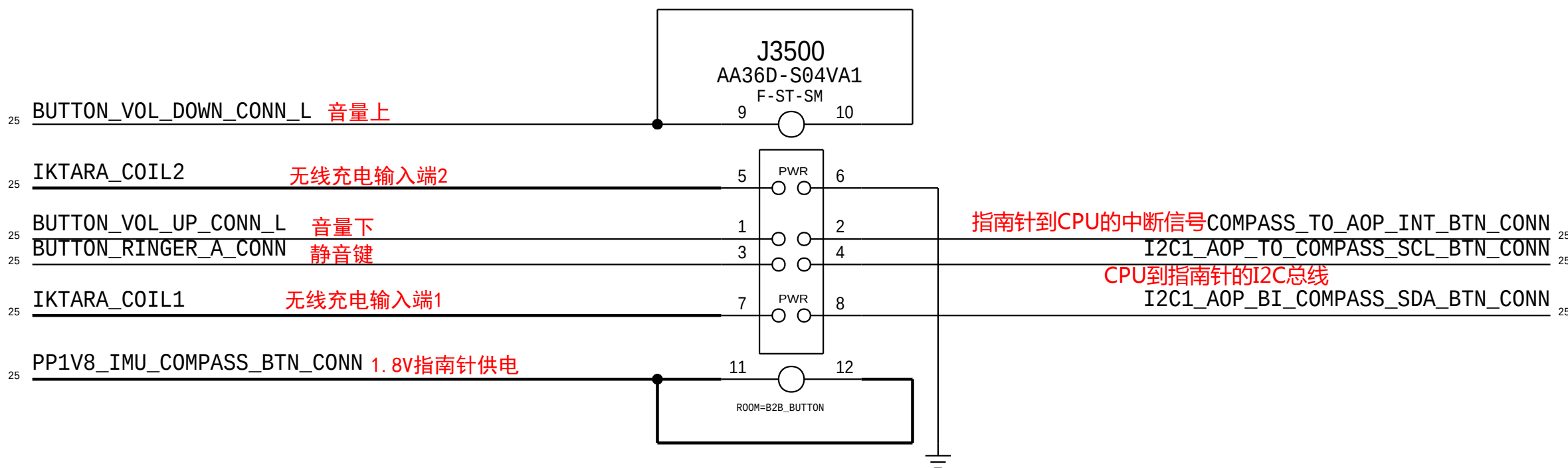
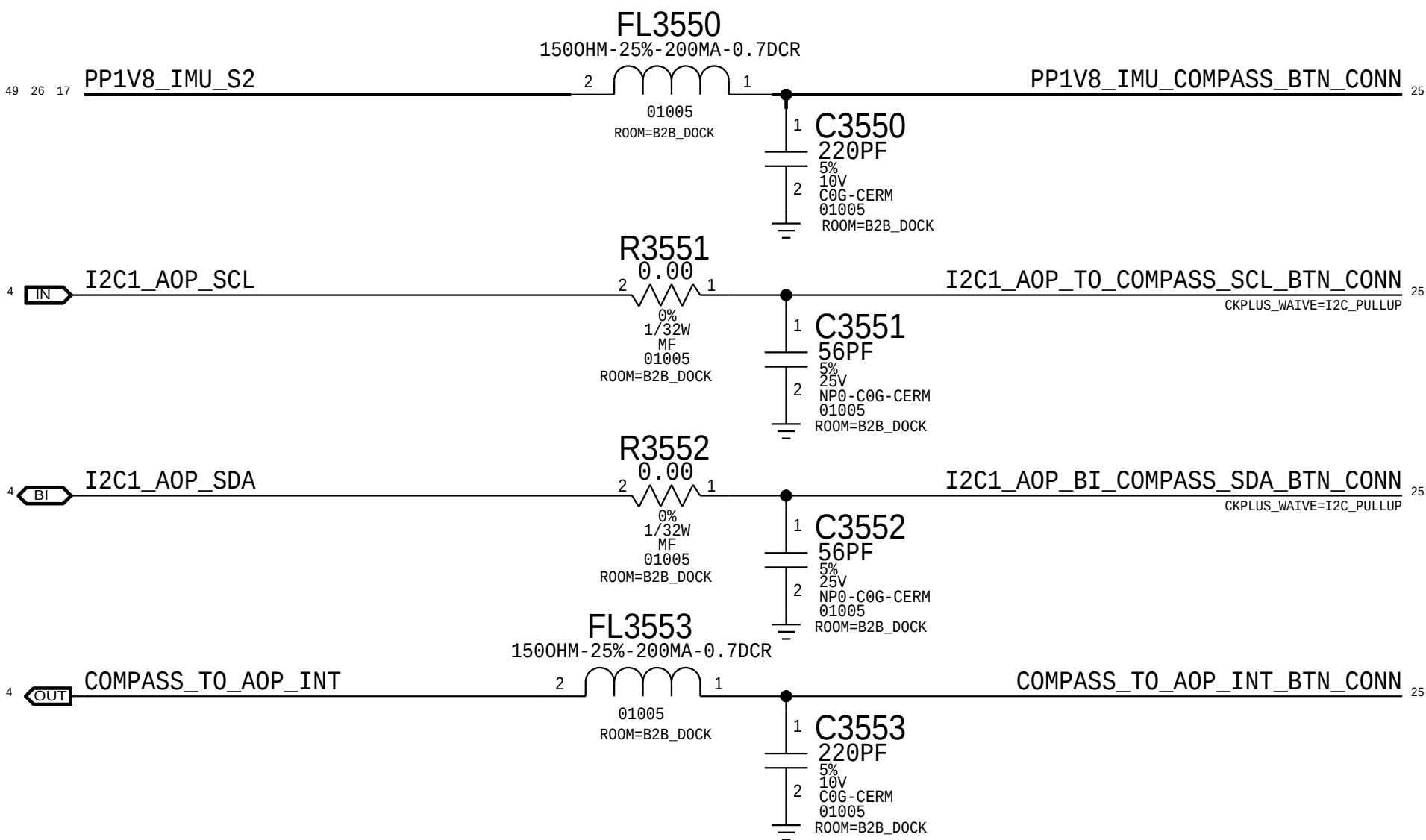
Cyclone Filtering



BUTTONS

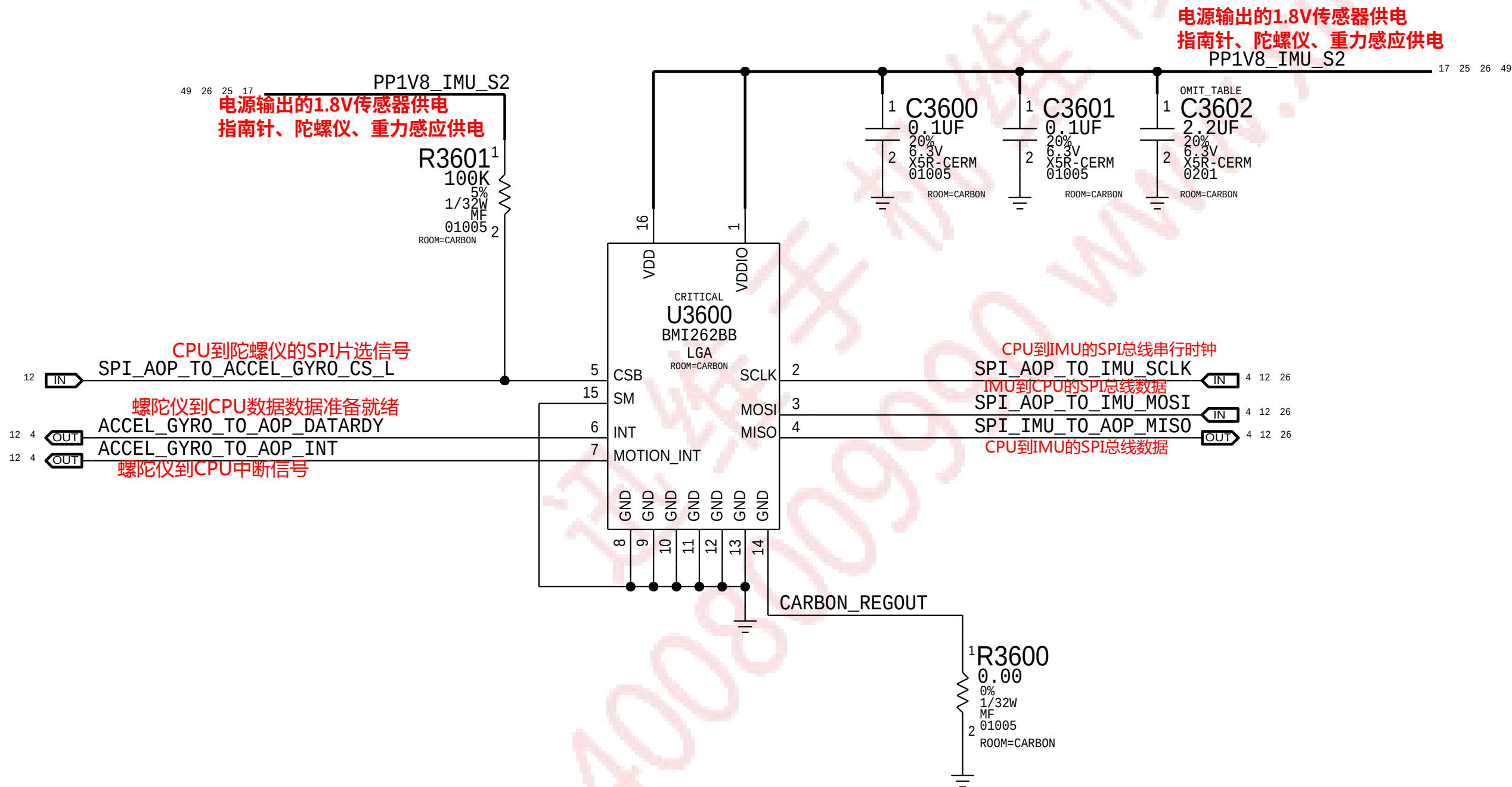


Compass (Button Flex Location)



Graphite - Accel & Gyro

APN: 338S00304



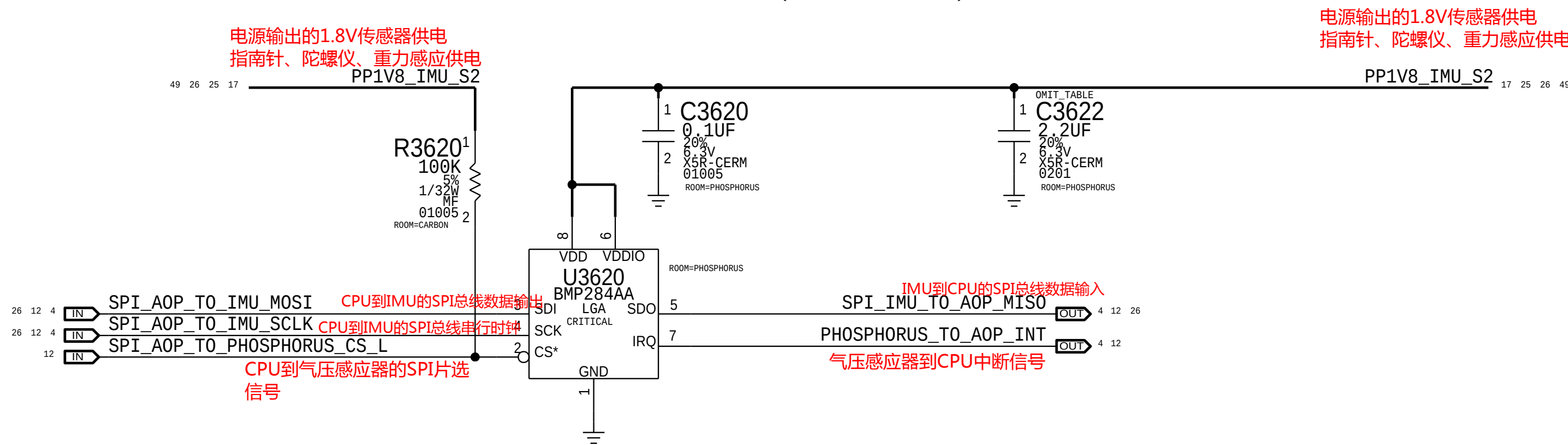
Magnesium - Compass

(On Dock or Button Flex)

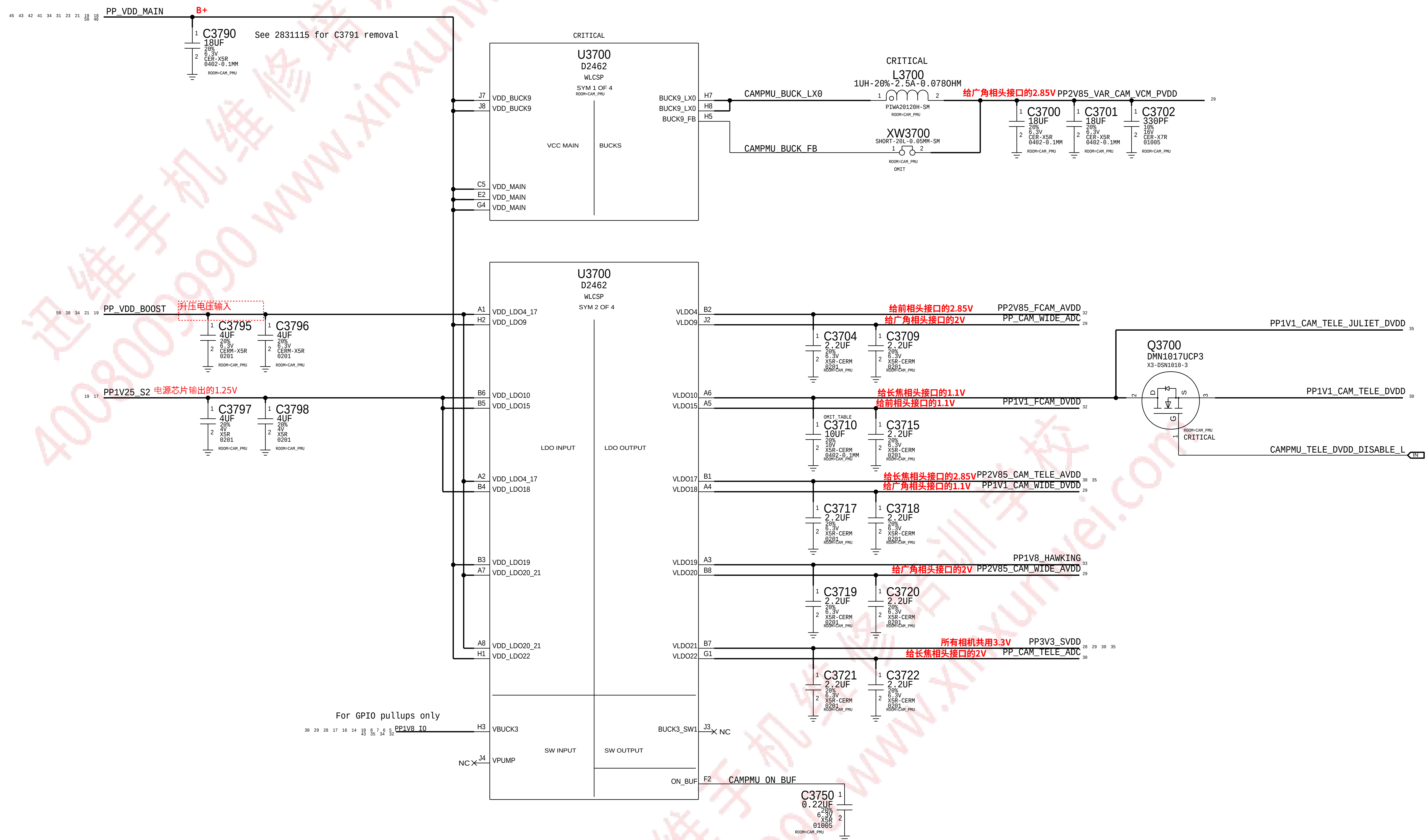
Phosphorus

BOSCH (APN: 338S00188)

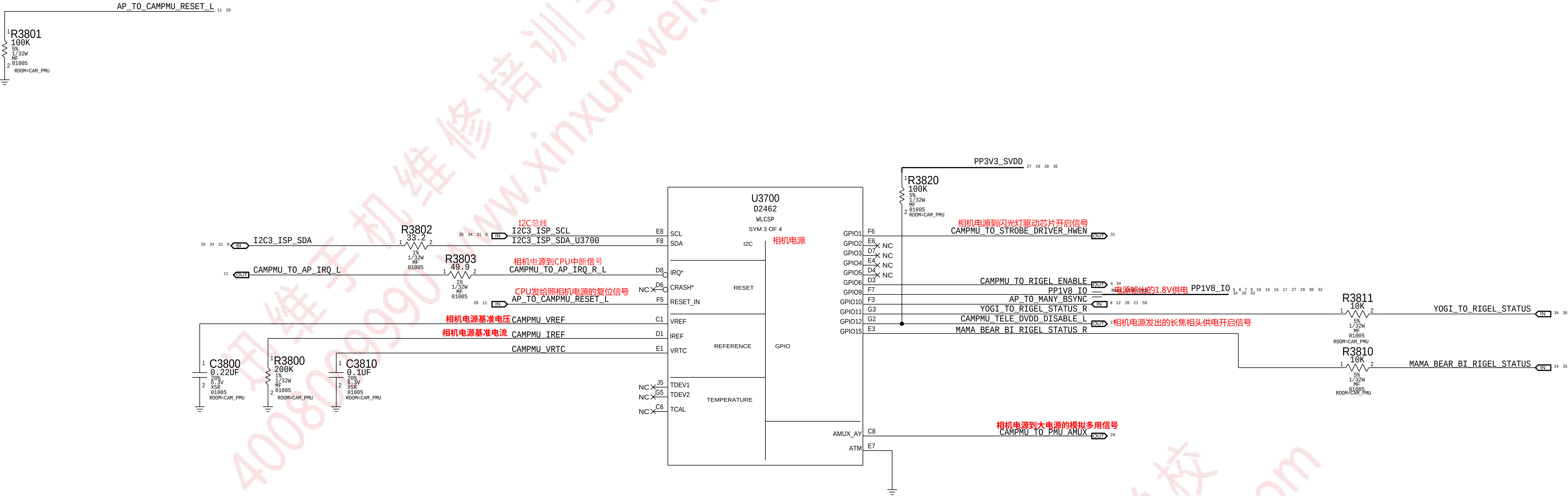
ST (APN: 338S00230)



相机电源



Pull Downs

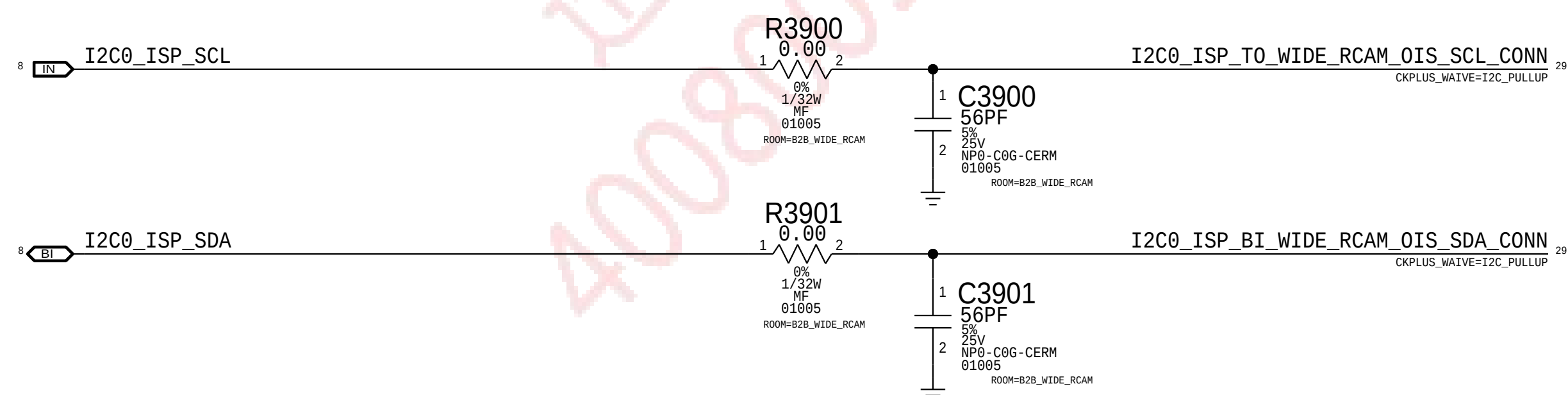


```
Rcpt: 516S00313      <-- This one on MLB
Plug: 516S00314
```

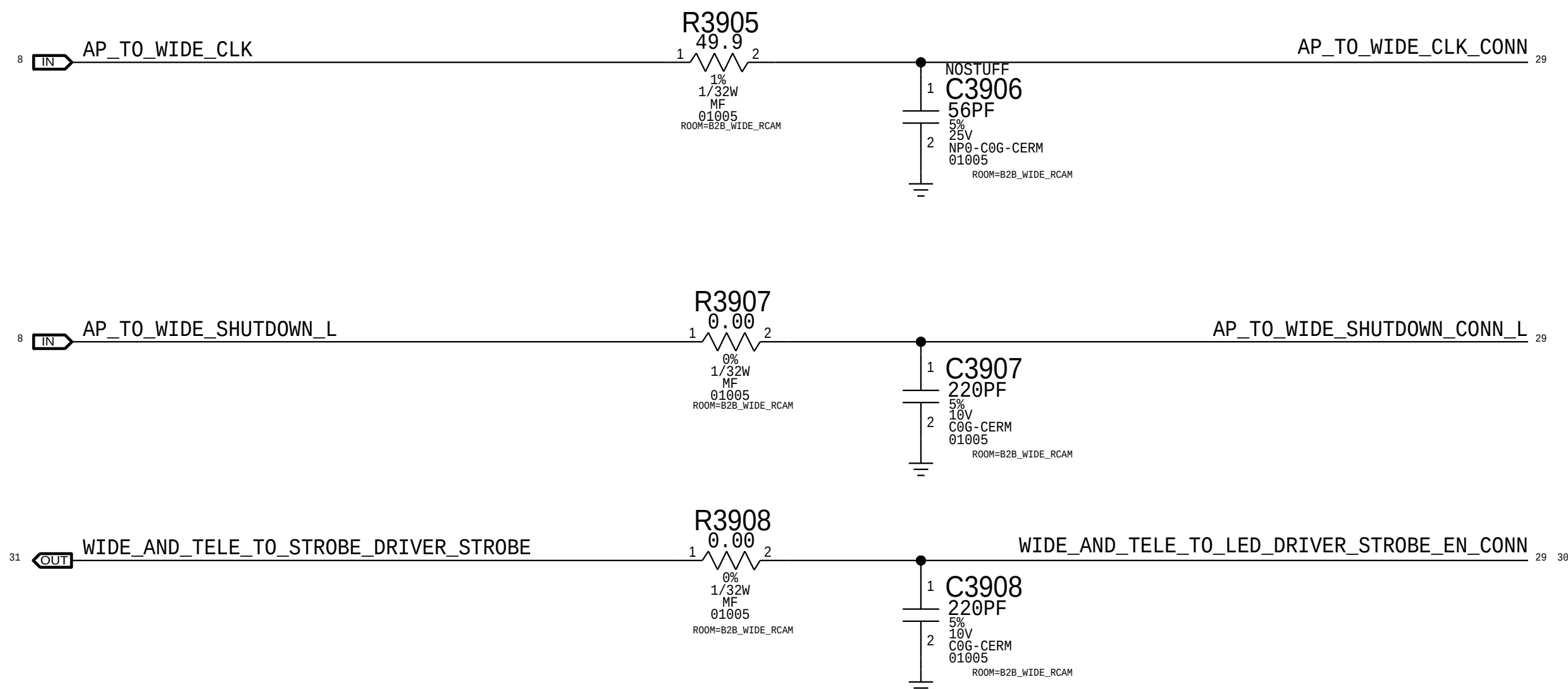
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Rcpt: 516S00313      <-- This one on MLB
Plug: 516S00314
```



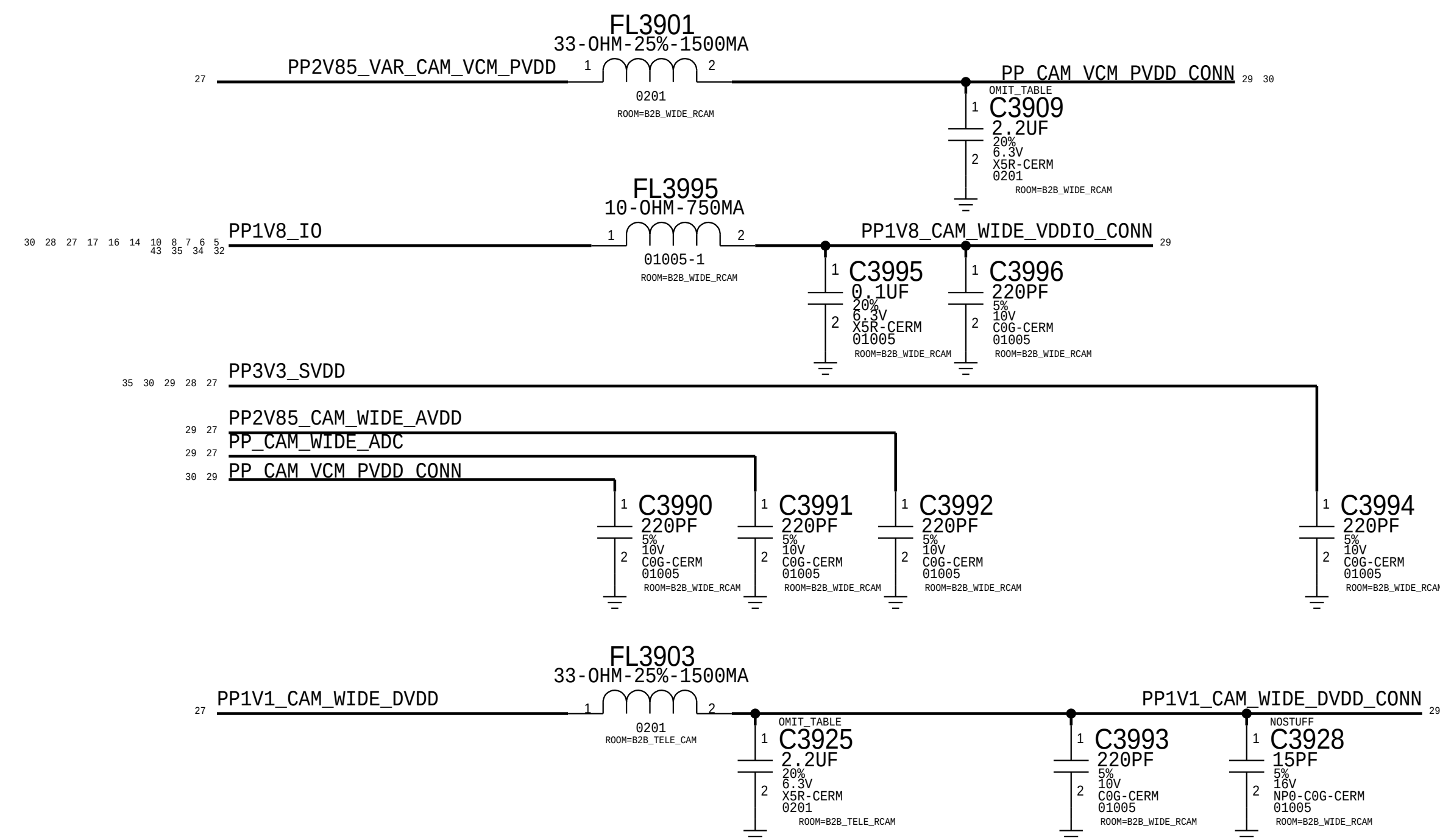
ISP I2C



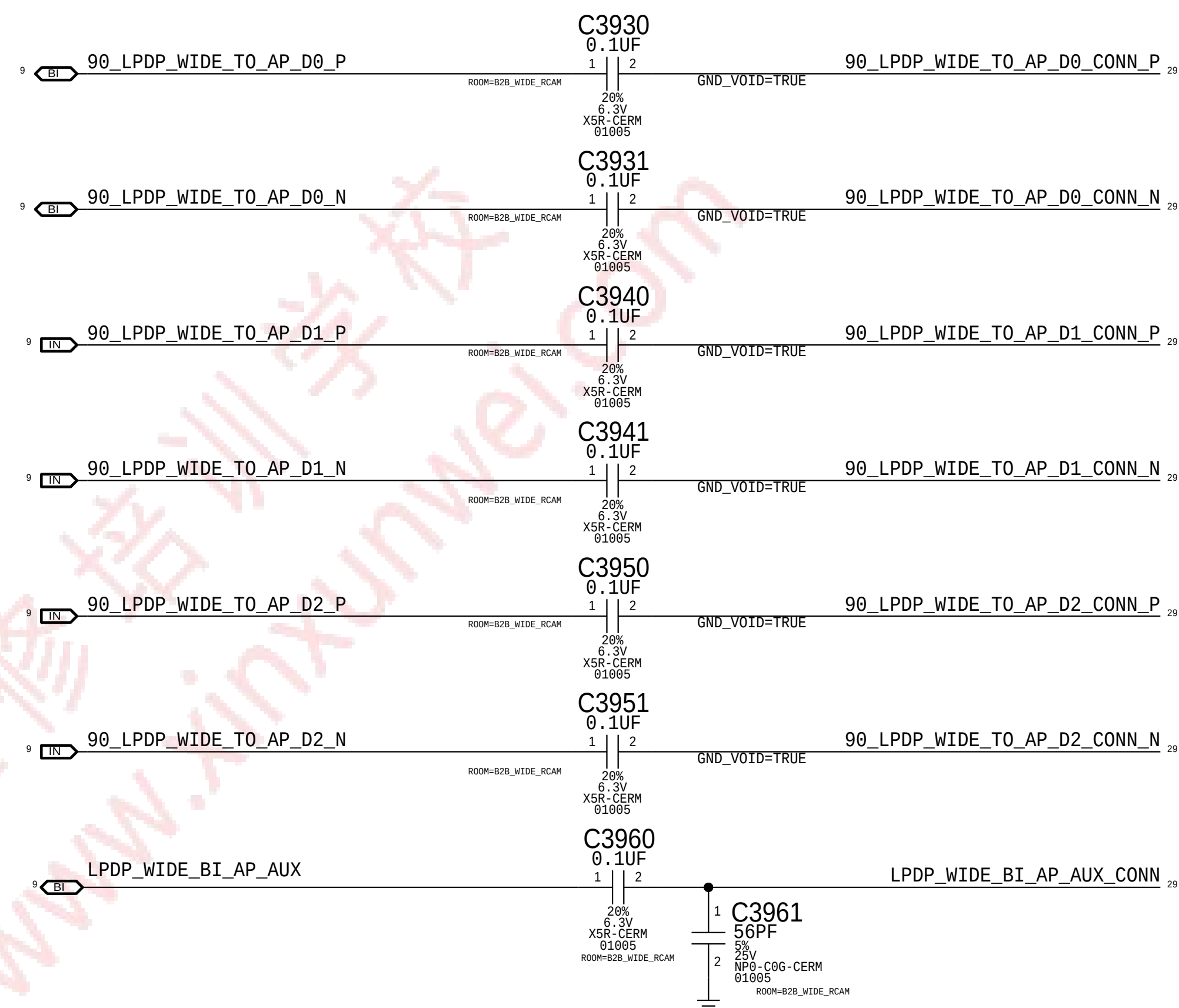
IO Filters



Power Filtering



LPDP Filters



90_LPDP_TELE_TO_AP_D0_P

ROOM-B2B, TELE_ICAM

END_VOID

90_LPDP_TELE_TO_AP_D0_CONN_P

90_LPDP_TELE_TO_AP_D0_N

ROOM-B2B, TELE_ICAM

END_VOID

90_LPDP_TELE_TO_AP_D1_P

ROOM-B2B, TELE_ICAM

END_VOID

90_LPDP_TELE_TO_AP_D1_CONN_P

90_LPDP_TELE_TO_AP_D1_N

ROOM-B2B, TELE_ICAM

END_VOID

90_LPDP_TELE_TO_AP_D2_P

ROOM-B2B, TELE_ICAM

END_VOID

90_LPDP_TELE_TO_AP_D2_CONN_P

90_LPDP_TELE_TO_AP_D2_N

ROOM-B2B, TELE_ICAM

END_VOID

LPDP_TELE_BI_AP_AUX

ROOM-B2B, TELE_ICAM

LPDP_TELE_BI_AP_AUX_CONN

C4030
0.1UF
1 2
20%
6.3V
XSR-CERM
01005

C4031
0.1UF
1 2
20%
6.3V
XSR-CERM
01005

C4040
0.1UF
1 2
20%
6.3V
XSR-CERM
01005

C4041
0.1UF
1 2
20%
6.3V
XSR-CERM
01005

C4050
0.1UF
1 2
20%
6.3V
XSR-CERM
01005

C4051
0.1UF
1 2
20%
6.3V
XSR-CERM
01005

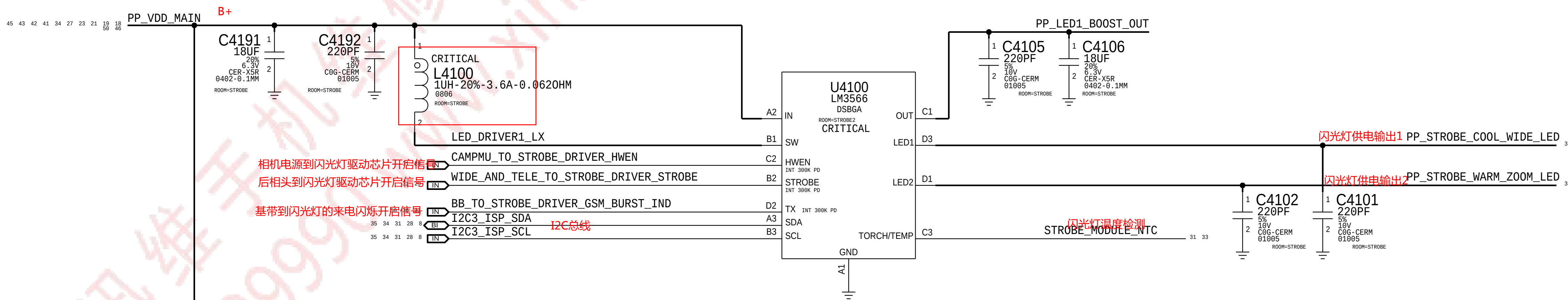
C4060
0.1UF
1 2
20%
6.3V
XSR-CERM
01005

C4061
56PF
1 2
25V
NPO-C0G-CERM
01005

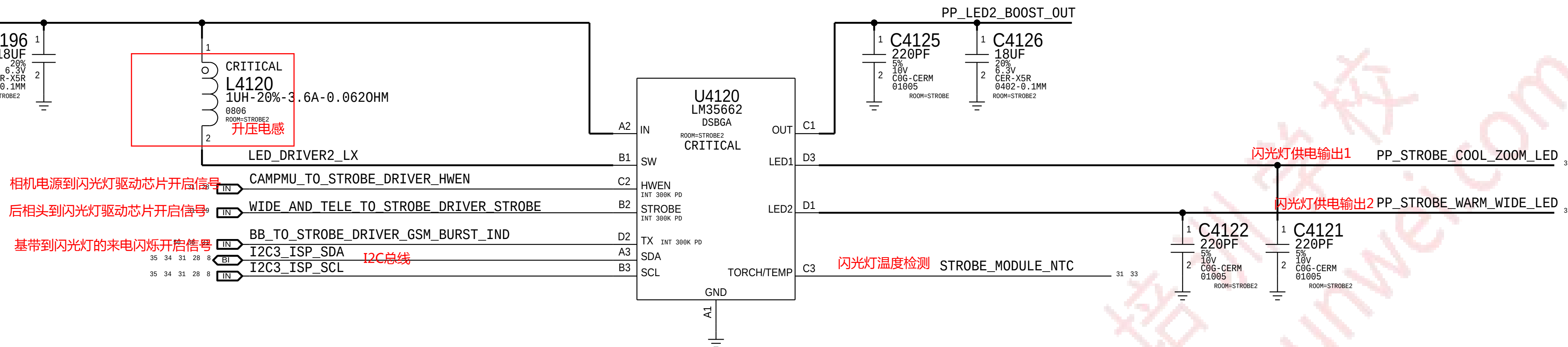
C4062
56PF
1 2
25V
NPO-C0G-CERM
01005

LED STROBE DRIVERS (NEON)

APN:353S00558
I2C Address (7-bit): 0x63



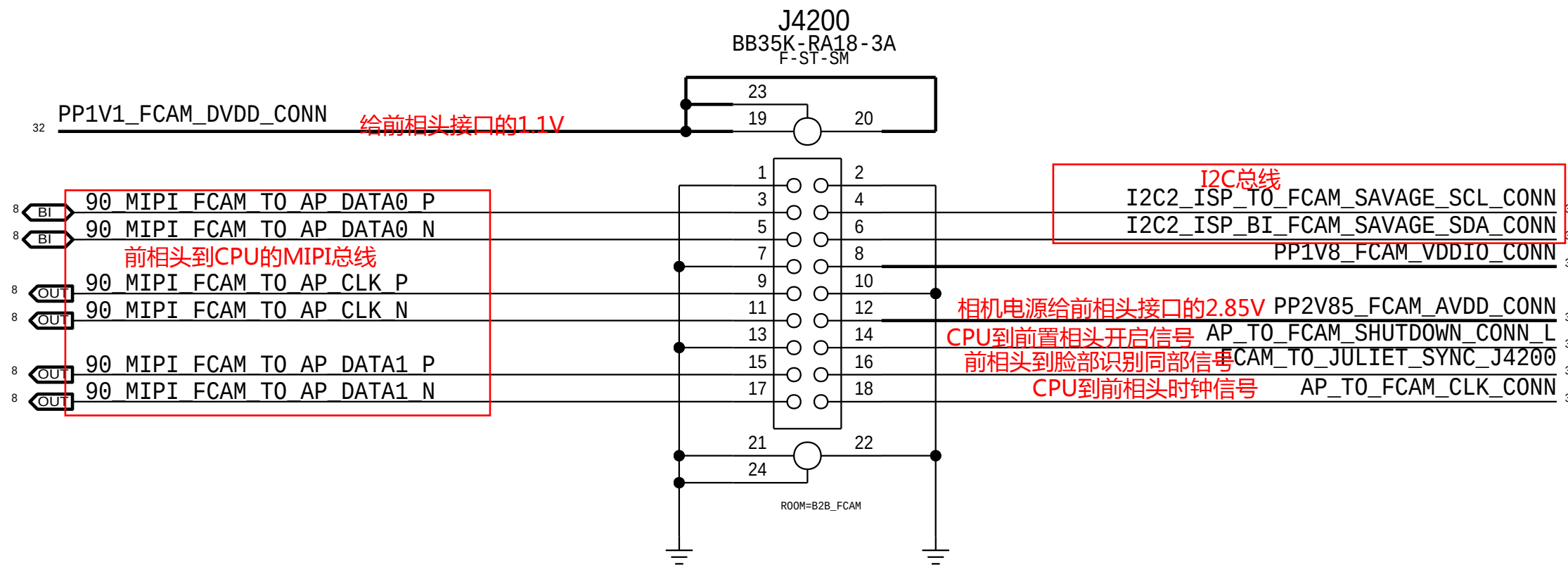
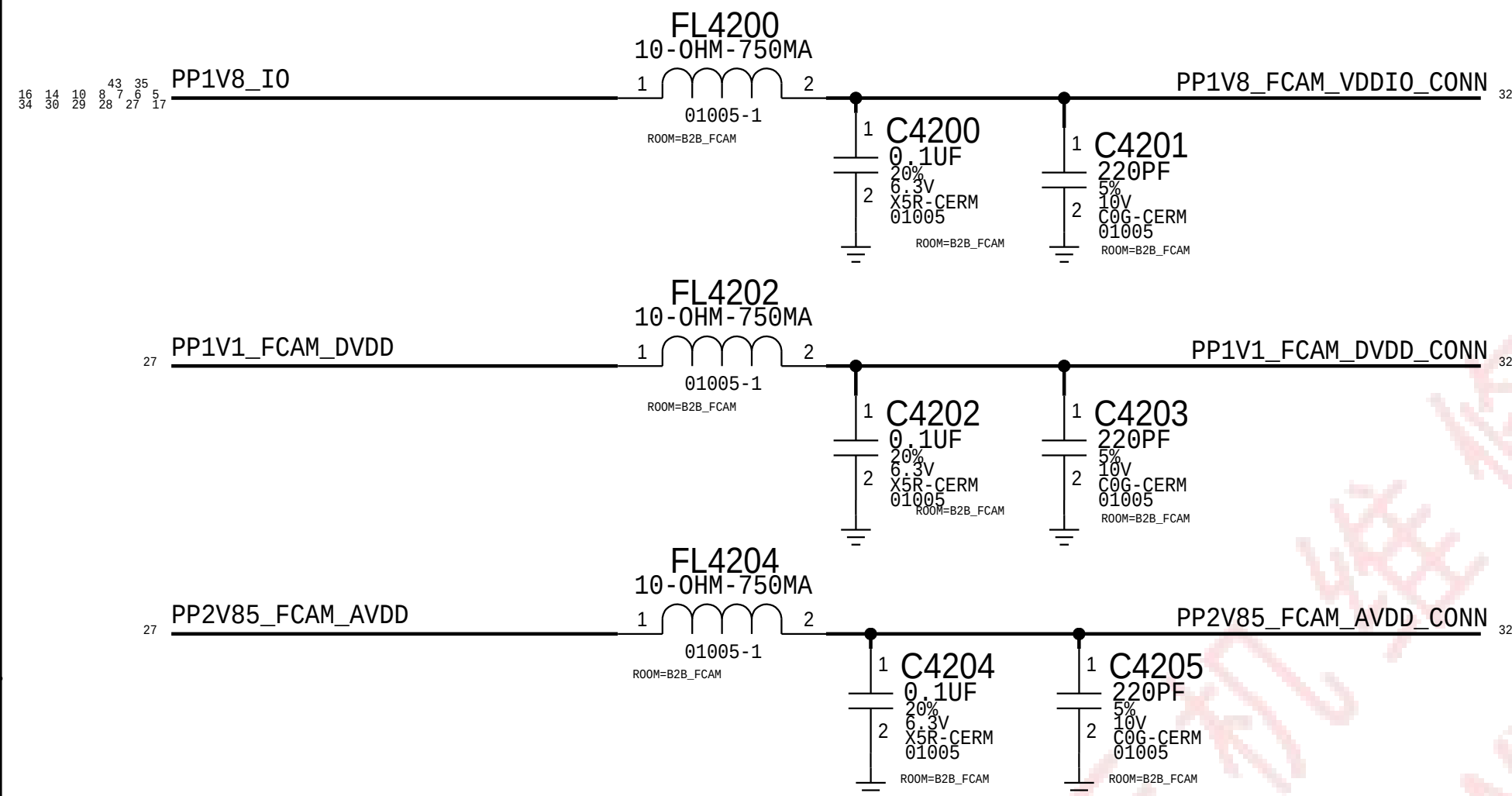
APN:353S00868
I2C Address (7-bit): 0x67



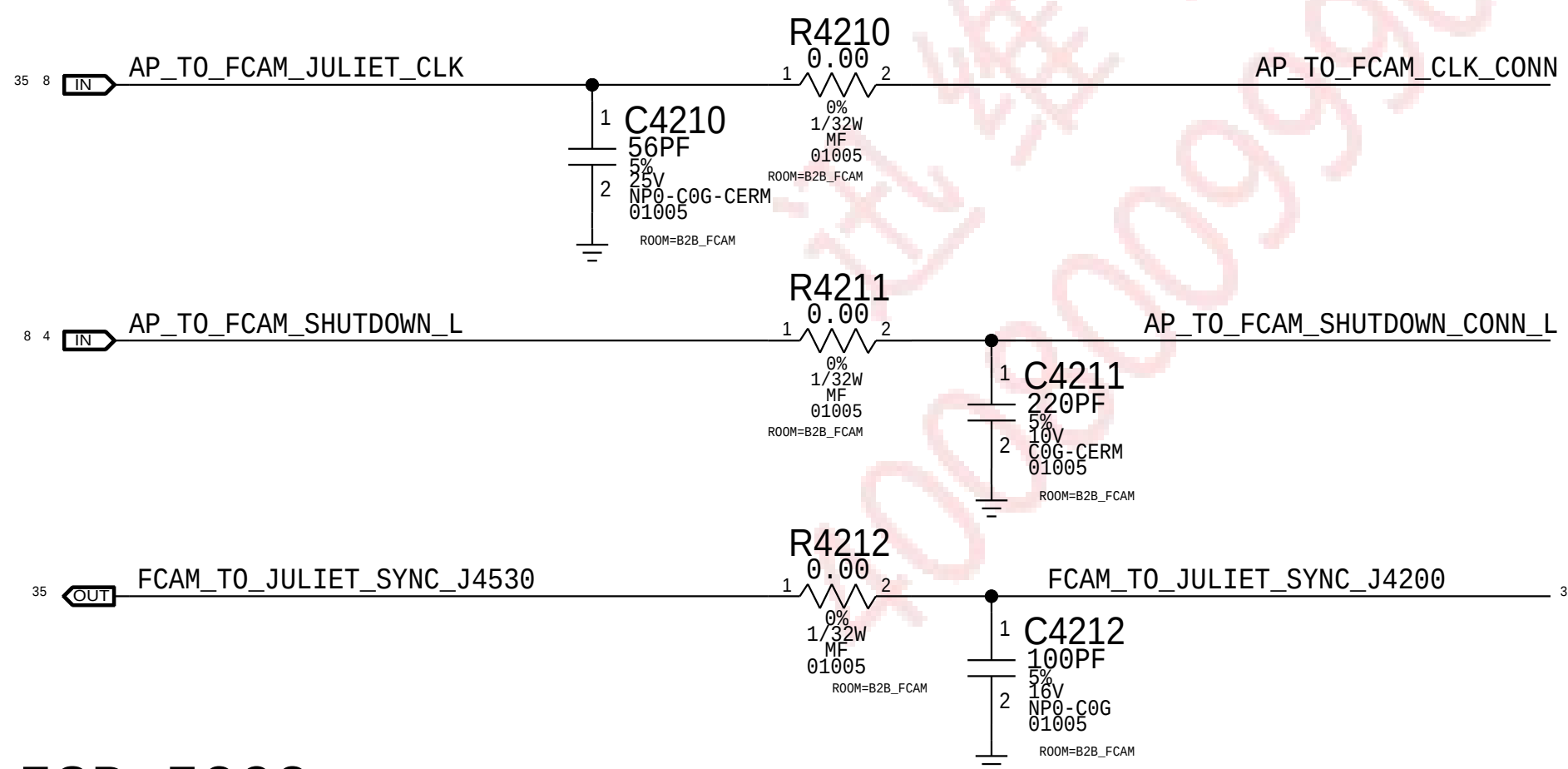
NEW HAMPSHIRE POWER

FCAM Connector

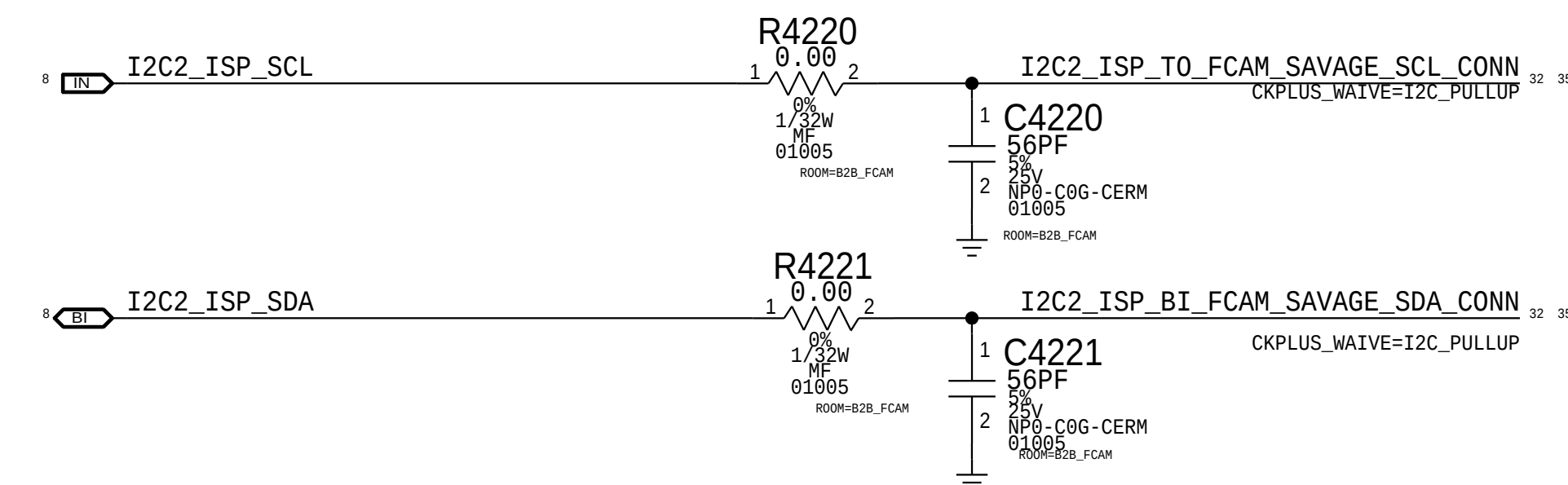
Rcpt: 516S00244 <-- This one on MLB
Plug: 516S00245



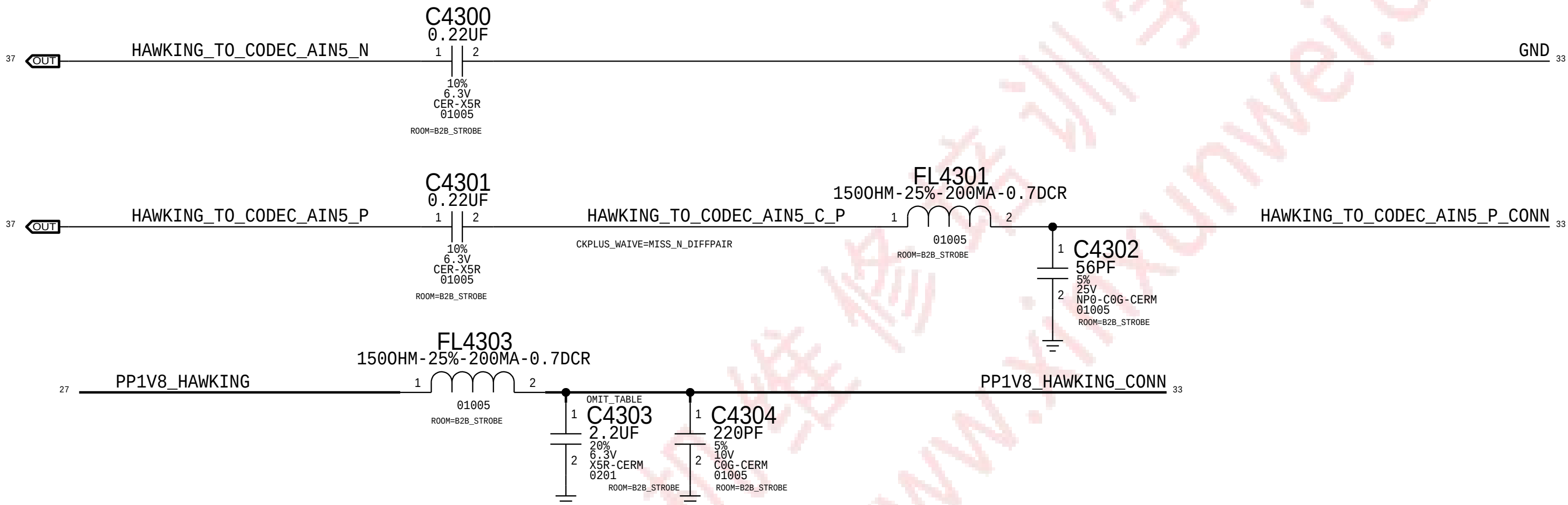
FCAM I/O



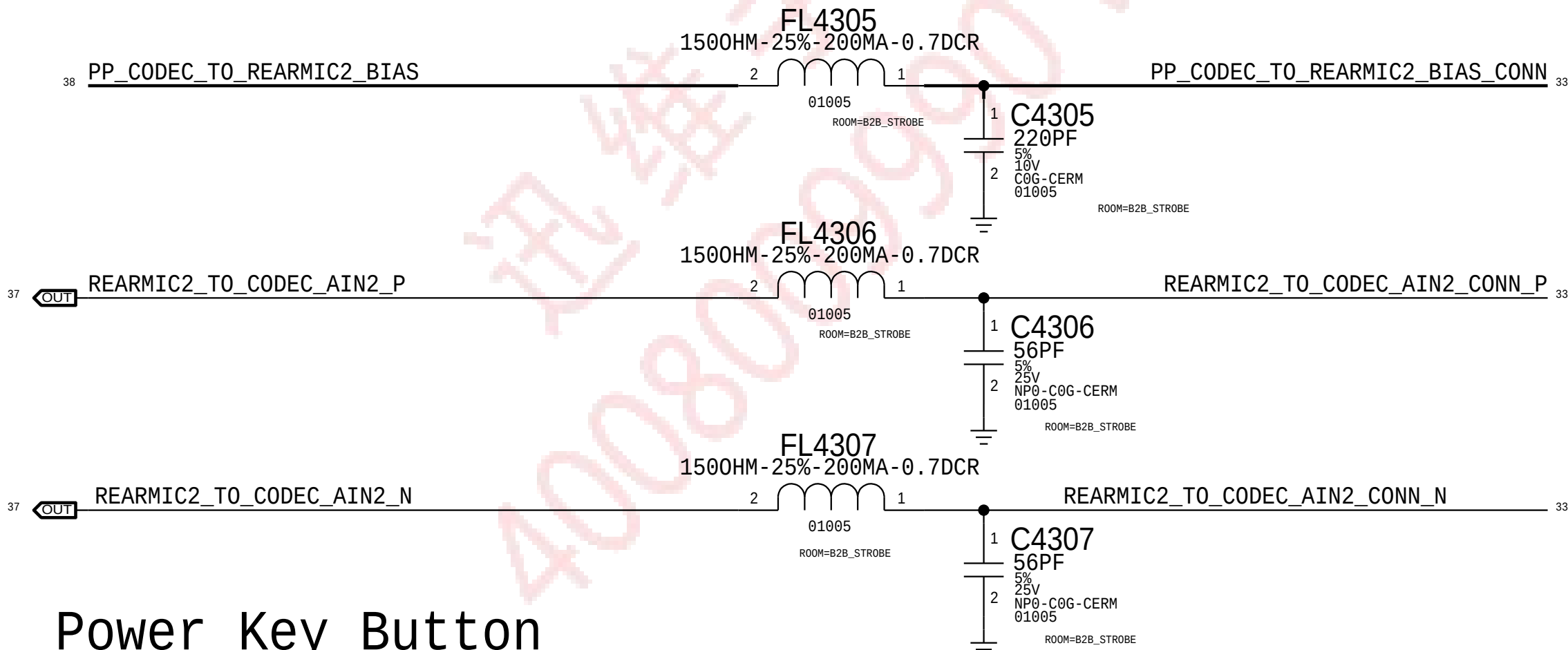
ISP I2C2



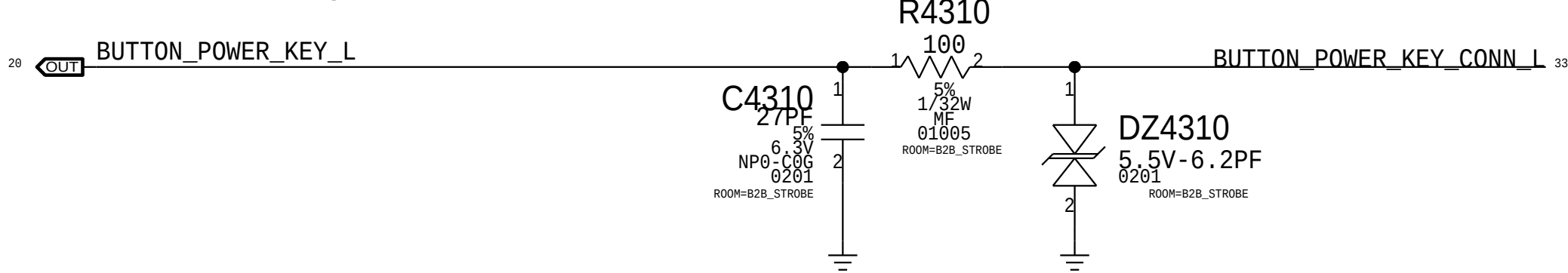
HAWKING



MIC2 (ANC REF)

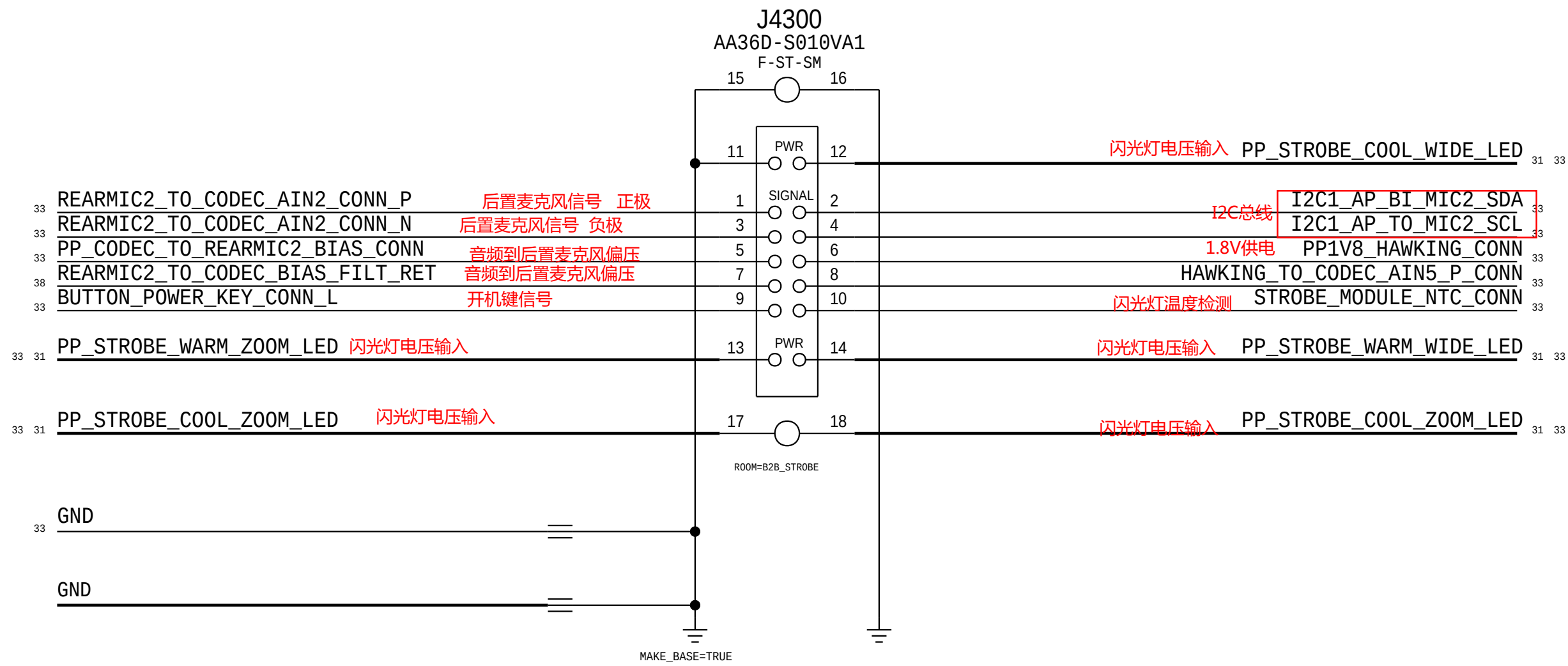


Power Key Button

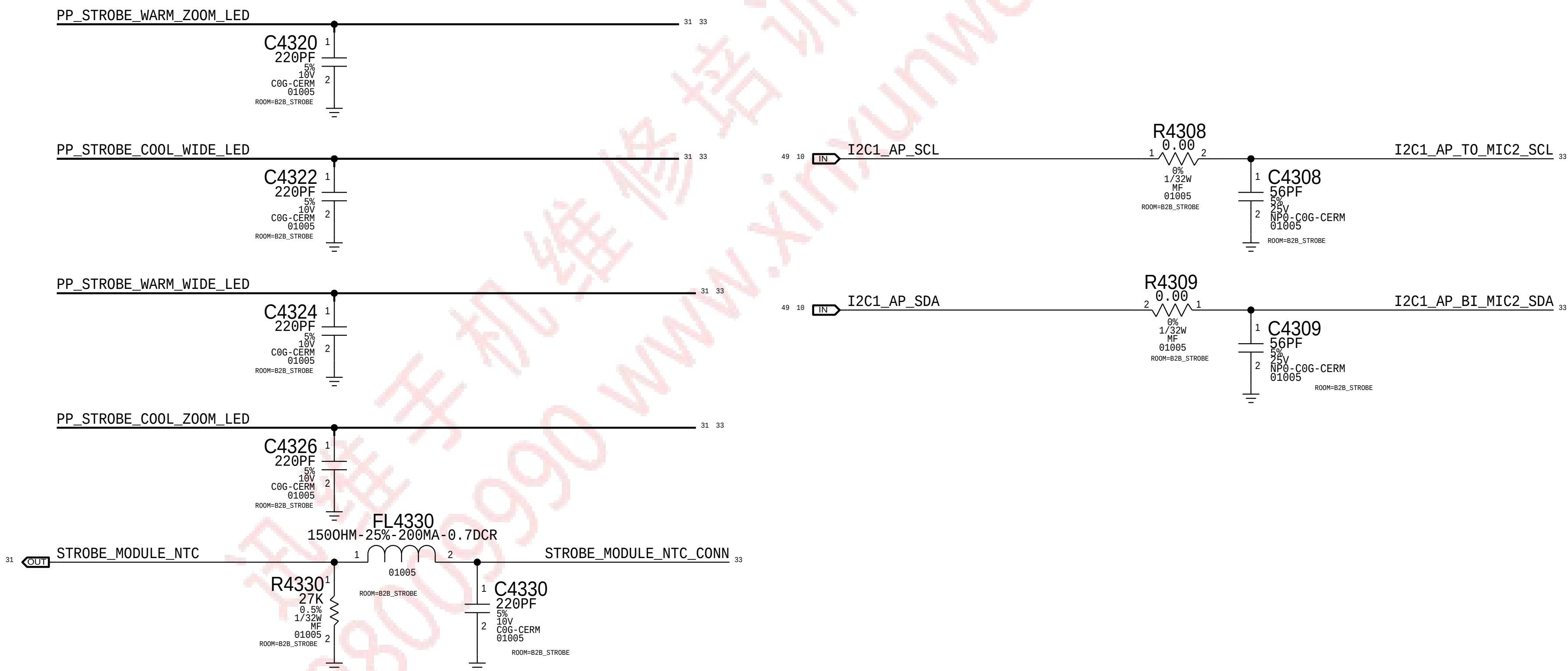


Strobe Connector

Rcpt: 516S00267 <-- This one on MLB
Plug: 516S00268
开机键、音量键闪光灯接口



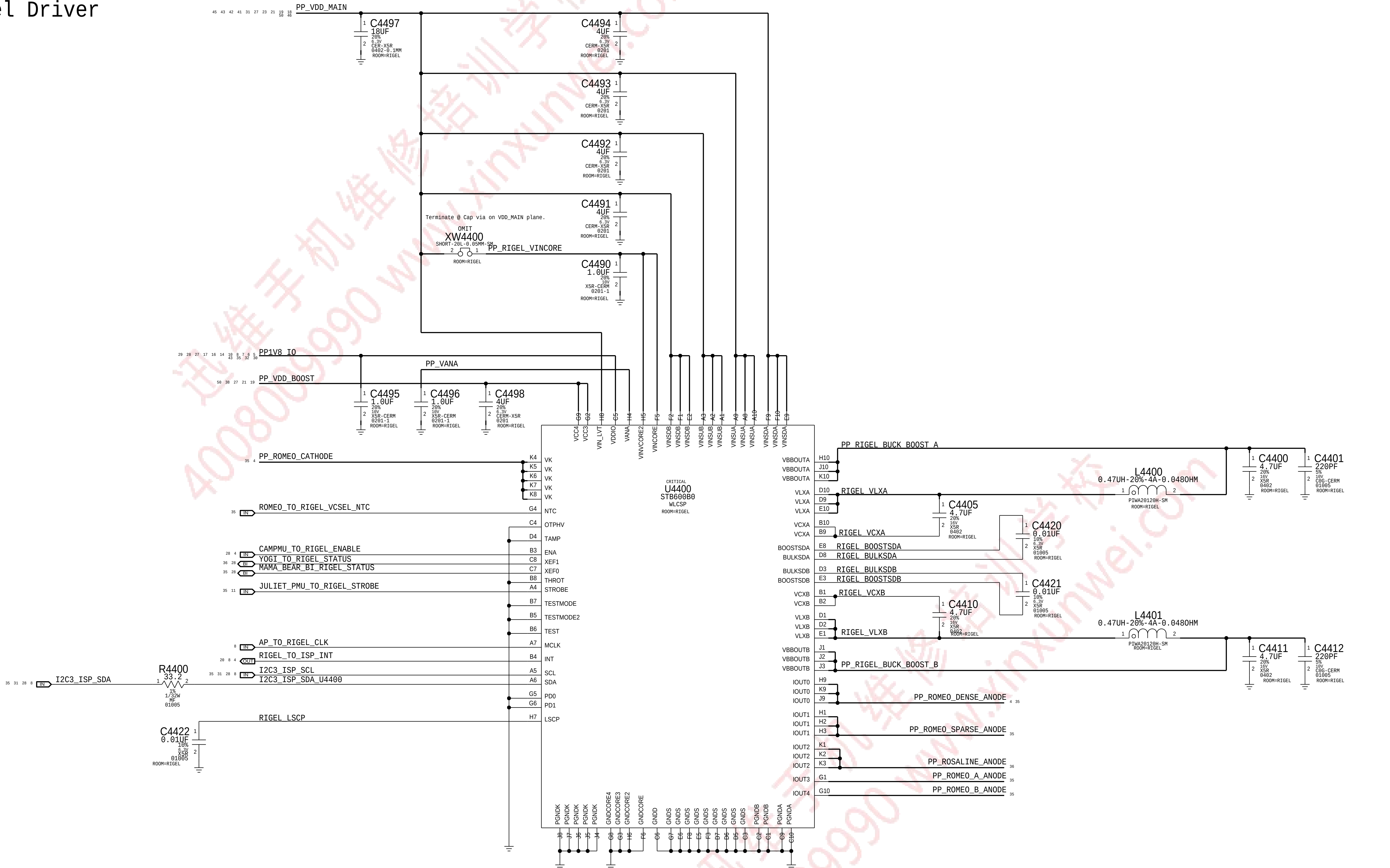
Strobe Filtering





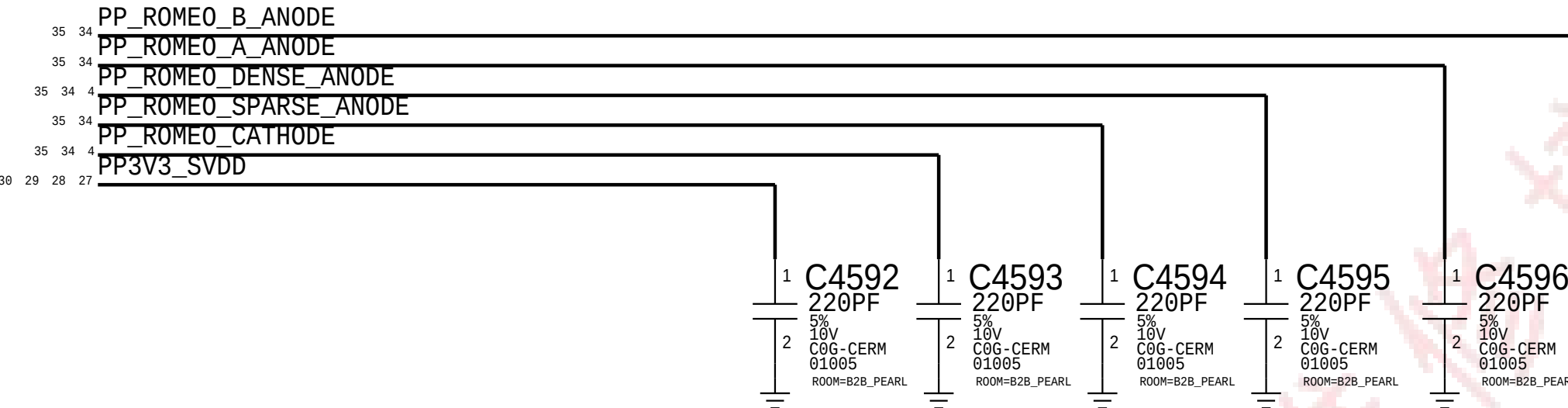
Rigel Driver

8 7 6 5 4 3 2 1

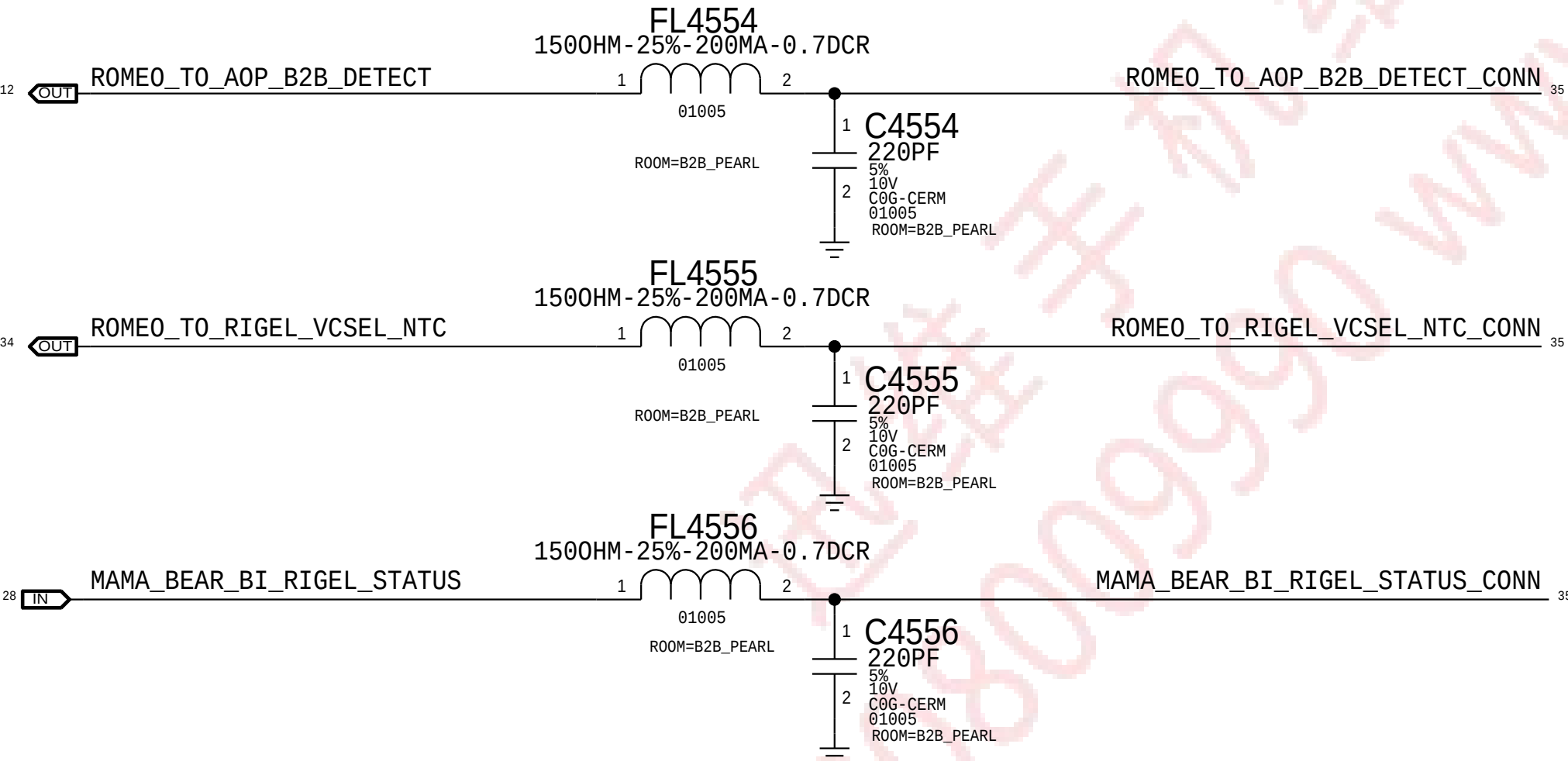


8 7 6 5 4 3 2 1

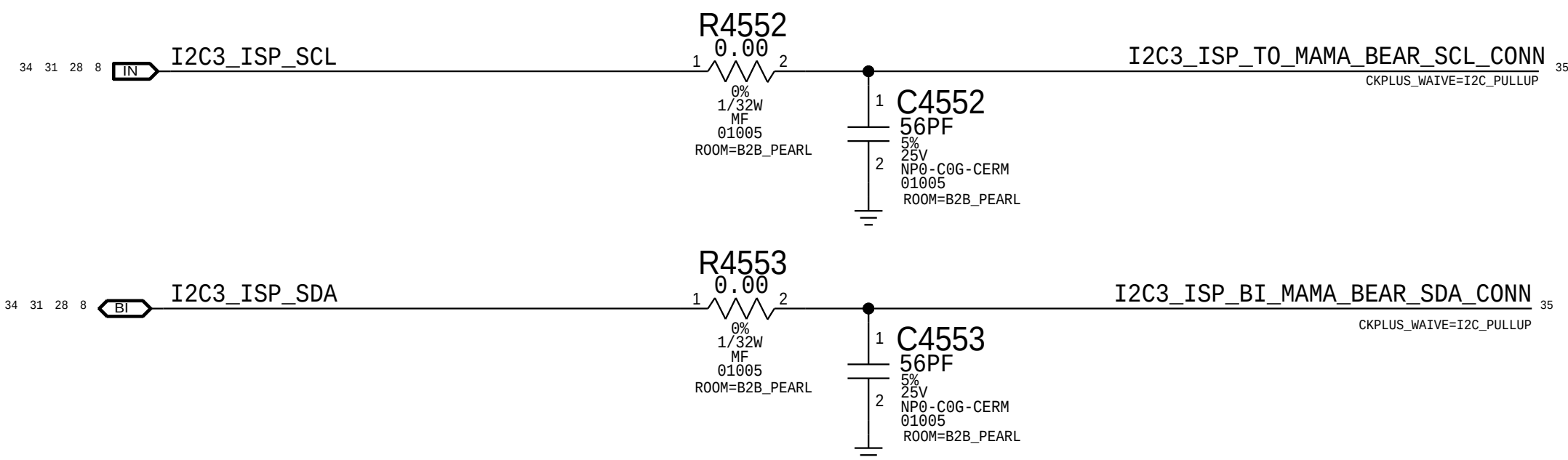
Romeo Power Filtering



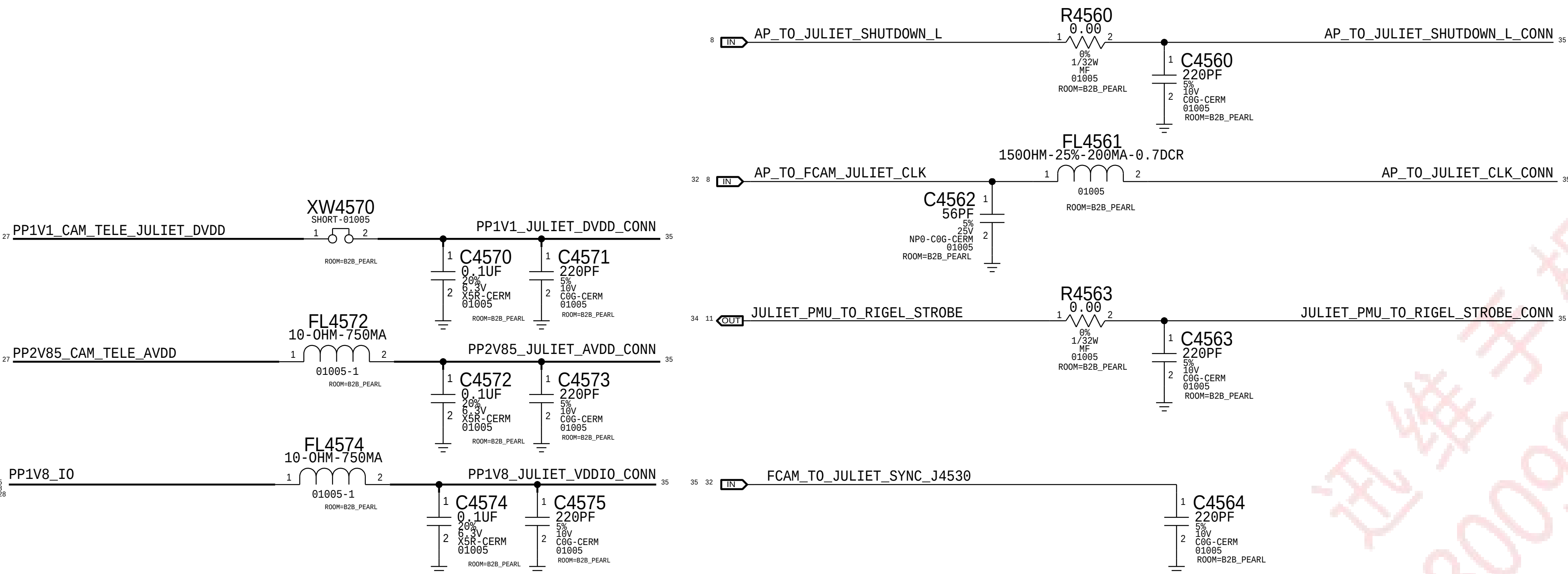
Romeo I/O



ISP I2C3



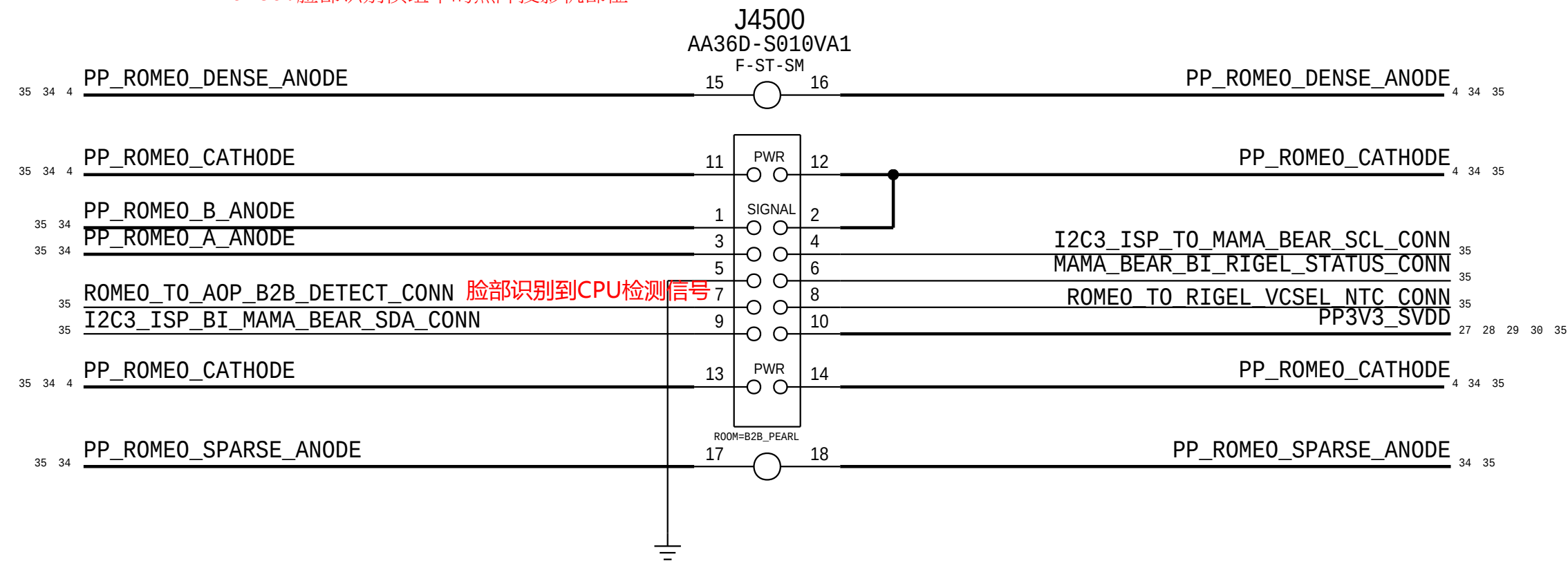
Juliet Power and I/O



Romeo Connector

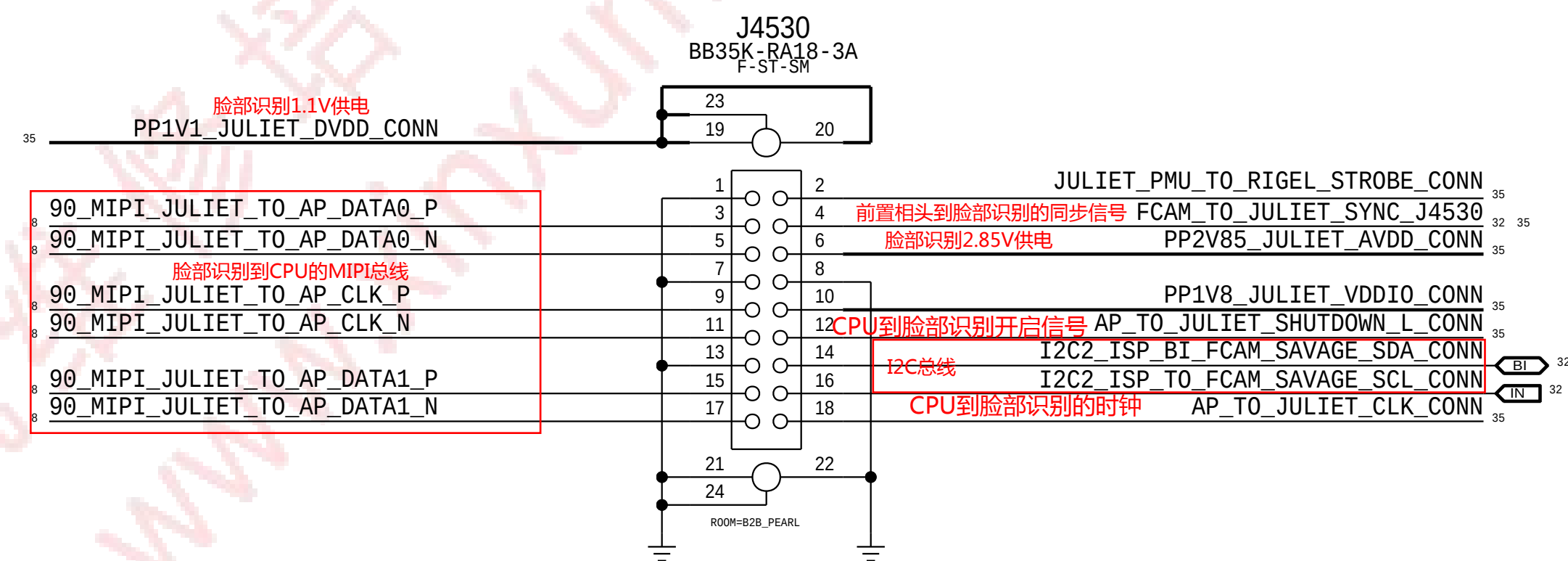
Rcpt: 516S00267 <-- This one on MLB
Plug: 516S00268

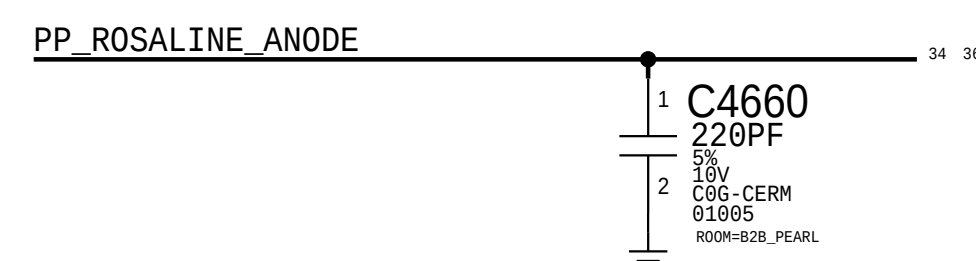
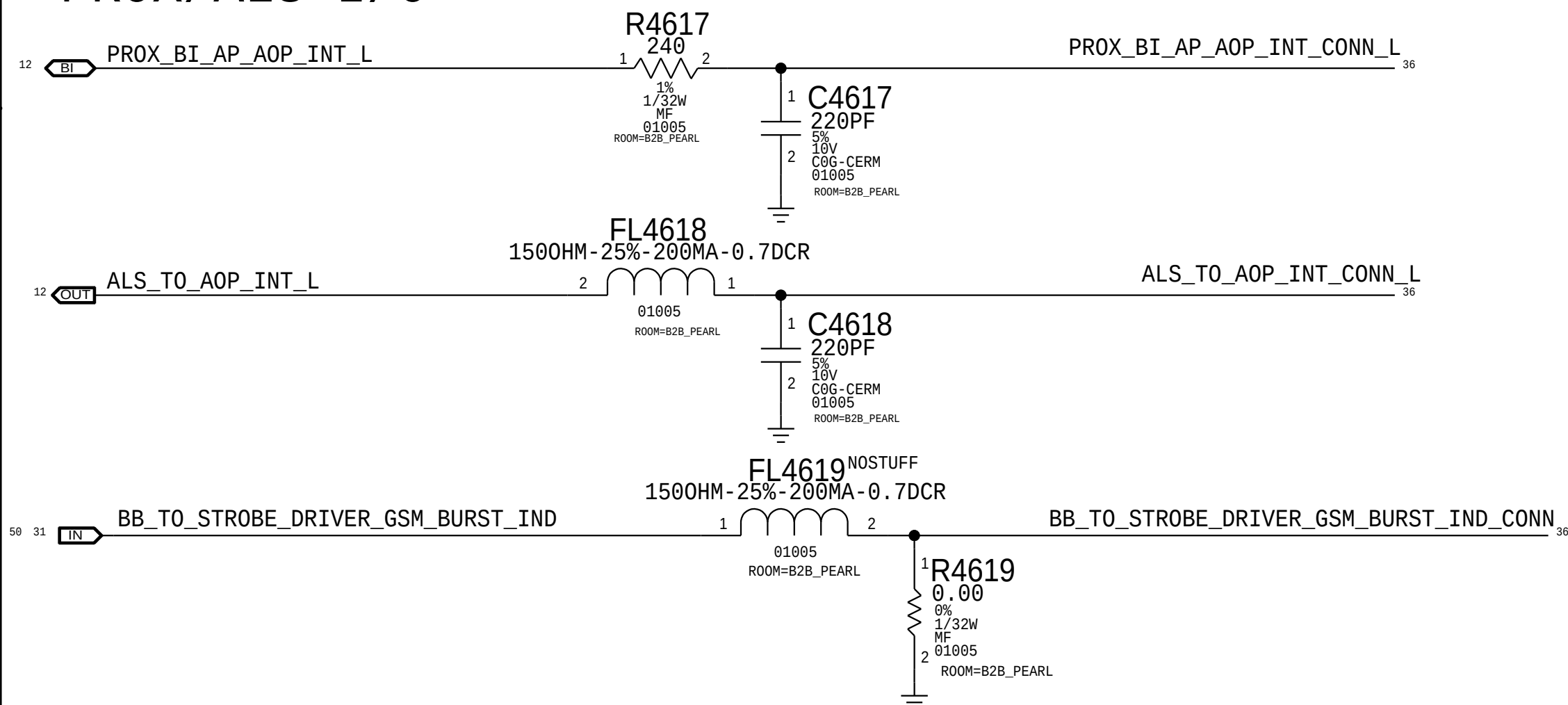
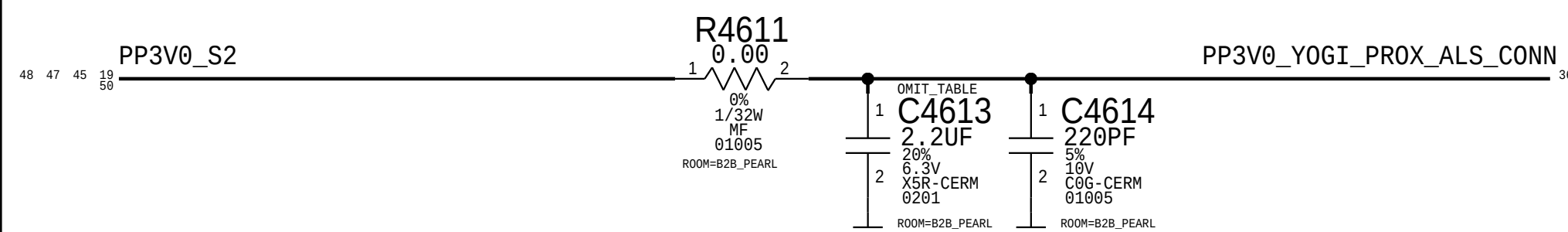
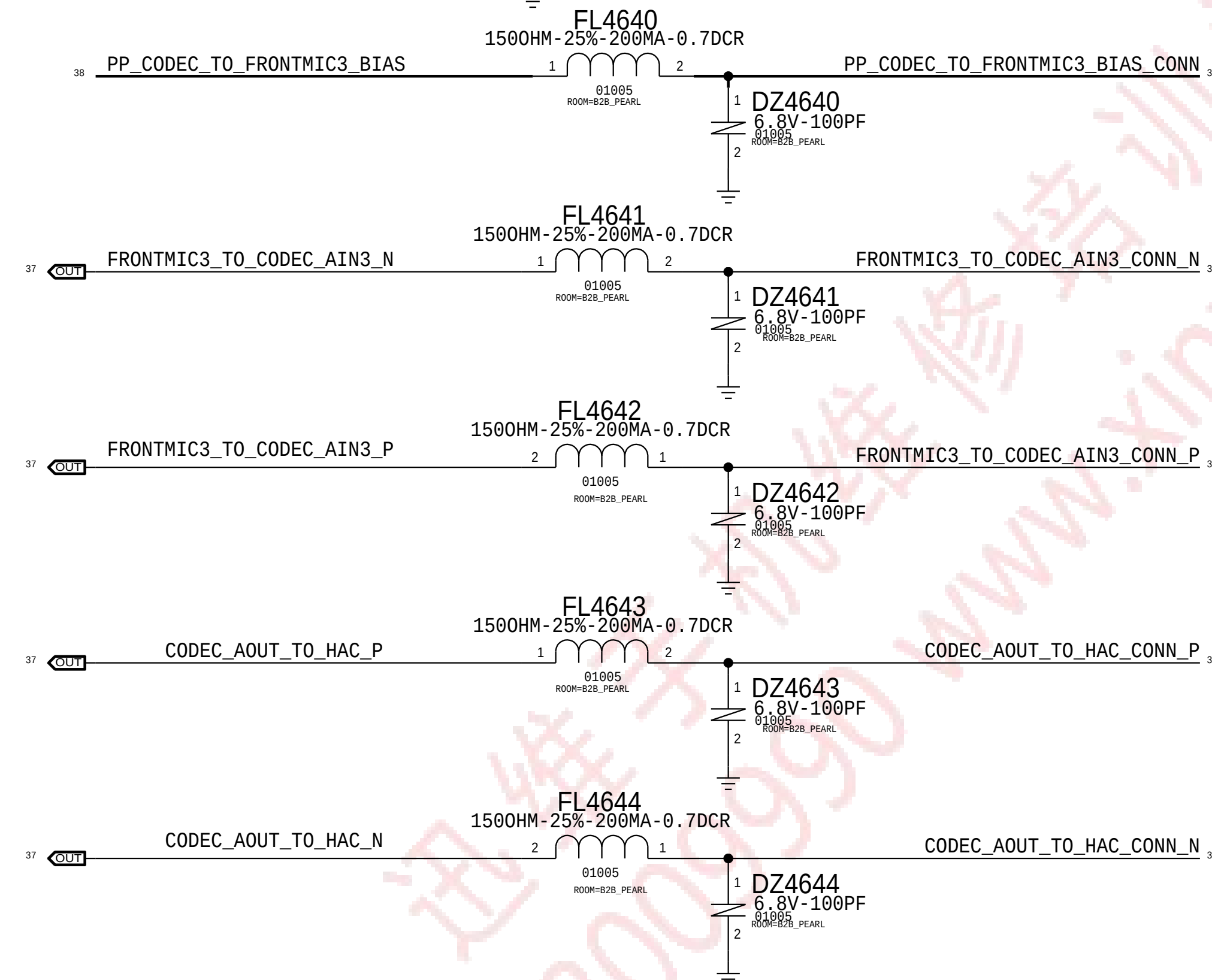
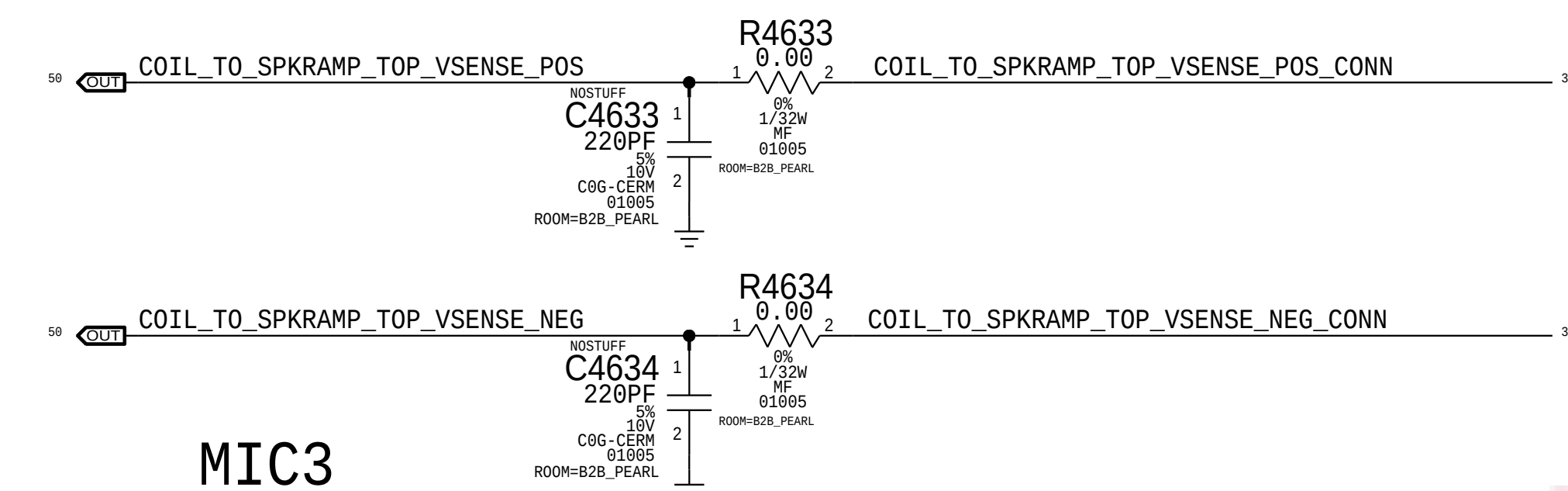
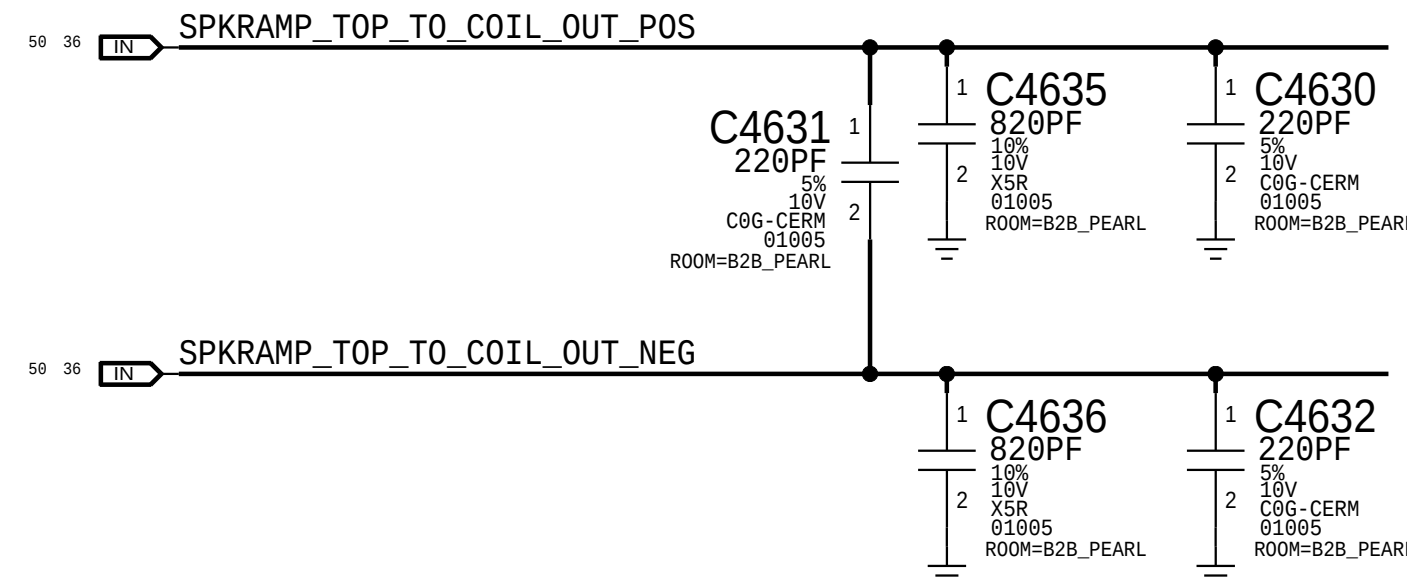
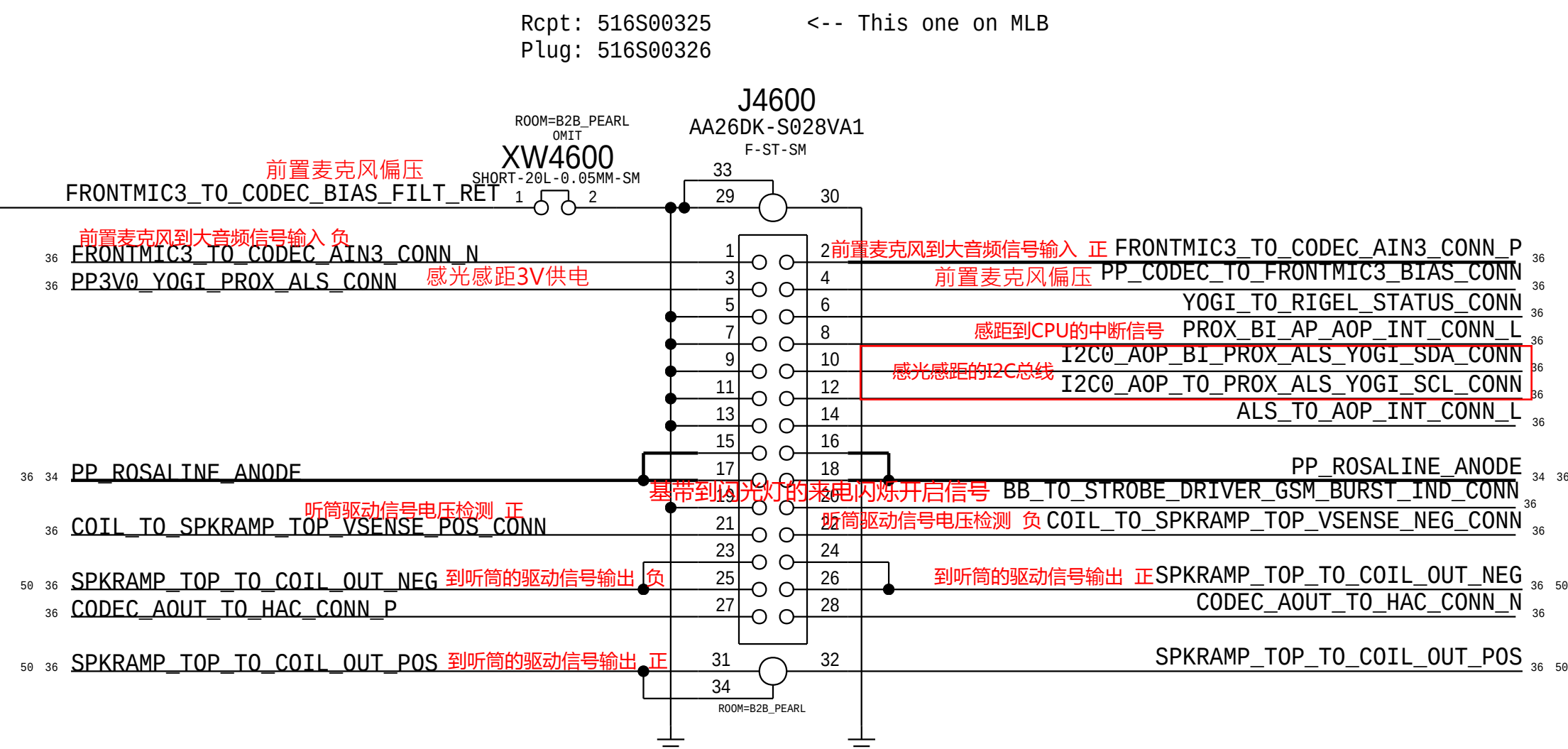
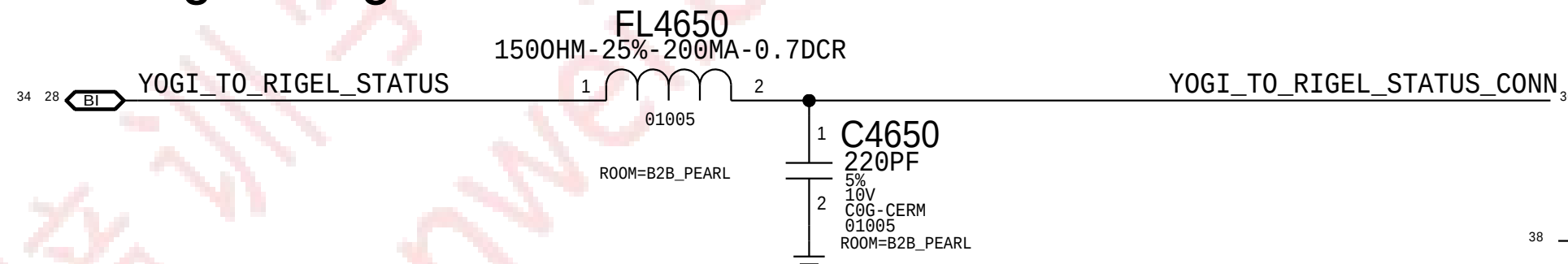
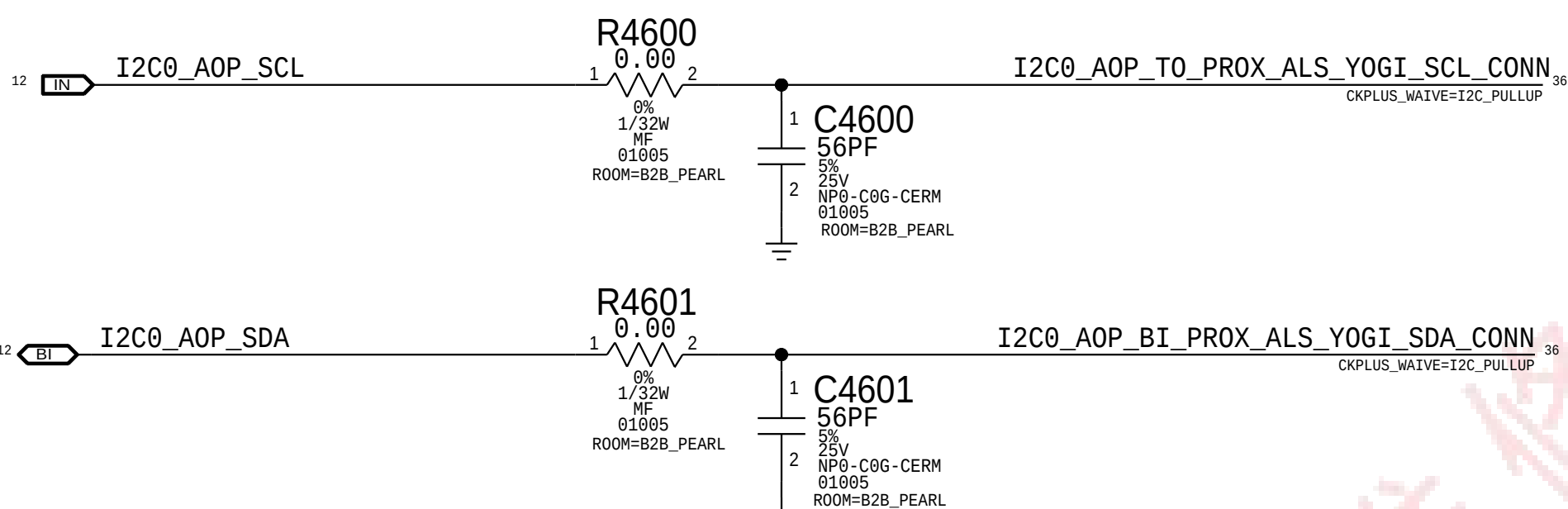
Romeo: 脸部识别模组中的点阵投影机部位



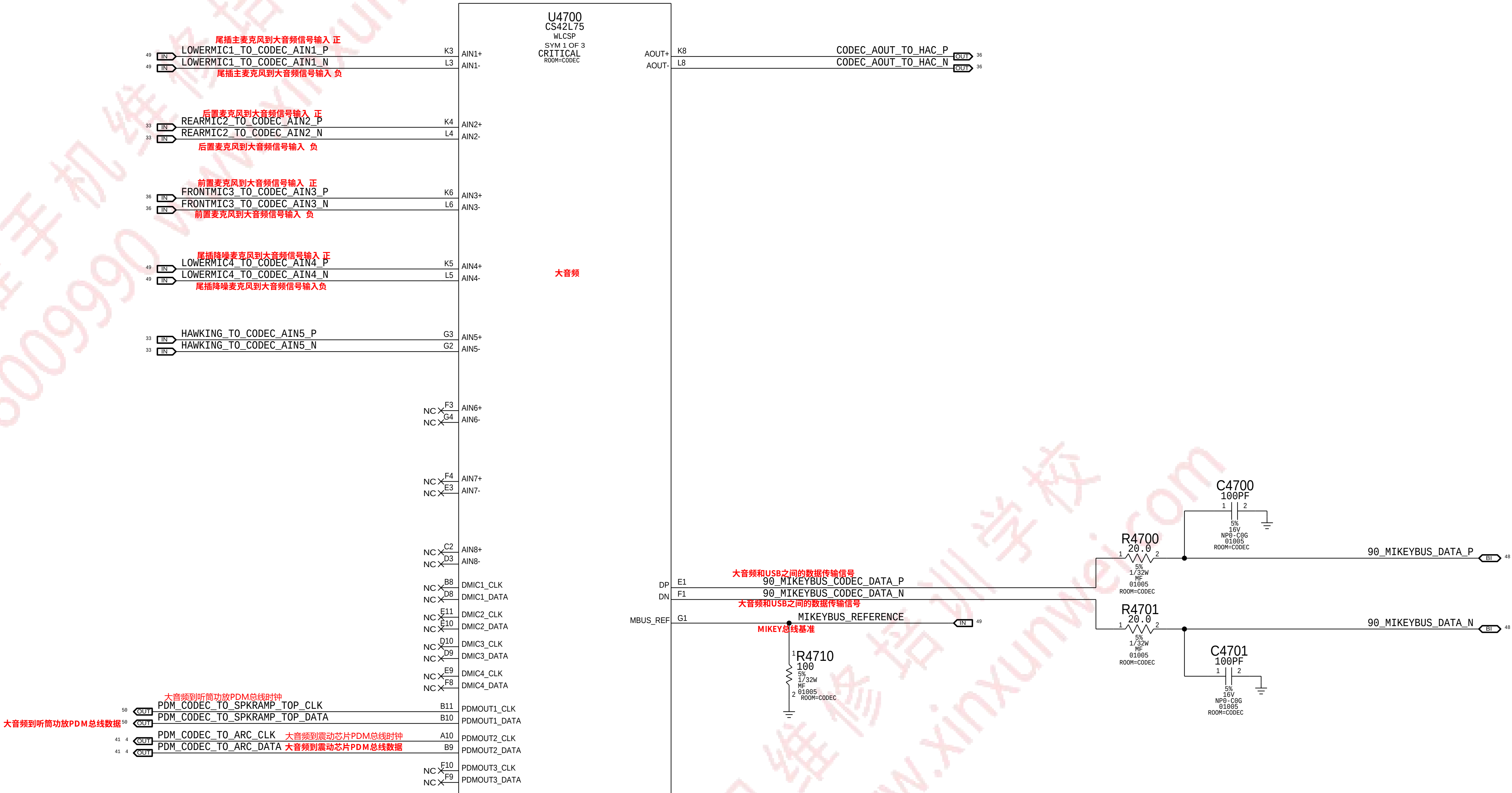
Juliet Connector

Rcpt: 516S00244 <-- This one on MLB
Plug: 516S00245
Juliet: 脸部识别功能中读取人脸数据的红外线相机镜头

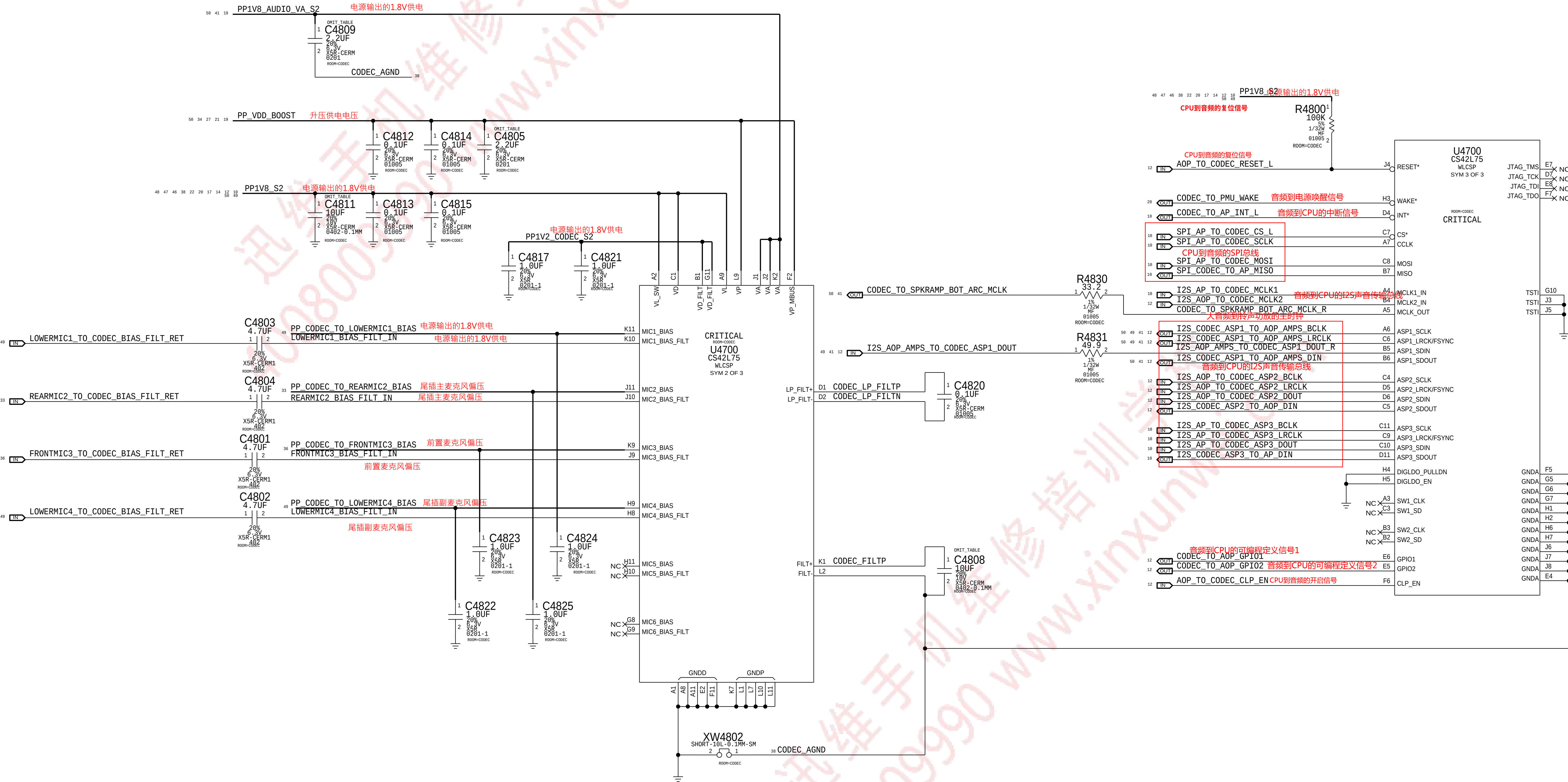




CALLAN AUDIO CODEC (ANALOG INPUTS & OUTPUTS)



CALLAN AUDIO CODEC (POWER & I/O)





8

7

6

5

4

3

2

1

D

C

B

A

D

C

B

South Speaker Amplifier on MLB Bottom

8

7

6

5

4

3

2

1



8

7

6

5

4

3

2

1

D

C

B

A

D

C

B

North Speaker Amplifier on MLB Bottom

8

7

6

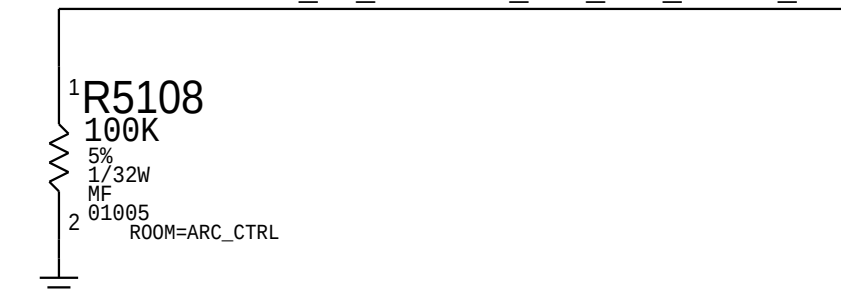
5

4

3

2

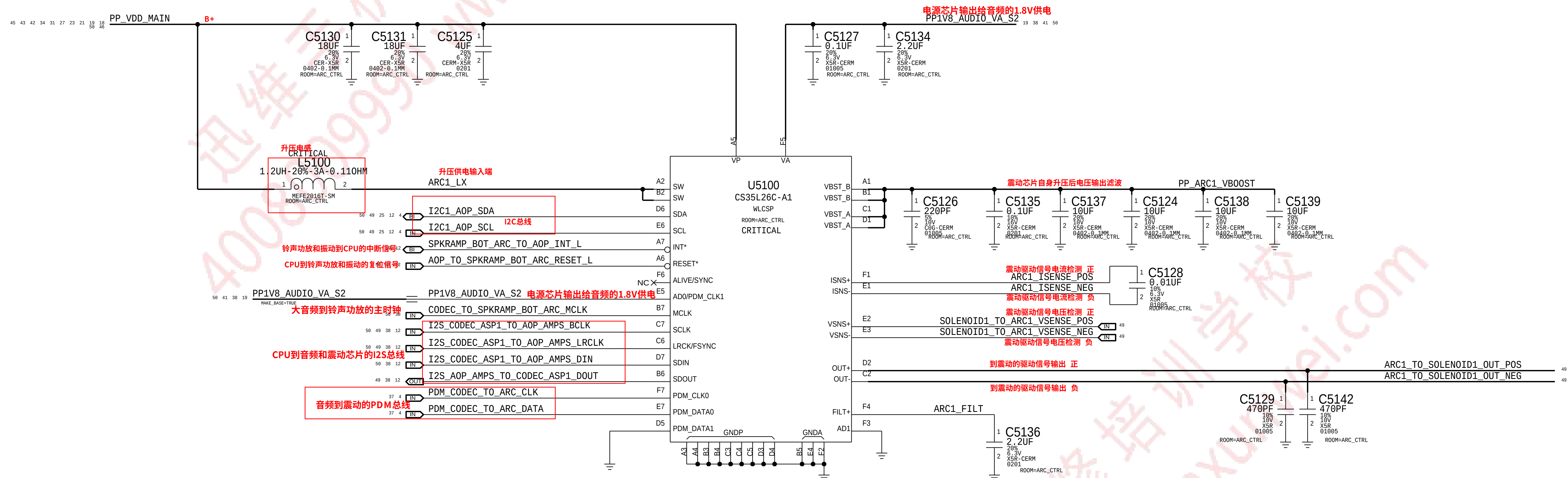
1



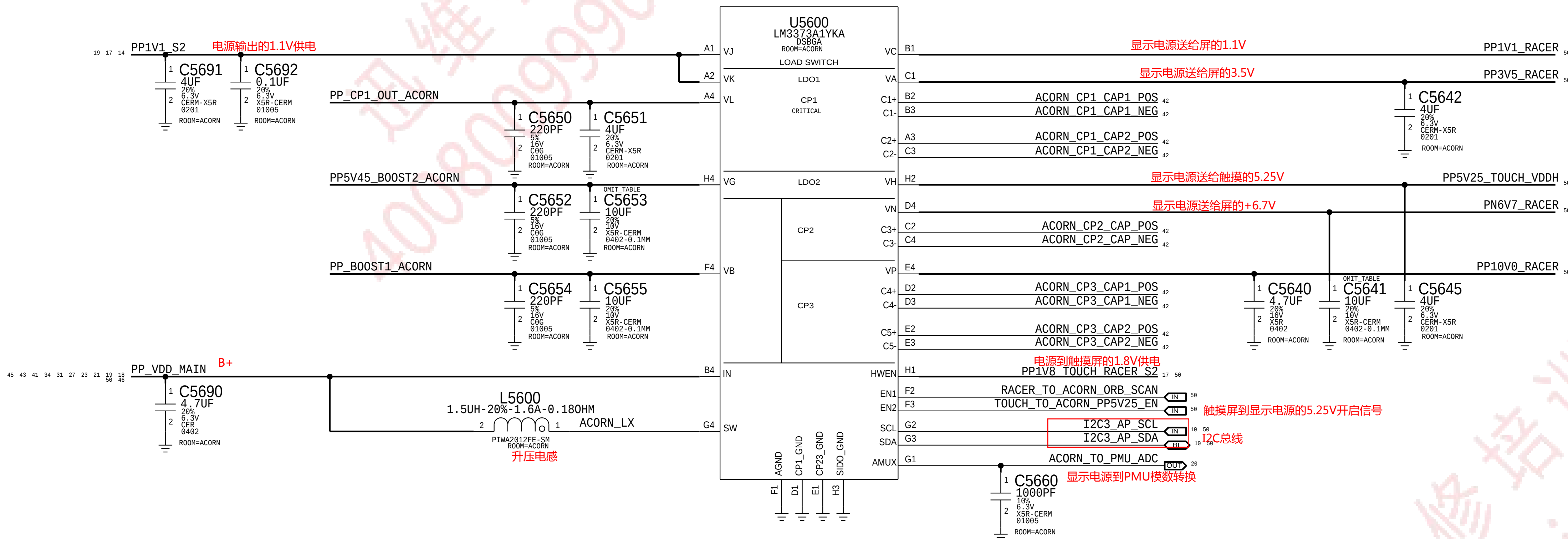
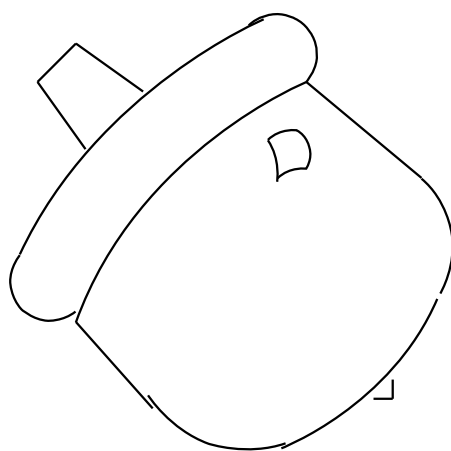
ARC DRIVER

APN: 338S00296

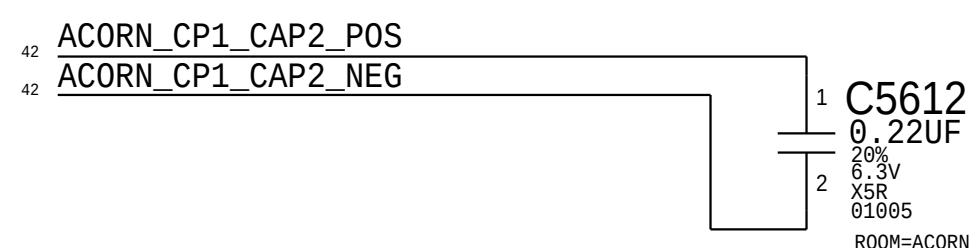
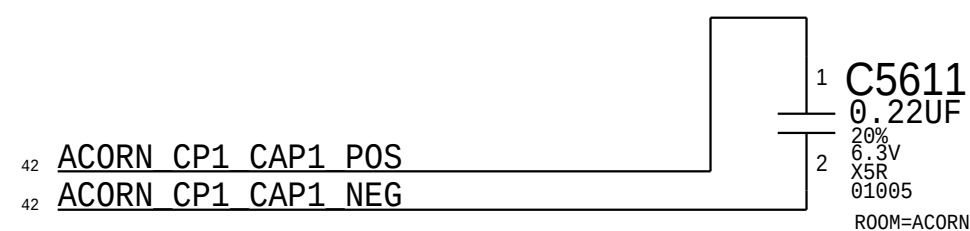
```
I2C ADDRESS: 1000 001x
                0x82
```



Acorn PMU



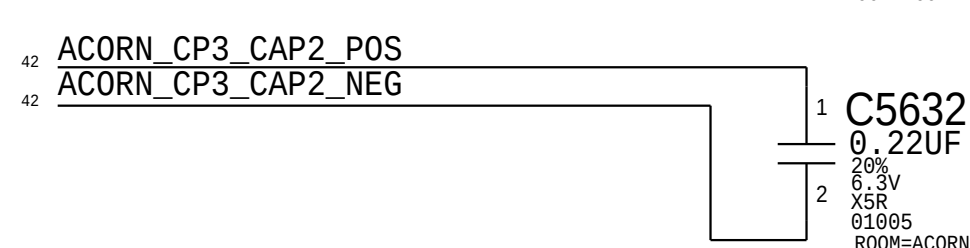
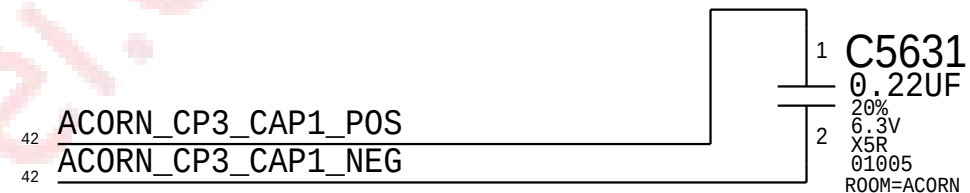
Charge Pump 1 Caps



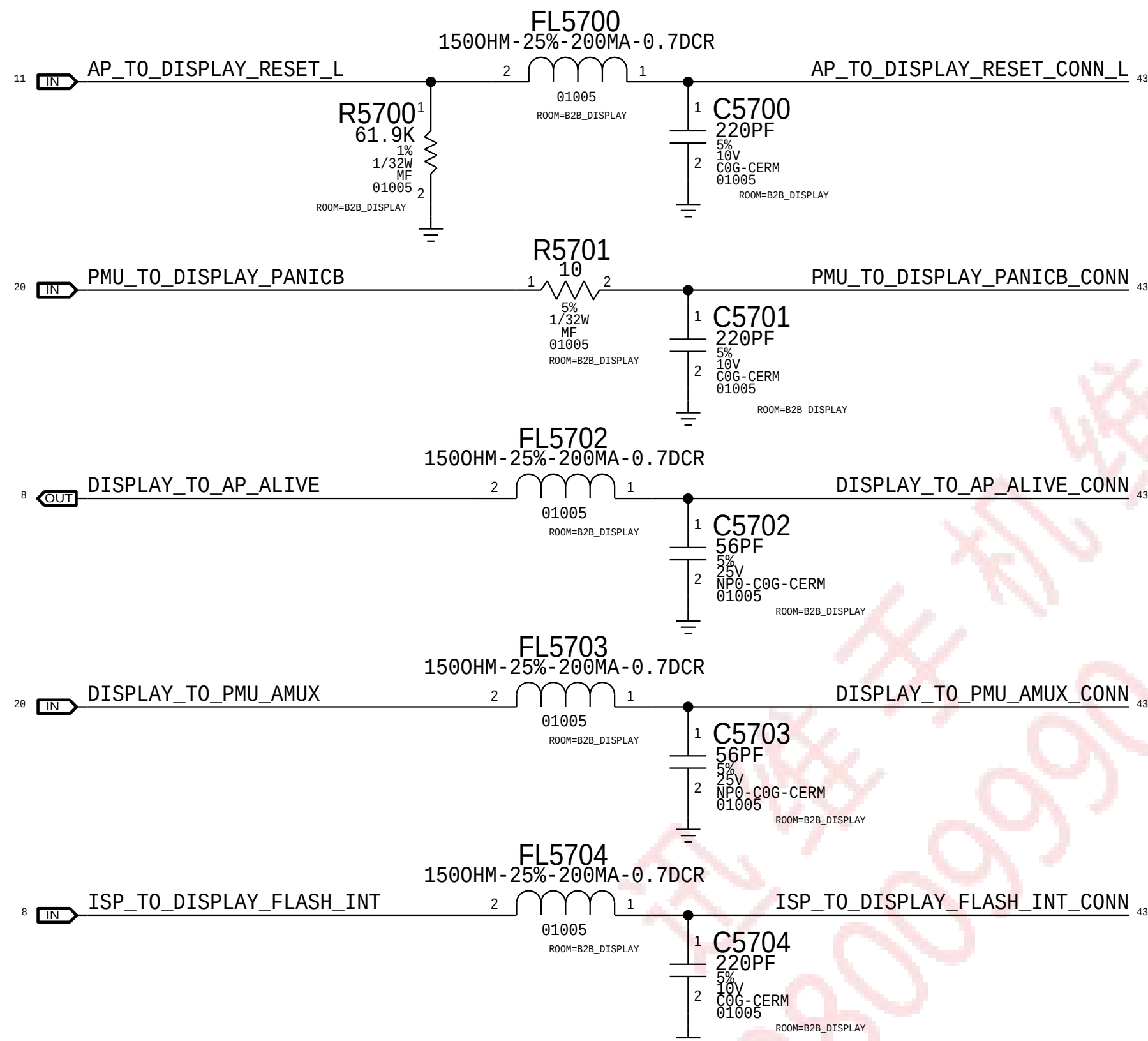
Charge Pump 2 Caps



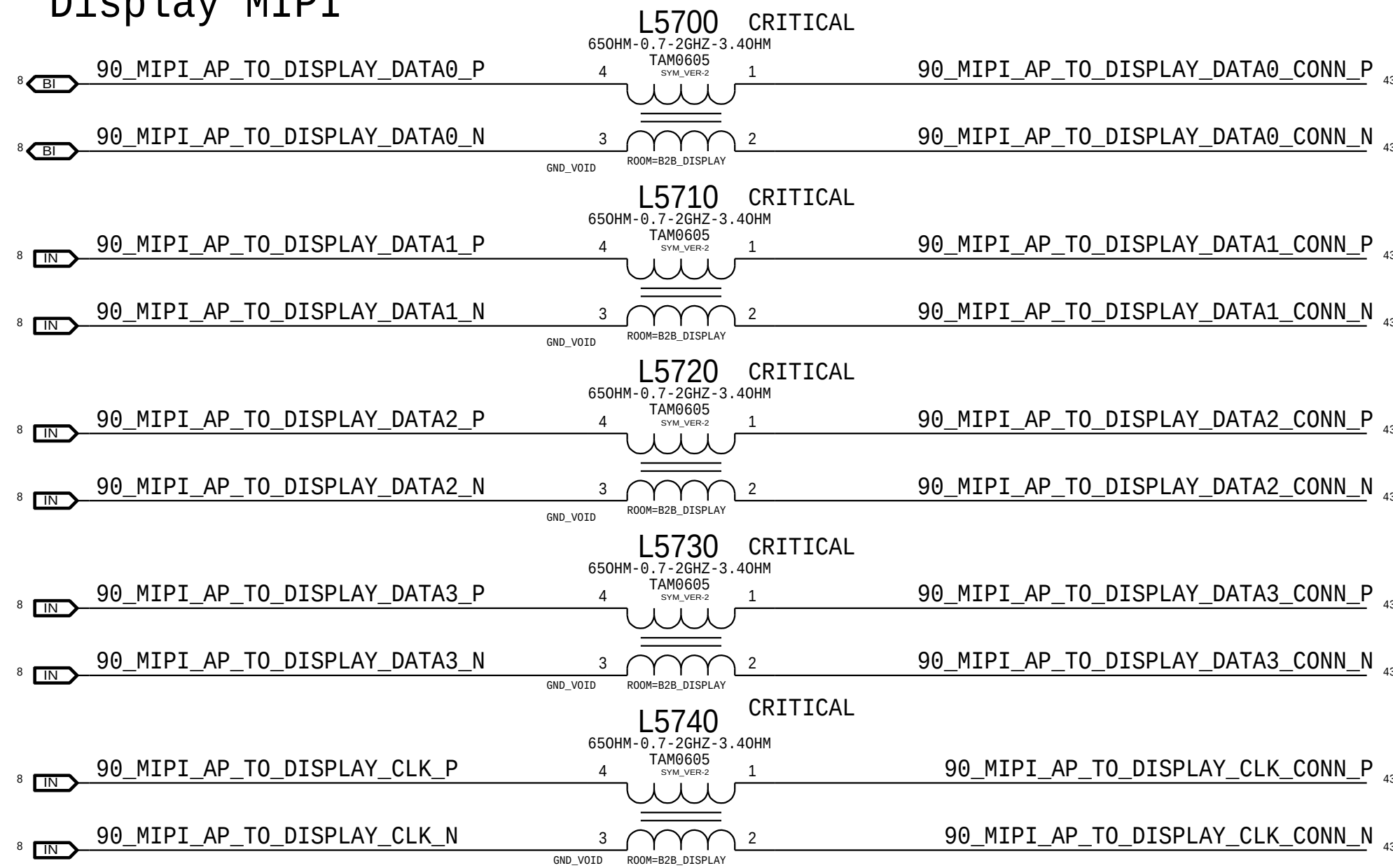
Charge Pump 3 Caps



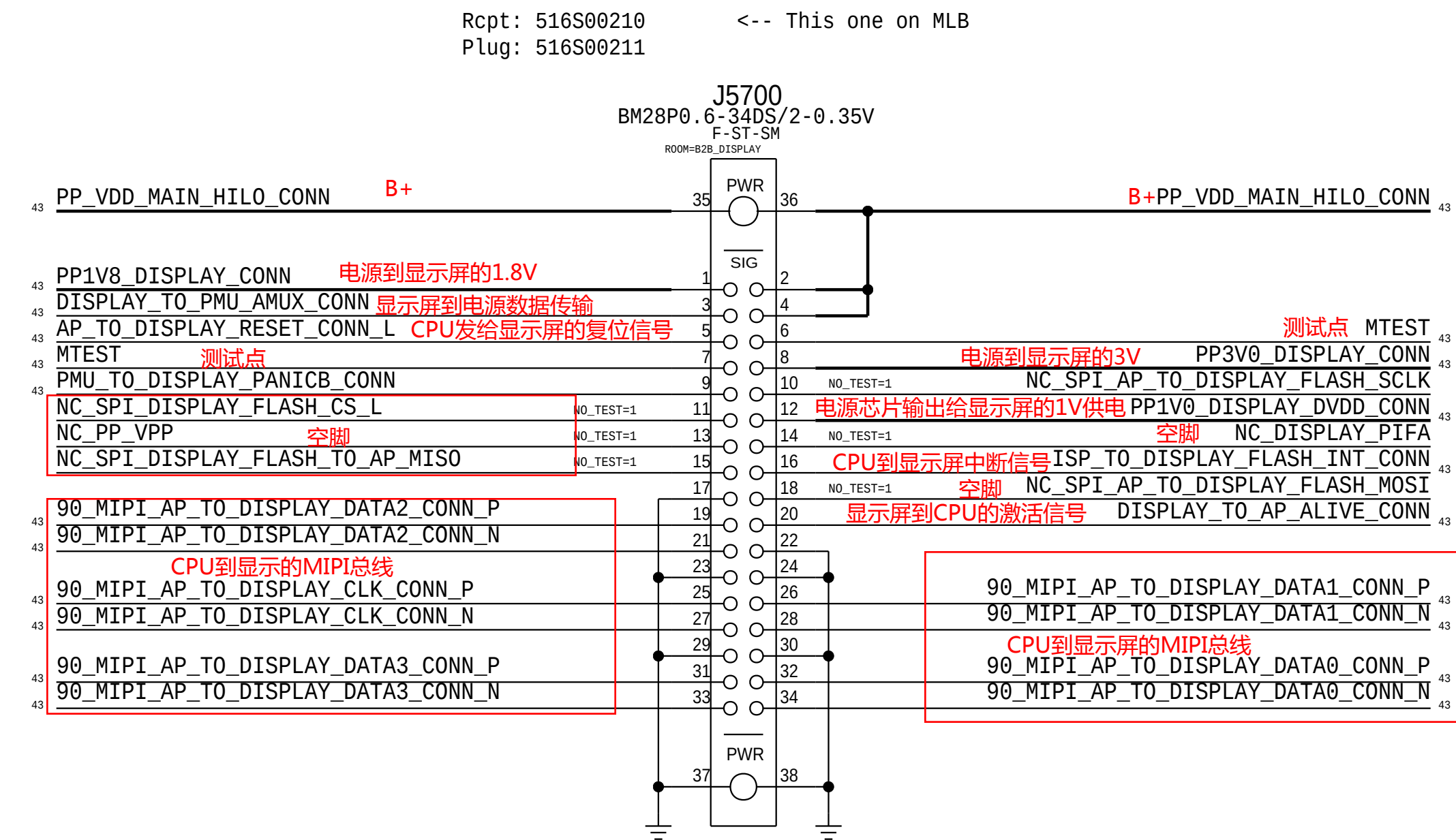
Display Control Signals



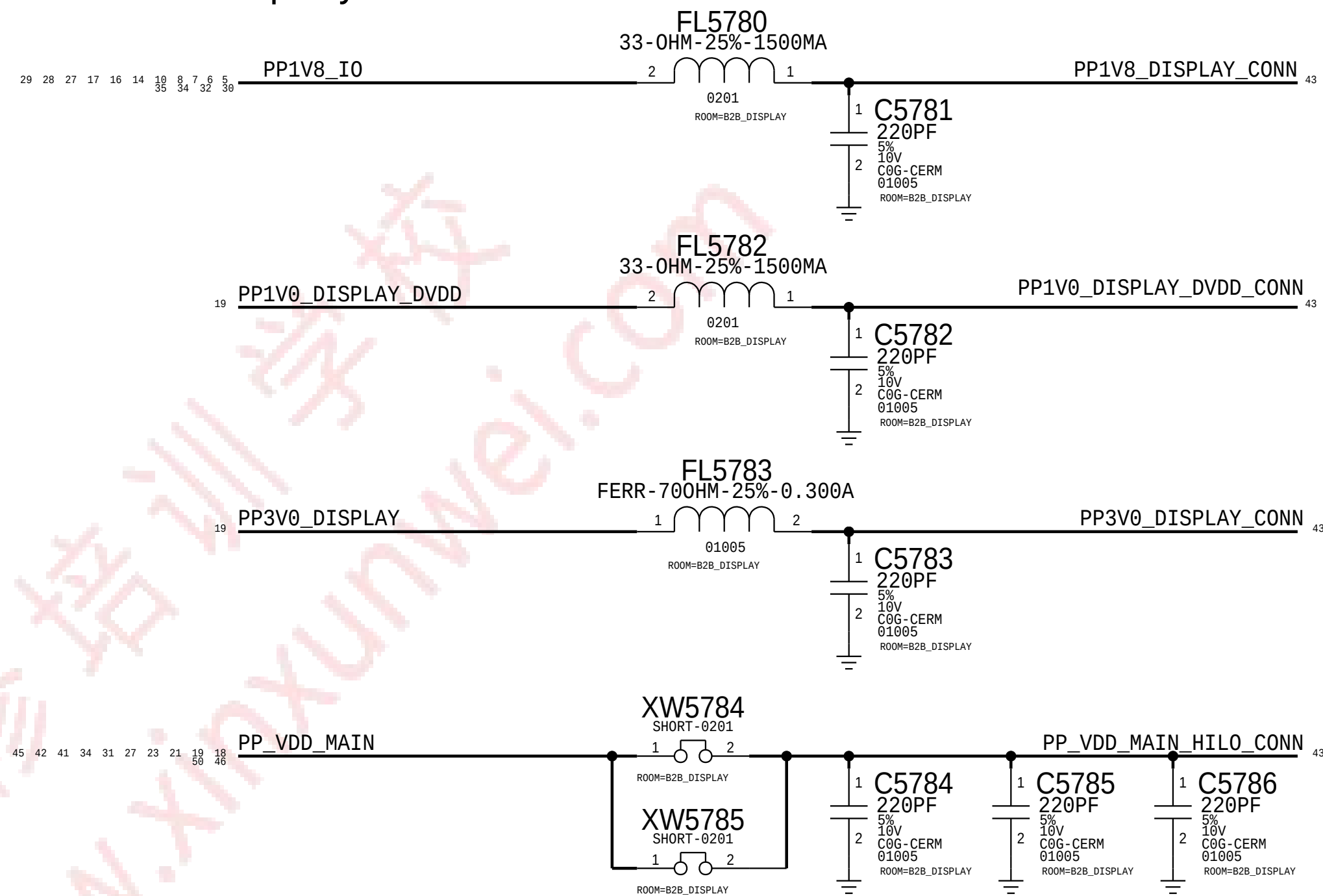
Display MIPI



Display Flex Connector



Display Power





D

C

B

1

2

3

4

5

6

7

8

D

C

B

A

1

2

3

4

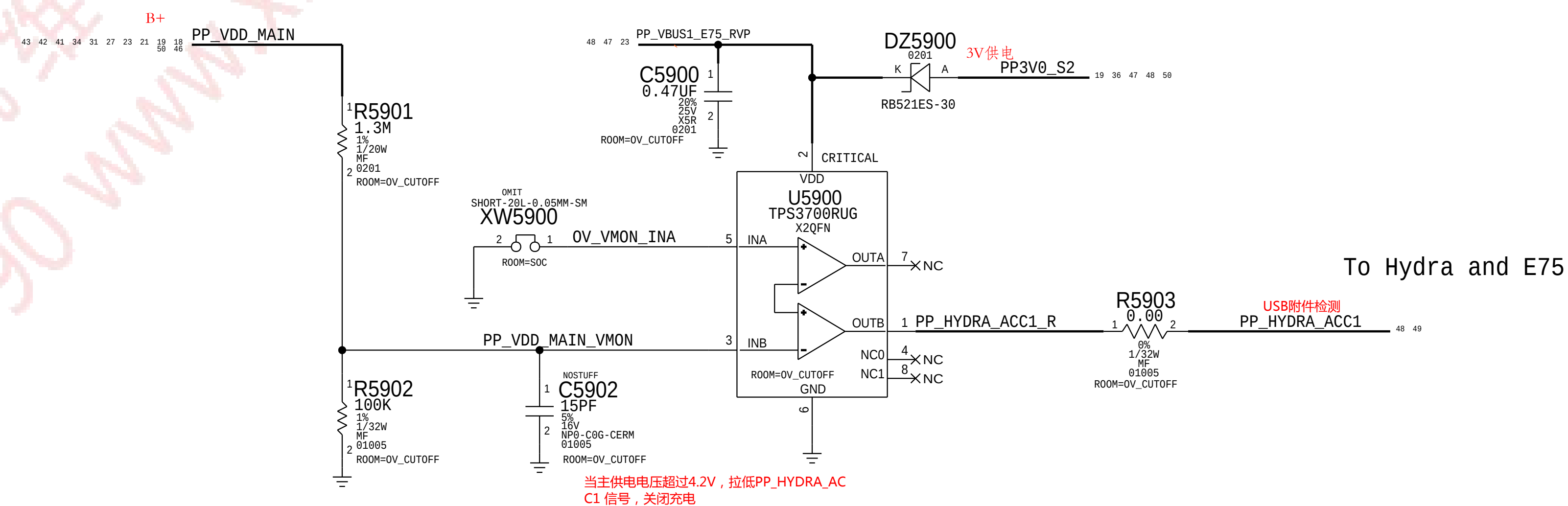
5

6

7

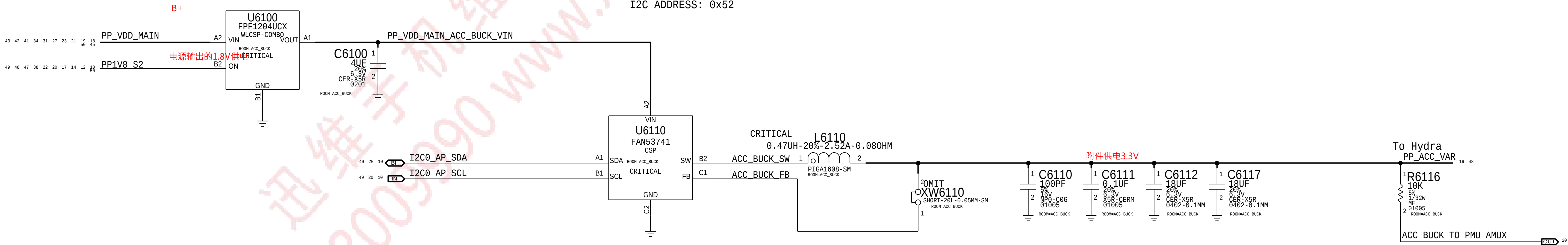
8

Orb + Touch Connector on MLB Bottom

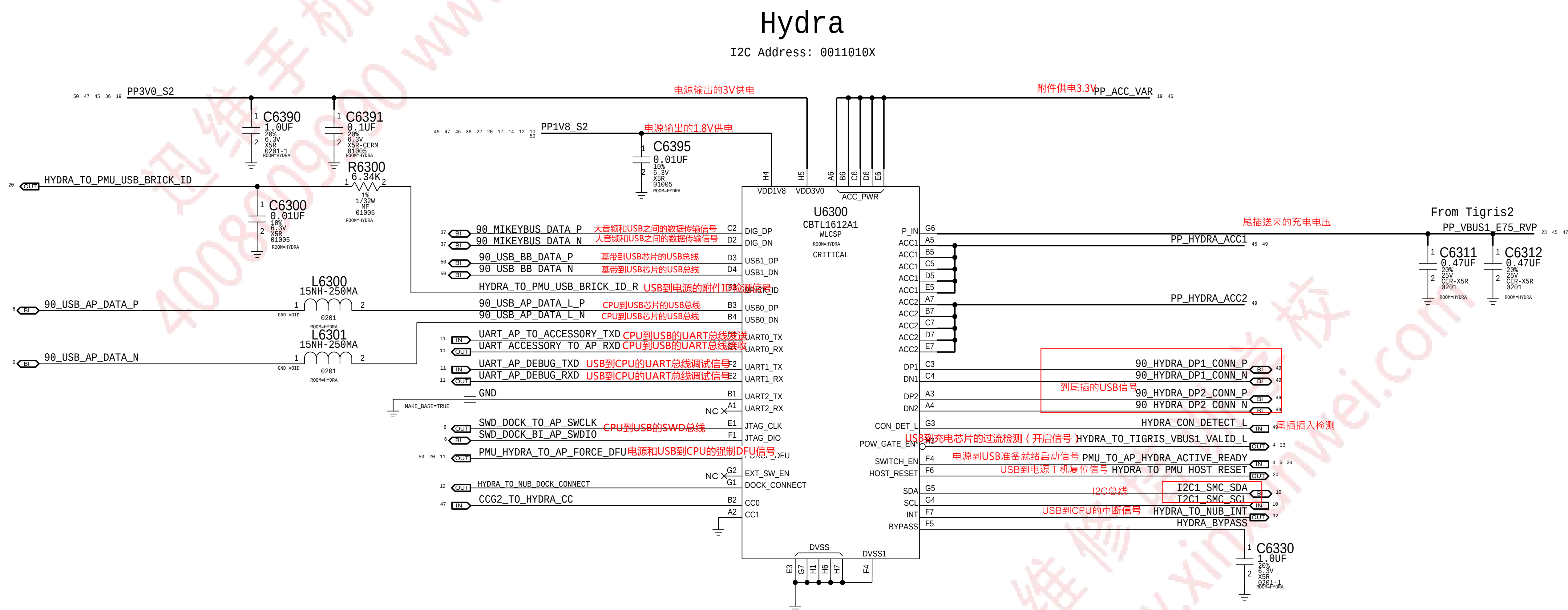


ACCESSORY BUCK

I2C ADDRESS: 0x52



	4	3	2
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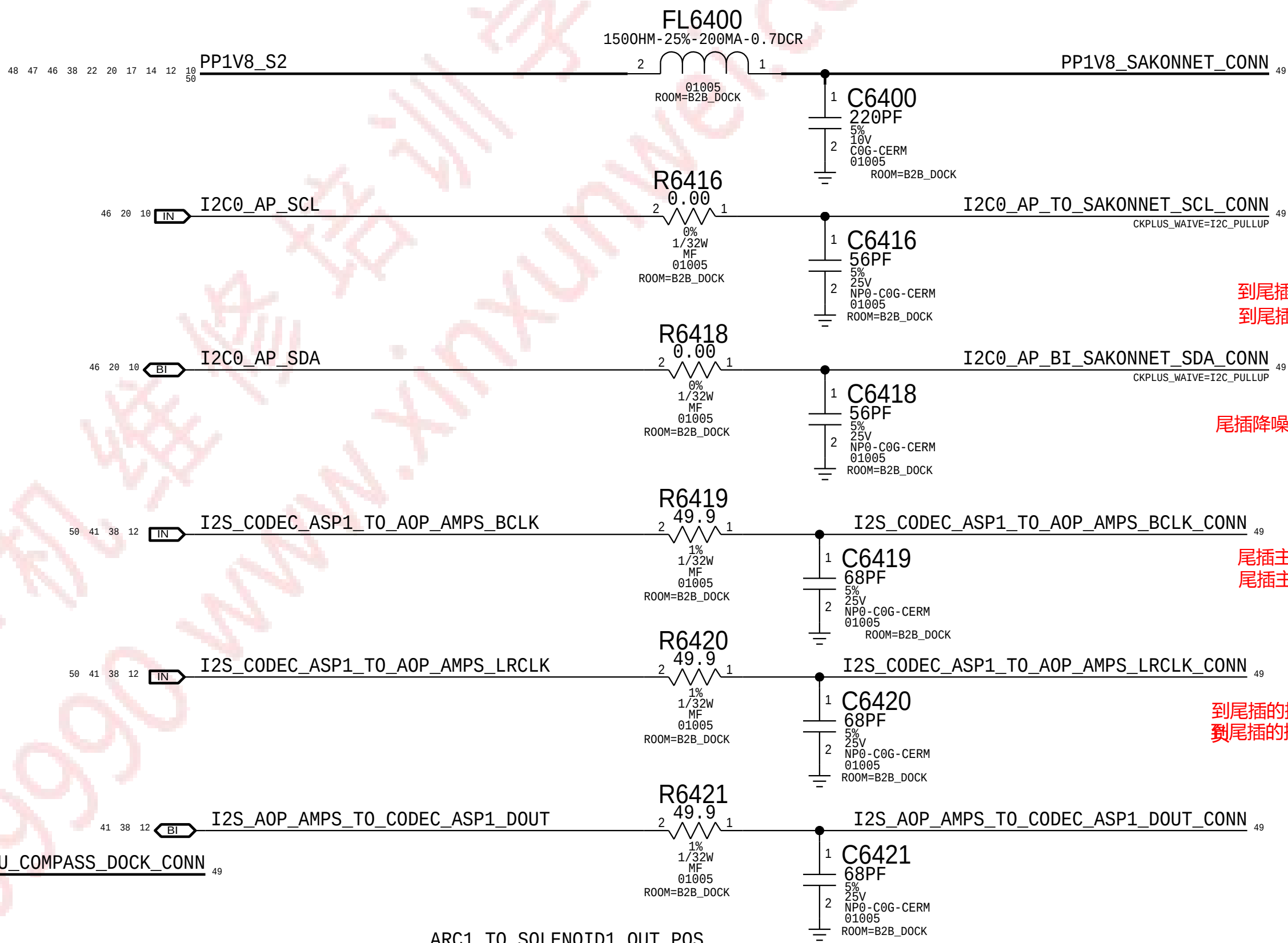


DOCK FLEX CONNECTOR

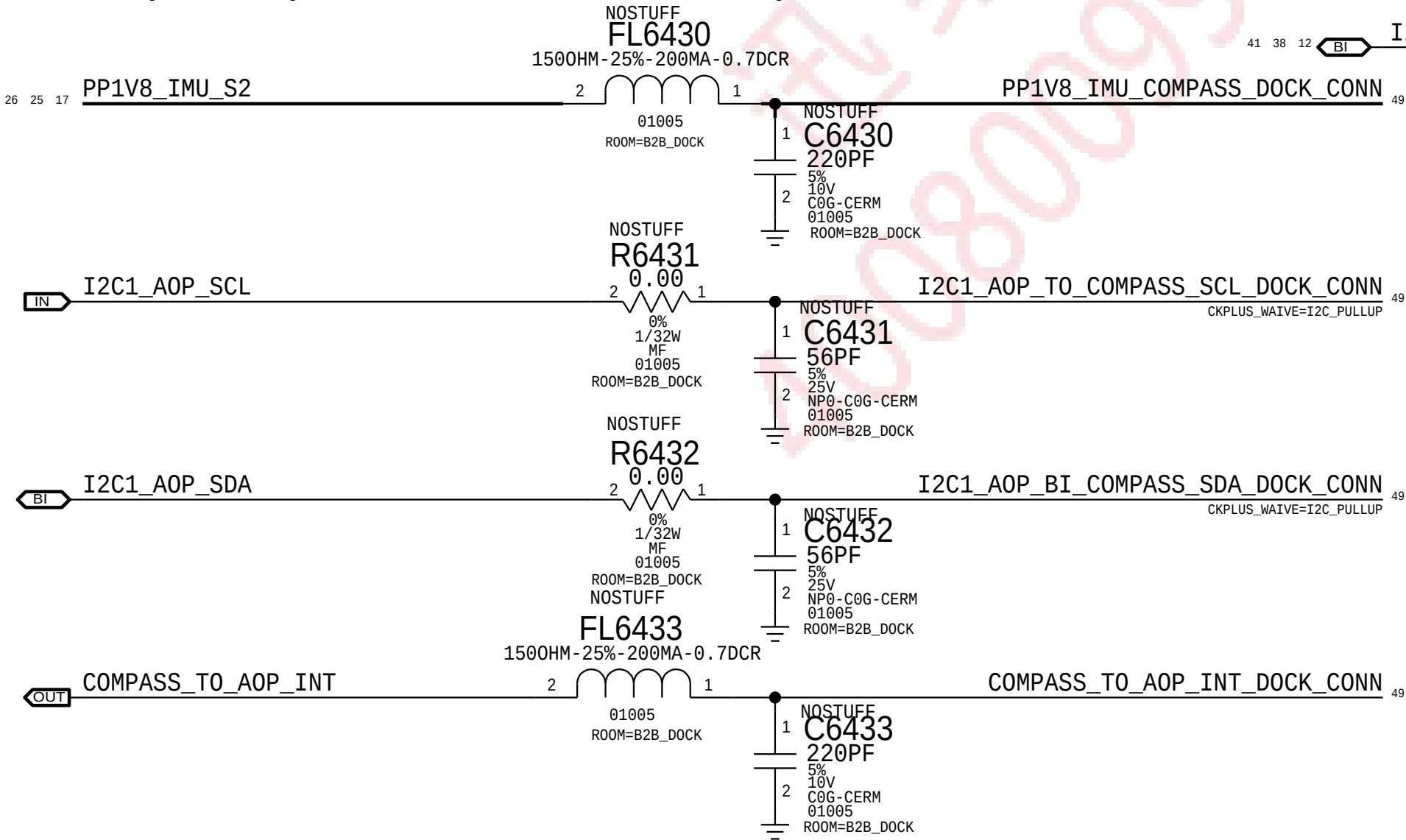
Rcpt: 516S00038
Plug: 516S00037

<-- This one on MLB

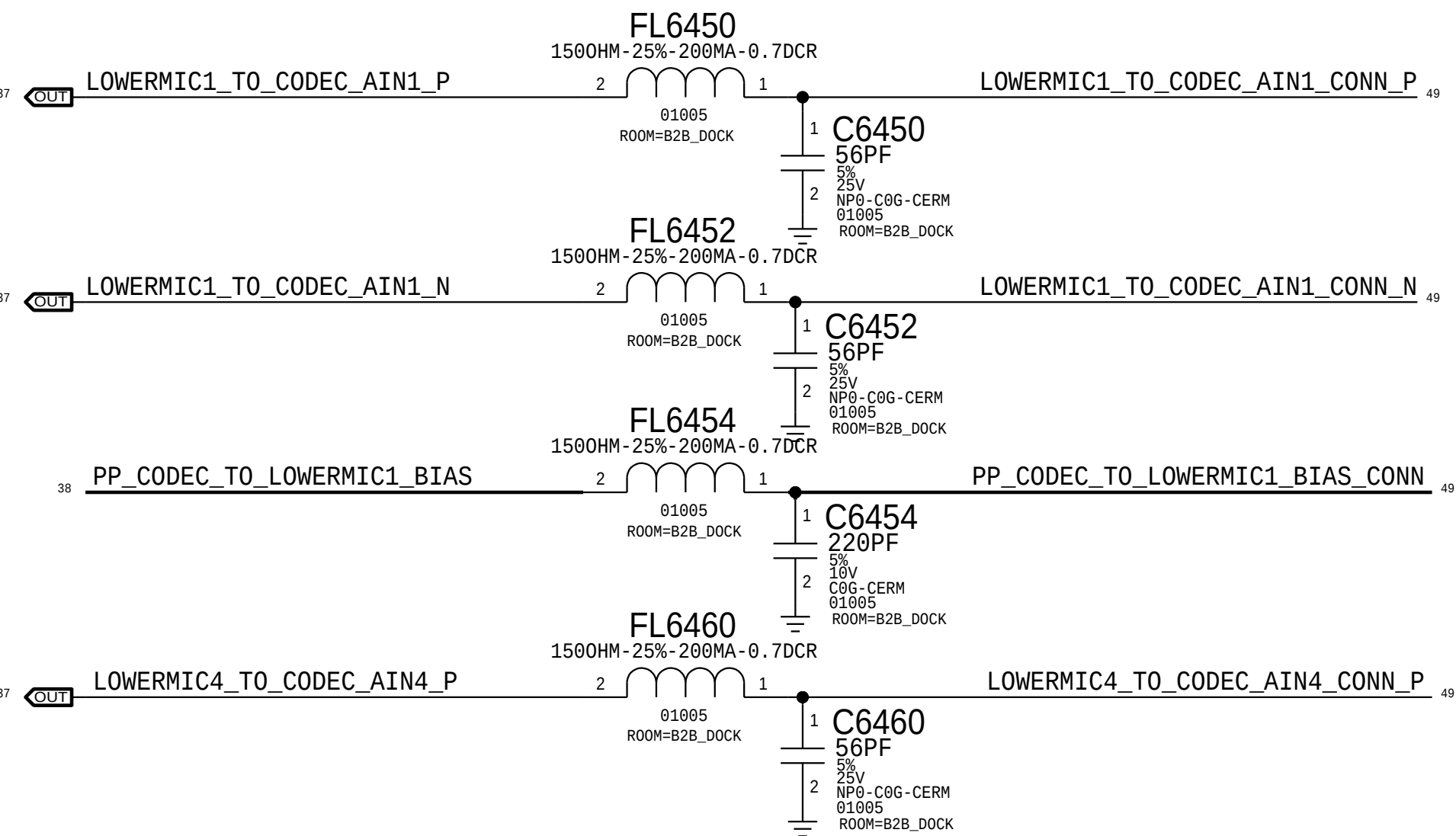
ARC



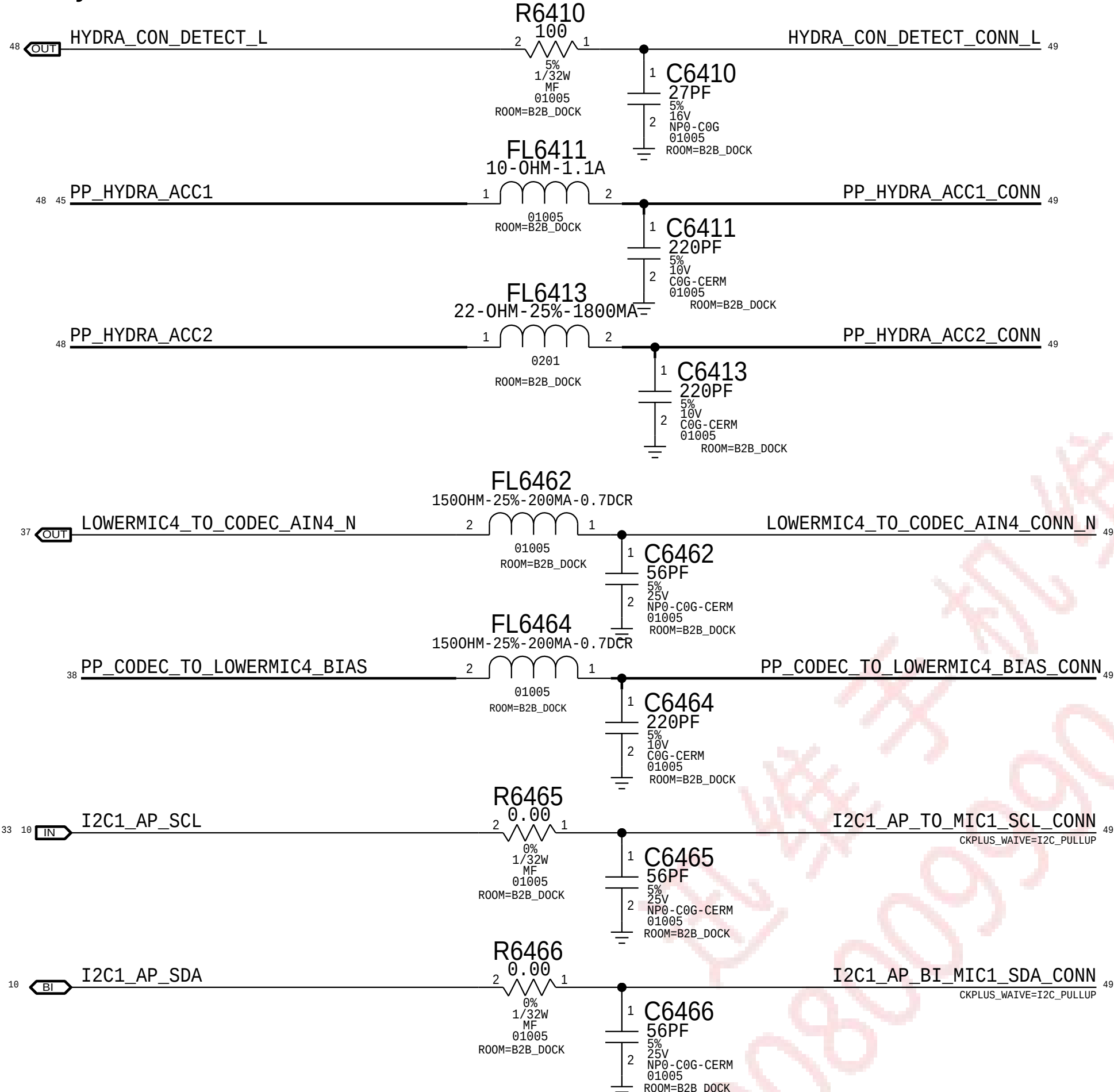
Compass (Dock Flex Location)



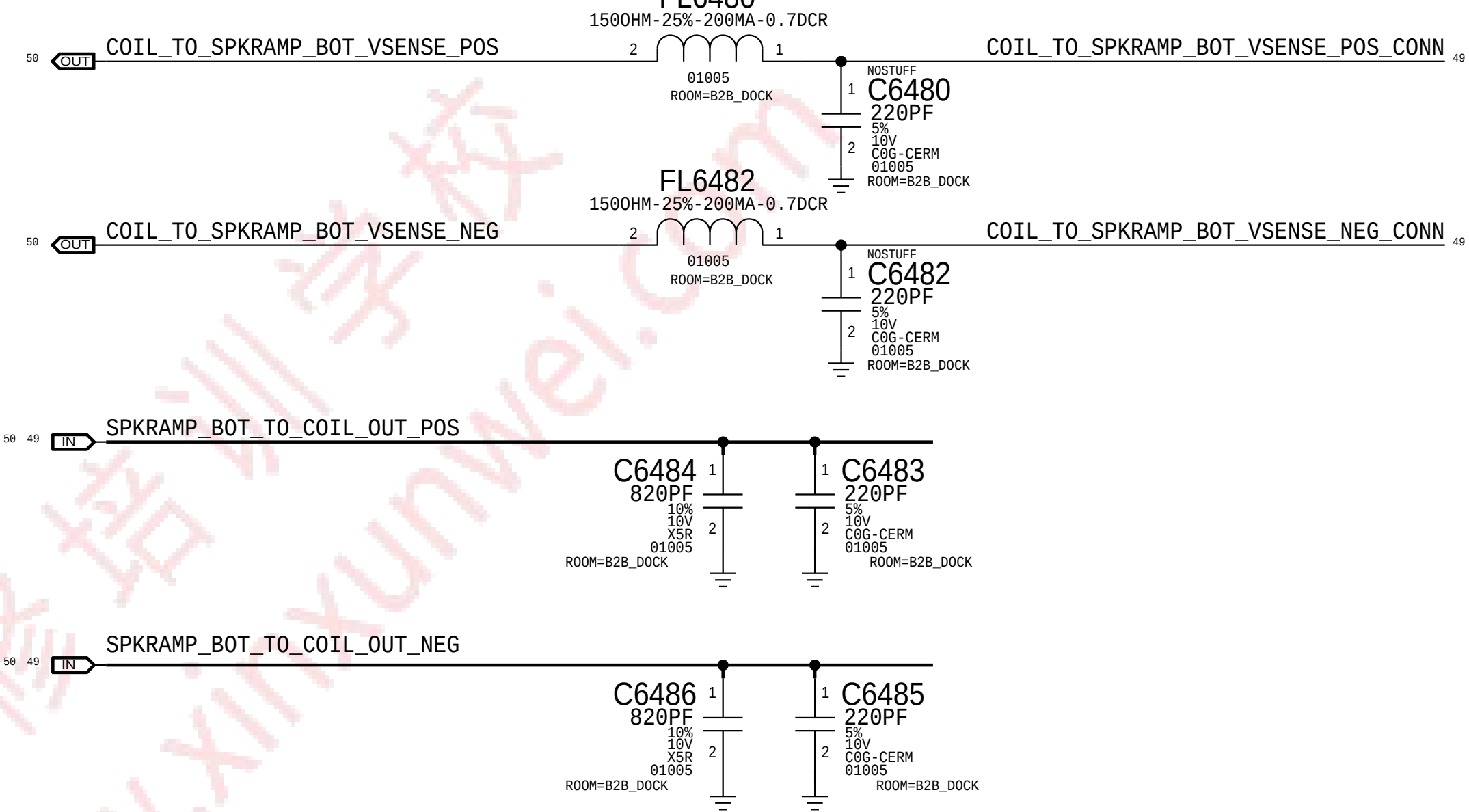
LOWER MIC1 + LOWER MIC4

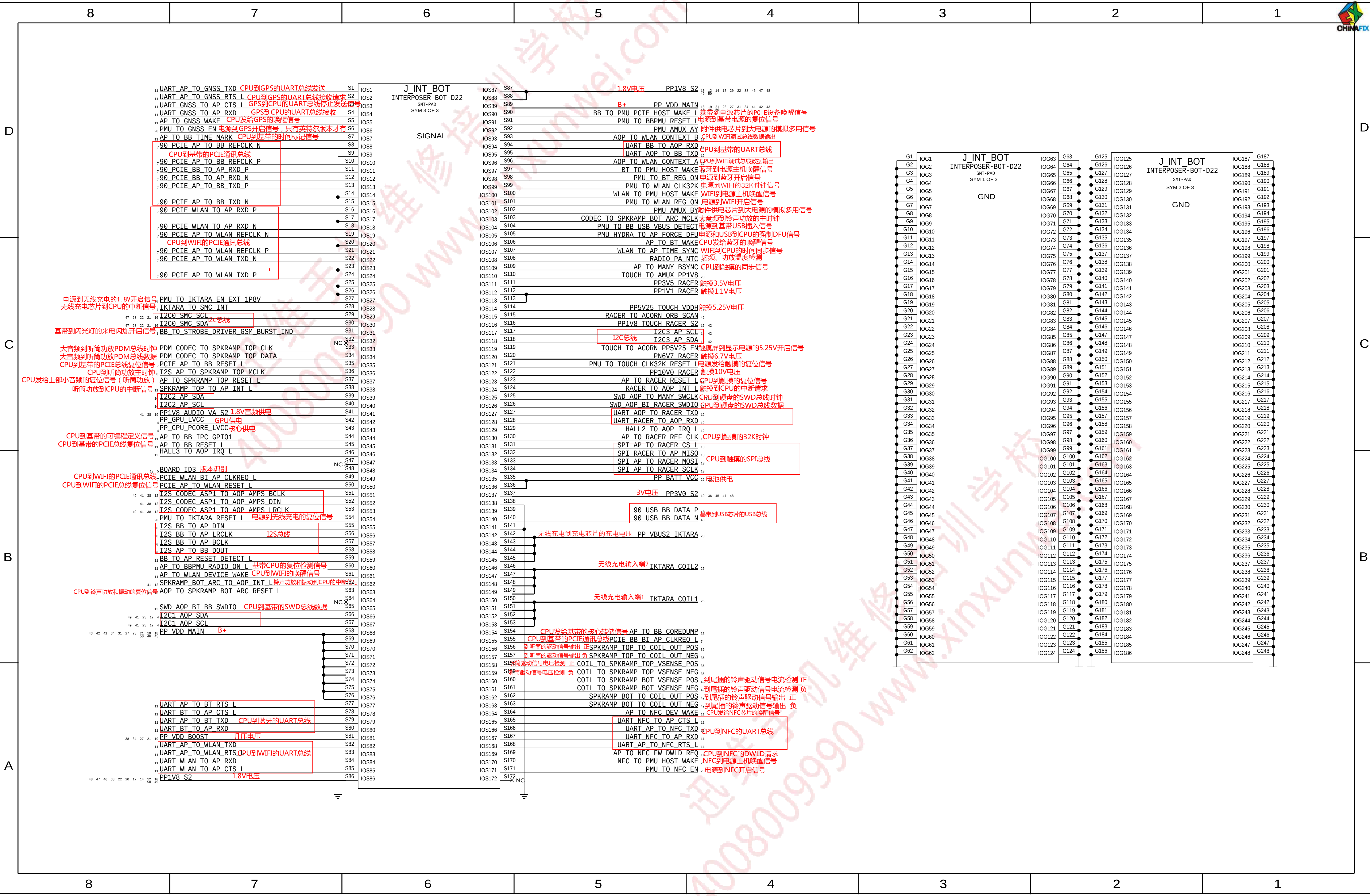


Hydra



SOUTH SPEAKER







D

C

B

A

1

2

3

4

5

6

7

8

Radios on MLB Bottom

1

2

3

4

5

6

7

8

8		7		6		5		4		3		2		1	
1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.												REV	ECN	DESCRIPTION OF REVISION	CK APPD
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.															DATE
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.												6	0008416016	ENGINEERING RELEASED	2017-04-06
X891 MLB Bottom: EVT															
LAST_MODIFICATION=Wed Apr 5 18:45:28 2017															
PAGE CSA		CONTENTS		SYNC		DATE		PAGE CSA		CONTENTS		SYNC		DATE	
1	1	TABLE OF CONTENTS													
2	2	SYSTEM: BOM Tables				11/03/2016									
3	4	SYSTEM: Mechanical Components				11/03/2016									
4	6	BOOTSTRAPPING				11/03/2016									
5	7	SYSTEM: Testpoints (Bottom) 测试点		test_mlb		11/09/2016									
6	34	SYSTEM POWER: Iktara 无线充电芯片		test_mlb		11/09/2016									
7	49	AUDIO: Speaker Amp Bottom 小音频(铃声功放)		test_mlb		11/09/2016									
8	50	AUDIO: Speaker Amp Top 小音频(听筒功放)		test_mlb		11/09/2016									
9	55	HALL EFFECT 霍尔传感器(合盖开关)		test_mlb		03/23/2017									
10	58	CG: B2B Luna & Touch 触摸座子		test_mlb		11/09/2016									
11	66	I/O: Interposer (Top) 上下主板连接接口		test_mlb		11/10/2016									
12	80	RADIOS AP电路到基带电路信号名称转换列表		test_mlb		08/19/2016									
13	1	page1													
14	2	BOM_OMIT_TABLE													
15	3	PMIC: BUCKS & LDOS 基带电源供电输出													
16	4	PMIC: CLOCKS & CONTROL 基带电源时钟、信号													
17	5	BB: POWER 基带CPU供电部分													
18	6	BB: CONTROL & HS PERIPHERALS 基带芯片													
19	7	BB: GPIOs & QLINK 基带芯片													
20	8	XCVR 射频芯片													
21	9	XCVR 射频芯片													
22	10	QPOET MODULE 功放供电													
23	11	LB PAD 低频功放													
24	12	HB PAD 高频功放													
25	13	UHB PAD 超高频功放													
26	14	2G PA 2G功放													
27	15	LB DSM 低频分集接收器													
28	16	HB DSM 高频分集接收器													
29	17	COUPLER2 耦合器													
30	18	GNSS GPS													
31	19	METROCIRC													
32	20	LOWER ANTENNA 下天线													
33	21	UPPER ANTENNA 上天线													
34	22	SIM, EUICC, DEBUG CONN SIM卡, 诊断接口		SIM TRAY		07/05/2016									
35	23	TEST POINTS		PROBE POINTS		07/06/2016									
36	24	RFFE TO GPO TRANSLATOR													
37	1	page1													
38	75	NFC NFC													
39	1	SymbolPorts													
40	4	Guinness		WIFI		01/30/2014									
41	5	WiFIAntFeeds WIFI													
BOM: 639-03222 (ROW)															
BOM: 639-03410 (JPN)															
MCO: 056-04079															
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION										
051-02222	1	SCH, MLB_BOT, X891	SCH	NO	COMMON										
820-00864	1	PCB, MLB_BOT, X891	PCB	NO	COMMON										
8		7		6		5		4		3		2		1	
Sub Designs															
SOURCE PROJECT	SUB-DESIGN NAME			VERSION	HARD/ SOFT	SYNC_DATE/TIME									
D22	RADIO_MLB			0.149.0	S	2017_04_05_18:45:19									
D22	NFC_MLB			0.22.0	S	2017_03_22_22:10:24									
D22	WIFI_MLB			0.20.0	S	2017_04_04_08:01:02									
8		7		6		5		4		3		2		1	



8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

EEEE Codes

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE FOR (MLB_BOT, 639-83418, JPN)	EEEE_HP27	NO	JPN
825-7691	1	EEEE FOR (MLB_BOT, 639-83222, ROW)	EEEE_HL39	NO	ROW

Global Capacitors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00148	138S00149	BOM_TABLE_ALTS	ALL	0402-3T,10.5uF@1V, Kyocera
138S00150	138S00149	BOM_TABLE_ALTS	ALL	0402-3T,10.5uF@1V, SEMCO
138S00151	138S00149	BOM_TABLE_ALTS	ALL	0402-3T,10.5uF@1V, TY

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00143	138S00144	BOM_TABLE_ALTS	ALL	0402,16uF@1V, Kyocera
138S00163	138S00144	BOM_TABLE_ALTS	ALL	0402,16uF@1V, Taiyo

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00138	138S00139	BOM_TABLE_ALTS	ALL	0201,3uF@1V, Kyocera
138S00164	138S00139	BOM_TABLE_ALTS	ALL	0201,3uF@1V, Taiyo

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00145	138S00146	BOM_TABLE_ALTS	ALL	0402,5.1uF@3V, Kyocera
138S00165	138S00146	BOM_TABLE_ALTS	ALL	0402,5.1uF@3V, Taiyo

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00140	138S00141	BOM_TABLE_ALTS	ALL	0201,1.1uF@3V, Kyocera
138S00142	138S00141	BOM_TABLE_ALTS	ALL	0201,1.1uF@3V, SEMCO
138S00166	138S00141	BOM_TABLE_ALTS	ALL	0201,1.1uF@3V, Taiyo

CRITICAL PART#	COMMENT
138S00149	0402-3T,10.5uF@1V

CRITICAL PART#	COMMENT
138S00144	0402,16uF@1V

CRITICAL PART#	COMMENT
138S00139	0201,3uF@1V

CRITICAL PART#	COMMENT
138S00146	0402,5.1uF@3V

CRITICAL PART#	COMMENT
138S00141	0201,1.1uF@3V

Iktara

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
132S00021	8	CAP,CER,0.082UF,10%,50V,X7R,0402	C3451,C3462,C3463,C3464,C3465,C3466,C3467,C3468	NO	COMMON
132S0423	2	CAP,CER,0.022UF,50V,X7R,10%,0402	C3455,C3465	CRITICAL	COMMON
132S0423	2	CAP,CER,0.022UF,50V,X7R,10%,0402	C3456,C3466	CRITICAL	NOSTUFF

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00183	685-00182	BOM_TABLE_ALTS	SUBBOM_DS	SUBBOM_MLB_BOT_100001,00001,X891

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00182	1	SUBBOM_MLB_BOT,DIODES,DIODES,X891	SUBBOM_DS	CRITICAL	COMMON
371S00133	4	DIODES,SHOTTKY DIODE,30V,2A,B6683	D3400,D3401,D3402,D3403	CRITICAL	DIODES_DS
371S00132	4	ONSEMT,SHOTTKY DIODE,30V,2A,B6683	D3400,D3401,D3402,D3403	CRITICAL	ONSEMT_DS

Soft-Term Cap Sub BOMs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00157	1	SUBBOM_MLB_BOT,CAP,TYPICAL,X891	SUBBOM_CAP	CRITICAL	COMMON

Touch/Luna B2B

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	1	CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA	C5899	CRITICAL	SOFT_CAP
138S0031	1	CAP,TYPICAL,2.2UF,6.3V,0201,MURATA	C5899	CRITICAL	TYPICAL_CAP

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00158	685-00157	BOM_TABLE_ALTS	SUBBOM_CAP	SUBBOM_MLB_BOT,CAP,SOFT,X891

Global Ferrites

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00558	152S00557	BOM_TABLE_ALTS	ALL	IND_MLD,0.47UH,20%,2.5A,0000,1608
155S00194	155S0010	BOM_TABLE_ALTS	ALL	FERR BD,150 OHM,25%,200MA,0.7 DCR,01005
155S00200	155S0010	BOM_TABLE_ALTS	ALL	FERR BD,150 OHM,25%,200MA,0.7 DCR,01005

CRITICAL PART#	COMMENT
152S00557	IND_MLD,0.47UH,20%,2.5A,0000,1608
155S0010	FERR BD,150 OHM,25%,200MA,0.7 DCR,01005

Global R/C Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0648	138S0652	BOM_TABLE_ALTS	ALL	CAP,XSR,4.7UF,6.3V,0.65MM,0402,TAIYO
138S00024	138S0986	BOM_TABLE_ALTS	ALL	CAP,CER,3-TERM,7.5UF,20%,4V,0402,TAIYO/TDK
138S0706	138S0739	BOM_TABLE_ALTS	ALL	CAP,CER,1UF,20%,10V,XSR,0201,MURATA
138S0945	138S0739	BOM_TABLE_ALTS	ALL	CAP,CER,1UF,20%,10V,XSR,0201,KOYOGBA
138S0739	138S0706	BOM_TABLE_ALTS	ALL	CAP,CER,XSR,0.22UF,20%,6.3V,20%
132S0436	132S0400	BOM_TABLE_ALTS	ALL	CAP,CER,XSR,0.22UF,20%,6.3V,01005
138S00049	138S0031	BOM_TABLE_ALTS	ALL	CAP,CER,XSR,2.2UF,20%,6.3V,0201

CRITICAL PART#	COMMENT
138S0652	CAP,XSR,4.7UF,6.3V,0.65MM,0402,TAIYO
138S0986	CAP,CER,3-TERM,7.5UF,20%,4V,0402,TAIYO/TDK
138S0739	CAP,CER,1UF,20%,10V,XSR,0201,MURATA

138S0706	CAP,CER,XSR,0.22UF,20%,6.3V,20%
132S0400	CAP,CER,XSR,0.22UF,20%,6.3V,01005
138S0031	CAP,CER,XSR,2.2UF,20%,6.3V,0201

Global Inductors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00653	152S00651	BOM_TABLE_ALTS	ALL	IND,1.2UH,3A,2016,0.6SZ
152S00654	152S00652	BOM_TABLE_ALTS	ALL	IND,1.2UH,3A,2016,0.8Z

CRITICAL PART#	COMMENT
152S00651	IND,1.2UH,3A,2016,0.6SZ
152S00652	IND,1.2UH,3A,2016,0.8Z

Multi-Vendor Criticals

CRITICAL PART#	COMMENT	CRITICAL PART#	COMMENT
138S0979	CAP,CER,XSR,10UF,20%,10V,0402,H=0.65MM	132S00008	CAP,CER,0.1UF,10%,50V,X7R,0402
138S0603	CAP,CER,XSR,1UF,10%,25V,0402	131S00004	CAP,CER,Z7PF,5%,C06,25V,0201
132S0663	CAP,CER,XSR,1UF,10%,25V,0402	131S0307	CAP,CER,NP0/C06,100PF,5%,10V,01005
132S0288	CAP,CER,XSR,0.1UF,10%,10V,0201	131S00053	CAP,CER,C06,220PF,5%,10V,01005
132S0275	CAP,CER,XSR,470PF,10%,10V,01005	117S00055	RES,RF,1/20W,2M OHM,5,0201,SMD
132S0245	CAP,CER,XSR,0.01UF,10%,6.3V,01005	107S0257	THERMISTOR,NTC,10K OHM,1%,B=3435,01005

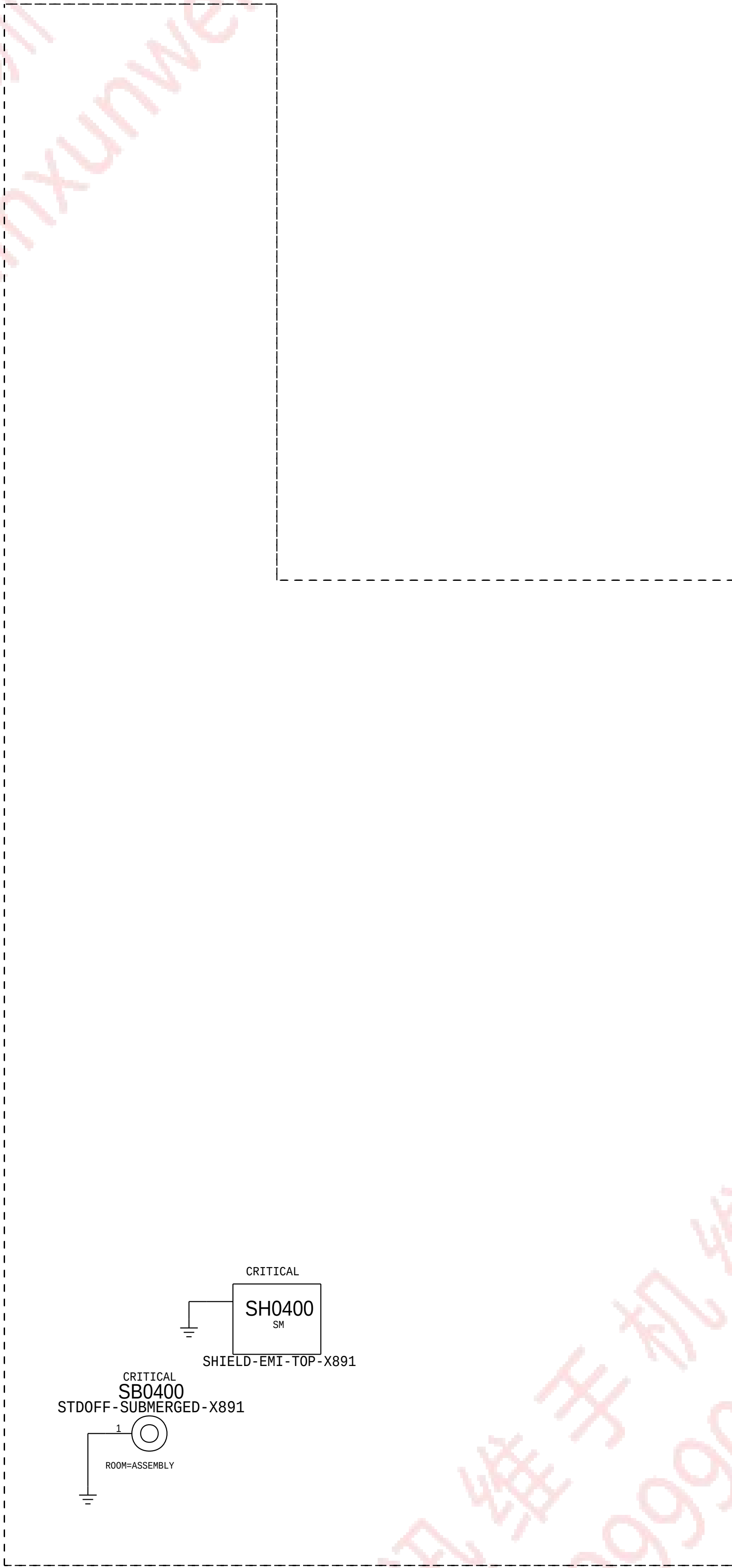
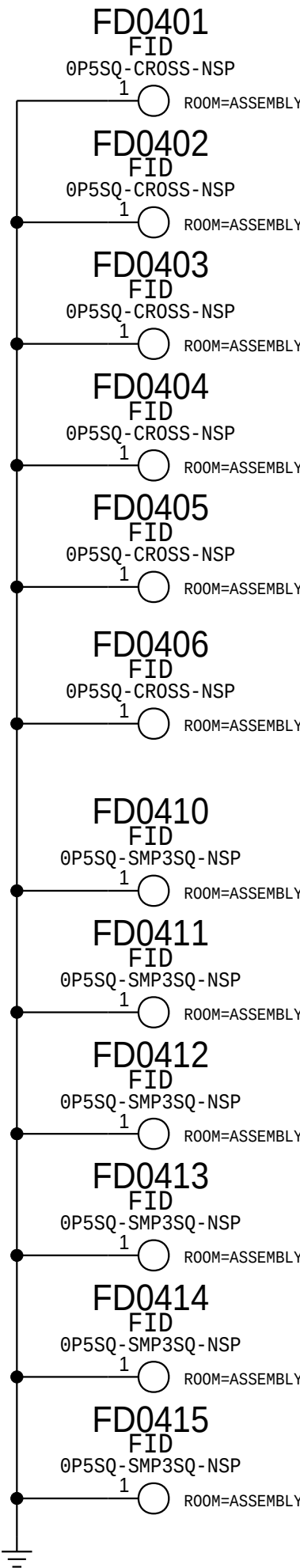
8	7	6	5	4	3	2	1
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D

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FIDUCIALS



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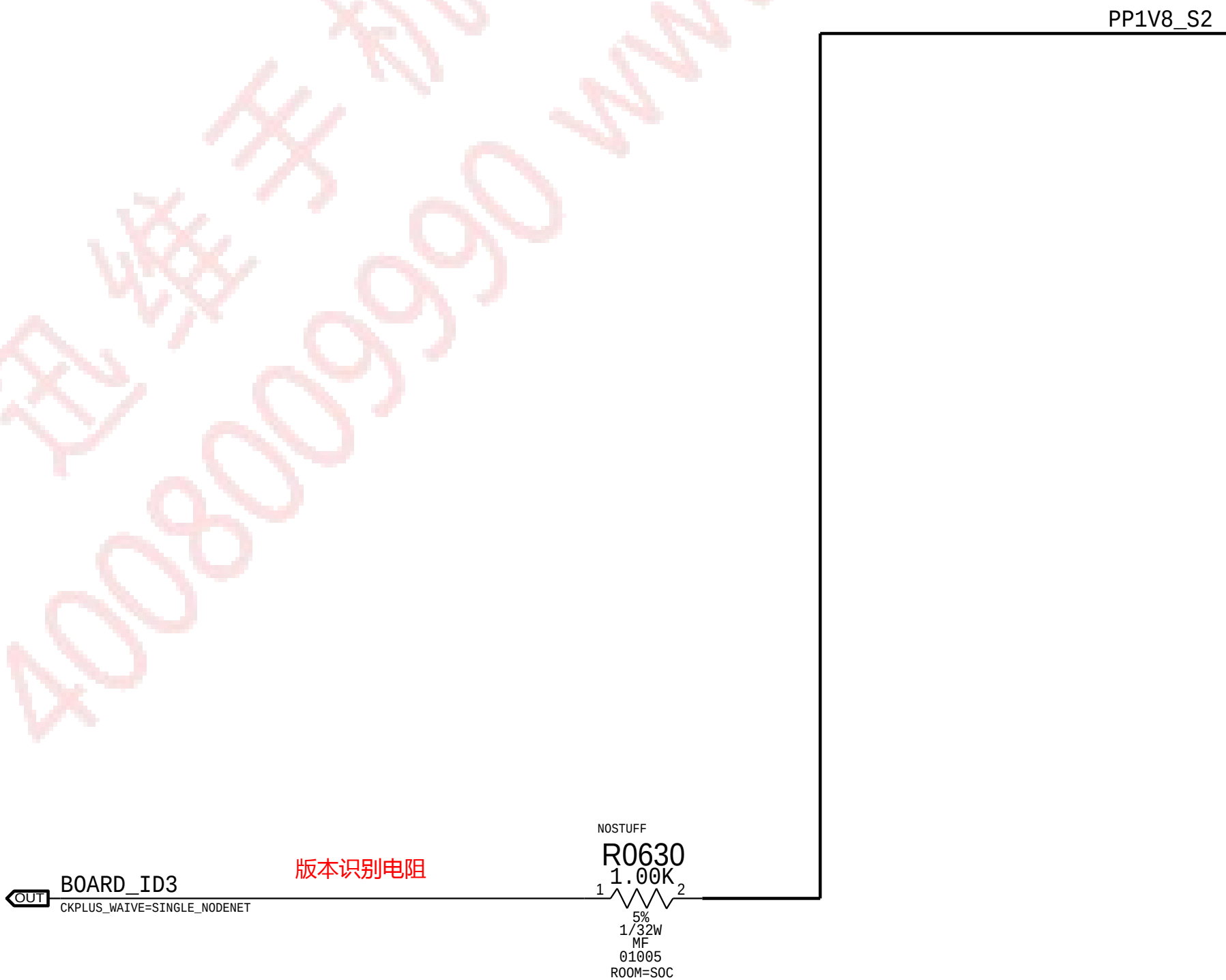
D

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B

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BOOTSTRAPPING:BOARD ID[3]



D22 Selected (BOARD_ID3) ->

Board Rev[3:0]				
Float=Low I PU=High				
	3	2	1	0
	Build Major		Build Minor	
Proto 1	1	1	1	1
Proto 2	1	1	1	0
Proto 3	1	1	0	1
(Spare)	1	1	0	0
EVT	1	0	1	1
(Spare)	1	0	1	0
Carrier	0	1	1	1
(Spare)	0	1	1	0
DVT	0	0	1	1
(Spare)	0	0	1	0
(Spare)	0	0	0	1
PVT	0	0	0	0

Board ID[4:0]					
Float=Low I PU=High					
	4	3	2	1	0
	X	Baseband	Form Factor	MLB/Dev	
D20 MLB	0	0	0	1	0
D20 Dev	0	0	0	1	1
D21 MLB	0	0	1	0	0
D21 Dev	0	0	1	0	1
D22 MLB	0	0	1	1	0
D22 Dev	0	0	1	1	1
D201 MLB	0	1	0	1	0
D201 Dev	0	1	0	1	1
D211 MLB	0	1	1	0	0
D211 Dev	0	1	1	0	1
D221 MLB	0	1	1	1	0
D221 Dev	0	1	1	1	1

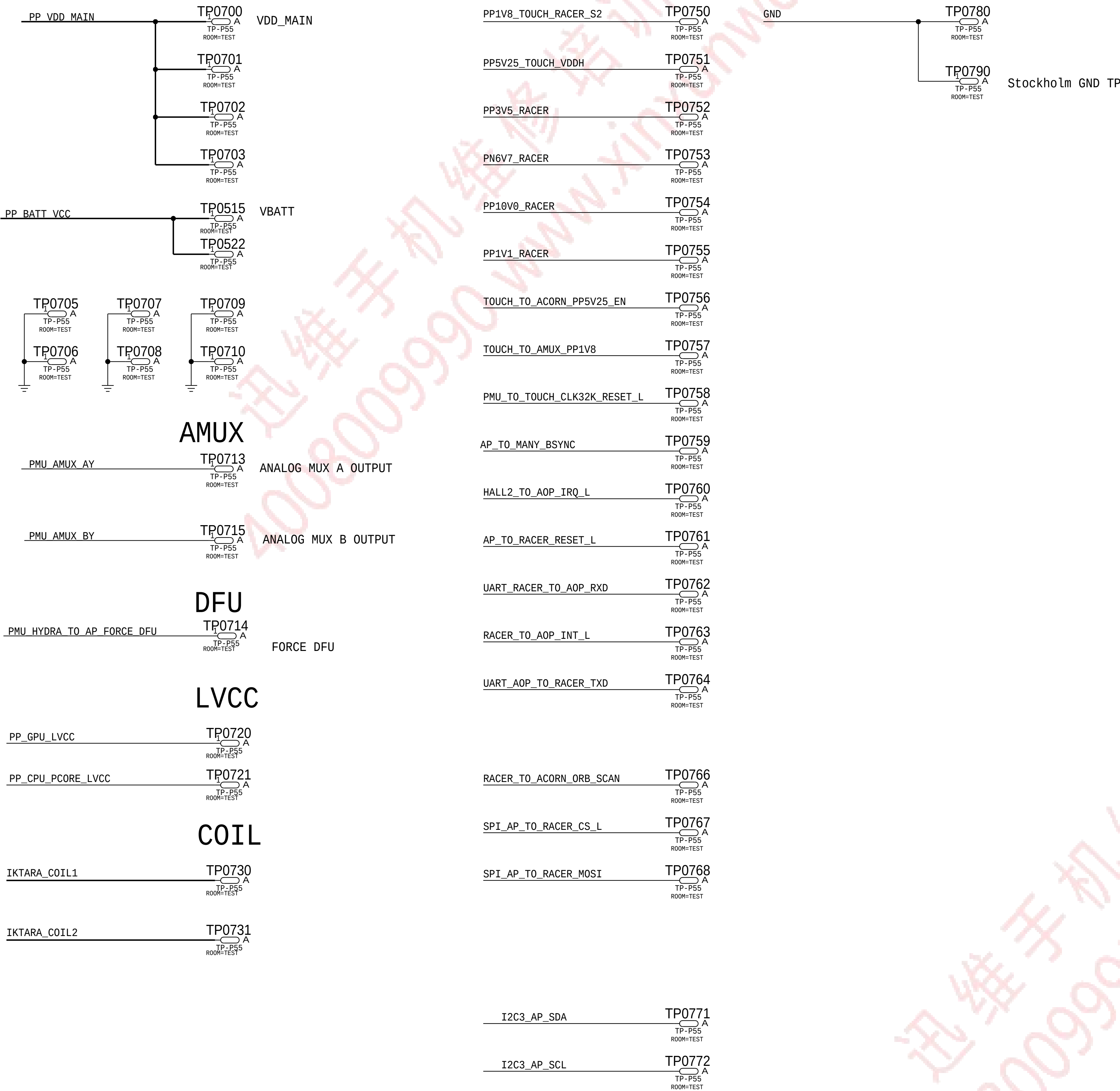
Boot Config[2:0]			
Float=Low I PU=High			
	2	1	0
SPI0	0	0	0
SPI0 Test Mode	0	0	1
SPI0 NAND	0	1	0
SPI0 NAND Test	0	1	1
Slow SPI0 Test	1	1	0



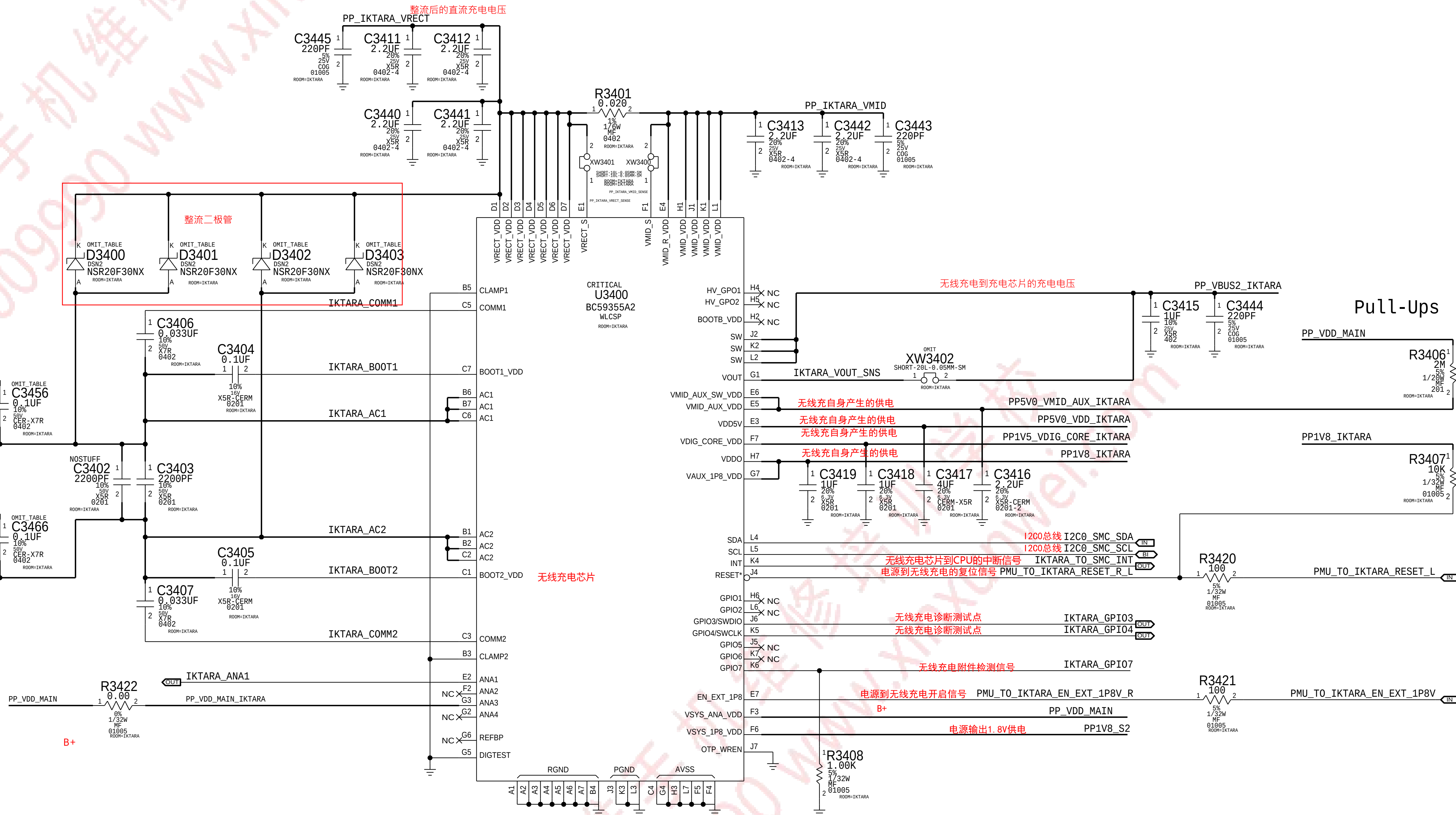
Test Points

Probe Points

Iktara Debug



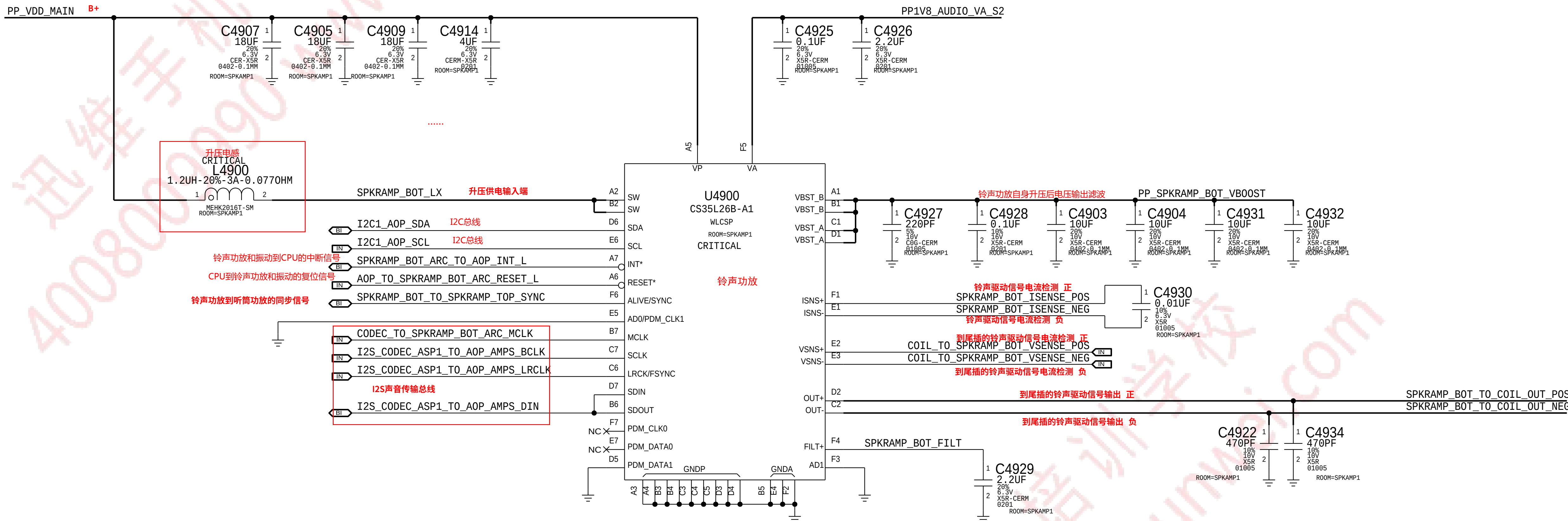
Iktara

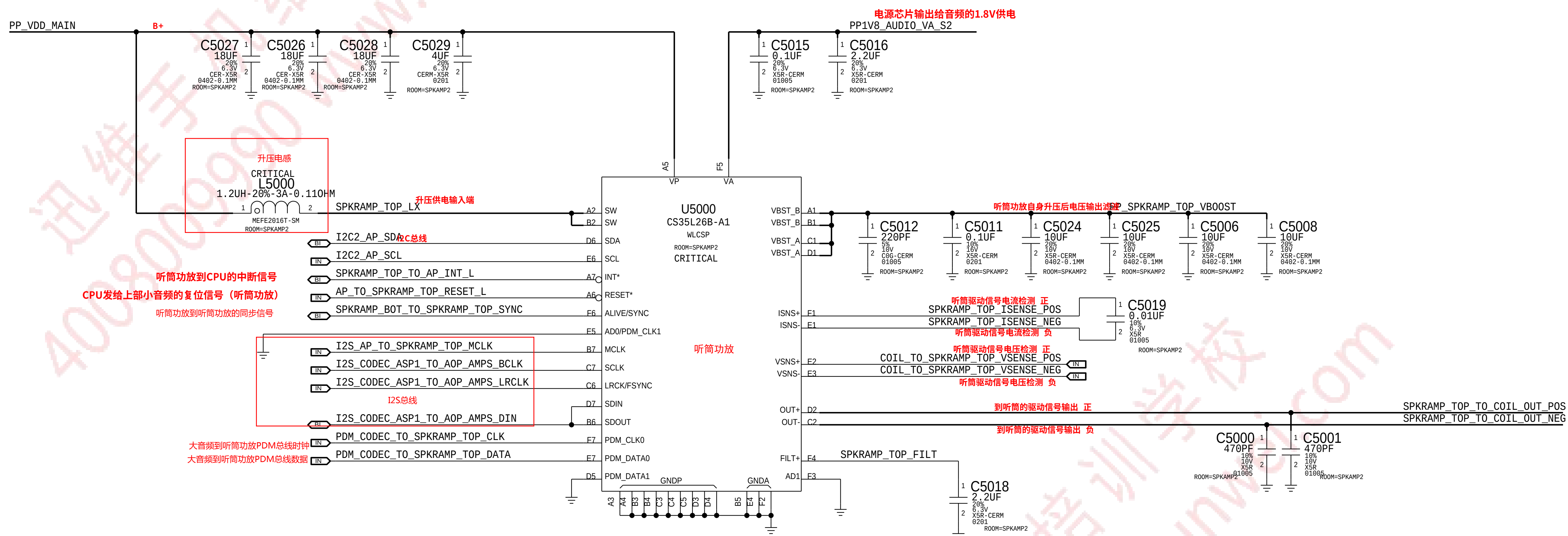
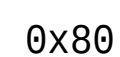


South Speaker Amplifier

APN: 338S00295

I2C ADDRESS: 1000 000x
0x80





D

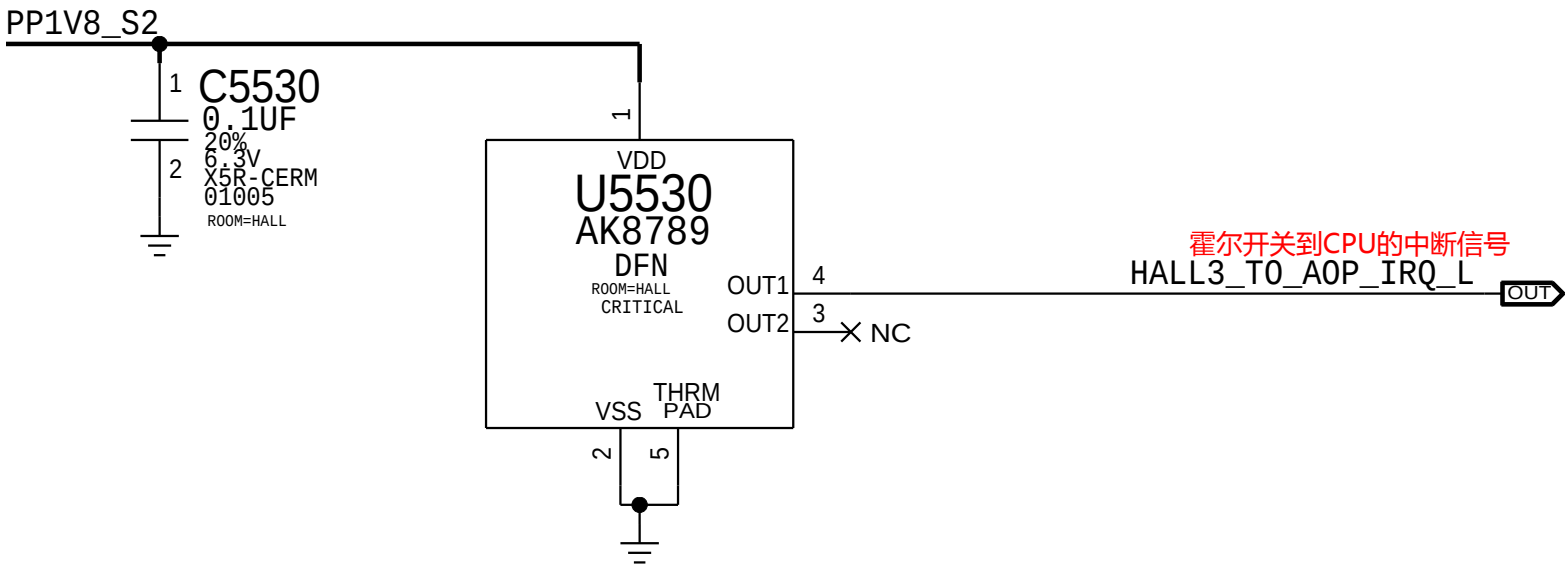
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Hall Effect

APN: 353S3697 霍尔磁控开关 (合盖开关)



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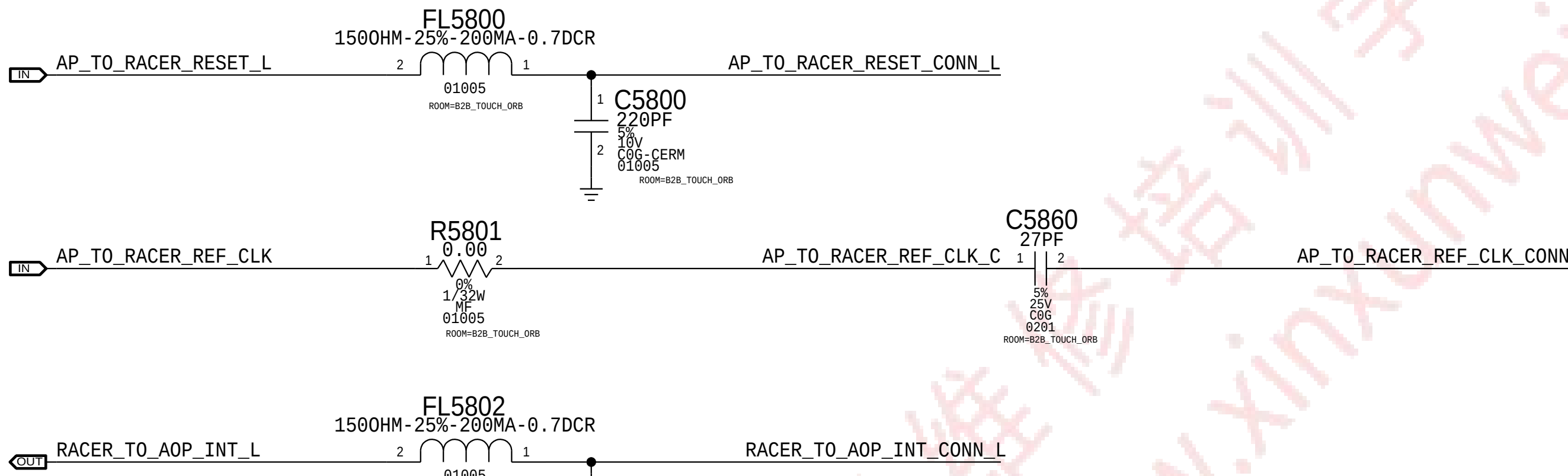
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3

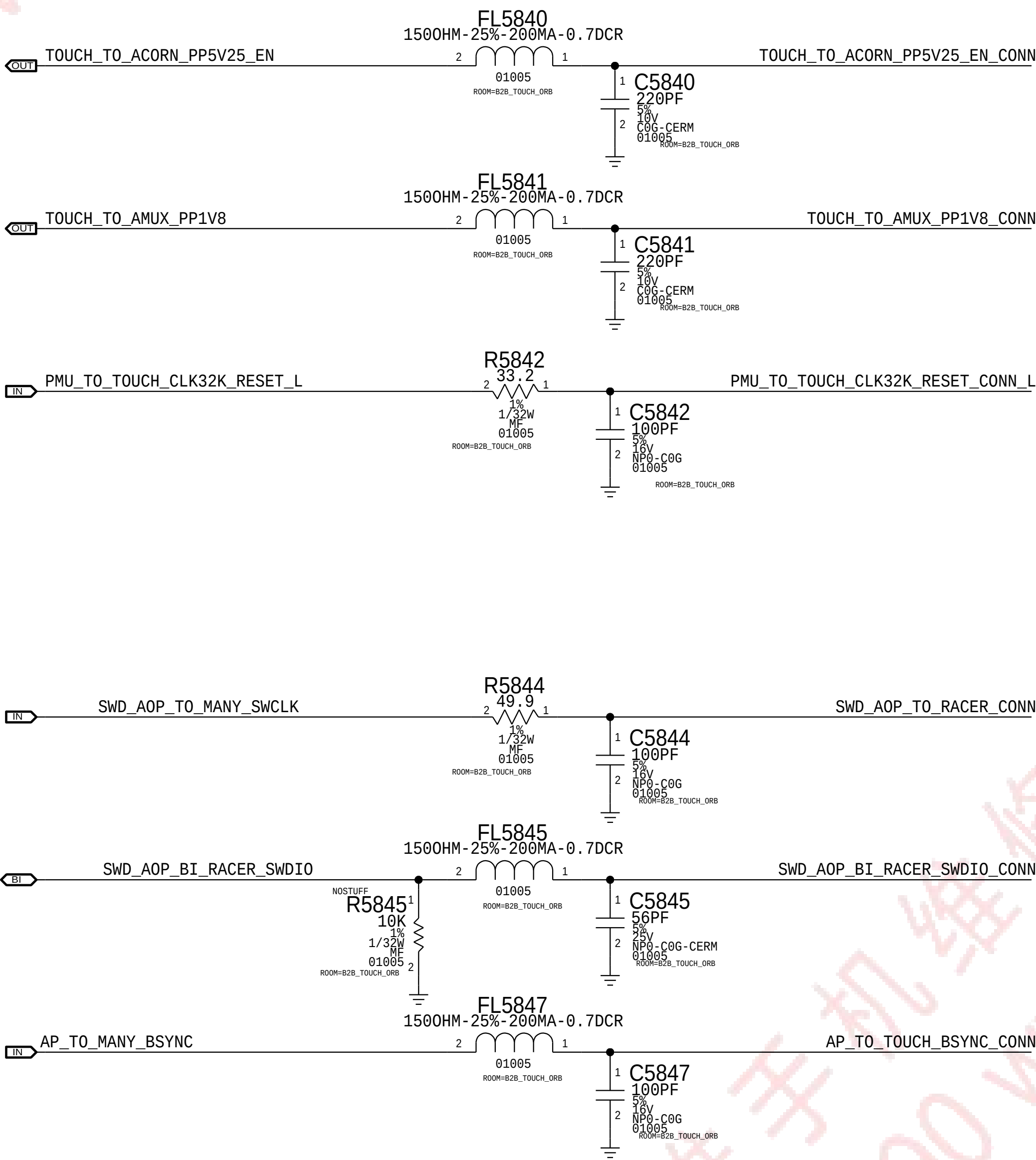
2

1

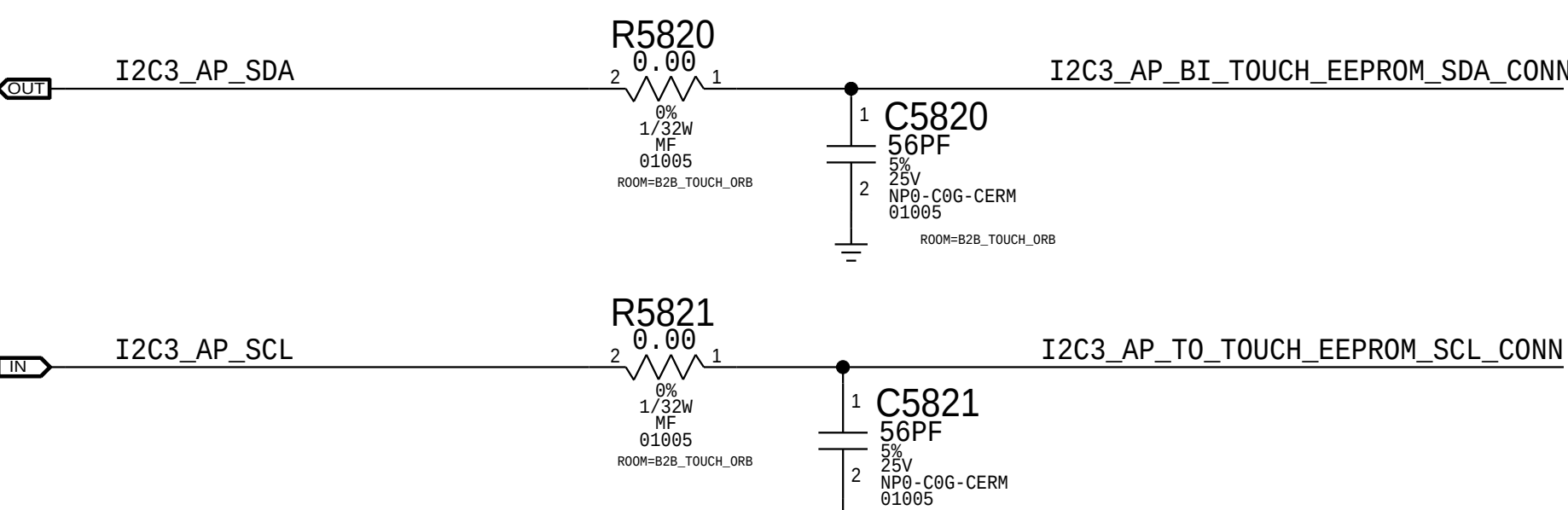
Racer I/O



Touch and Misc I/O



AP I2C Filters



Luna + Touch Connector

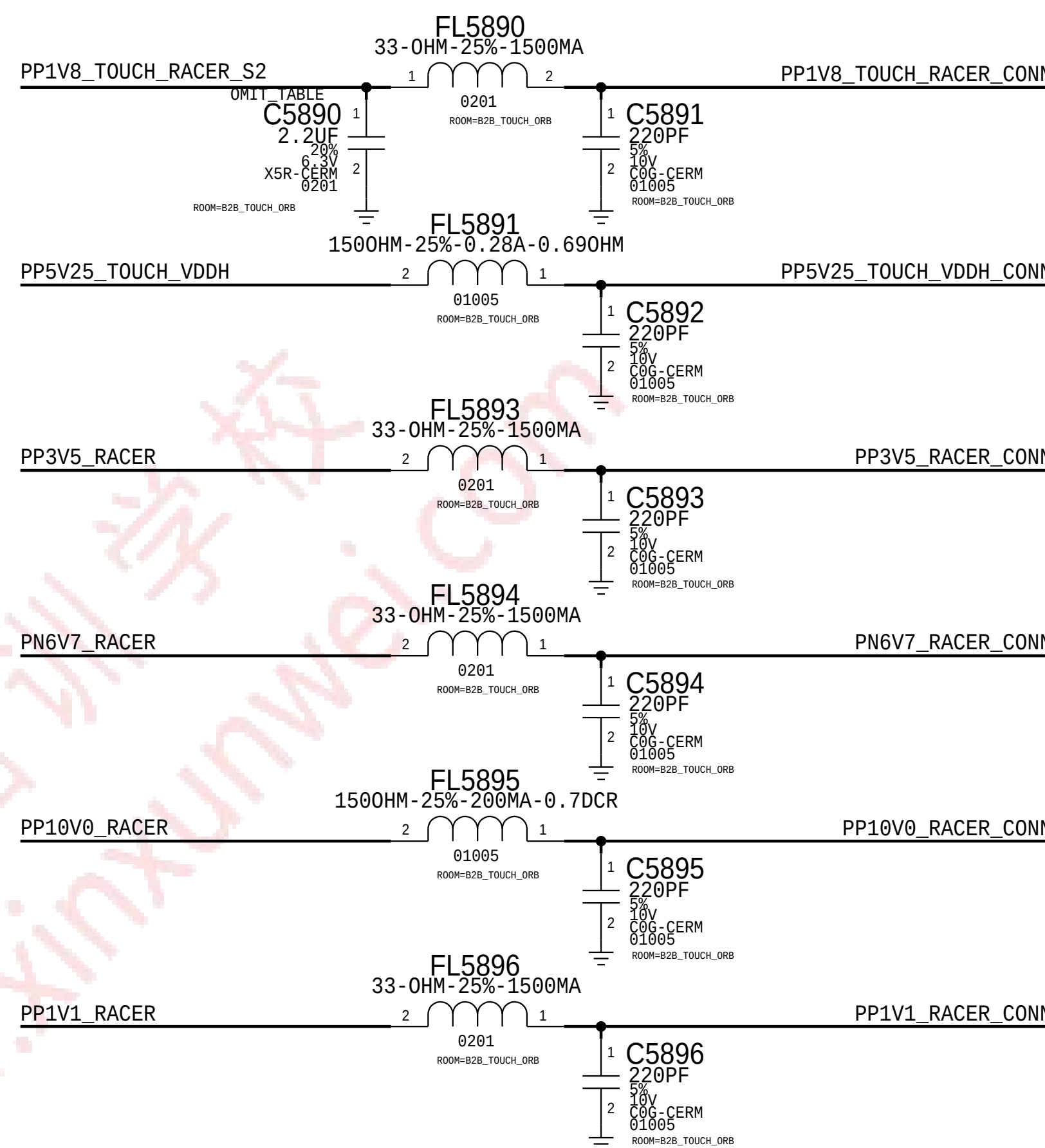
Rcpt: 516S00325
Plug: 516S00326

触摸座子

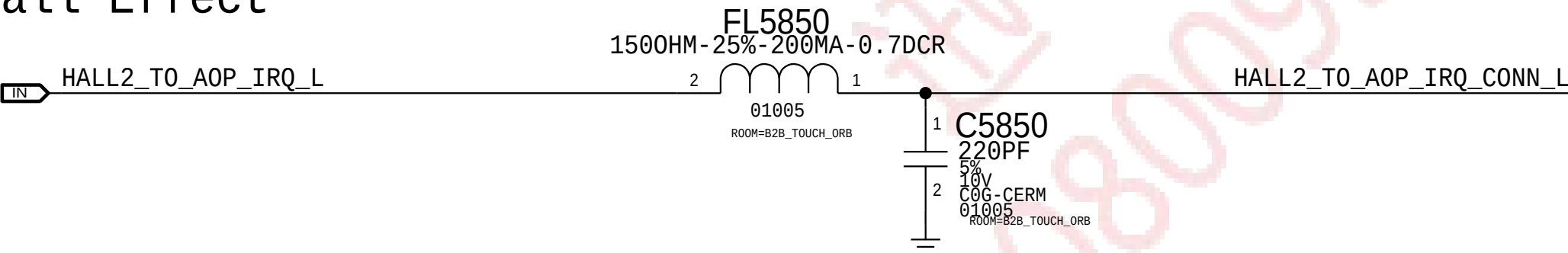
J5800
AA26DK-S028VA1
F-ST-SM

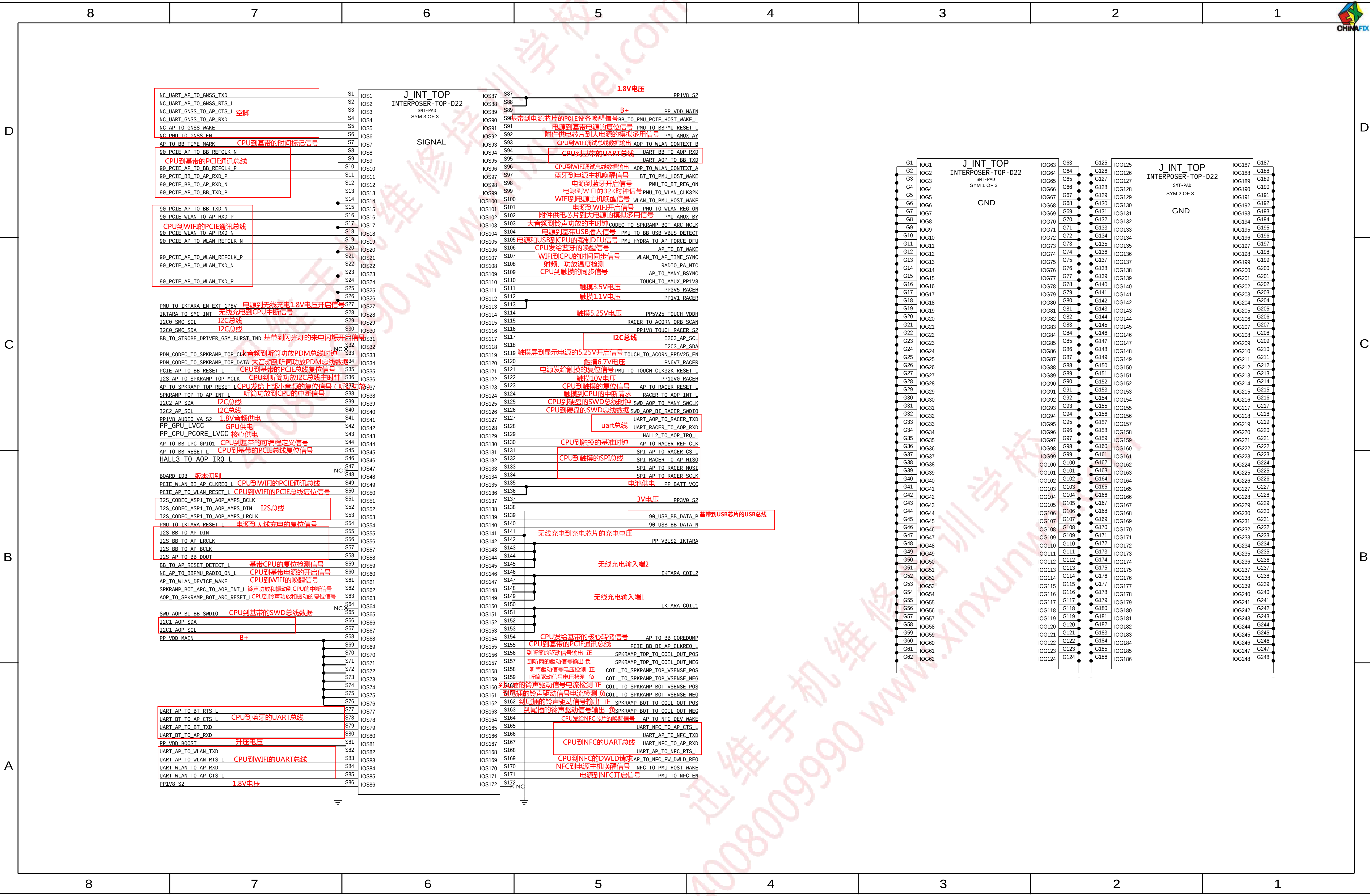
SPI_AP_TO_RACER_SCLK_CONN	CPU到触摸的SPI总线串行时钟	1	2	触摸SPI总线数据传输	SPI_AP_TO_RACER_MOSI_CONN
SPI_AP_TO_RACER_CS_CONN_L	CPU到触摸的SPI片选信号	3	4	触摸SPI总线数据传输	SPI_RACER_TO_AP_MISO_CONN
AP_TO_RACER_RESET_CONN_L	CPU到触摸的复位信号	5	6	触摸到CPU中断信号	RACER_TO_AOP_INT_CONN_L
RACER_TO_ACORN_ORB_SCAN_CONN		7	8		PP10V0_RACER_CONN
SWD_AOP_TO_RACER_CONN	CPU发出的SWD总线时钟	9	10	触摸到CPU的UART总线	UART_RACER_TO_AOP_RXD_CONN
SWD_AOP_BI_RACER_SWDIO_CONN	CPU发出的SWD总线数据	11	12	触摸到CPU的UART总线	UART_AOP_TO_RACER_TXD_CONN
HALL2_TO_AOP_IRQ_CONN_L	霍尔开关到CPU的中断信号	13	14		PP1V8_TOUCH_RACER_CONN
AP_TO_RACER_REF_CLK_CONN	CPU到触摸的基准时钟	15	16		PP1V1_RACER_CONN
PMU_TO_TOUCH_CLK32K_RESET_CONN_L	电源发给触摸的复位信号	17	18	触摸1.1V电压	PP6V7_RACER_CONN
I2C3_AP_BI_TOUCH_EEPROM_SDA_CONN	I2C总线	19	20	触摸6.7V电压	PP3V5_RACER_CONN
I2C3_AP_TO_TOUCH_EEPROM_SCL_CONN		21	22	触摸3.5V电压	PP3V5_RACER_CONN
TOUCH_TO_AMUX_PP1V8_CONN	1.8V供电	23	24	CPU到触摸同步信号	AP_TO_TOUCH_BSYN_CONN
		25	26	触摸到显示触摸电源5.25V电压	TOUCH_TO_ACORN_PP5V25_EN_CONN
		27	28	触摸5.25 V电压	PP5V25_TOUCH_VDDH_CONN

Touch And Racer Power

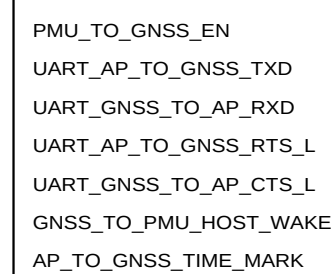
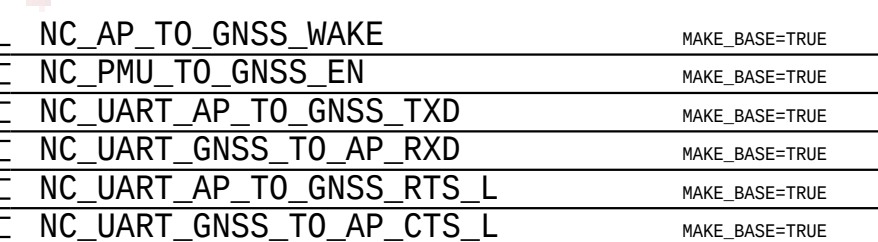
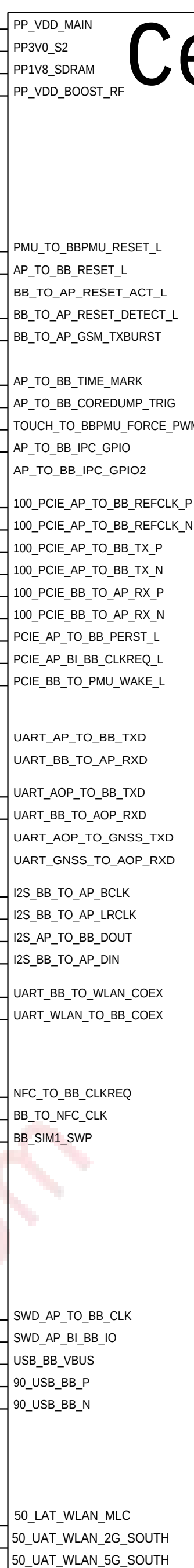


Hall Effect





Wifi/BT



8		7		6		5		4		3		2		1					
BOM OPTIONS:																			
D																		D	
C																		C	
B																		B	
A																			
8		7		6		5		4		3		2		1					

The diagram illustrates the PMIC section of the PDM9655, featuring a central U_PMIC_E PDM9655 component. The layout includes power and ground planes, decoupling capacitors, and various signal traces. Key components and their values are as follows:

- Power and Ground Planes:** PP_VDD_MAIN (B+), PP_GND, PP_VSS_MAIN (B-), and PP_GND are shown on the left. The right side shows PP_0V8_SMP51_E, PP_1V2_SMP2_E, PP_1V0_SMP3_E, PP_1V8_SMP4_E, PP_0V8_SMP5_E, and PP_0V8_SMP5_E.
- Decoupling Capacitors:** C317, C322, C318, C323, C319, C324, C320, C325, C327, C321, and C326 are all 26uF capacitors, with values ranging from 0.1mm to 0.1mm.
- Inductors:** L304, L301, L305, L302, and L303 are all 2.2uH inductors, with values ranging from 0.0350H to 0.090H.
- Resistors:** R301, R302, R303, R304, R305, R306, R307, R308, R309, R310, R311, R312, R313, R314, R315, R316, R317, R318, R319, R320, R321, R322, R323, R324, R325, R326, R327, R328, R329, R330, R331, R332, R333, R334, R335, R336, R337, R338, R339, R340, R341, R342, R343, R344, R345, R346, R347, R348, R349, R350, R351, R352, R353, R354, R355, R356, R357, R358, R359, R360, R361, R362, R363, R364, R365, R366, R367, R368, R369, R370, R371, R372, R373, R374, R375, R376, R377, R378, R379, R380, R381, R382, R383, R384, R385, R386, R387, R388, R389, R390, R391, R392, R393, R394, R395, R396, R397, R398, R399, R400, R401, R402, R403, R404, R405, R406, R407, R408, R409, R410, R411, R412, R413, R414, R415, R416, R417, R418, R419, R420, R421, R422, R423, R424, R425, R426, R427, R428, R429, R430, R431, R432, R433, R434, R435, R436, R437, R438, R439, R440, R441, R442, R443, R444, R445, R446, R447, R448, R449, R450, R451, R452, R453, R454, R455, R456, R457, R458, R459, R460, R461, R462, R463, R464, R465, R466, R467, R468, R469, R470, R471, R472, R473, R474, R475, R476, R477, R478, R479, R480, R481, R482, R483, R484, R485, R486, R487, R488, R489, R490, R491, R492, R493, R494, R495, R496, R497, R498, R499, R500, R501, R502, R503, R504, R505, R506, R507, R508, R509, R510, R511, R512, R513, R514, R515, R516, R517, R518, R519, R520, R521, R522, R523, R524, R525, R526, R527, R528, R529, R530, R531, R532, R533, R534, R535, R536, R537, R538, R539, R540, R541, R542, R543, R544, R545, R546, R547, R548, R549, R550, R551, R552, R553, R554, R555, R556, R557, R558, R559, R560, R561, R562, R563, R564, R565, R566, R567, R568, R569, R570, R571, R572, R573, R574, R575, R576, R577, R578, R579, R580, R581, R582, R583, R584, R585, R586, R587, R588, R589, R590, R591, R592, R593, R594, R595, R596, R597, R598, R599, R600, R601, R602, R603, R604, R605, R606, R607, R608, R609, R610, R611, R612, R613, R614, R615, R616, R617, R618, R619, R620, R621, R622, R623, R624, R625, R626, R627, R628, R629, R630, R631, R632, R633, R634, R635, R636, R637, R638, R639, R640, R641, R642, R643, R644, R645, R646, R647, R648, R649, R650, R651, R652, R653, R654, R655, R656, R657, R658, R659, R660, R661, R662, R663, R664, R665, R666, R667, R668, R669, R670, R671, R672, R673, R674, R675, R676, R677, R678, R679, R680, R681, R682, R683, R684, R685, R686, R687, R688, R689, R690, R691, R692, R693, R694, R695, R696, R697, R698, R699, R700, R701, R702, R703, R704, R705, R706, R707, R708, R709, R710, R711, R712, R713, R714, R715, R716, R717, R718, R719, R720, R721, R722, R723, R724, R725, R726, R727, R728, R729, R730, R731, R732, R733, R734, R735, R736, R737, R738, R739, R740, R741, R742, R743, R744, R745, R746, R747, R748, R749, R750, R751, R752, R753, R754, R755, R756, R757, R758, R759, R760, R761, R762, R763, R764, R765, R766, R767, R768, R769, R770, R771, R772, R773, R774, R775, R776, R777, R778, R779, R780, R781, R782, R783, R784, R785, R786, R787, R788, R789, R790, R791, R792, R793, R794, R795, R796, R797, R798, R799, R800, R801, R802, R803, R804, R805, R806, R807, R808, R809, R810, R811, R812, R813, R814, R815, R816, R817, R818, R819, R820, R821, R822, R823, R824, R825, R826, R827, R828, R829, R830, R831, R832, R833, R834, R835, R836, R837, R838, R839, R840, R841, R842, R843, R844, R845, R846, R847, R848, R849, R850, R851, R852, R853, R854, R855, R856, R857, R858, R859, R860, R861, R862, R863, R864, R865, R866, R867, R868, R869, R870, R871, R872, R873, R874, R875, R876, R877, R878, R879, R880, R881, R882, R883, R884, R885, R886, R887, R888, R889, R890, R891, R892, R893, R894, R895, R896, R897, R898, R899, R900, R901, R902, R903, R904, R905, R906, R907, R908, R909, R910, R911, R912, R913, R914, R915, R916, R917, R918, R919, R920, R921, R922, R923, R924, R925, R926, R927, R928, R929, R930, R931, R932, R933, R934, R935, R936, R937, R938, R939, R940, R941, R942, R943, R944, R945, R946, R947, R948, R949, R950, R951, R952, R953, R954, R955, R956, R957, R958, R959, R960, R961, R962, R963, R964, R965, R966, R967, R968, R969, R970, R971, R972, R973, R974, R975, R976, R977, R978, R979, R980, R981, R982, R983, R984, R985, R986, R987, R988, R989, R990, R991, R992, R993, R994, R995, R996, R997, R998, R999, R1000, R1001, R1002, R1003, R1004, R1005, R1006, R1007, R1008, R1009, R1010, R1011, R1012, R1013, R1014, R1015, R1016, R1017, R1018, R1019, R1020, R1021, R1022, R1023, R1024, R1025, R1026, R1027, R1028, R1029, R1030, R1031, R1032, R1033, R103

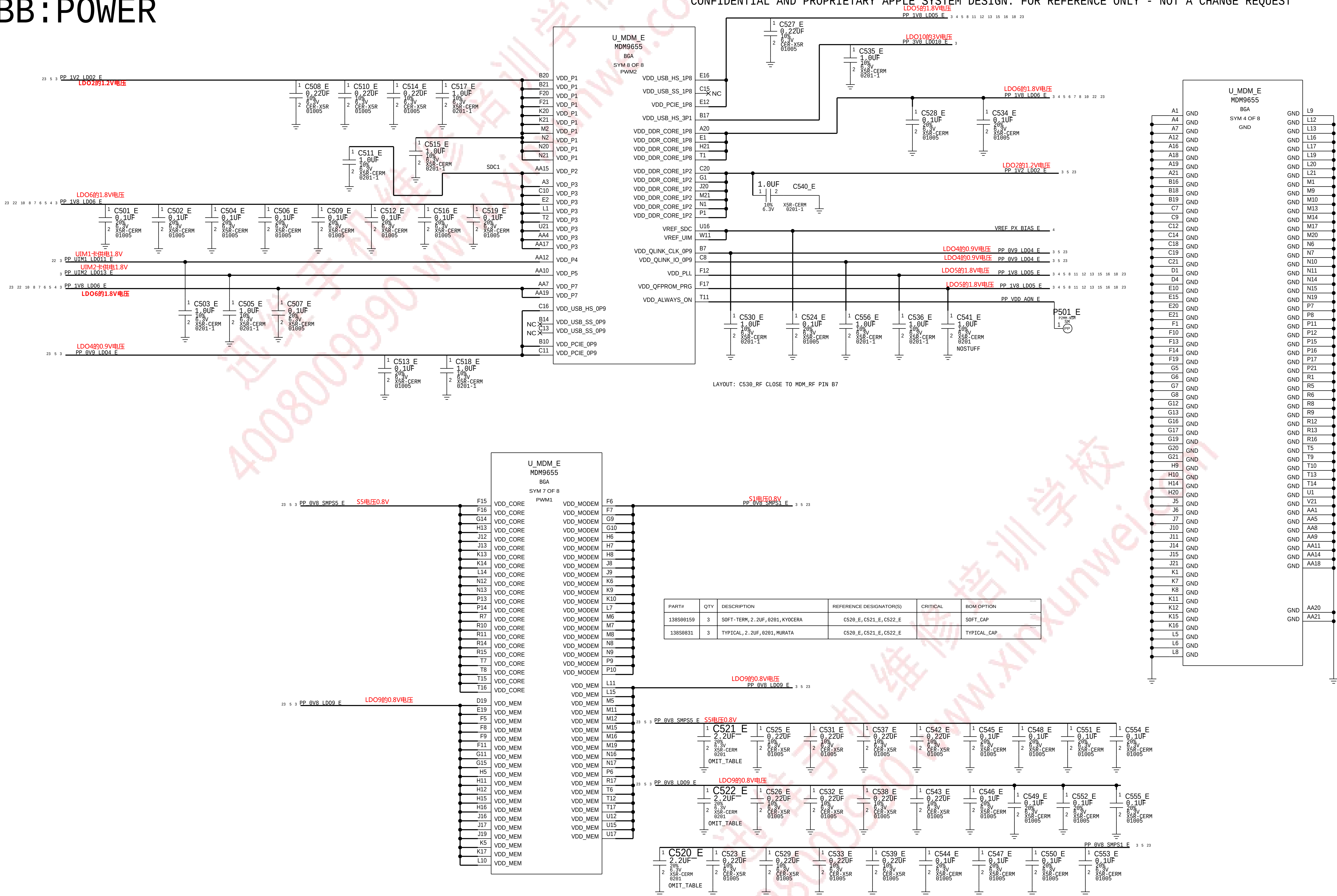
The image displays five decoupling capacitor footprints on a PCB layout, each with its component value and connection details:

- C302 E**: 180UF, 20% tolerance, 6.3V rating, 0603 package, 0.1MM pitch. Connected to PP_VDD_MAIN (pin 1) and GND (pin 2). MAKE_BASE=TRUE.
- C303 E**: 180UF, 20% tolerance, 6.3V rating, 0603 package, 0.1MM pitch. Connected to PP_VDD_MAIN (pin 1) and GND (pin 2). MAKE_BASE=TRUE.
- C304 E**: 180UF, 20% tolerance, 6.3V rating, 0603 package, 0.1MM pitch. Connected to PP_VDD_MAIN (pin 1) and GND (pin 2). MAKE_BASE=TRUE.
- C301 E**: 180UF, 20% tolerance, 6.3V rating, 0603 package, 0.1MM pitch. Connected to PP_VDD_MAIN (pin 1) and GND (pin 2). MAKE_BASE=TRUE.
- C305 E**: 180UF, 20% tolerance, 6.3V rating, 0603 package, 0.1MM pitch. Connected to PP_VDD_MAIN (pin 1) and GND (pin 2). MAKE_BASE=TRUE.

Timing diagram for PP_1V8_LD06_E MAKE_BASE=TRUE. The diagram shows three signals: XW306_E, XW307_E, and XW308_E, all labeled as SHORT-19L-07-1MM-SM. Each signal has a pulse width of 1 unit and a period of 11 units. The signals are connected to ROOM=PMU. A capacitor C1114_E (47PF, 5%, 36V, CERM, 01805) is connected to the ground of the signals.

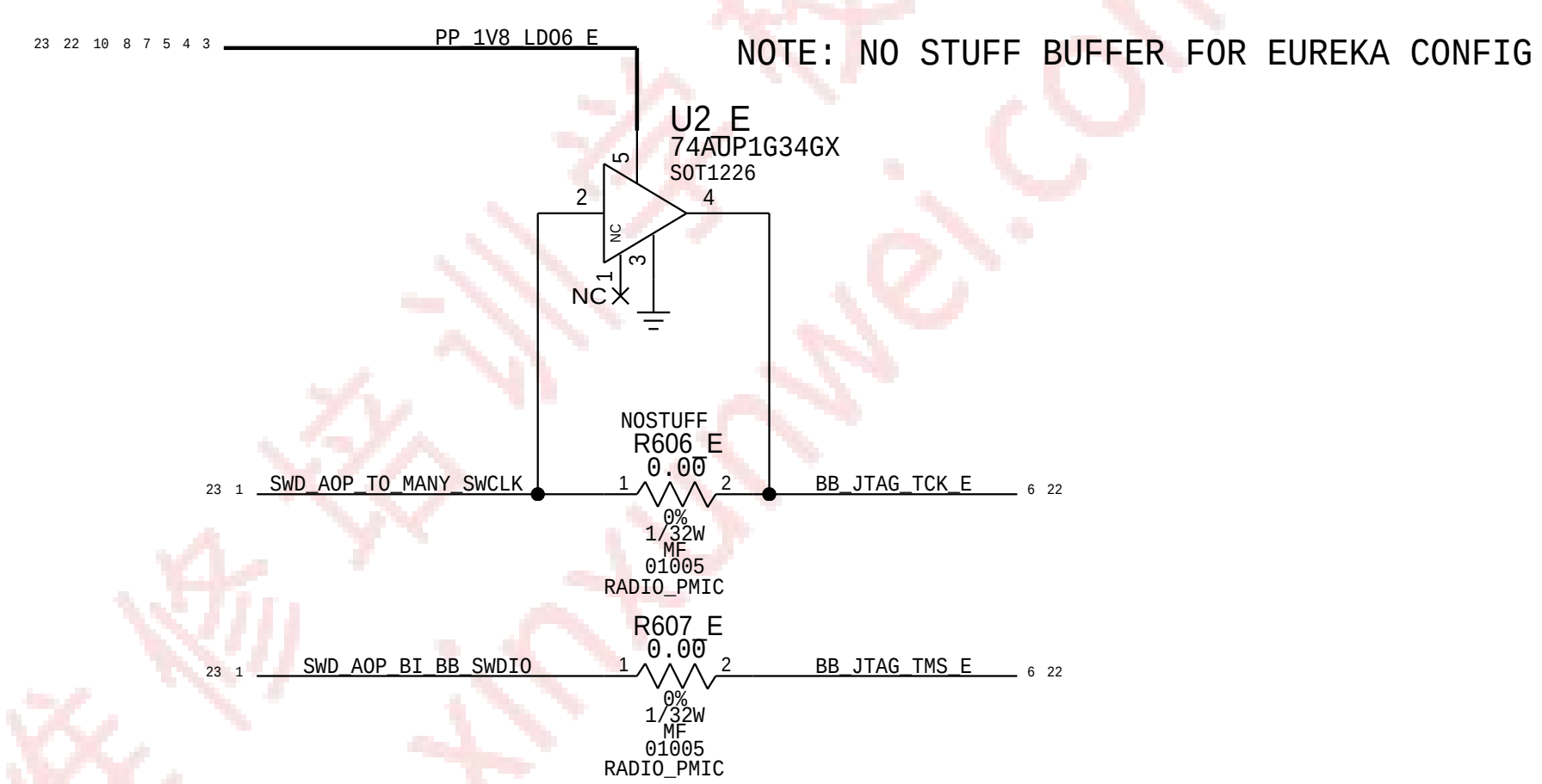
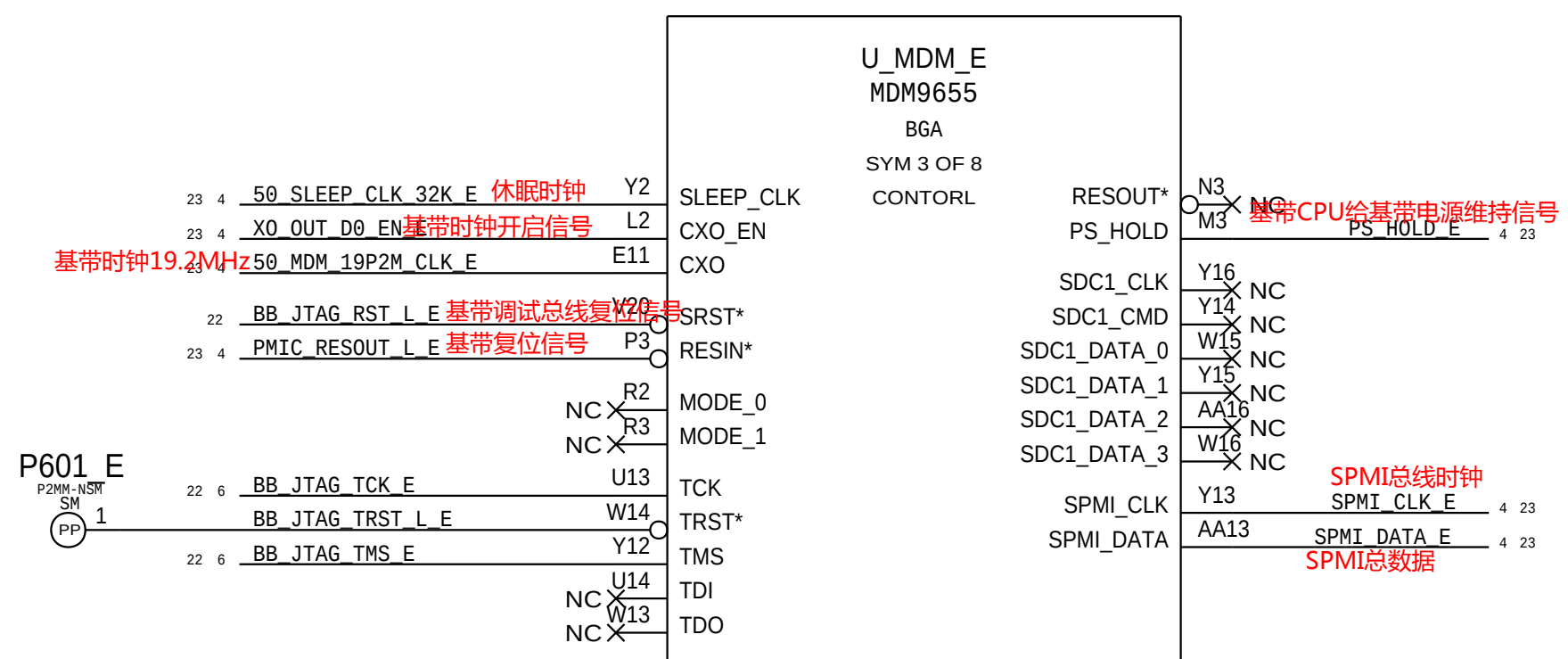
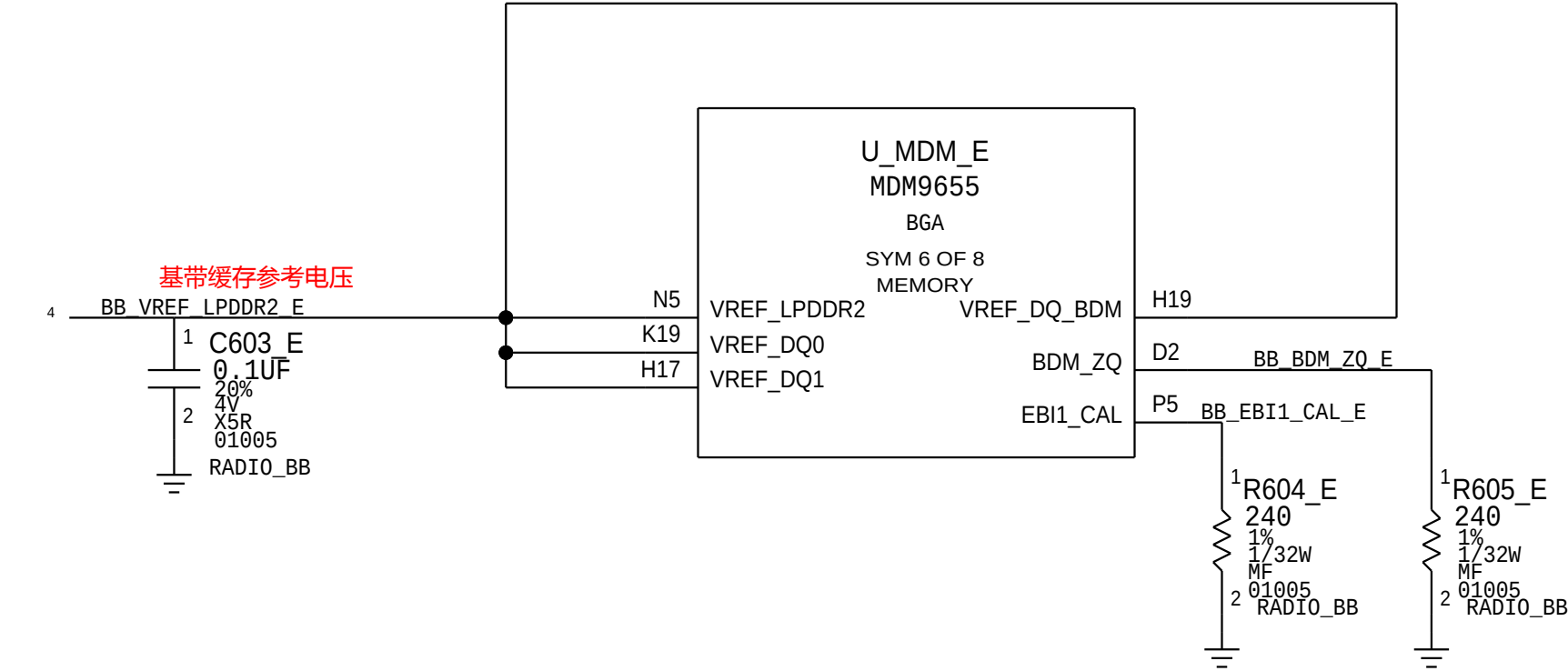
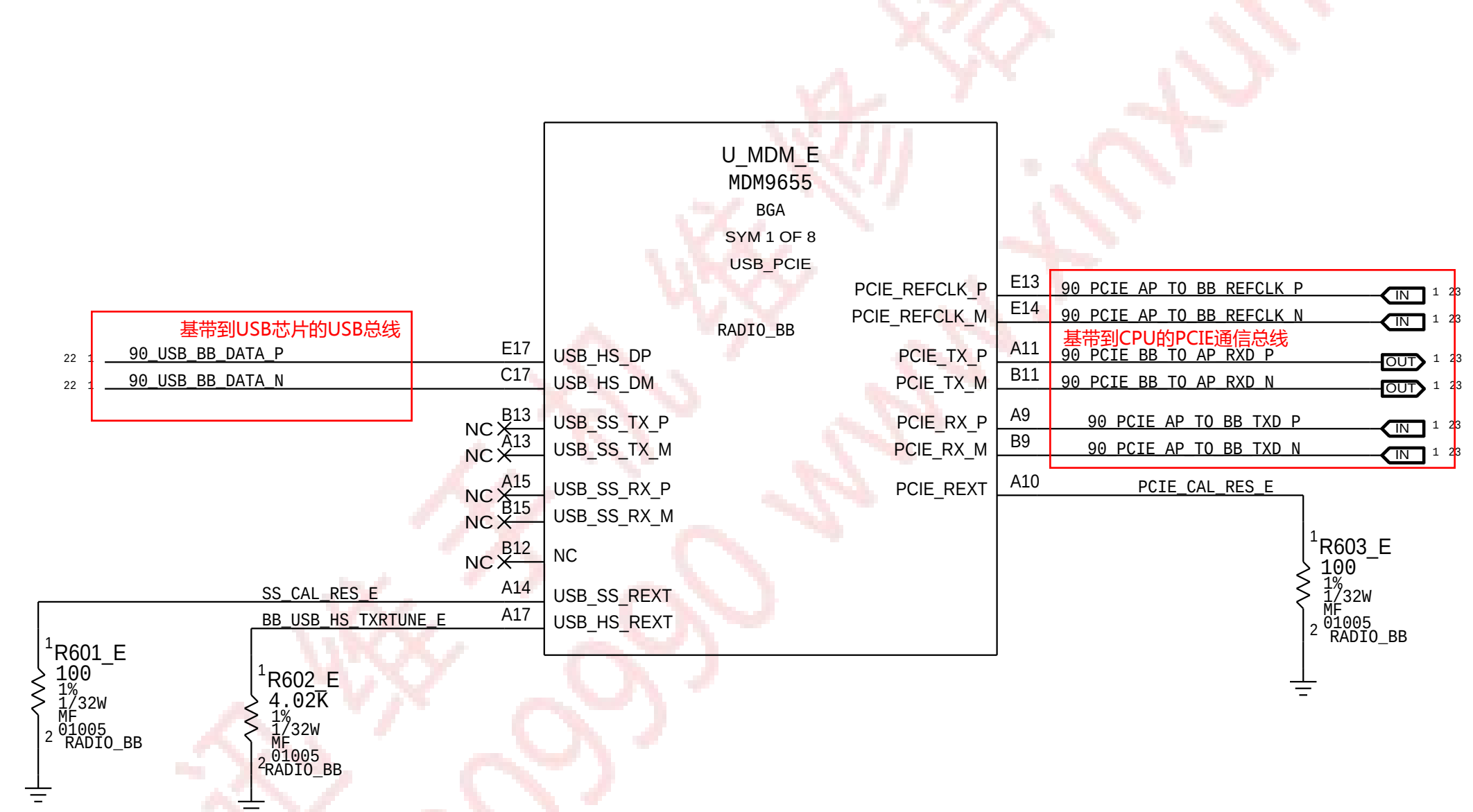
1.20V/1.5MA
1.20V/2MA
1V/431MA
0.90V/128MA
1.80V/272.4MA
1.80V/221.6MA
2.7V/5MA
1.00V/368.3MA
0.80V/1045MA
3.00V/30MA
1.8V/60MA
2.7V/69MA
1.8V/30MA

BB : POWER



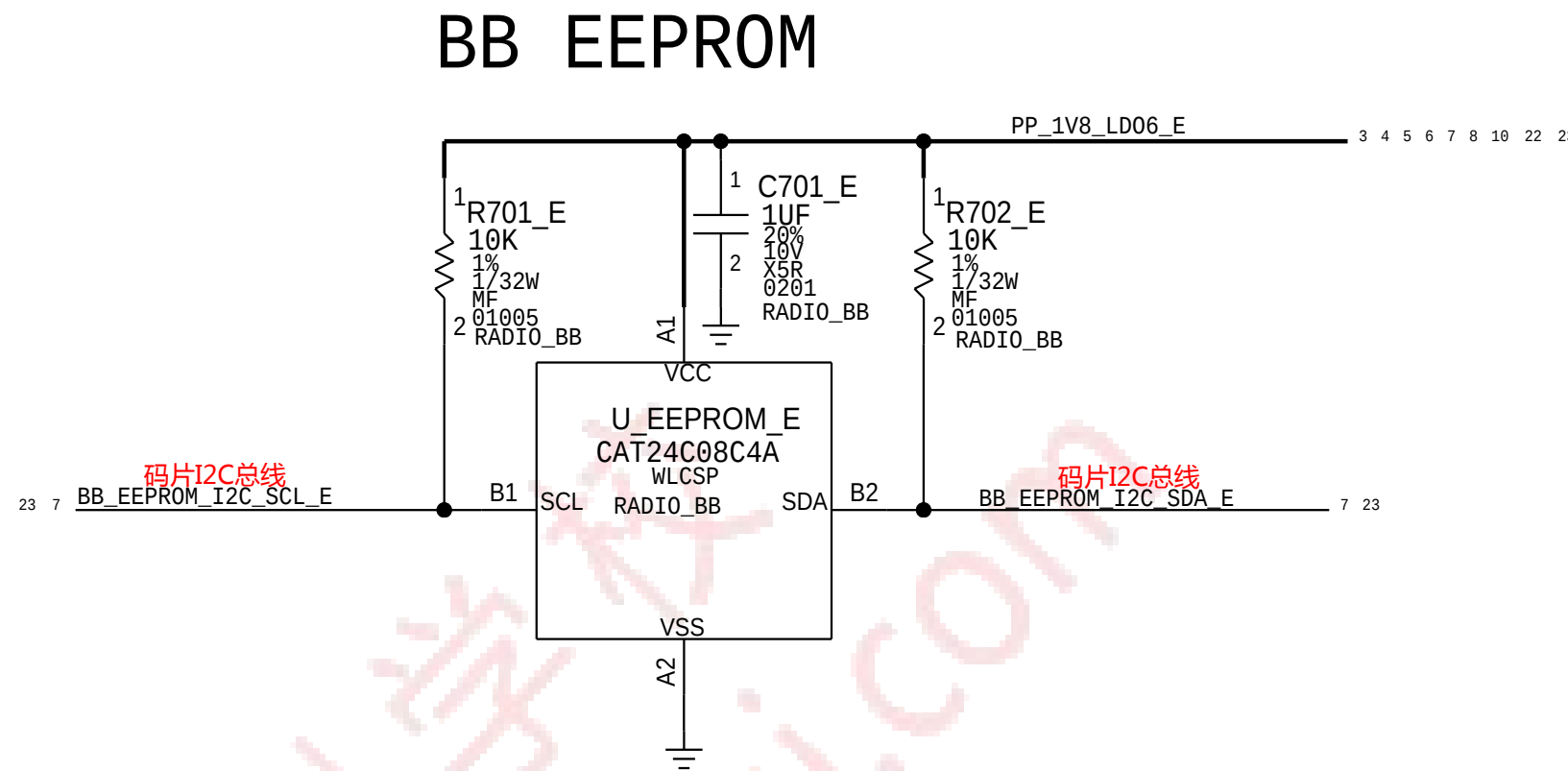
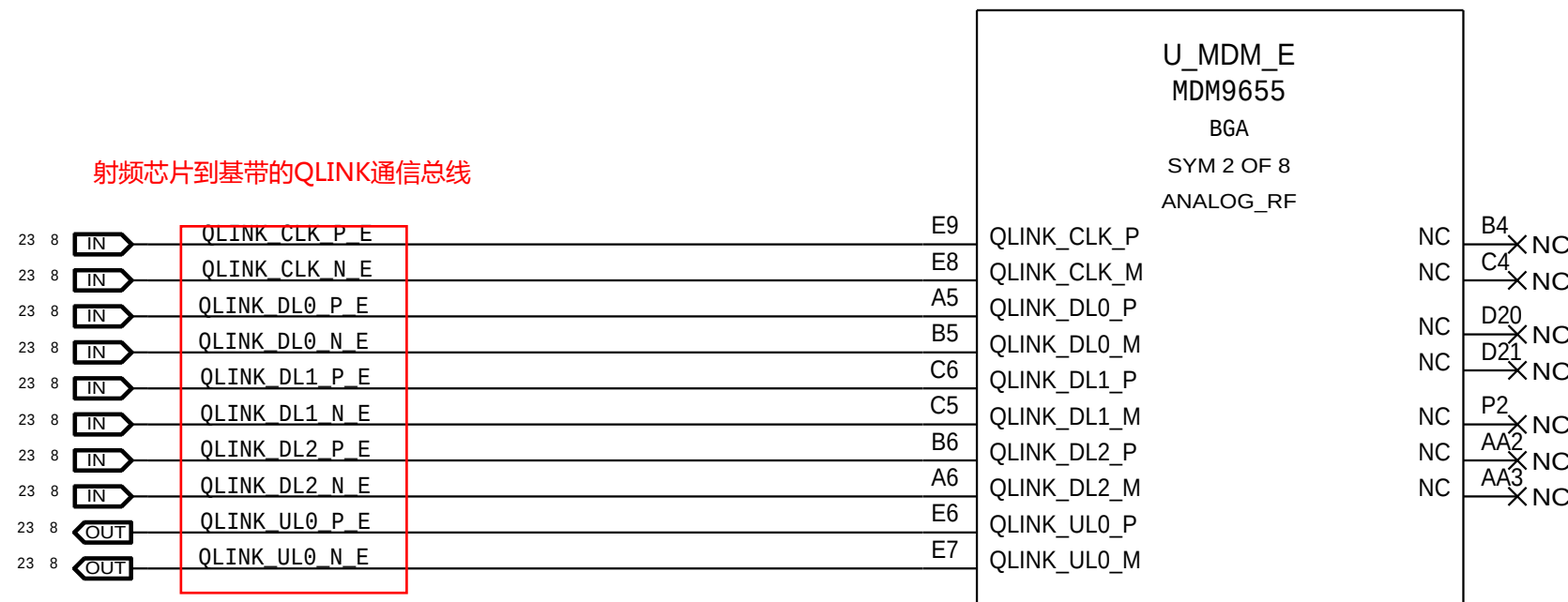
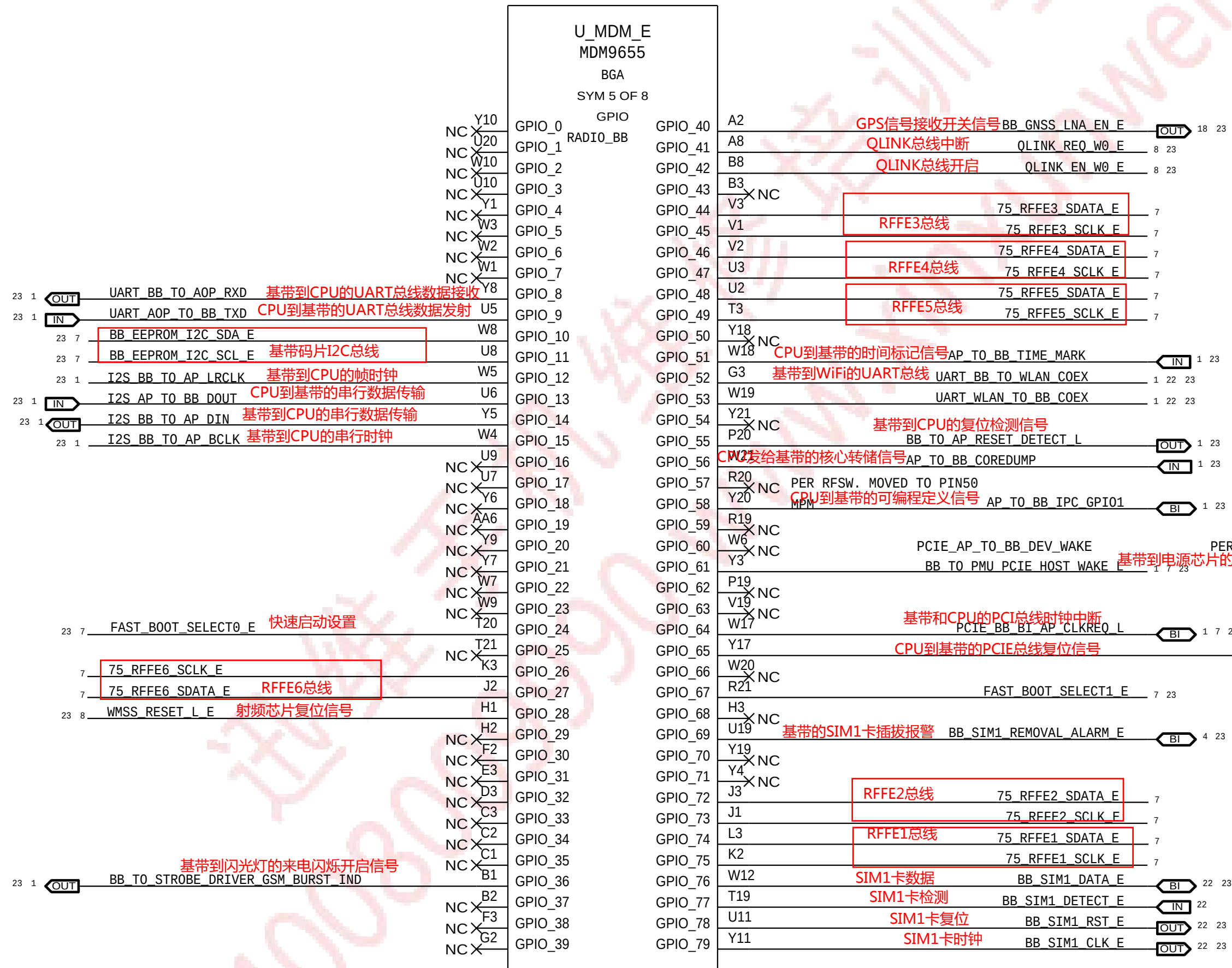


BB:CONTROL & HS PERIPHERALS



BB:GPIOS & QLINK

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RFFE MAP

RFFE1 - WTR, LATGPO, LATUNE

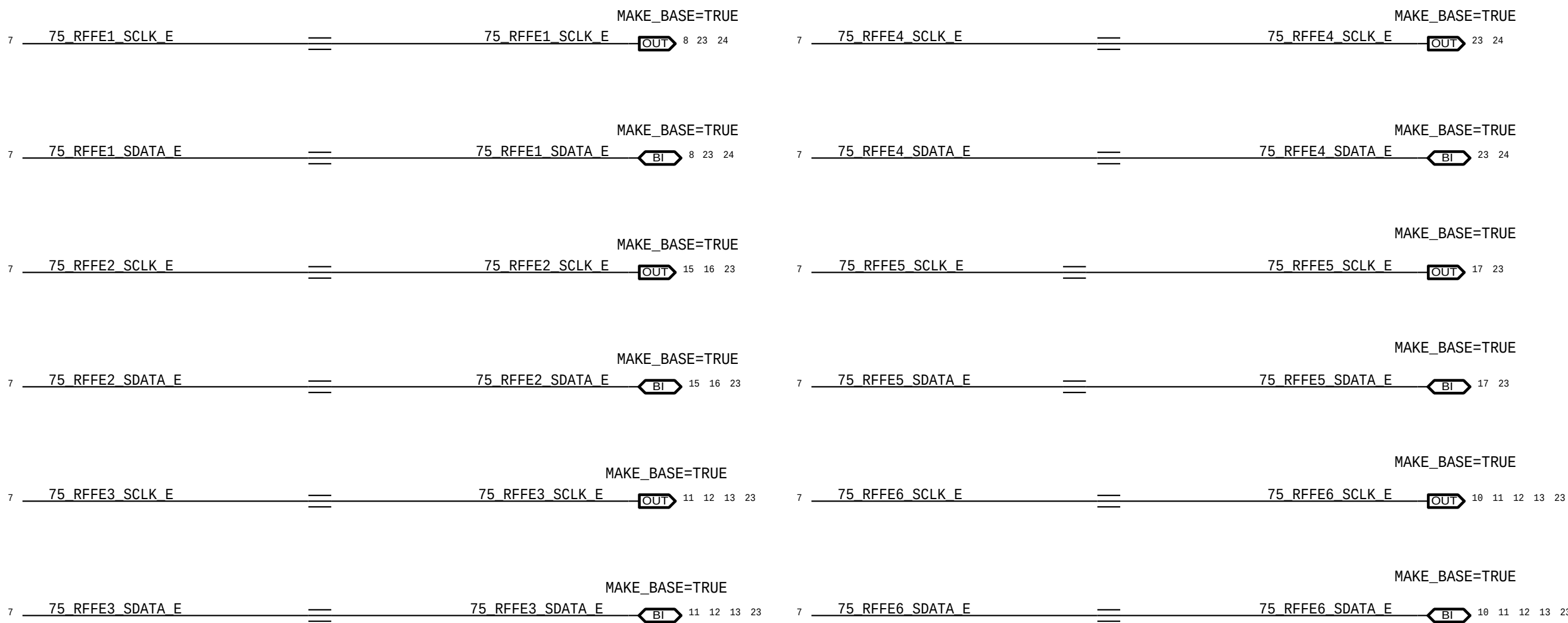
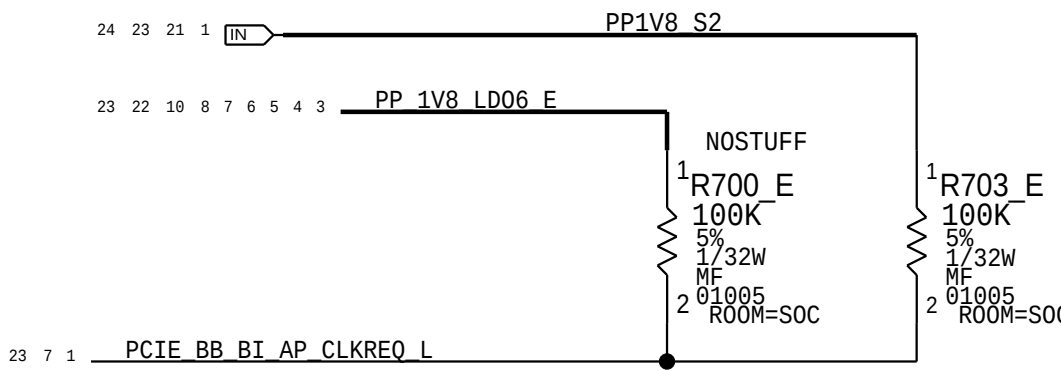
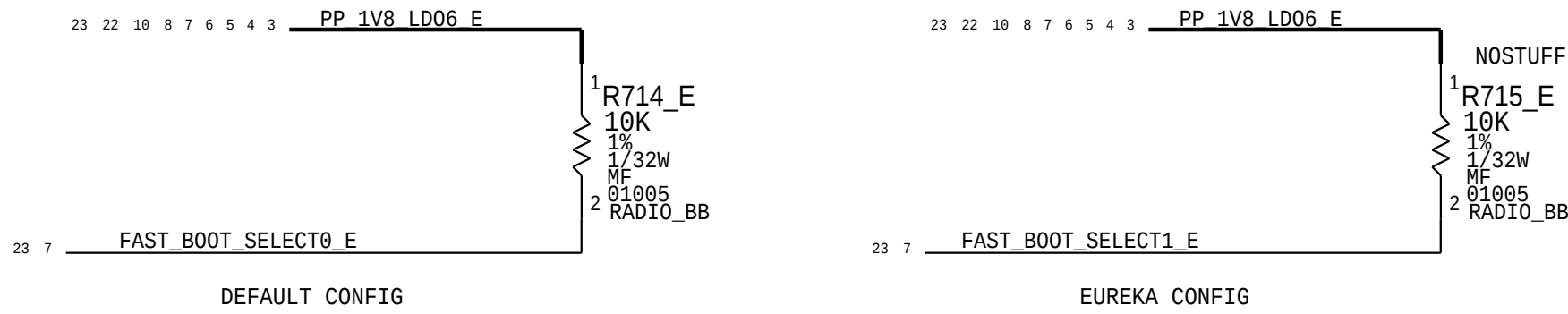
RFFE2 - HB DSM, LB DSM

RFFE3 - UHB, HB, LB PAD LNA

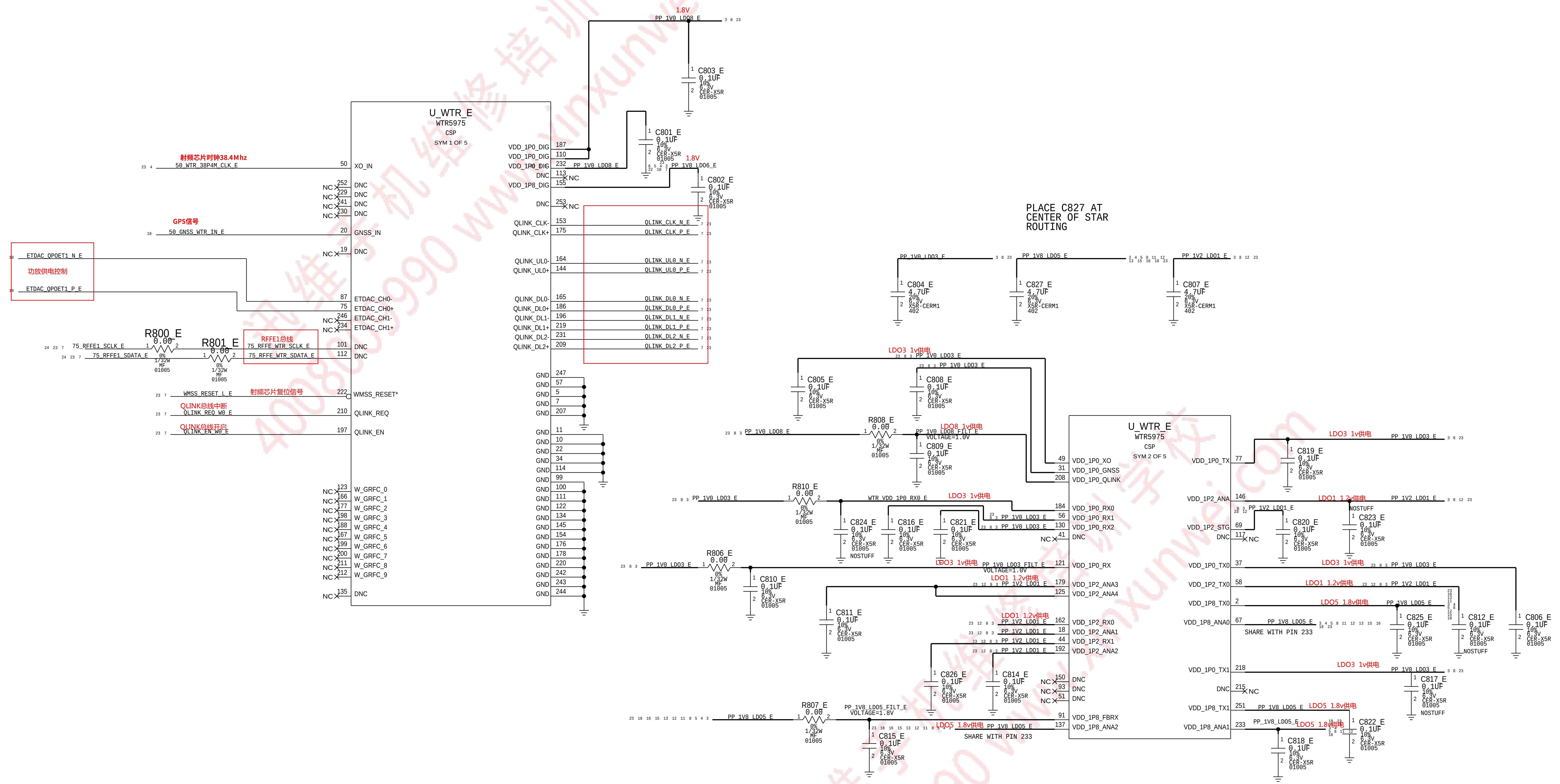
RFFE4 - UATGPO, UATUNE

RFFE5 - CPL

RFFE6 - QET, 2G PA/UHB, HB, LA PAD



TRANSCEIVER: QLINK & POWER

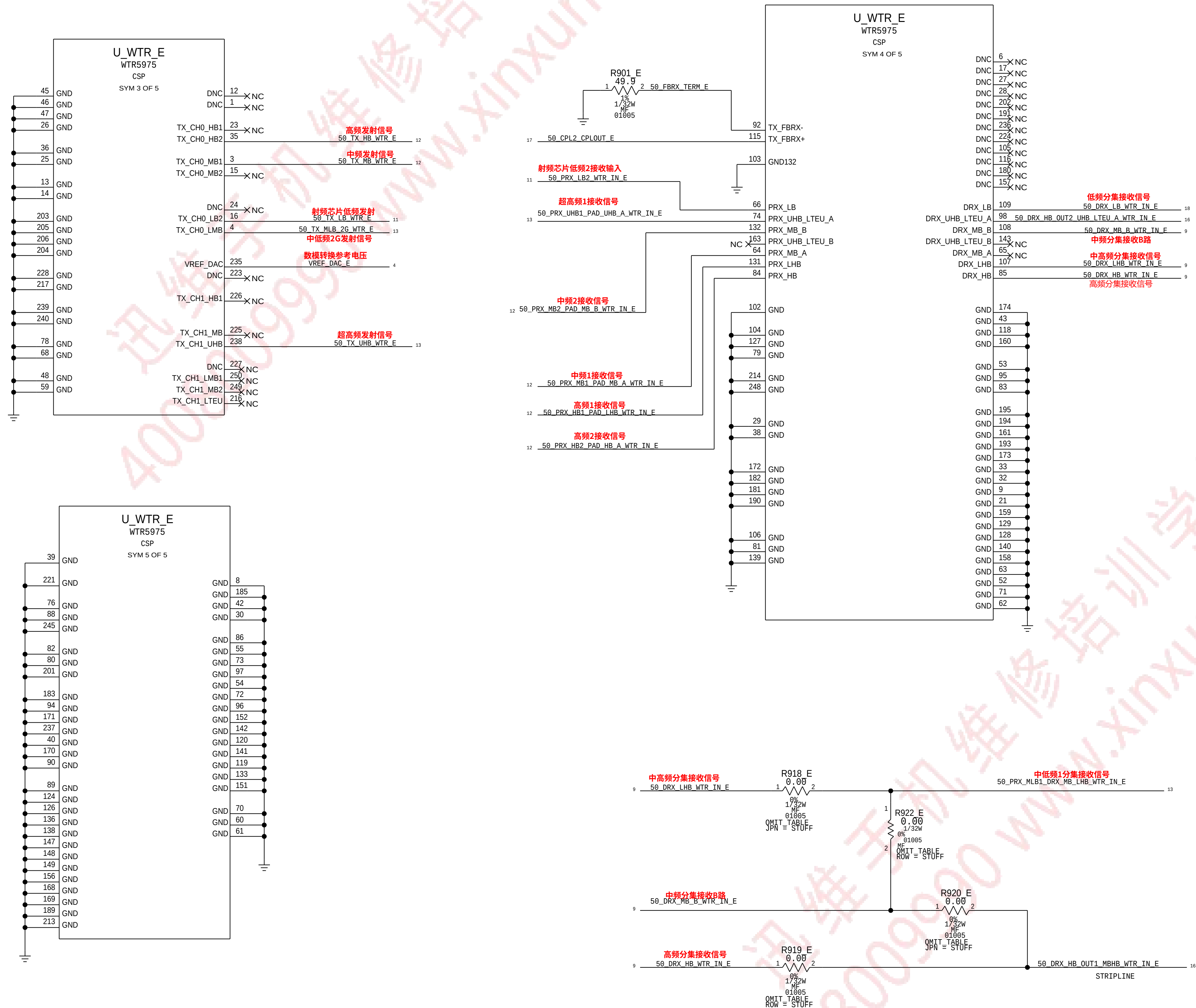


TRANSCEIVER: TX & RX

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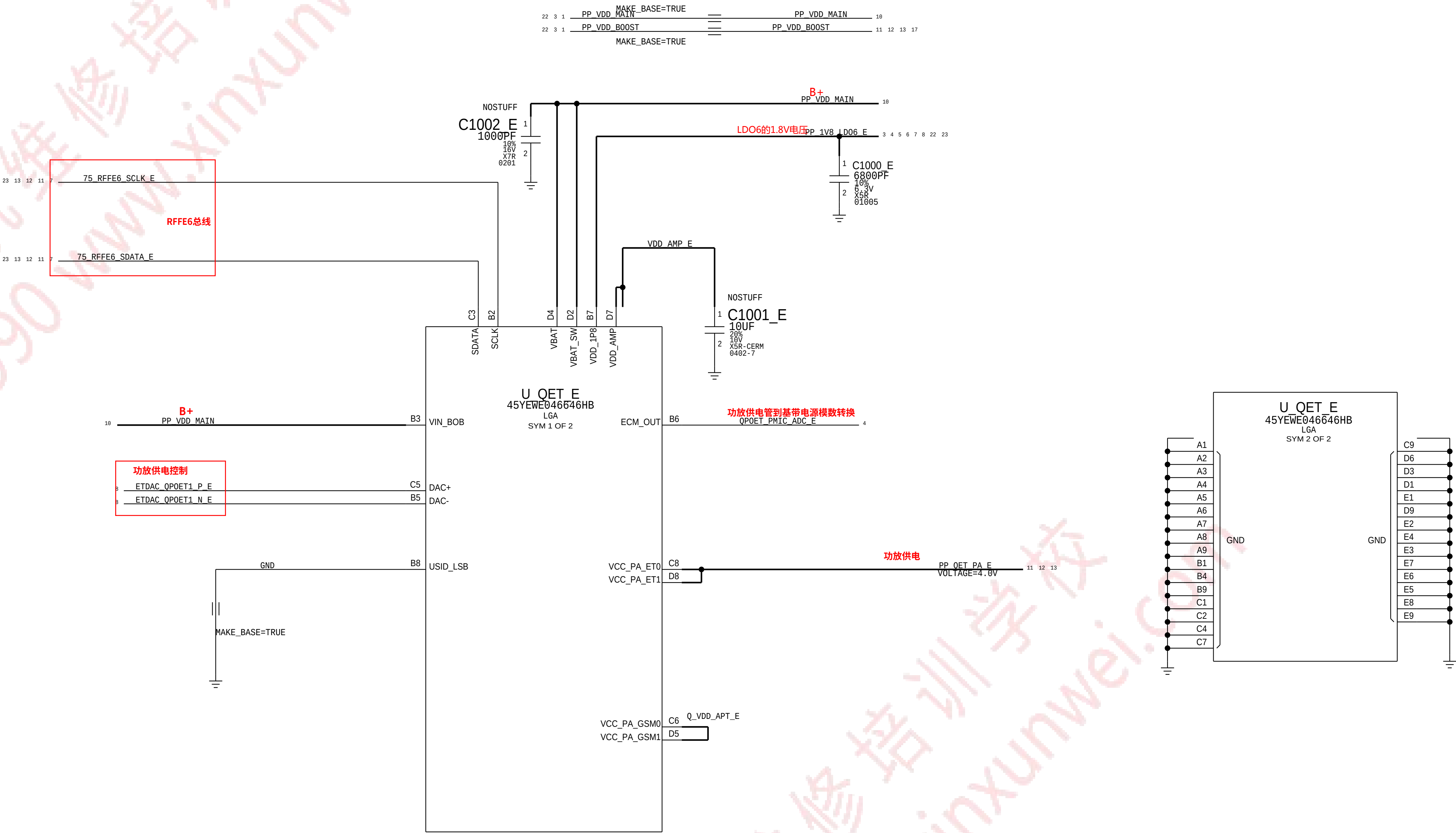
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 0R	R919_E	CRITICAL	ROW
117S0161	1	01005 0R	R922_E	CRITICAL	ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 0R	R018_E	CRITICAL	JPN
117S0161	1	01005 0R	R020_E	CRITICAL	JPN



QET:MODULE

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D





PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
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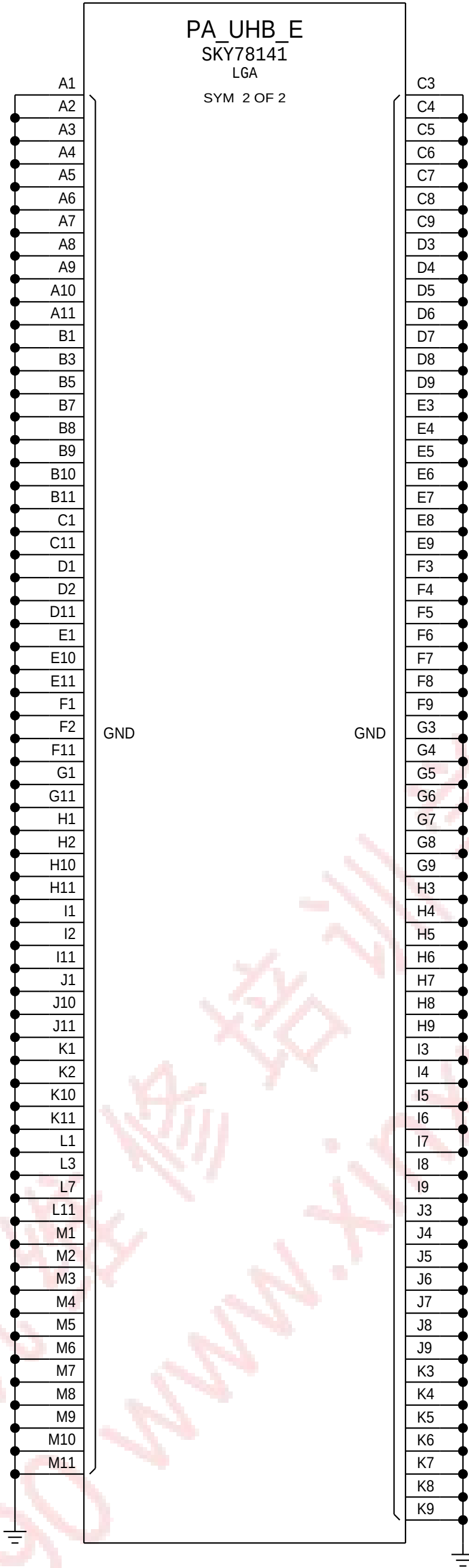
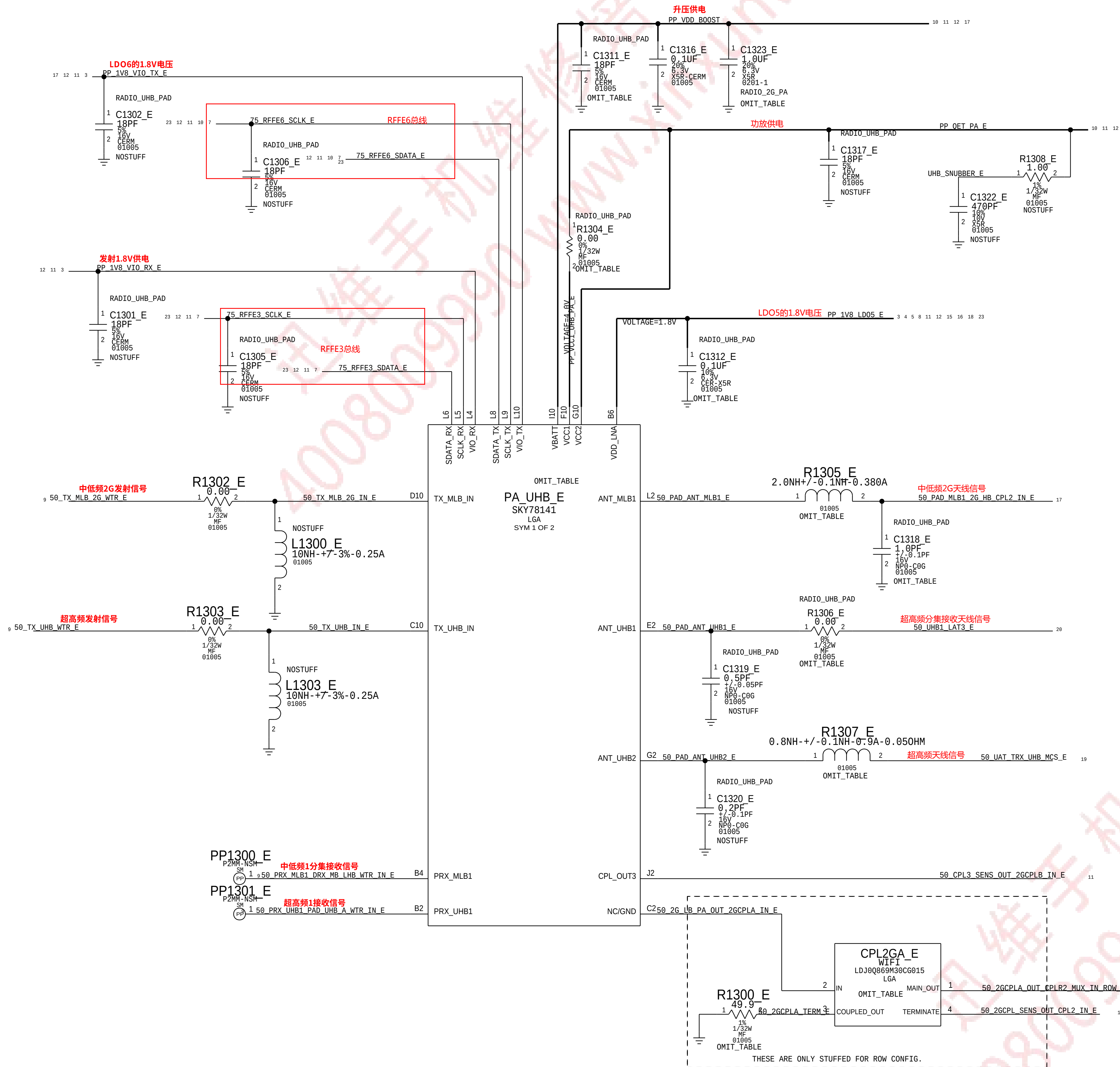




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UHB/2G PAD - JPN

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01243	1	UHB PAD MP	PA_UHB_E	CRITICAL	JPN
11750161	1	MLB/2GTX IN MCH2, 0 OHMS	R1302_E	CRITICAL	JPN
11750161	1	UHB INPUT MCH1, 0 OHMS	R1303_E	CRITICAL	JPN
11750161	1	MLB/2G MB OUT MCH1 01005 0R	R1305_E	CRITICAL	JPN
11750161	1	UHB1 OUTPUT MCH1, 0 OHMS	R1306_E	CRITICAL	JPN
152500494	1	UHB2 OUT/DRX BP MCH1, 0.8NH 0.9A	R1307_E	CRITICAL	JPN
13250603	1	UHB LNA DECOUPLING, 0.1UF 6.3V 10%	C1312_E	CRITICAL	JPN
13150214	1	VBAT DECOUPLING, 18PF	C1311_E	CRITICAL	JPN
13850692	1	VBAT DECOUPLING, 1UF	C1323_E	CRITICAL	JPN
11750161	1	VCC2/VCC1 RESISTOR OPTION, 0 OHM	R1304_E	CRITICAL	JPN
13150375	1	MLB OUTPUT MCH UAT, 01005 1.0PF	C1318_E	CRITICAL	JPN



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CHINAFOX

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D

2G PA - ROW

C

B

A

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S01247	1	2G PA MP	PA_UHB_E	CRITICAL	ROW
117S0161	1	MLB/26TX IN MCH2, 0 OHMS	R1302_E	CRITICAL	ROW
117S0161	1	UHB INPUT MCH1, 0 OHMS	R1303_E	CRITICAL	ROW
152S1917	1	01005 2.0NH 0.6A	R1305_E	CRITICAL	ROW
117S0161	1	UHB2 OUT/DRX BP MCH1, 0 OHMS	R1307_E	CRITICAL	ROW
131S0214	1	VBAT DECOUPLING,18PF	C1311_E	CRITICAL	ROW
138S0692	1	VBAT DECOUPLING, 1UF	C1323_E	CRITICAL	ROW
117S0161	1	VCC2/VCC1 RESISTOR OPTION, 0 OHM	R1304_E	CRITICAL	ROW
155S00316	1	ROW CLPA	CPL2GA_E	CRITICAL	ROW
118S0652	1	01005 49.9R	R1300_E	CRITICAL	ROW

8

7

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5

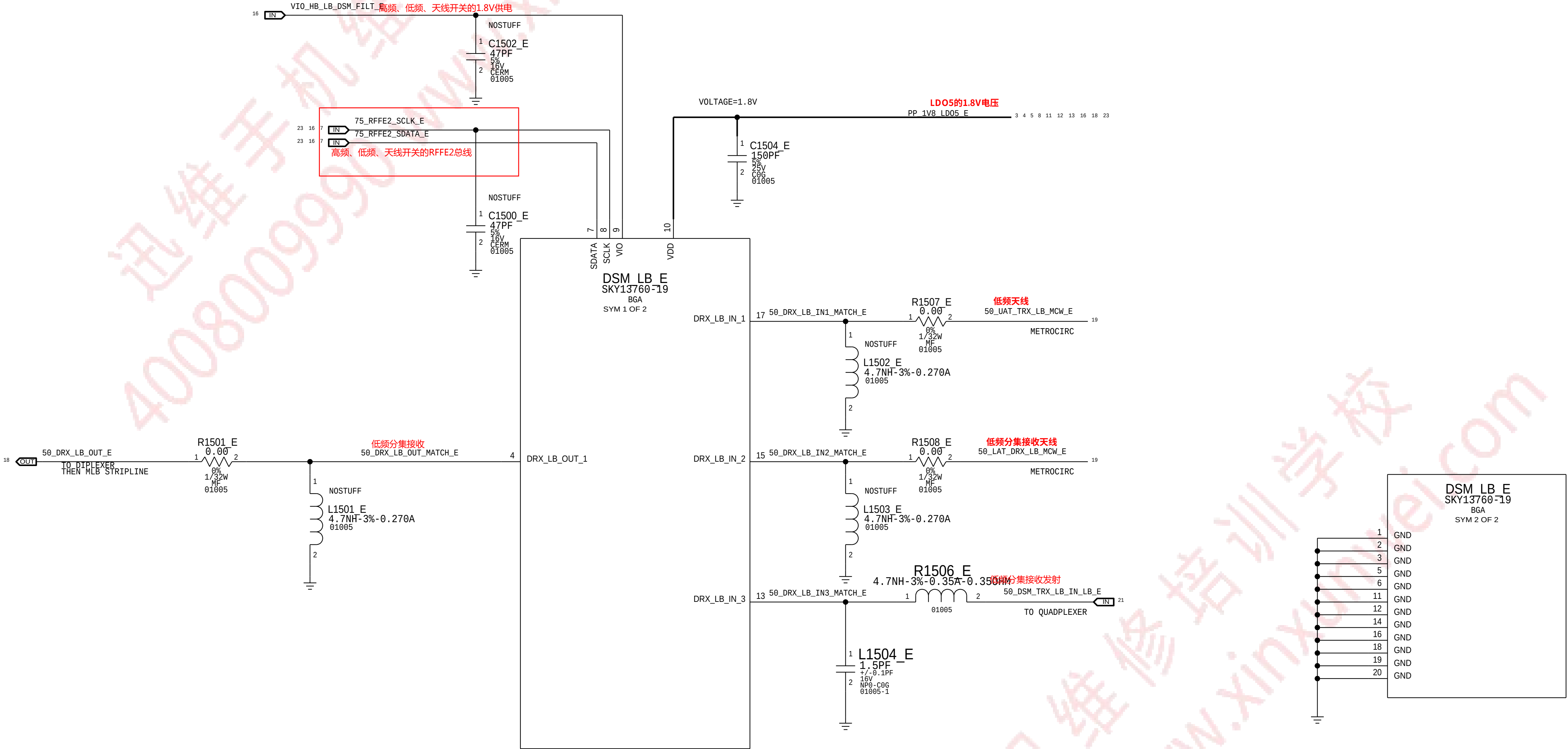
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3

2

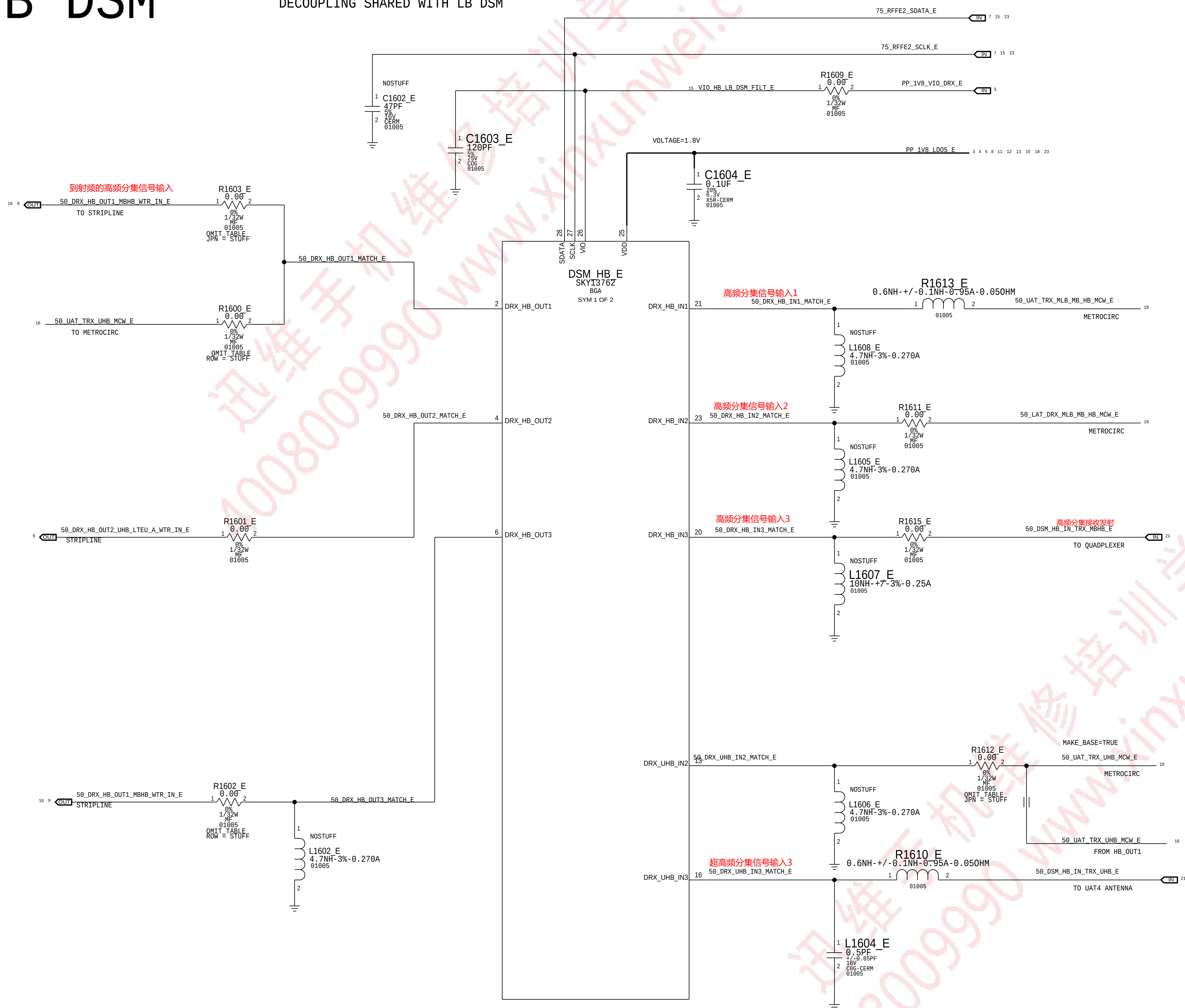
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LB DSM



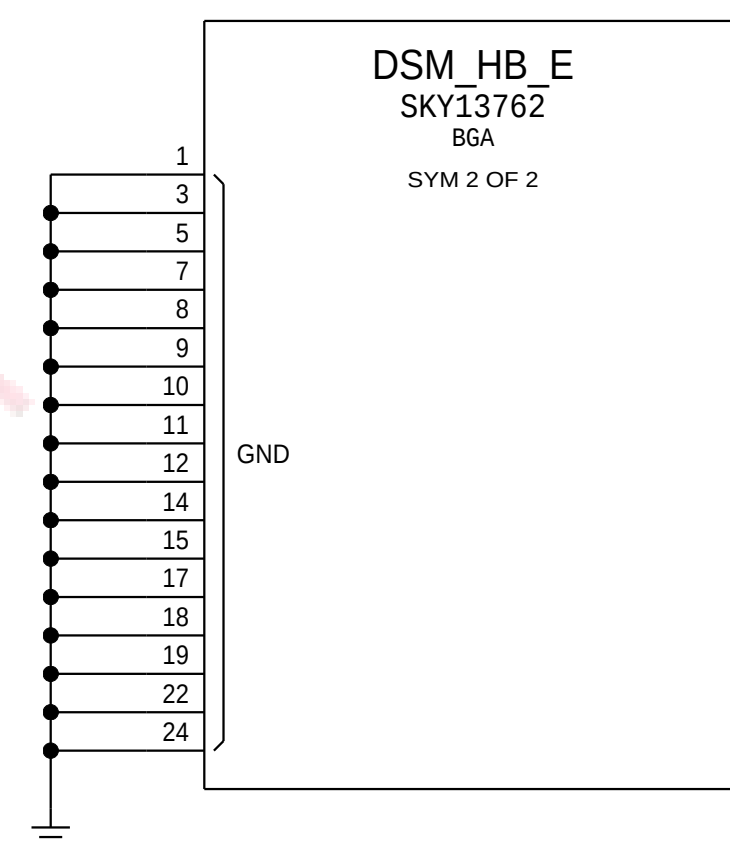
HB DSM

DECOUPLING SHARED WITH LB DSM

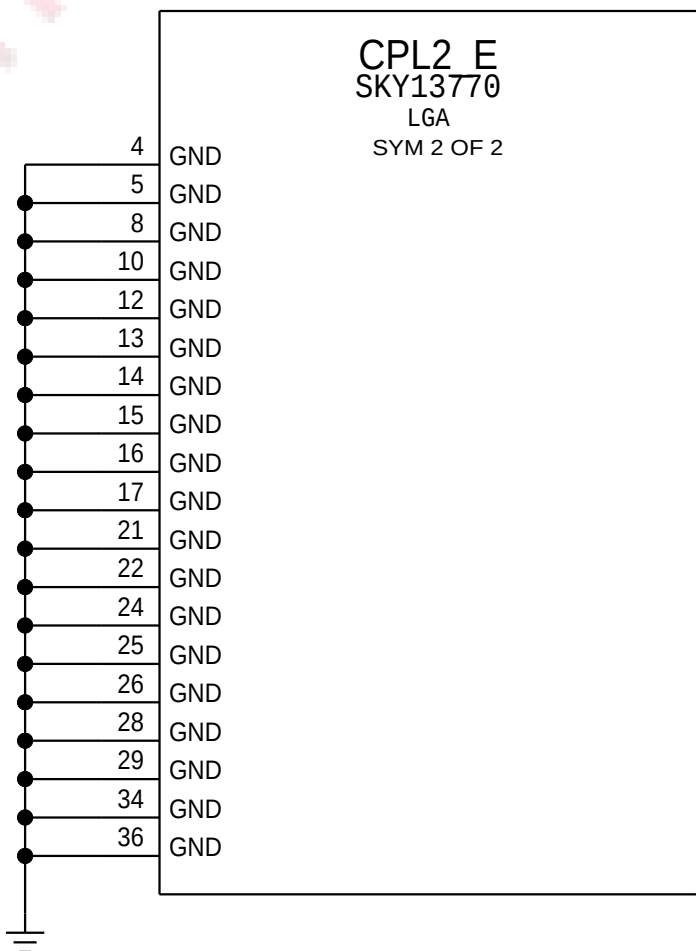
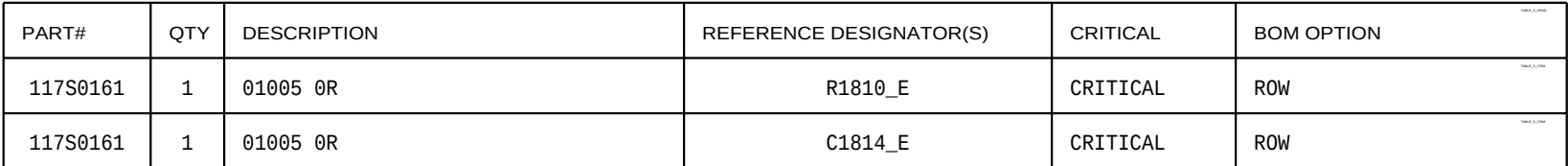


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
117S0161	1	01005 0R	R1600_E	CRITICAL	ROW
117S0161	1	01005 0R	R1602_E	CRITICAL	ROW

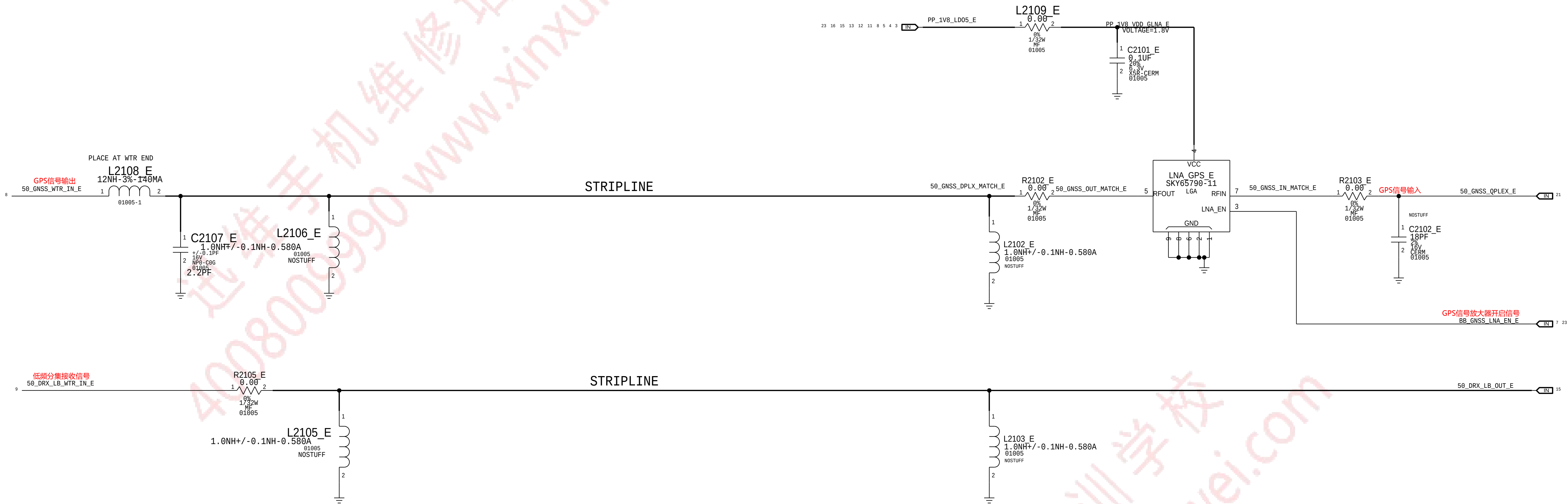
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
11750161	1	01005 0R	R1603_E	CRITICAL	JPN
11750161	1	01005 0R	R1612_E	CRITICAL	JPN



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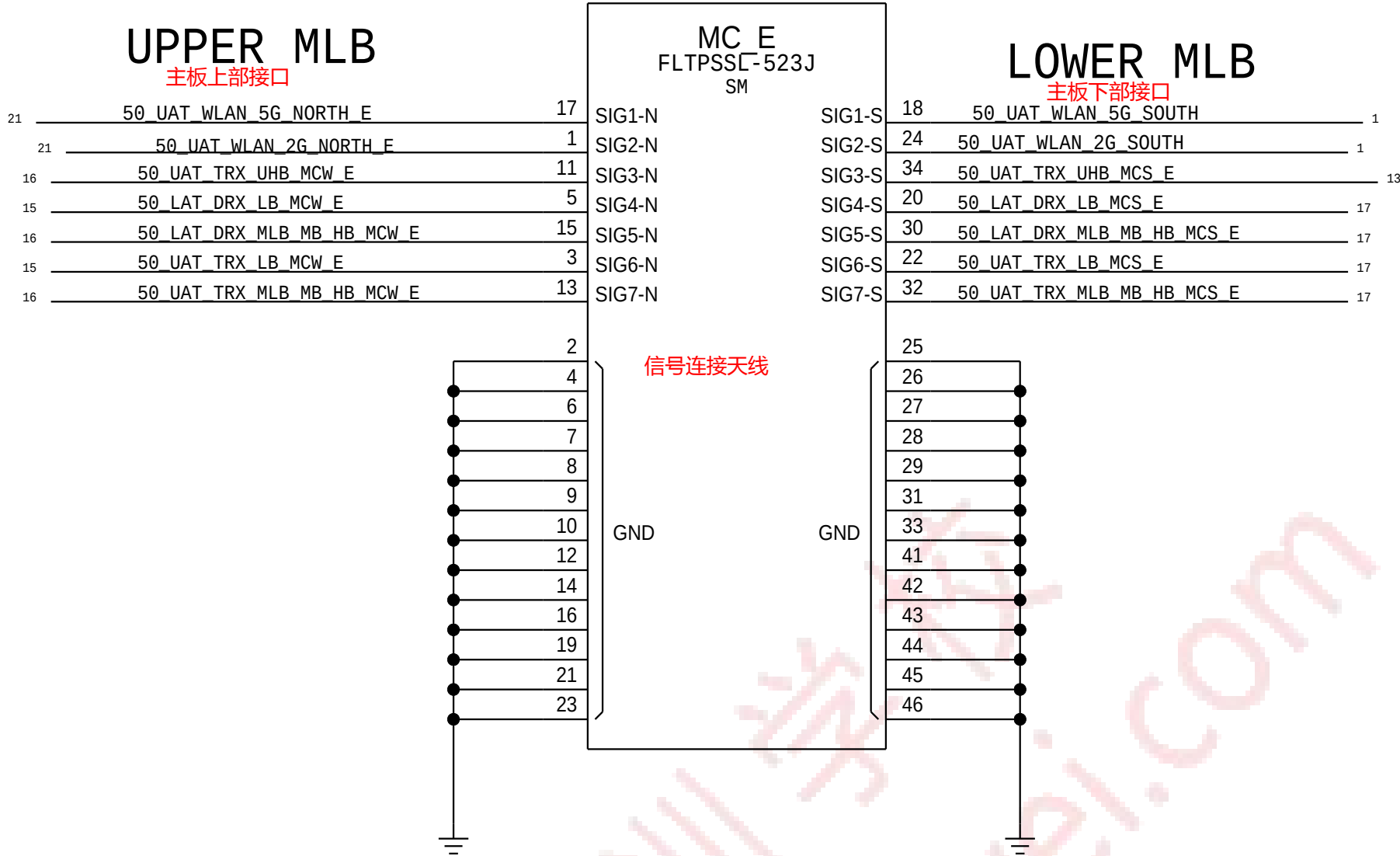


GPS & GPS/LB DIPLEXER

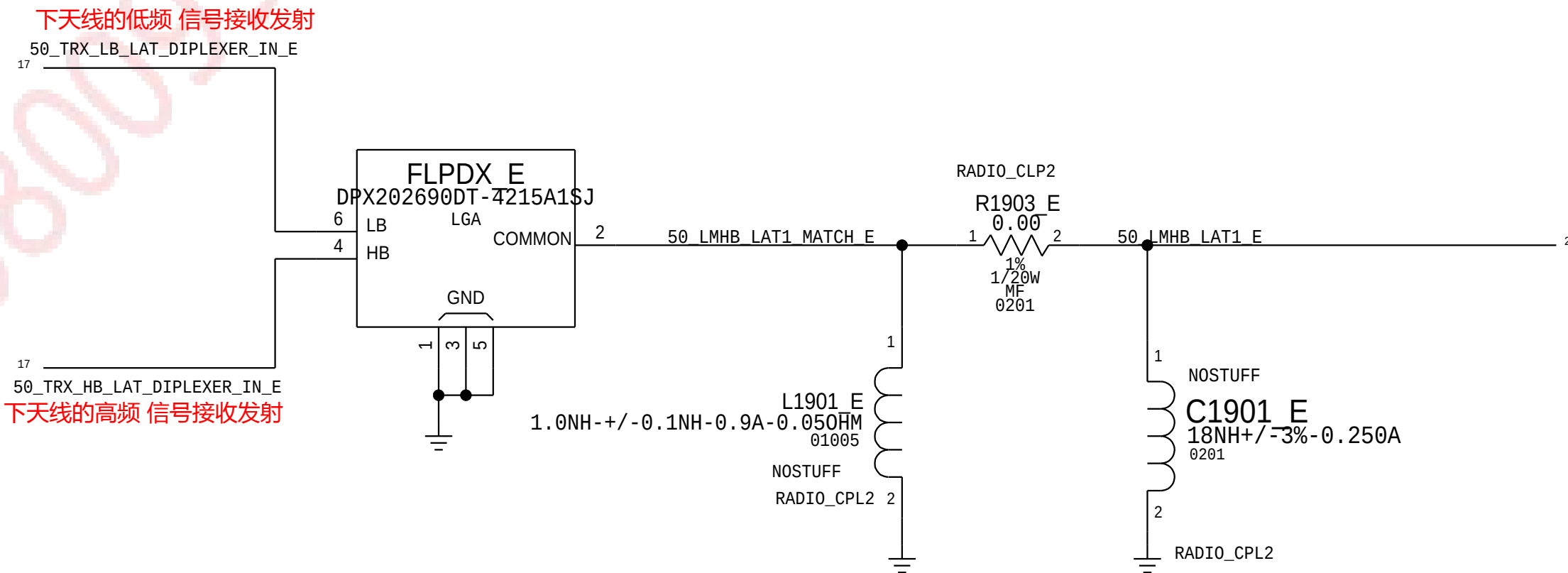


METROCIRC

METROCIRC



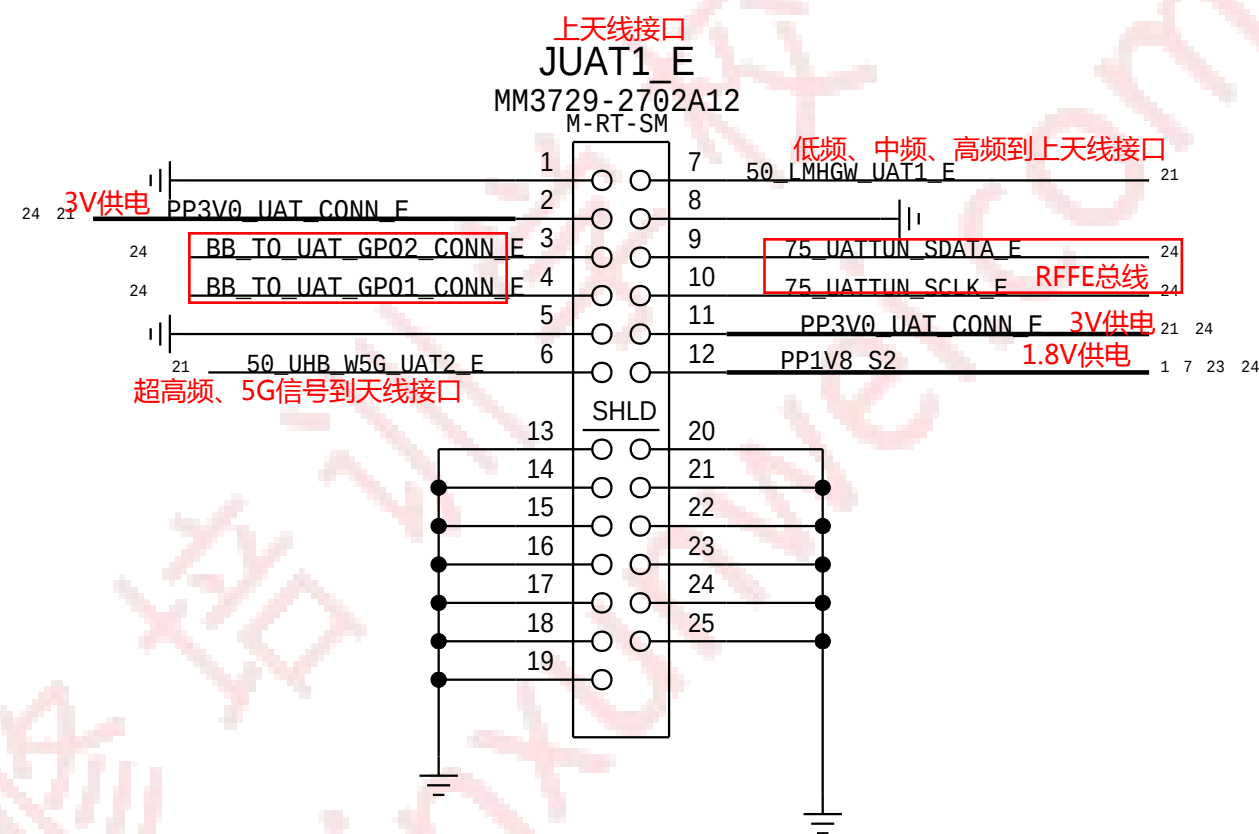
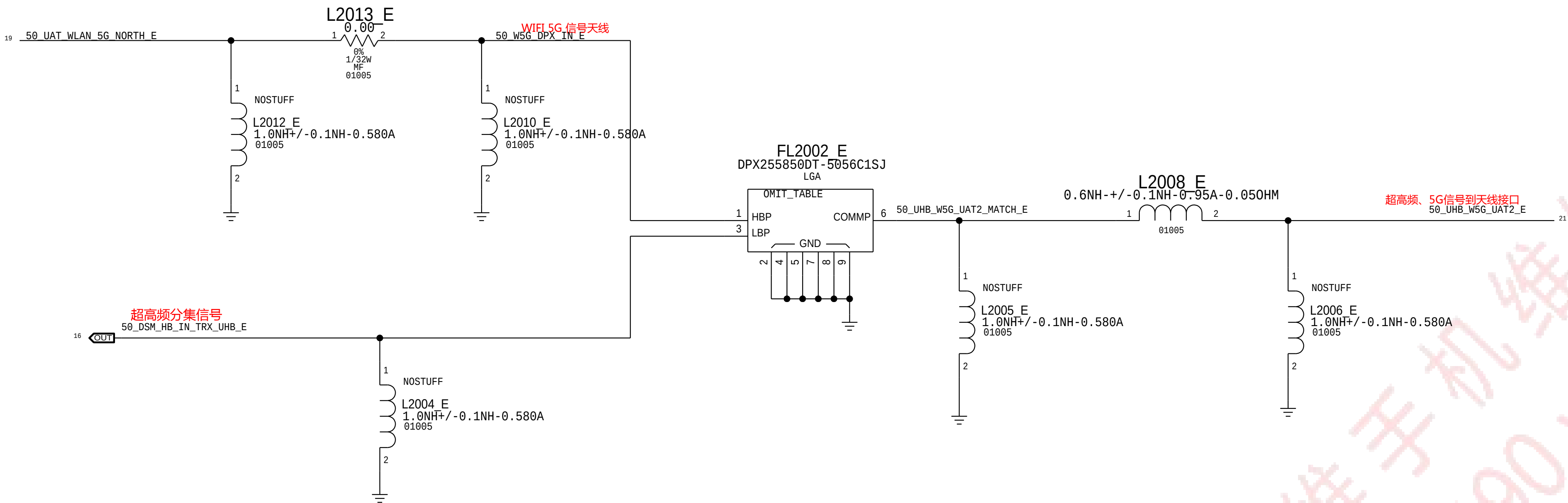
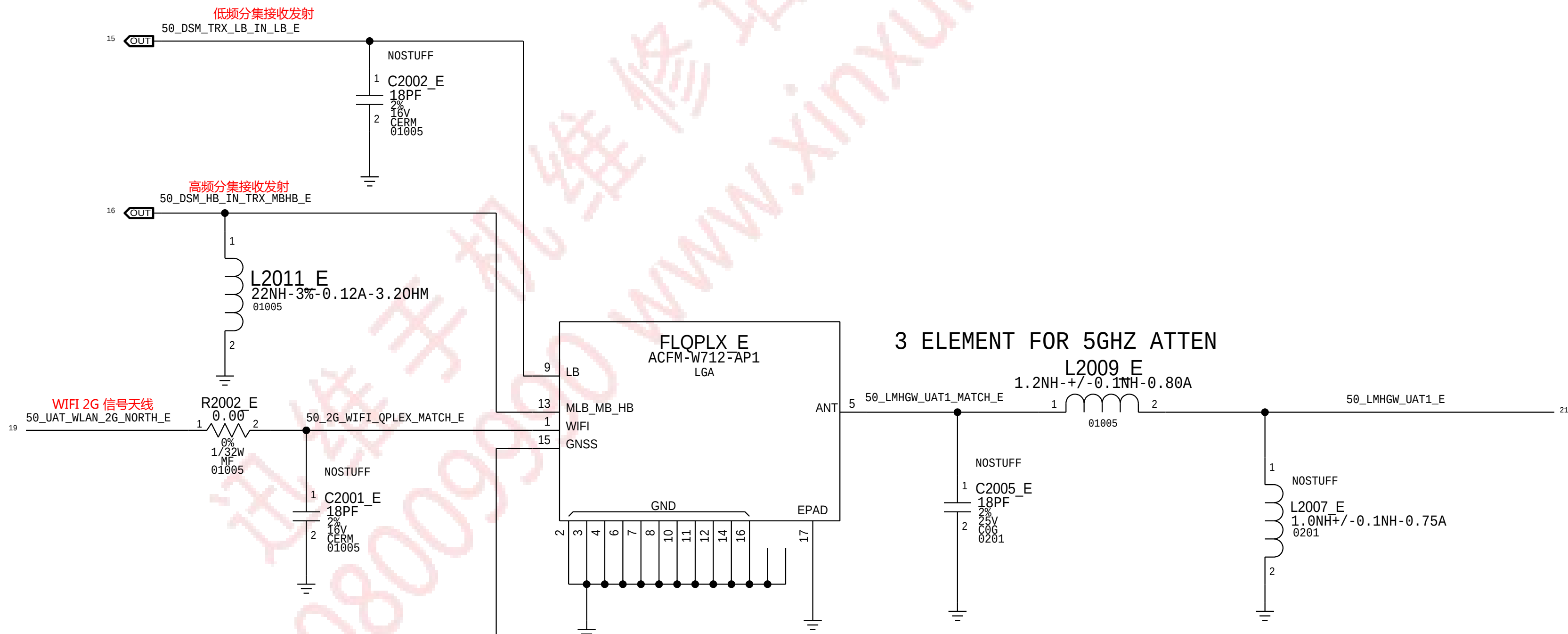
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UAT & LB/UHB DIPLEXER

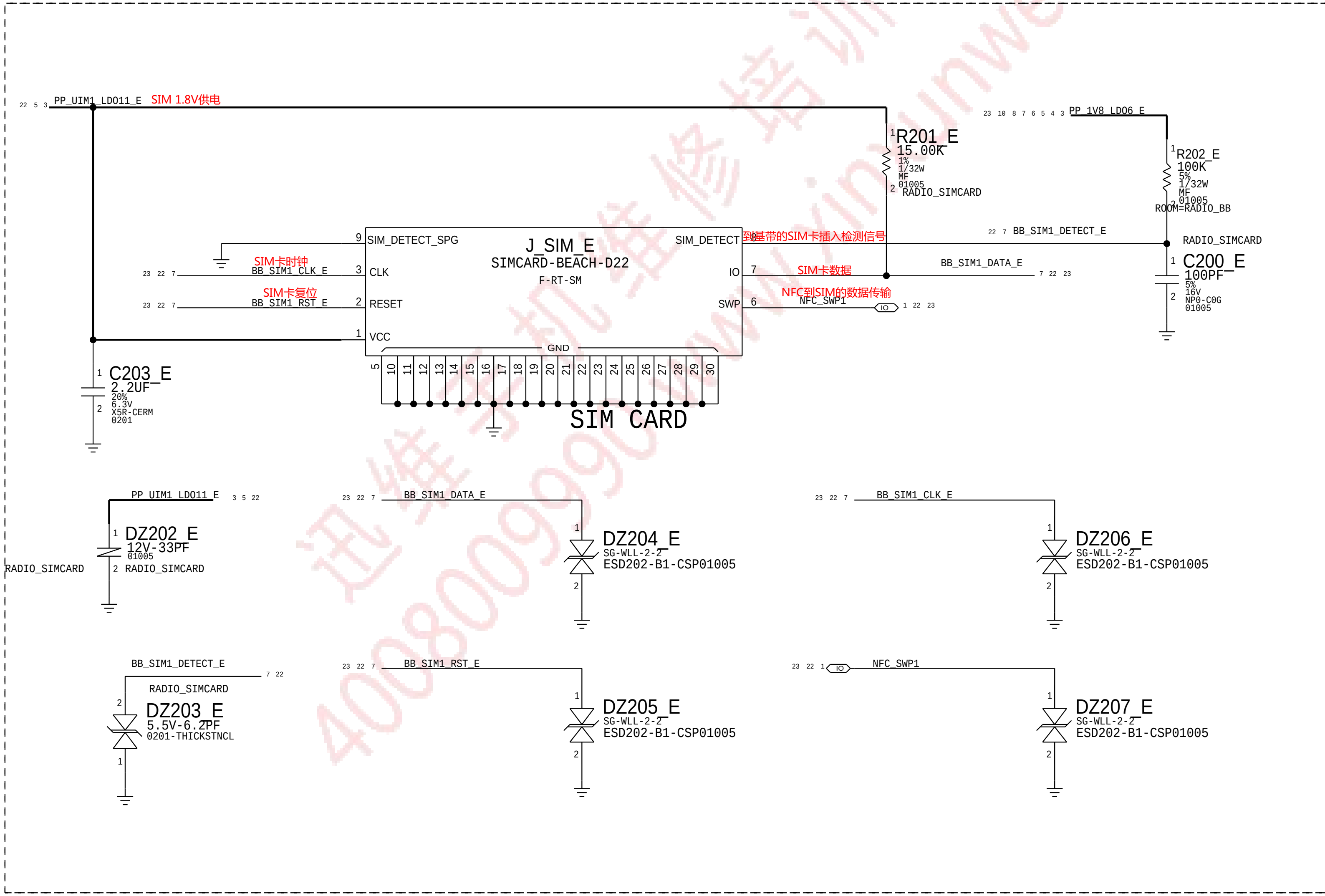
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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
155S00200	1	B42_WIFI5_DPX	FL2002_E	CRITICAL	JPN
155S00290	1	B42_WIFI5_DPX_PASSTHRU	FL2002_E	CRITICAL	ROW

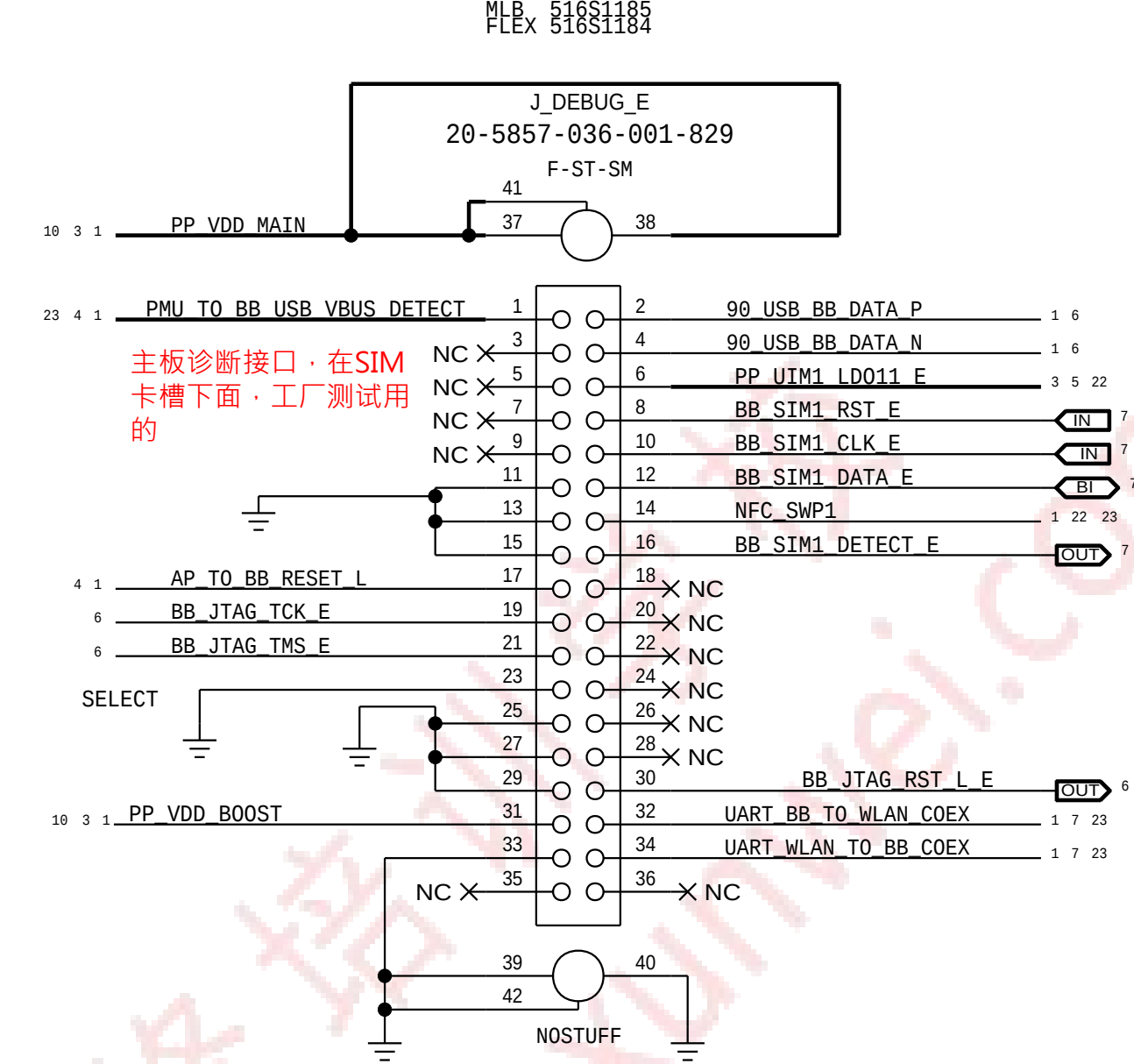


SIM CONN EUICC & DEBUG CONN

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DEBUG CONNECTOR

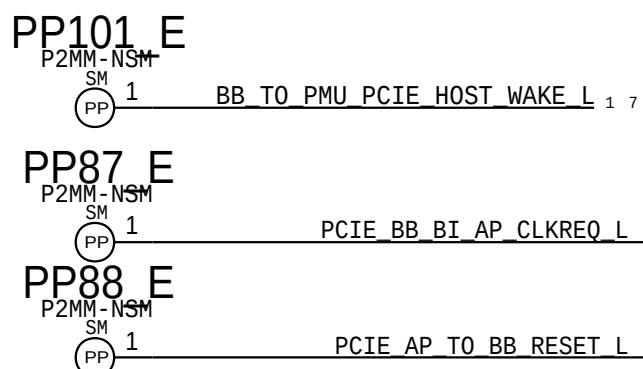




BB PROBE POINTS

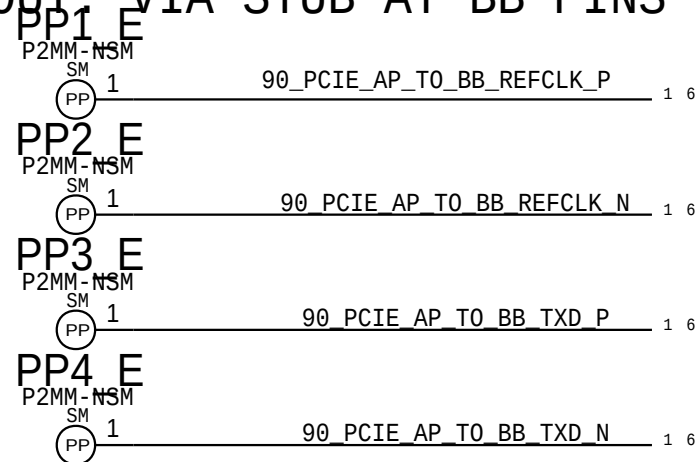
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D

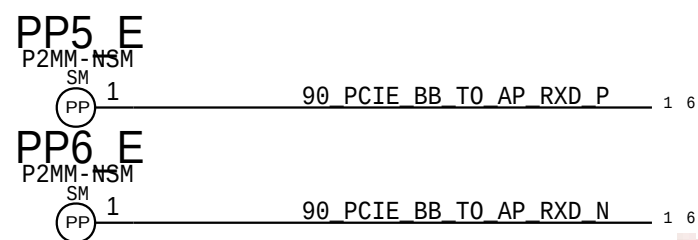


PCIE

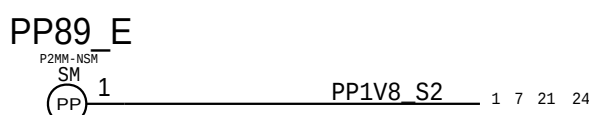
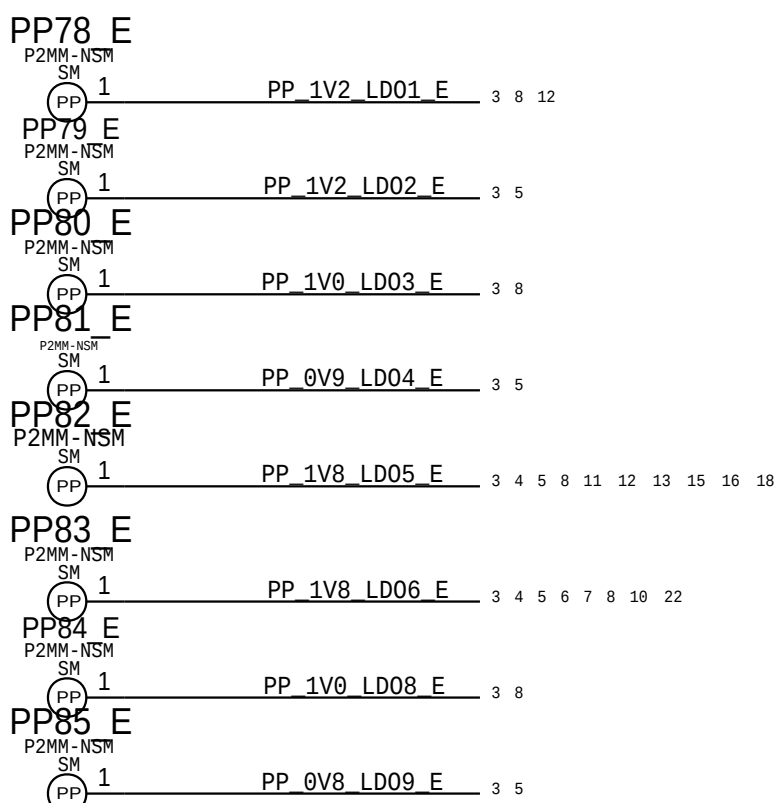
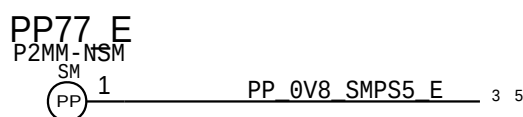
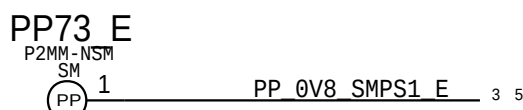
LAYOUT: VIA STUB AT BB PINS



LAYOUT: VIA STUB AT PADS

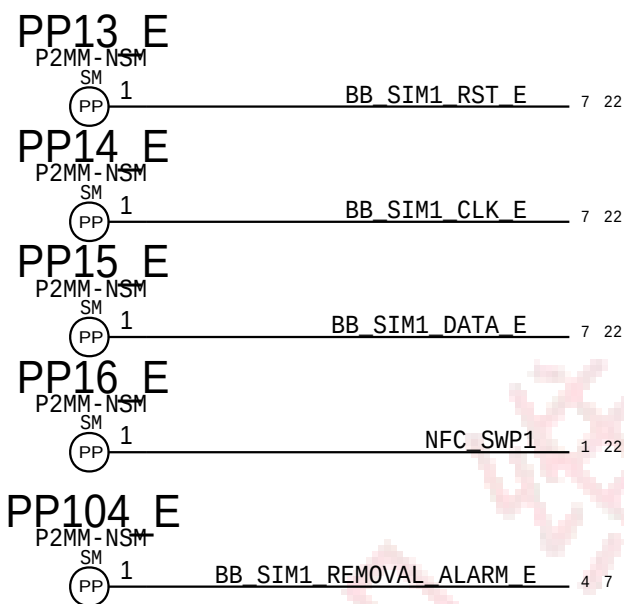


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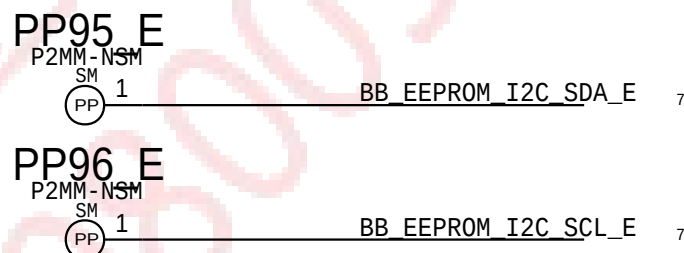


A

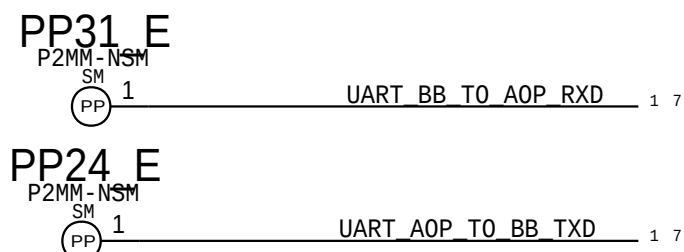
SIM



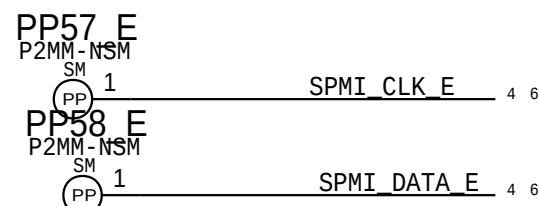
I2C



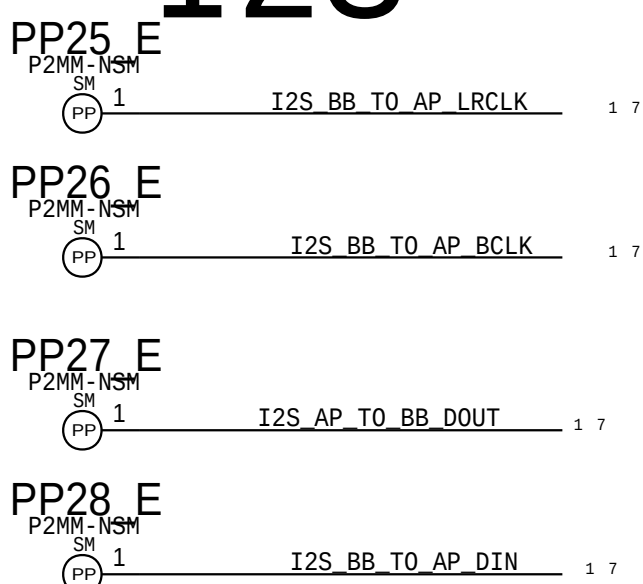
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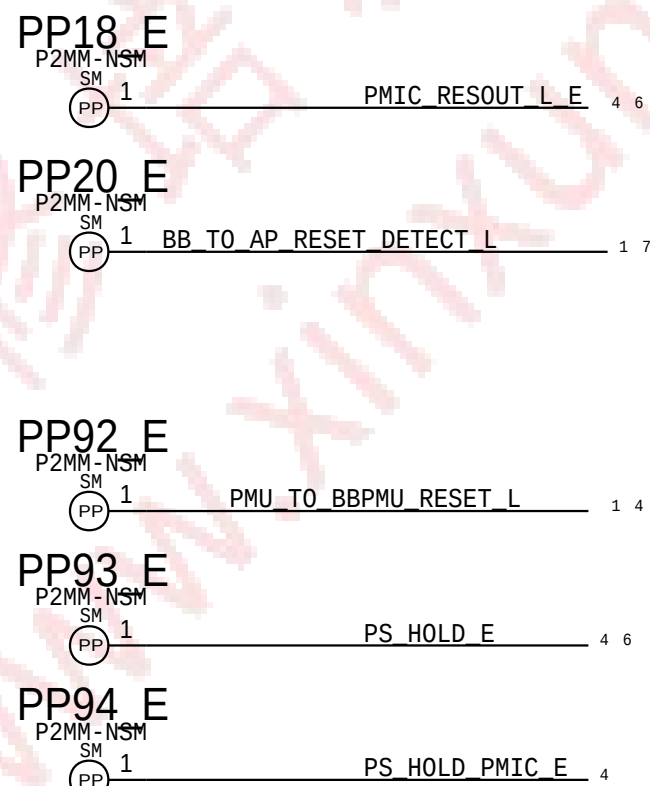
SPMI



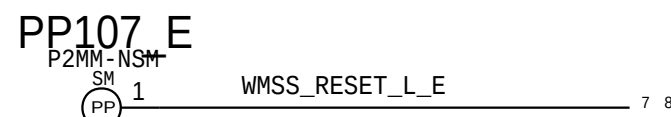
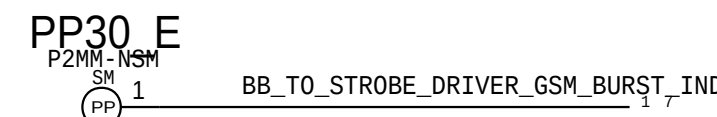
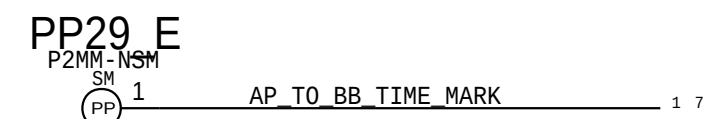
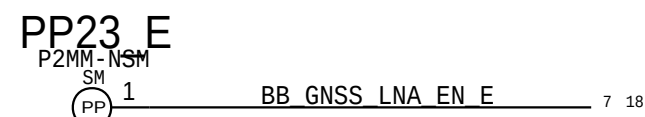
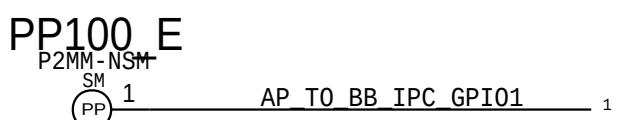
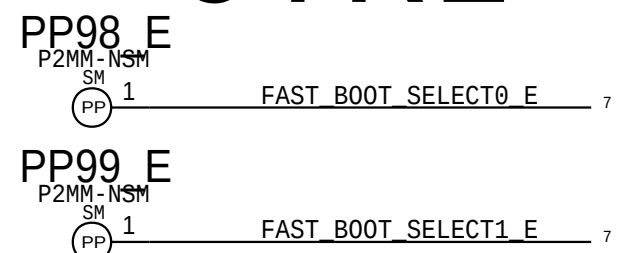
I2S



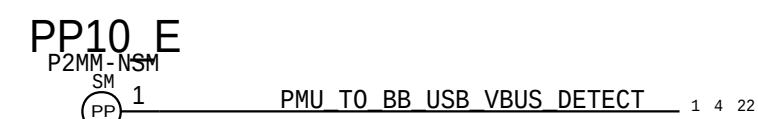
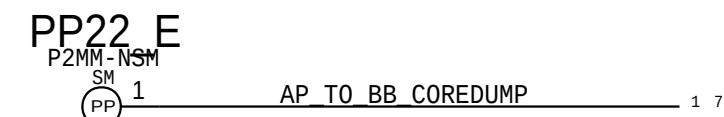
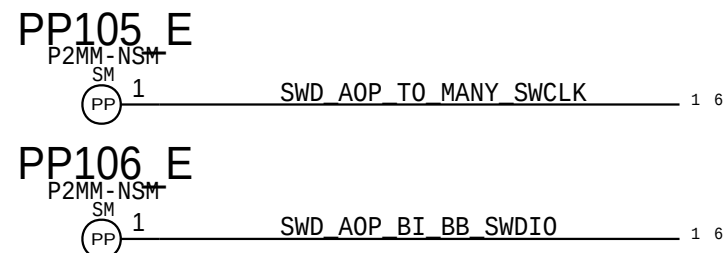
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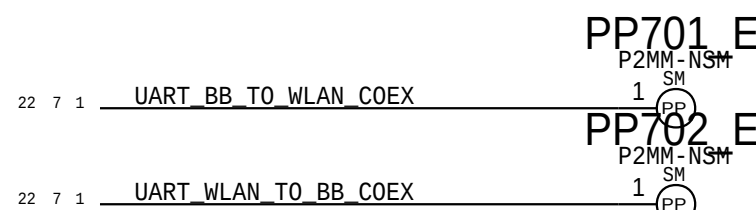
CTRL



DEBUG

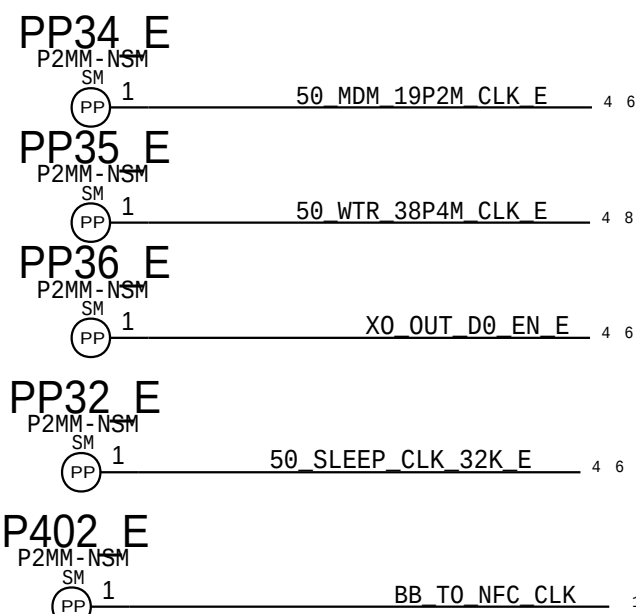


COEX

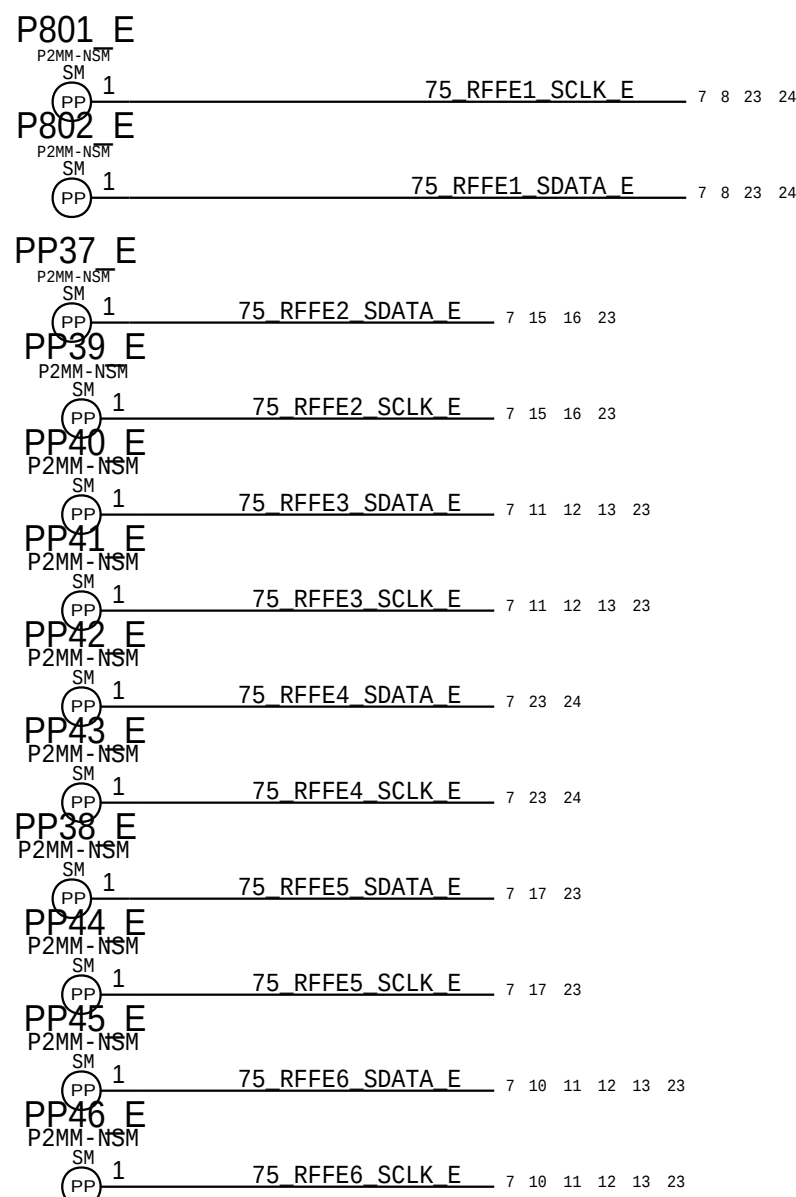


BASEBAND

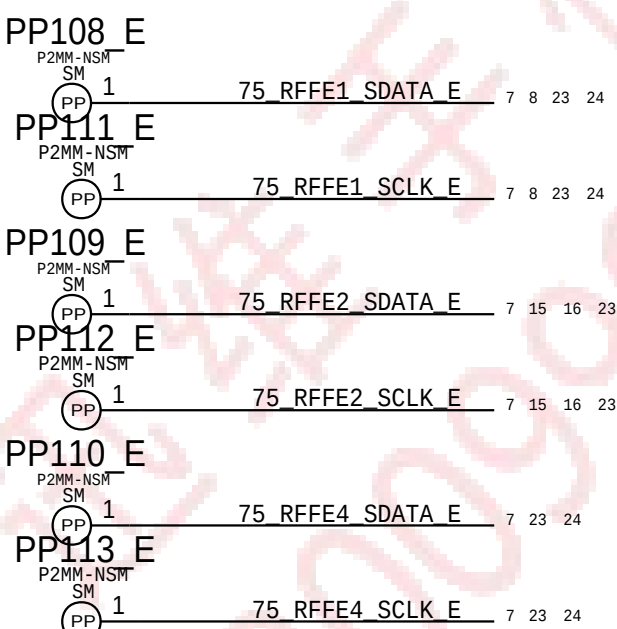
CLK



RFFE

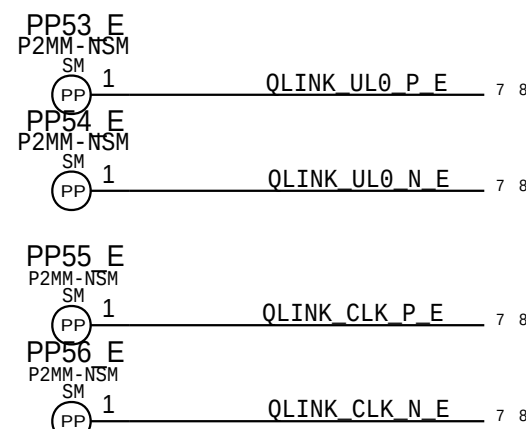


RFFE BB

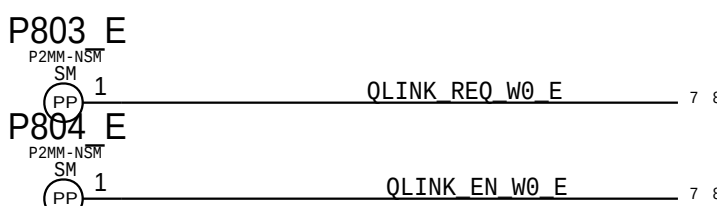
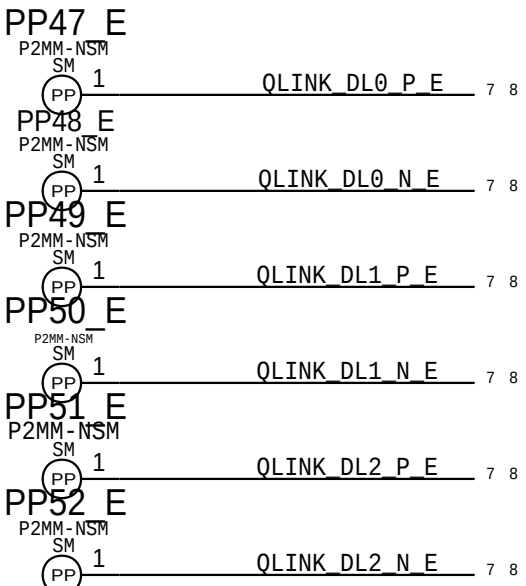


QLINK

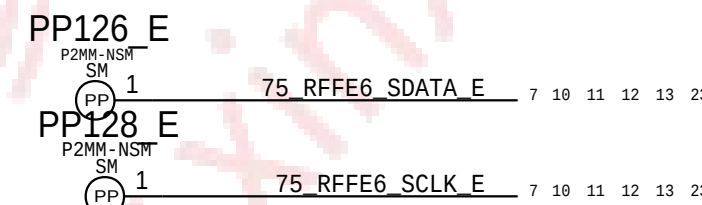
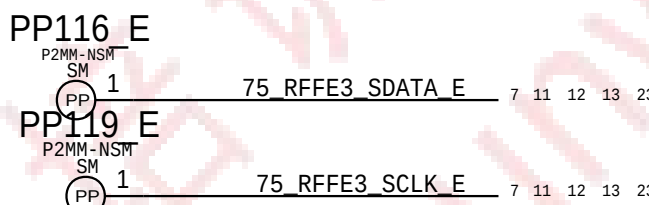
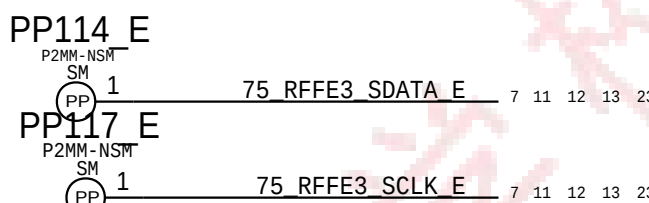
LAYOUT: VIA STUB AT WTR PINS



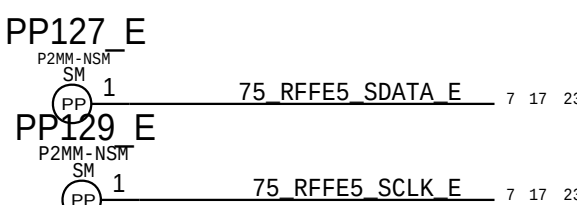
LAYOUT: VIA STUB AT BB PINS

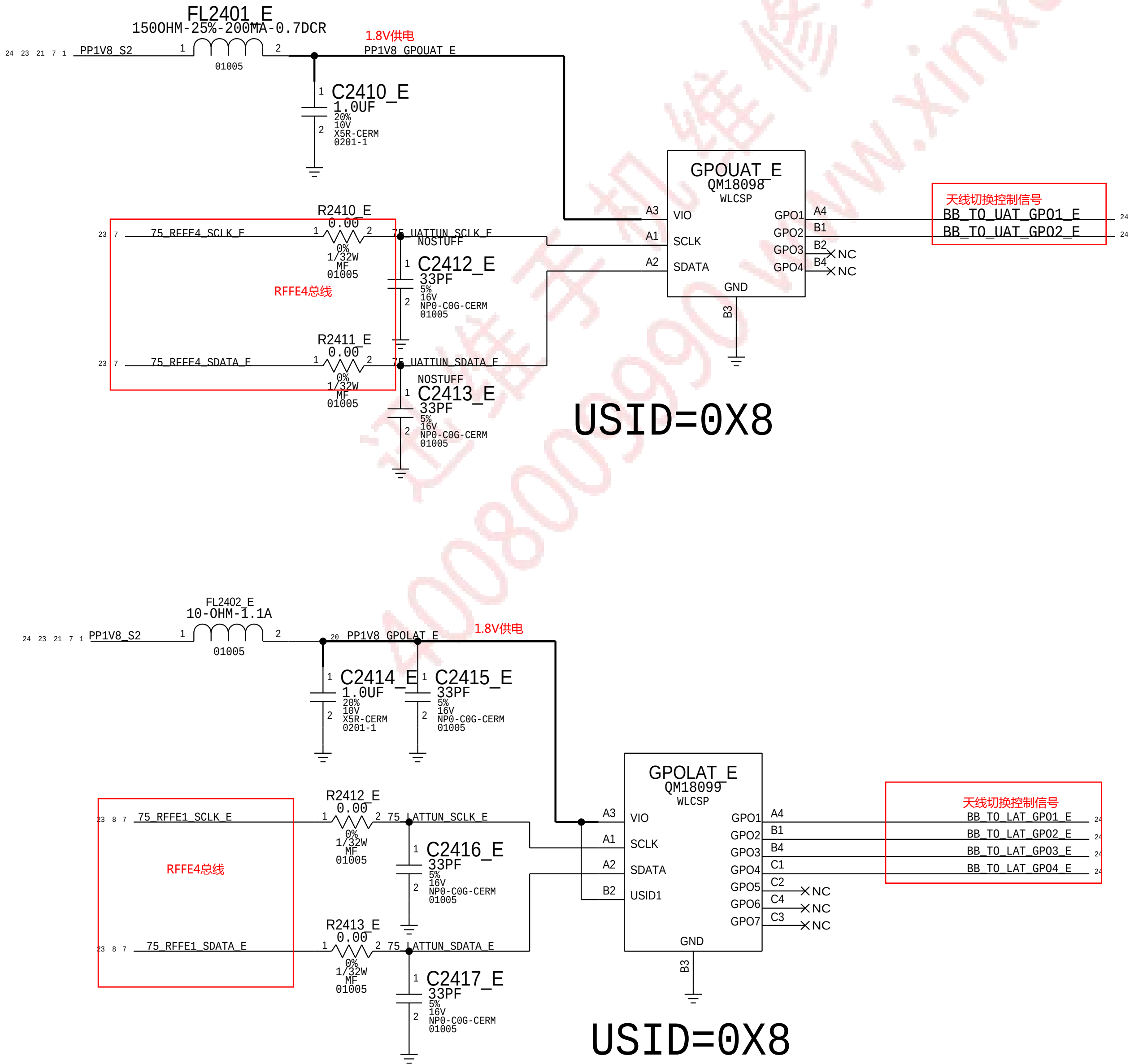
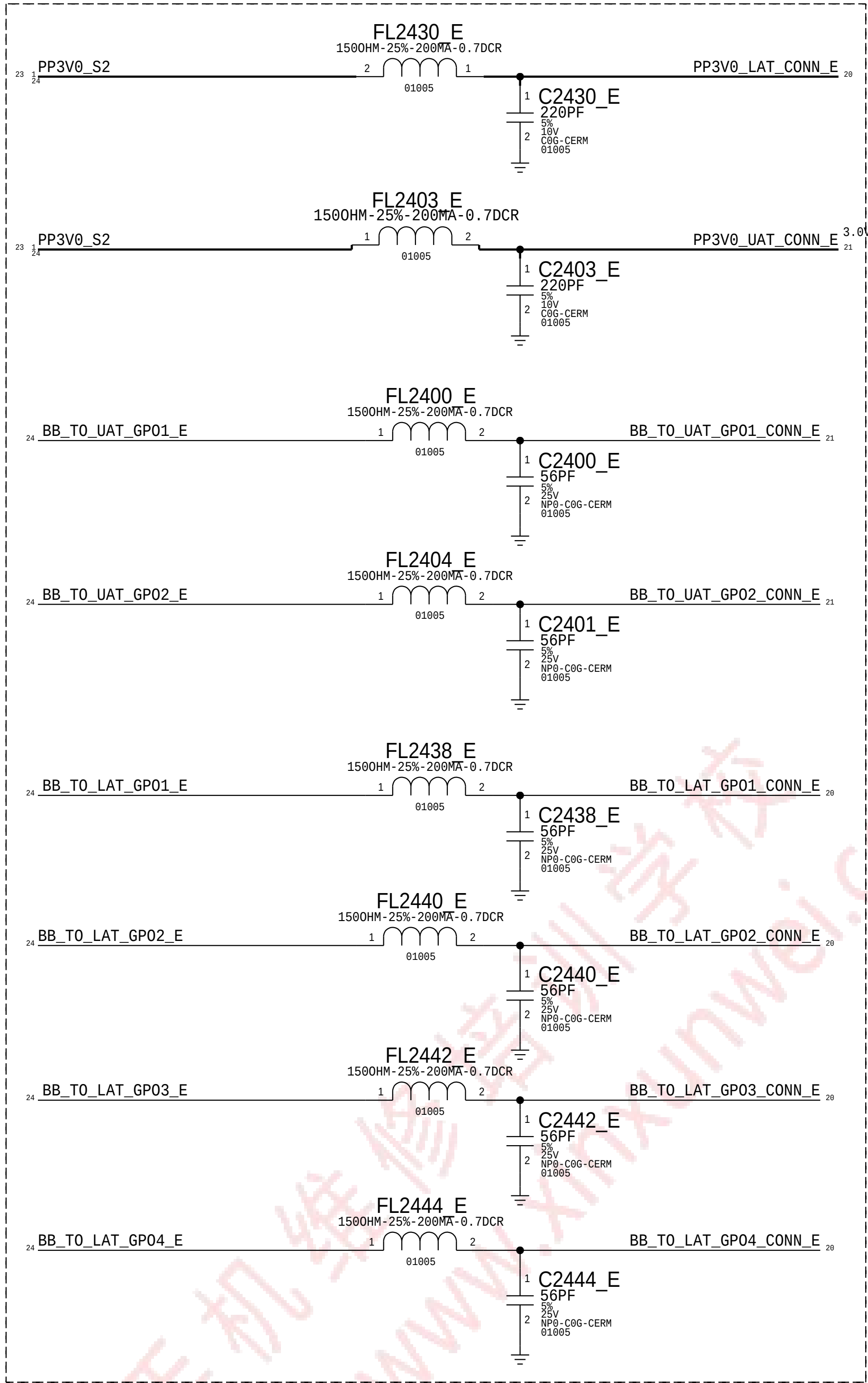


RFFE PA/LNA



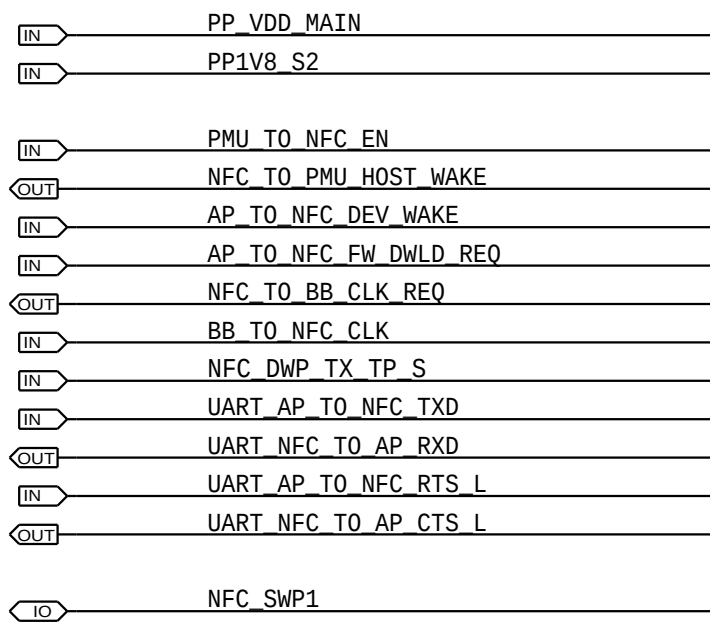
RFFE CPL





D22 NFC_MLB

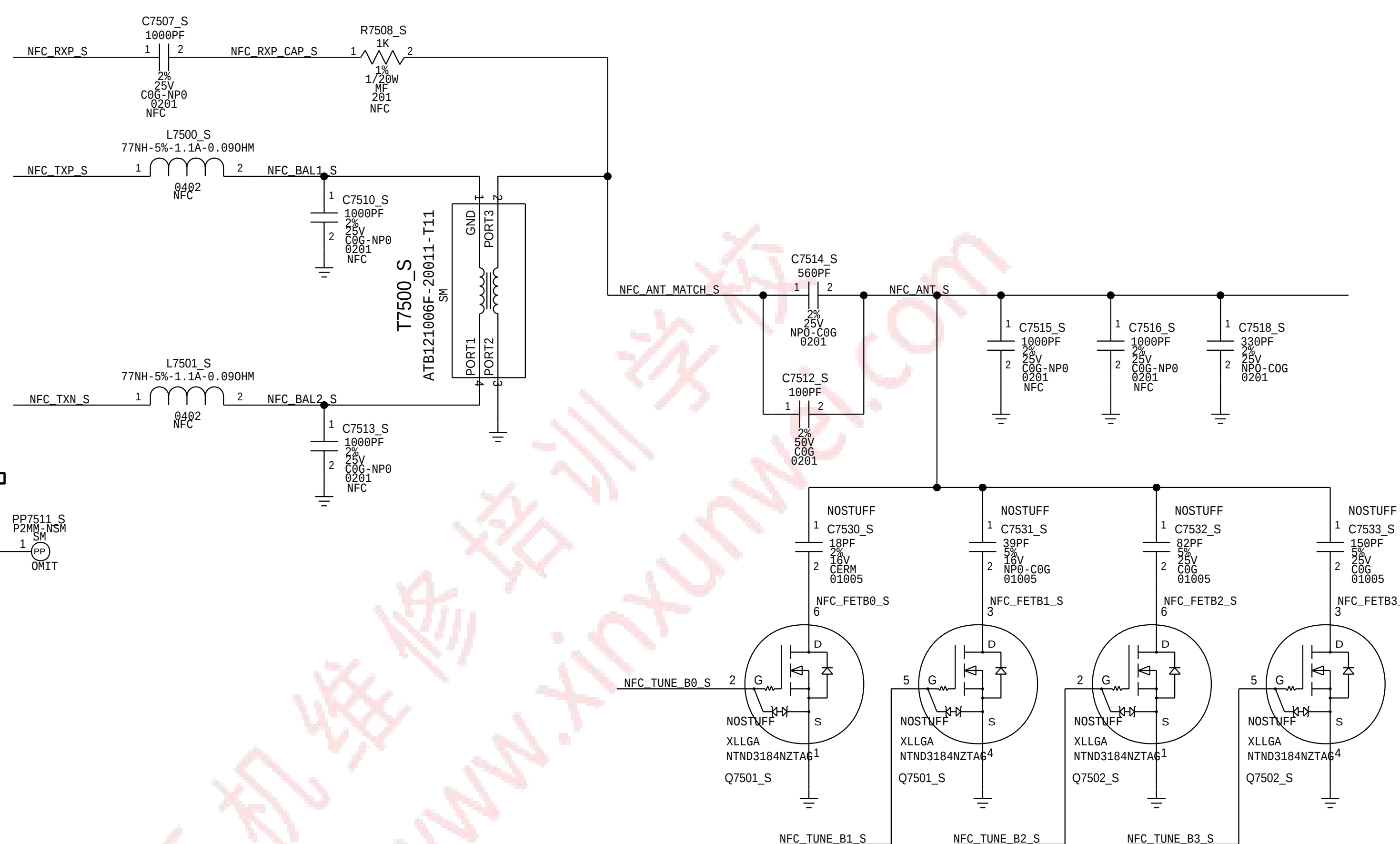
MARCH 22, 2017



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
138S00159	3	CAP, SOFT-TERM, 2.2UF, 6.3V, 0201	C7500_S, C7502_S, C7505_S	CRITICAL	SOFT_CAP
138S0031	3	CAP, TYPICAL, 2.2UF, 6.3V, 0201	C7500_S, C7502_S, C7505_S	CRITICAL	TYPICAL_CAP

5V BOOSTER

NFC CONTROLLER



8		7		6		5		4		3		2		1																
1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%. 2. ALL CAPACITANCE VALUES ARE IN MICROFARADS. 3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ. D22/D221 WIFI_MLB (GUINNESS) APR 03, 2017 <table><tr><td>PDF</td><td>PAGE</td><td>CSA</td><td>PAGE</td><td>CONTENTS</td></tr><tr><td>2</td><td></td><td>76</td><td></td><td>GUINNESS</td></tr><tr><td>3</td><td></td><td>77</td><td></td><td>WIFI FRONT-END</td></tr></table>												PDF	PAGE	CSA	PAGE	CONTENTS	2		76		GUINNESS	3		77		WIFI FRONT-END	REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
												PDF	PAGE	CSA	PAGE	CONTENTS														
2		76		GUINNESS																										
3		77		WIFI FRONT-END																										
												6	0008416016	ENGINEERING RELEASED	2017-04-06															

CPU的PCIE总线

AN_B1_AP_CLKREQ_L CPU到WIFI的复位信号

WIFI的PCIE总线

AP_TO_WLAN_TXD_P IN 1 2

AP_TO_WLAN_TXD_N IN 1 2

WLAN_TO_AP_RXD_P OUT 1 2

WLAN_TO_AP_RXD_N OUT 1 2

AP_TO_WLAN_REFCLK_P IN 1 2

AP_TO_WLAN_REFCLK_N IN 1 2

WLAN_CONTEXT_A CPU到WIFI测试总数据输入

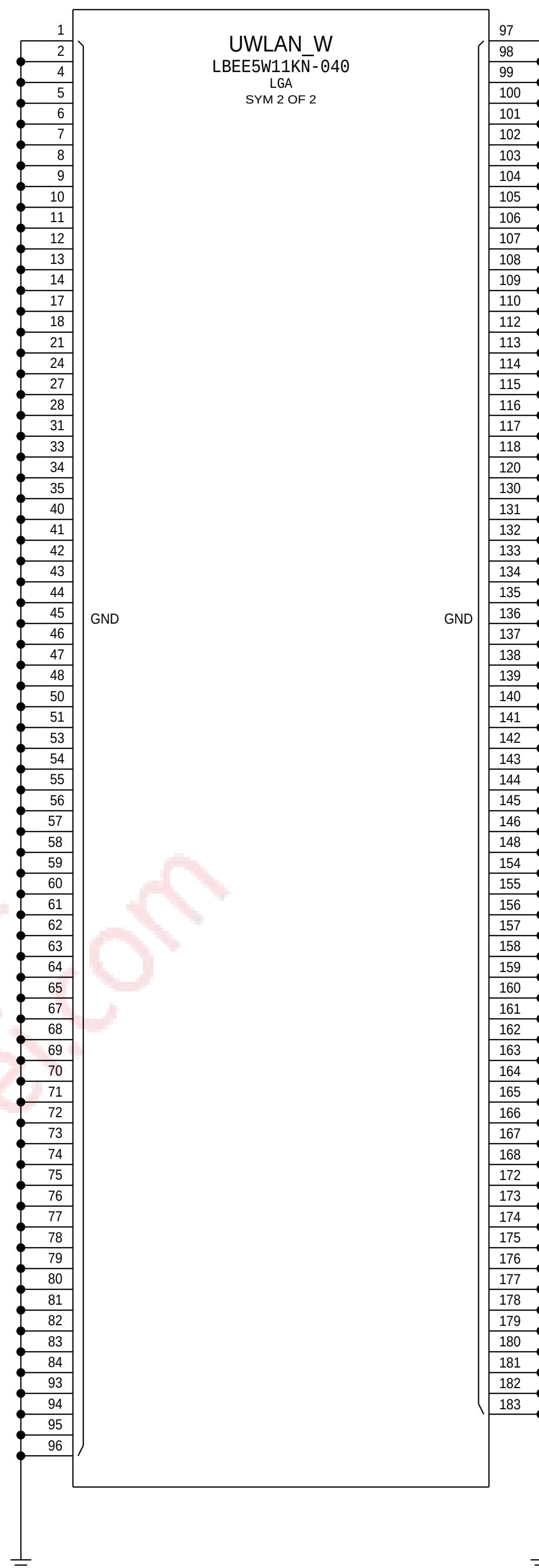
WLAN_CONTEXT_B CPU到WIFI测试总数据输入

1 C7603 W
100PF
2 18V
NP0-CGG
01005
WLAN

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
998-18376	998-18147	BOM_TABLE_ALTS	UWLAN_W	USI ES6.1 WIFI MODULE

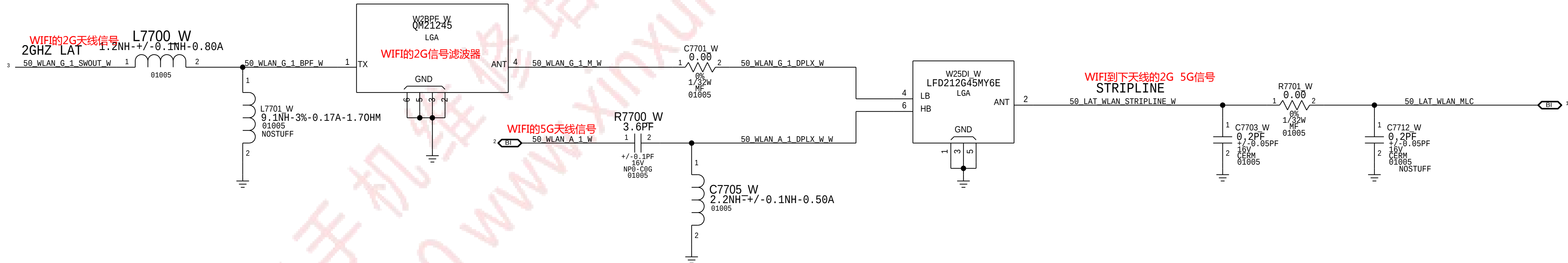
Pin List (Left Column): 59, 60, 61, 62, 63, 64, 65, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 93, 94, 95, 96.

Pin List (Right Column): 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183.

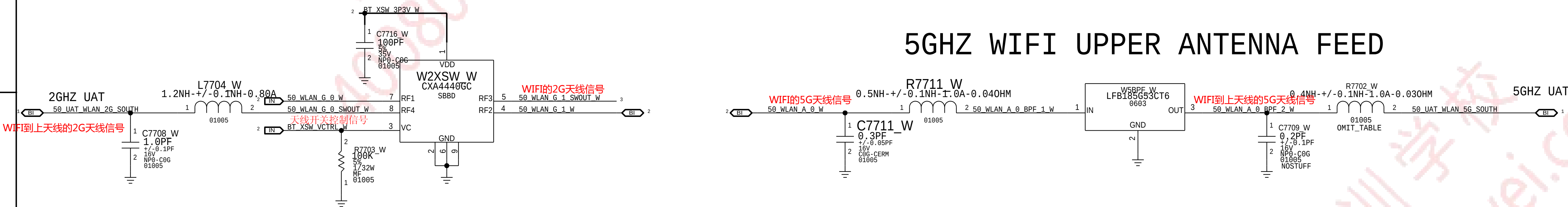


PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
998-10376	998-10147	BOM_TABLE_ALTS	UWLAN_W	USI ES6.1 WIFI MODULE

WIFI LOWER ANTENNA FEED



5GHZ WIFI UPPER ANTENNA FEED



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
11750161	1	5GHZ C0 BPF OUTPUT MATCH	R7702_W	CRITICAL	D221
152S00498	1	5GHZ C0 BPF OUTPUT MATCH	R7702_W	CRITICAL	ROW
152S00498	1	5GHZ C0 BPF OUTPUT MATCH	R7702_W	CRITICAL	JPN